# Semiconductor & semiconductor materials

Based on the electrical conductivity, all the materials in the nature are classified as:

1. Insulator
2. Conductor
3. Semiconductor

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## 1. Insulator

An insulator is the material that are characterized by poor electrical conductivity. for eg: Glass, Quartz, rubber, etc. The resistivity level of an insulator is of the order of $10^{10}$ to $10^{12}$ Ωm. The energy band structure of an insulator is shown below:

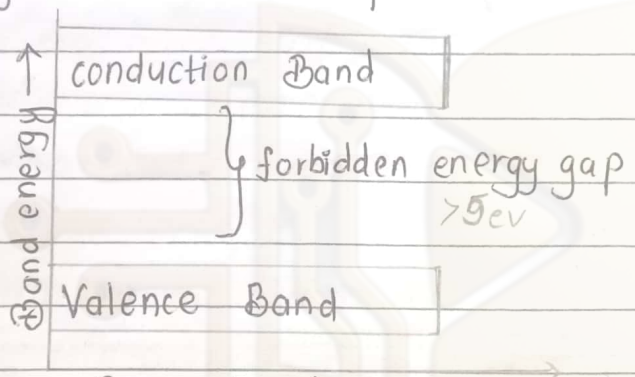


fig: Energy level diagram of insulator

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Band structure of the material defines the band of energy level that an electron can occupy. Valence band is the range of electron energy where the electron remain bonded to the atom and donot contribute to the electric current.

conduction band is the range of electron energies higher than valence band where electrons are free to accelerate under the influence of external voltage source resulting flow of charge. The energy band betⁿ the valence band and the conduction band is called forbidden band gap or forbidden energy gap. It is the energy required by an electron to move from valence band to the conduction band.

Note: $1eV = 1.6 \times 10^{-19} J$

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There is the large forbidden gap ≥ 5ev. Because of the large gap, there are very few electrons in the conduction band and hence conductivity of an insulator is poor.

## 2. Conductor

Conductor or conducting materials are good conductor of electricity and characterized by a large conductivity and small resistivity. For example: Aluminium, silver, etc. The resistivity of conductor is about $10^{-8}$ Ωm.

In conductor, there is no forbidden energy gap betⁿ Valence band and conduction band i.e ~~they~~ they overlap each other. Electrons randomly move through the solid. So, electrons in conductors are called free electrons. Therefore, at room temperature (i.e 25°C) when electric field is applied, an electric current flow through a conductor.

* The resistance of the conductor increases with the increase of temp. (Temp ↑, collision ↑, resistance ↑ ∴ flow of current ↓)

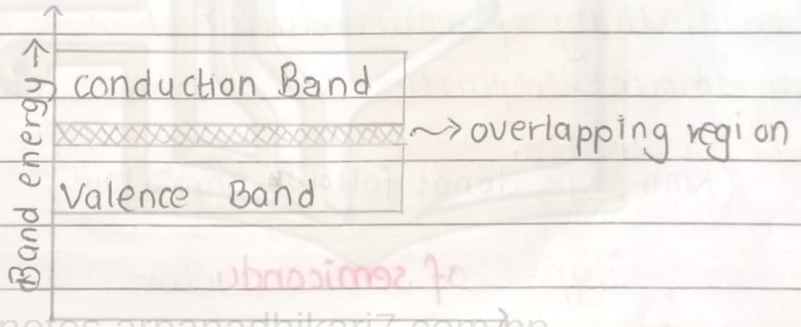


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fig: Energy level diagram of conductor.

## 3. Semiconductor (S|C)

A semiconductor is the material whose conductivity lies between insulator and conductor. The resistivity of the semiconductor is in the range of $10^{5}$ Ωm to $10^{-5}$ Ωm. Silicon, Germanium acts as insulator at absolute temp$^{r}$ (ie 0K or -273°C). However, at normal room temp$^{r}$ some electrons can easily jump to conduction band. Thus, semiconductor are capable of conducting small current even at room temperature.

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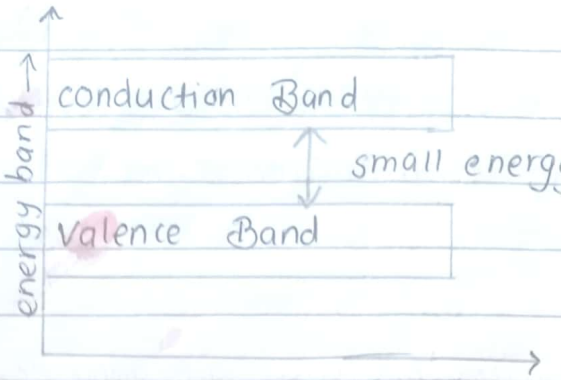


fig: Energy diagram of semiconductor

## Properties of semiconductor

1. The resistivity of semiconductor depends on illumination and decrease in brighter surrounding. (S|C negative temp. coefficient)
2. The resistivity of semiconductor depends on the magnitude of electric field.
3. semiconductor are non-linear elements.
4. The conductivity of semiconductor changes considerably when small amount impurities added to it so that, its conductivity can be controlled.

(Note: S|C donot follow Ohm's law)

## Types of semiconductor

1. Intrinsic semiconductor
2. Extrinsic semiconductor — N-type S|C
   P type S|C

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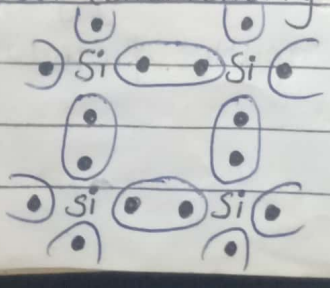
### 1. Intrinsic semiconductor

Intrinsic S|C is extremely pure form of S|C in which there are almost equal no. of electrons and holes in valence band and conduction band respectively. Pure germanium & silicon are widely used intrinsic S|C. poor conductivity

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## Production of Holes & Electrons

Let us consider, a structure of silicon atom. Silicon atomic no. is 14 and has four valence electrons. These 4 electrons are shared by 4 neighbouring atoms in the crystal structure by means of covalent bond.

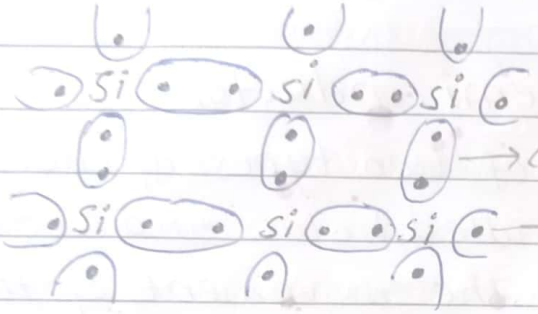


fig: crystal strc. of Si at absolute zero temp$^{r}$ (-273°C or 0K)

Thus, pure semiconductor acts as poor conductor (due to lack of free electrons at a low temp or at absolute 0 temp). At room temperature, some of the covalent bonds break up. As a result electrons are released and move to the conduction band; are called free electrons. The vacancy created in broken bond is called a hole (always +ve charge particle) Since, hole is created after removal of electron from the covalent bond, it is considered as a positive charge.

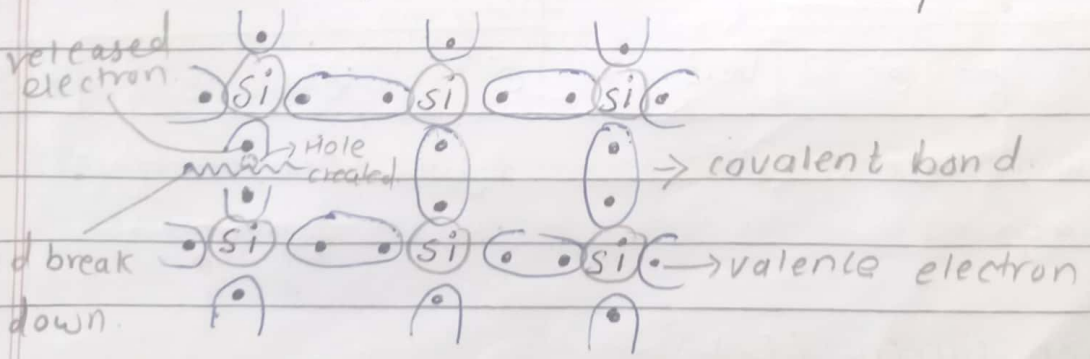


fig: Production of Hole & free electron.

Electrons and holes are created in pairs. Where the free electron approaches the holes, it gets attracted and falls

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into the hole. This process of merging of free electron and hole is called recombination. During this process, both the free electrons and hole disappear. However, an energy is released and causes another covalent bond to break down and generates a new electron pair.

## Conduction in semiconductor

Semiconductor consists of two types of charge carriers Positive charge carriers called holes and -ve charge carriers called electrons. The movement of these charge carrier give rise to the flow of current in semi-conductor. Thus, we can say that s/c possess two types of current.

① Electron current ($I_e$)
② Hole current ($I_h$)

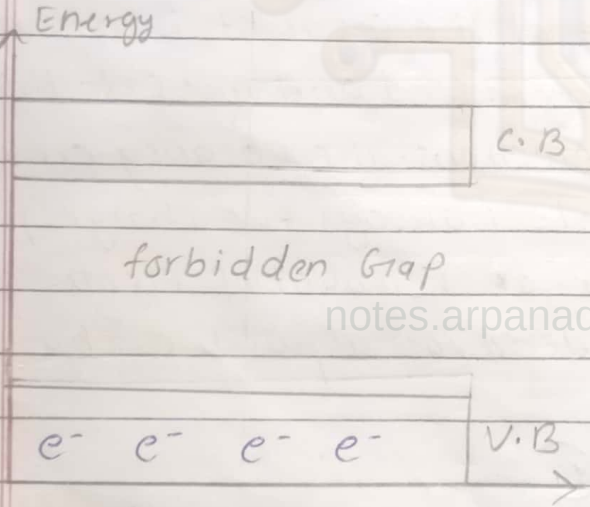


fig: ① when no energy is applied.

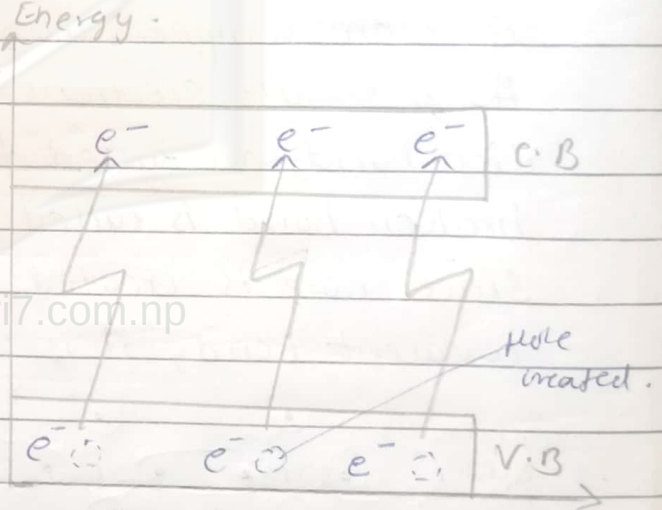


fig (ii) when energy is applied.

### ① Electron current ($I_e$)

When an energy is applied to a a semiconductor, the valence electron at valence band can gain sufficient energy and jump to the conduction band and become free electrons. The movement of these free electron, give

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rise to a current called electron current ($I_e$)

(2) Hole current ($I_h$)

When a valence electron gain sufficient energy and jump to conduction band, they leave a hole at a valence a hole at a valence band. Further valence electron can move to this hole. Thus, relative motion of valence electron at valence band through holes give rise to hole current ($I_h$). Thus, total current on the s/c, is the sum of electron current & hole current. i.e

$$I_{total(SC)} = I_e + I_h$$

## 2. Extrinsic semiconductor

An intrinsic s/c has a very limited application/significance as it conduct very small amount of current at room temperature. However, the electrical conductivity of intrinsic s/c can be increased significantly by adding small amount of impurities. This process of adding an impurity to a pure semiconductor is called dopping and the resulted s/c formed is called extrinsic semiconductor.

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There are two types of s/c on the basis of type of impurities added. They are:

1. N-type s/c
2. P-type s/c

### (1) N-type semiconductor.

When a small amount of pentavalent impurity is added such as arsenic (As), Antimony (Sb), Bismuth (Bi), Phosphorus (P) to a pure semiconductor. It results a crystal like structure called N-type s/c.

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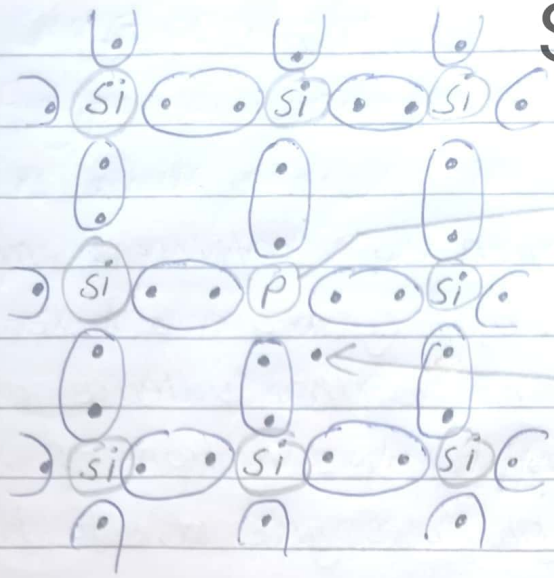


fig: Crystal structure of N-type S/C.

For each impurity added, one free electron is genera~~one~~ which then moves to conduction band. As the pentav electron donates electron, it is also called done impurit or donar agent.

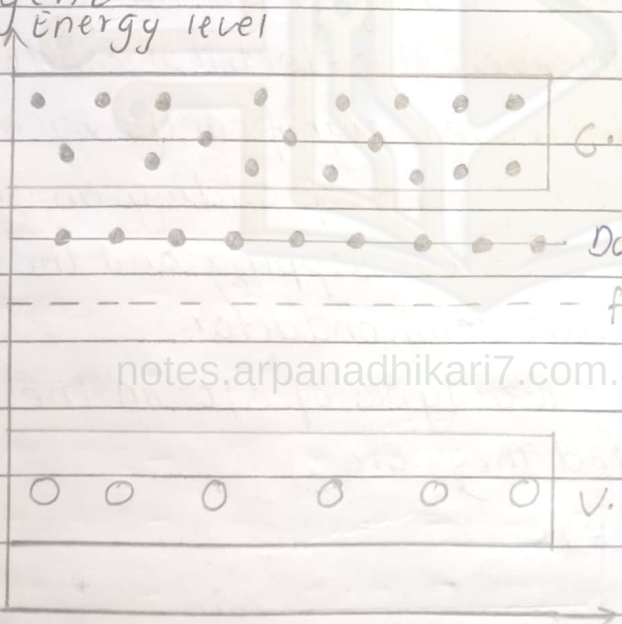


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fig: Energy level diagram of N-type S/C.

Note:

Fermi level: (1) The highest energy level that an electron can occupy at absolute zero temperature.

(2) Fermi energy level lies in betⁿ V.B & C.B.

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## 2. P-type semiconductor

When a small amount of trivalent impurity such as aluminium, Gallium, indium, Boron is added to a pure semiconductor, the P-type s/c is formed.

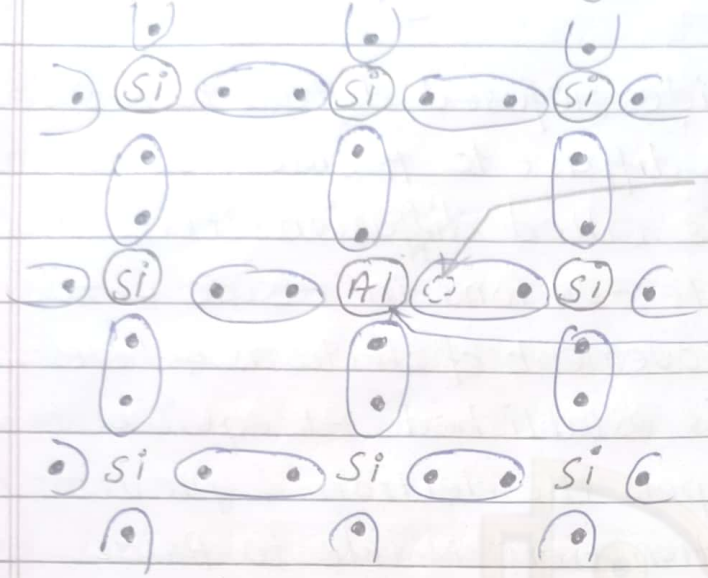


fig: Crystal struc. of P-type s/c.

The free valence electrons of the trivalent impurity forms 3 covalent bonds with the neighbouring atoms and a vacancy exists in the fourth bond giving rise to the hole which can accept an atom. So, trivalent impurities are also called accepter impurities or accepter agent.

In P-type s/c, holes are majority carries and electrons are minority carrier.

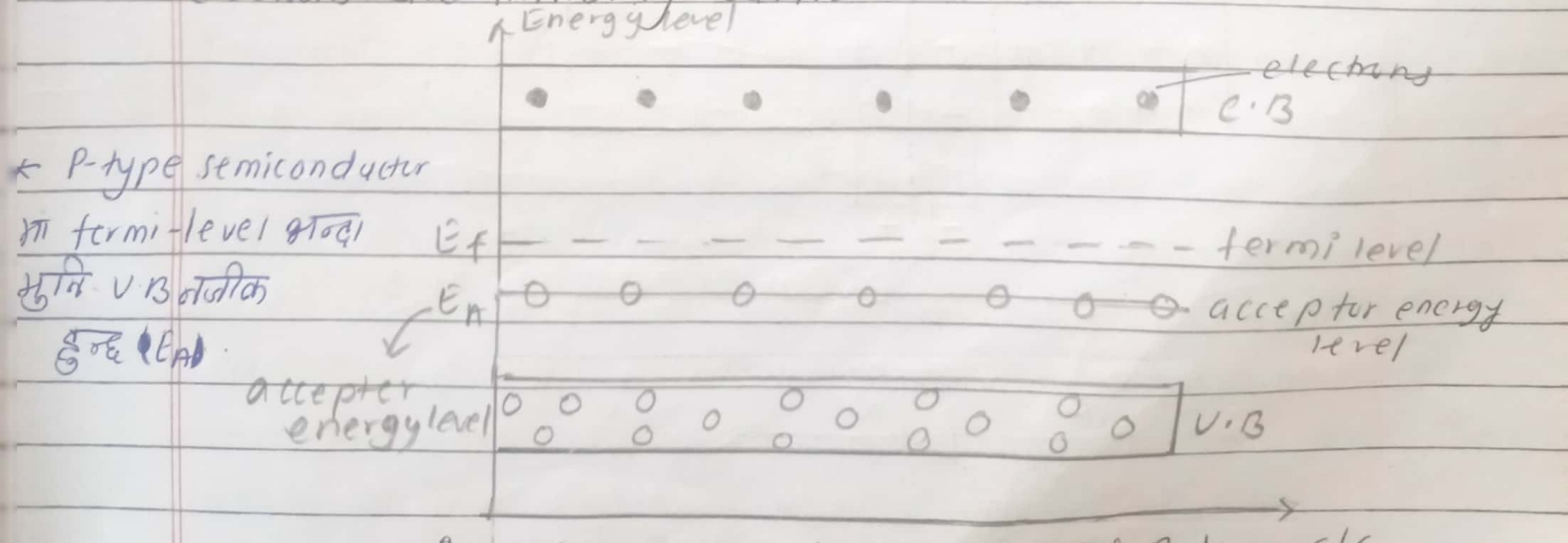


fig: Energy level diagram of P-type s/c

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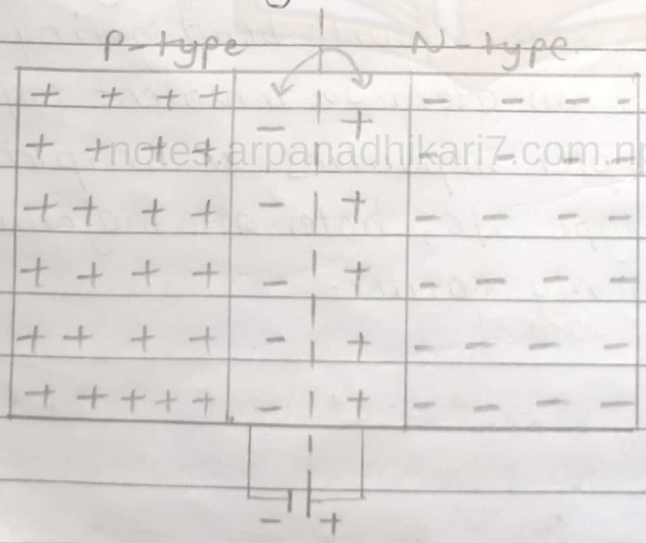
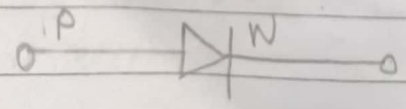
# PN Junction diode

P-type semiconductor have holes as majority charge carriers while N-type have electrons. When a layer of P type is placed next to the N-type s/c, a PN Junction is formed.

Holes from P-side ~~refused~~ diffuse to N-side & electrons from N-side diffuse to P-side due to density gradient. This process is called diffusion. Free electrons crossing junction create -ve ions in P-side & +ve ion in N-side. The movement of electrons & holes occur until an equilibrium is established. At equilibrium, -ve ions at P-side repels the electron & prevents it from crossing the junction from N-side to P-side.

Similarly, +ve ions prevents hole to cross the junction from P-side to N-side. Thus, diffusion of ~~carrier~~ charge carriers creates a barrier potential ~~for~~ or barrier voltage across a junction.




Symbol: PN Junction diode

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# Biasing of PN Junction Diode

The process of connecting the diode terminals (ie P and N) with the source terminals (voltage source) is known as biasing of diode. A PN junction diode can be biased in two ways;

1. Forward biasing &
2. Reverse biasing

## 1. Forward biasing

If the +ve terminal of a battery is connected to P-type material & -ve terminal of a battery is connected to N-type material then this process is called forward biasing.

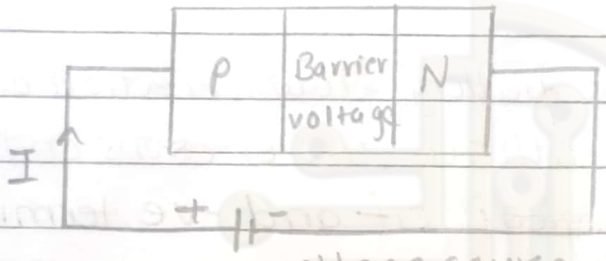


fig: forward biasing

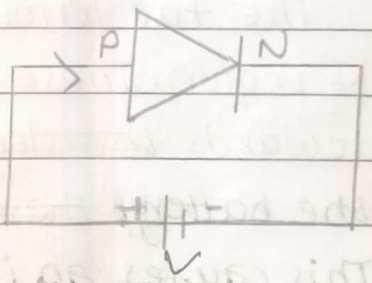


fig: symbolic ckt of forward biasing

The -ve terminal of battery flows electron on N-region. These electrons repels the electrons on N-region towards depletion region. If the electrons have sufficient energy to cross the barrier potential, they will transfer to P-region. For this, the battery should have potential difference more than barrier potential. In P-region, these electrons recombines with both. But due to attraction of +ve terminal of the battery, these electrons move towards +ve terminal through holes. Finally, electrons reached to the cell. The cell again pumps electrons to N-region and the process continues. So, we can conclude that current flows through diode during forward diode condition.

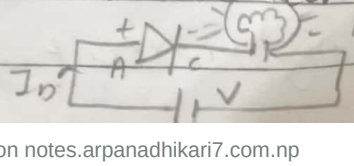


fig: Practical implementation

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## 1. Forward biasing

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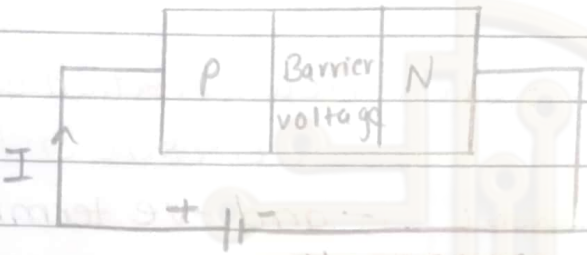


fig: forward biasing

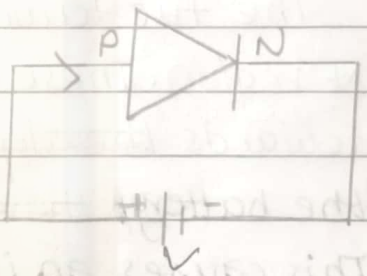


fig: symbolic ckt of forward biasing

The -ve terminal of battery flows electron on N-region. These electrons repels the electrons on N-region towards depletion region. If the electrons have sufficient energy to cross the barrier potential, they will transfer to P-region. For this, the battery should have potential difference more than barrier potential. In P-region, these electrons recombines with both. But due to attraction of +ve terminal of the battery, these electrons move towards +ve terminal through holes. Finally, electrons reached to the cell. The cell again pumps electrons to N-region and the process continues. So, we can conclude that current flows through diode during forward diode condition.

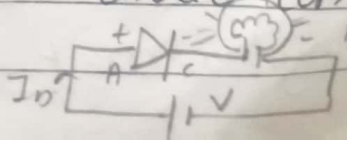


fig: Practical implementation ckt for [illegible]

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## Reverse biasing.

If the -ve terminal of the battery is connected to P type material and +ve terminal is connected to N-type material then this condition is called Reverse biasing.

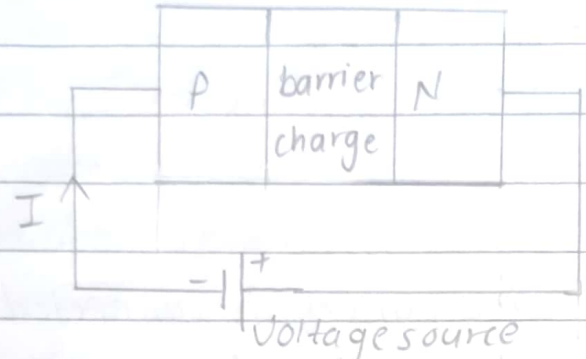


fig: Reverse biasing of diode

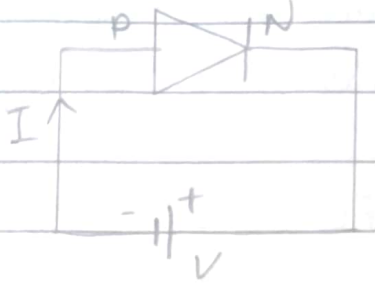


fig: Symbolic ckt of Reverse biasing

The +ve terminal of the battery flows electron on the N-region. These electrons attracts the electrons on N-region towards ~~the depletion region~~ itself and -ve terminal of the battery ~~the electrons~~ flows electrons on the P-region. This causes an increase in the thickness of the depletion layer and allows very less current to flow through the PN-junction.

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In reverse biasing, the +ve terminal of a source attracts the electrons from N-type s/c whereas the -ve terminal of a source attracts the holes from P-type s/c, away from the junction. Due to the movement of electrons and holes the depletion region widens & diode doesn't conduct current. When a diode is reverse biased, the majority charge carriers give rise to small current, which is negligible.

When reverse bias voltage is greater than a breakdown voltage, the diode breakdown and acts as a short circuit, it conducts high current

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# V-I Characteristics of PN Junction Diode (VI)

## 1. Ideal diode

The diode that acts like perfect conductor when forward biased & perfect insulator when reverse biased is called an ideal diode.

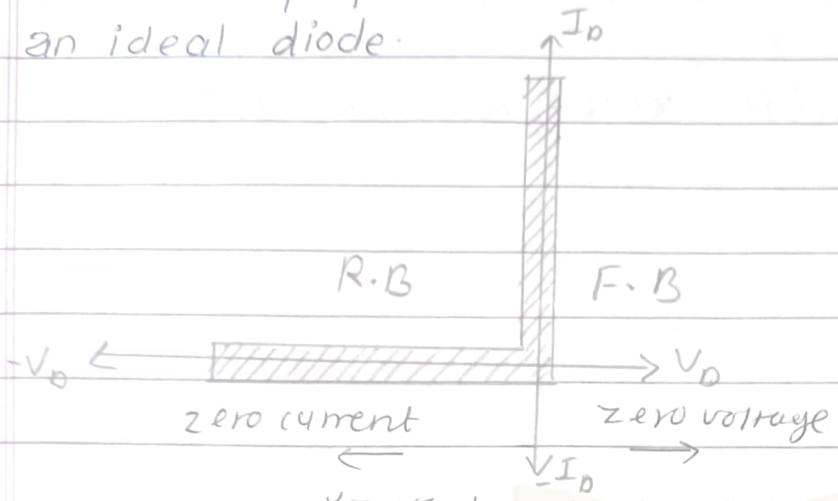


Fig: VI characteristics curve of ideal diode.

| Forward biasing | Reverse biasing |
|---|---|
| → zero voltage | → maximum voltage |
| → zero resistance | → infinite resistance |
| → maximum current | → zero current |
| → closed switch (—•—) | → open switch (—•  •—) |

## 2. Real / Practical Diode

If '$V_D$' is the voltage applied across the diode and '$I_D$' be the current flowing through it, then the relationship between the '$V_D$' and '$I_D$' is given by,

$$I_D = I_S\left(e^{\frac{V_D}{\eta V_T}} - 1\right) \quad \text{—— ①}$$

where,

$I_D$ = current through diode
$V_D$ = voltage across diode
$I_S$ = Reverse saturation current
$\eta$ = Emission coefficient ($1 \leq \eta \leq 2$) → numerical value (Either 1 or 2)
$V_T$ = Thermal voltage (0.026 V) at normal room temp$^{r}$

$V_T$ is given by,

$$V_T = KT$$

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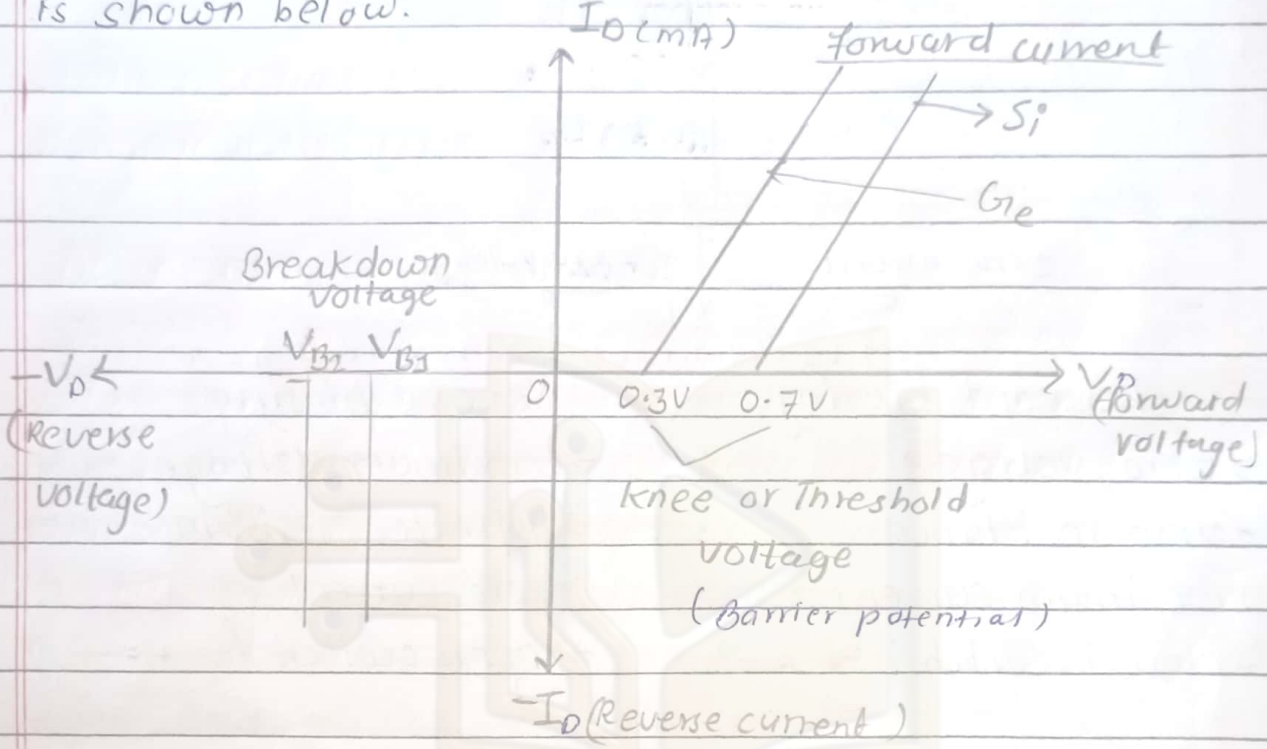
where,

K = Boltzmann's constant ($K = 1.38 \times 10^{-23} J/k$)

T = absolute room temp$^r$

q = electronic charge ($q = 1.6 \times 10^{-19} C$)

The VI characteristics curve of Real/practical diode is shown below:




## Operation

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When a diode is forward biased, small amount of current flows through it until the supply voltage is greater than the barrier potential.

When the supply voltage ~~exists~~ exceeds the barrier potential, the current rises gradually as shown in the figure.

Also when the diode is reverse biased, negligible amount of current flows through it. But if the supply voltage exceeds the breakdown voltage ($V_B$), the diode is broke down (destroyed) and it acts as a short circuit due to which the current

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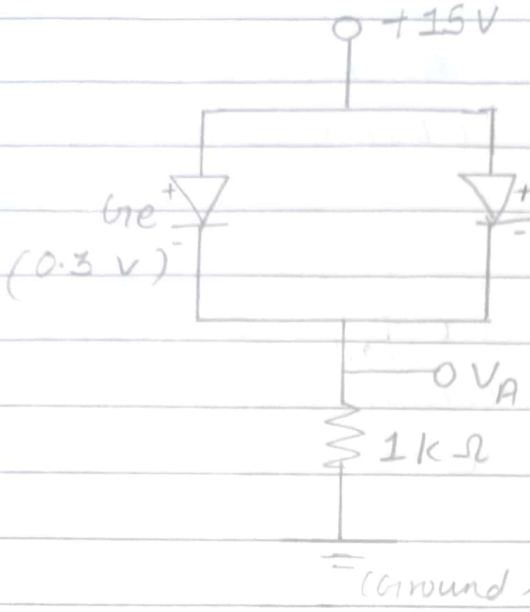
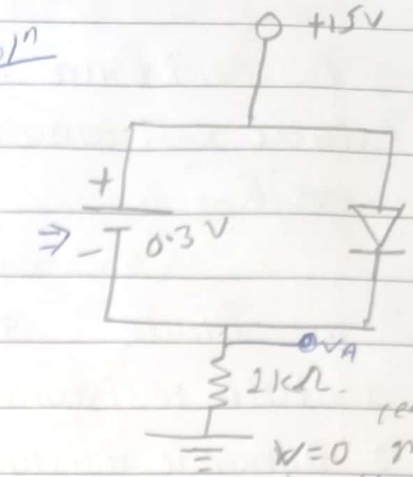
increases sharply as shown in fig.

Qn 1.




Soln




Note: ground voltage = 0V

* (+ve) to (-ve) = minus
* (-ve) to (+ve) = plus

* Ge activate

Find the value of $V_A$.

We know,

$$V_A = IR$$
$$= I \times 1K\Omega$$
$$= \frac{V}{R} \times 1K\Omega$$
$$= \frac{V}{1K\Omega} \times 1K\Omega$$
$$= V$$

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Then,

$$V = 15 - 0.3 - 0$$
$$= 14.7$$

# Terminal characteristics of Junction Diode.

from shockley's diode eqn we have.

$$I_D = I_S \left( e^{\frac{V_D}{\eta V_T}} - 1 \right) \quad \text{—①}$$

where; $I_D$ = diode current
$I_S$ = leakage saturation current
$V_D$ = voltage drop across diode.
$V_T$ = Thermal voltage.

where, $V_T$ is given by

$$V_T = \frac{kT}{q}$$

k = Boltzmann's constant [$k = 1.3 \times 10^{-23} J/K$]
T = absolute room Temp.
q = electronic charge
($q = e = 1.67 \times 10^{-19} C$)

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# Diode Resistance

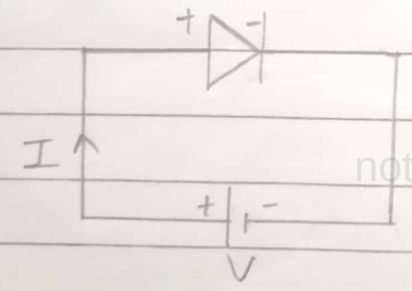
- DC Resistance (Static Resistance) → 'R_DC'
- AC Resistance. (Dynamic resistance) → 'r'

Diode shows two types of resistive effect: static resistance (DC resistance) & Dynamic resistance (AC resistance)

## DC Resistance.

If 'V' be the DC voltage applied to a diode and 'I' be the current flowing through it then the ratio of 'V' & 'I' is called static resistance of a diode. It is denoted by $R_{DC}$ and given by;

$$R_{DC} = \frac{V}{I}$$




Its value depends on operating point on V-I characteristics curve of diode.

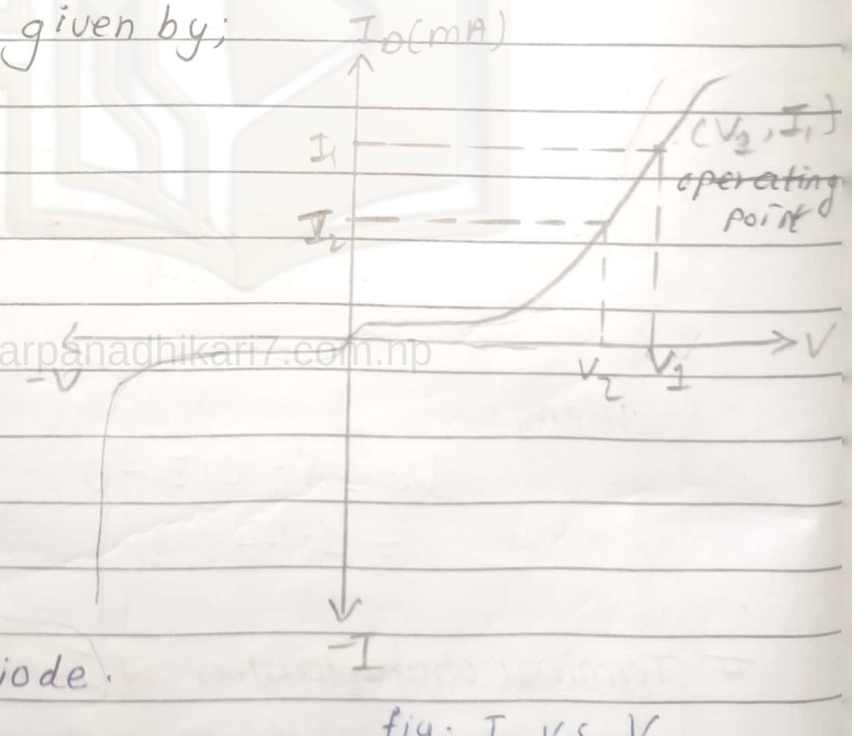



fig: I vs V

The value depends on the operating point on VI characteristic curve of diode.

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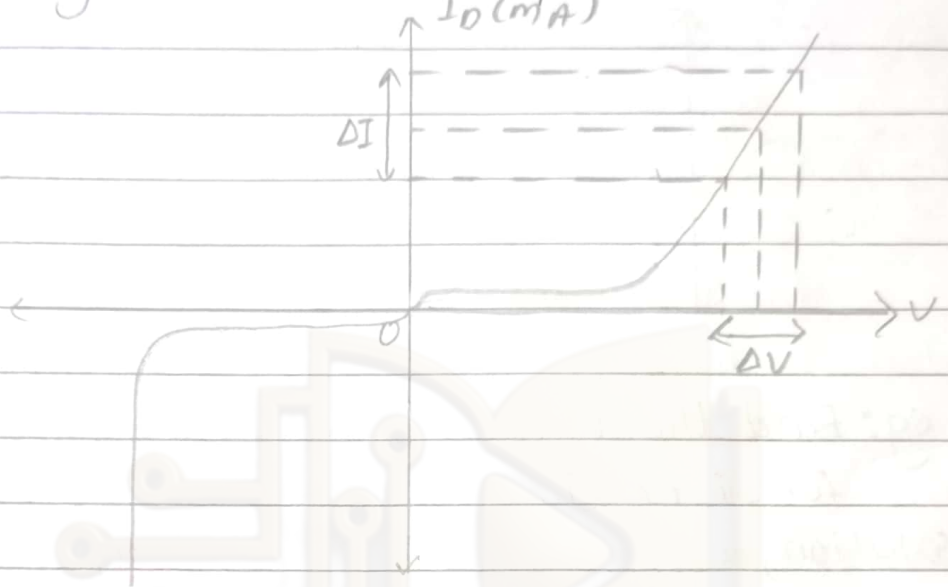
## 2 AC resistance

When the voltage applied across the diode is changed by $\Delta V$ then the current flowing through it will also be changed by $\Delta I$.

The ratio of change in voltage to change in current is called dynamic resistance of a diode (r).




$$\text{Dynamic Resistance } (r) = \frac{\Delta V}{\Delta I} = \frac{dV}{dI_D} \quad \text{---(i)}$$

We know that,

$$I_D = I_S\left(e^{\frac{V_D}{\eta V_T}} - 1\right) \approx I_S\left(e^{\frac{V_D}{\eta V_T}}\right) \quad \text{---(ii)}$$

Now,

$$\frac{dI_D}{dV_D} = \frac{d}{dV_D}\left[I_S\left(e^{\frac{V_D}{\eta V_T}}\right)\right]$$

$$\text{or, } \frac{dI_D}{dV_D} = \frac{I_S}{\eta V_T} e^{\frac{V_D}{\eta V_T}}$$

$$\text{or, } \frac{dI_D}{dV_D} = \frac{I_D}{\eta V_T} \quad [\because \text{from eq}^n \text{(ii)}]$$

$$\text{or, } \frac{1}{r} = \frac{I_D}{\eta V_T}$$

$$\text{or, } \boxed{r = \frac{\eta V_T}{I_D}}$$

Hence, it is clearly seen that the dynamic resistance 'r' varies inversely with current across diode.

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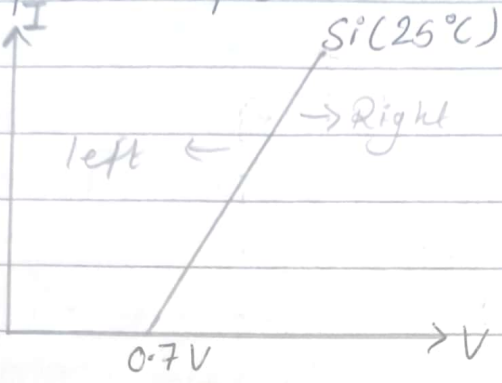
## Temperature dependence of V-I characteristics.

1. Forward Bias

For Si diode, the characteristics in forward bias shifted towards left side by the increase in temperature at the rate of 2.5 mV/°C.




For eg: Find the threshold voltage at 75°C and at -100°C for Si PN Junction Diode

Solution;

(i) At 75°C

New threshold voltage ($V_{Th}$) = $V_{Th(old)}$ - 2.5mV × (75-25°C

= $0.7 - 2.5 \times 10^{-3} \times 50$

= 0.575 V

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(ii) For -100°C

New threshold voltage ($V_{Th}$) = $V_{Th(old)}$ + 2.5mV × (-100-2

= $0.7 + 2.5 \times 10^{-3} \times 125$°C

= 0.7 + 0.3125

= 1.0125 V

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$\eta$ for germanium diode = 1

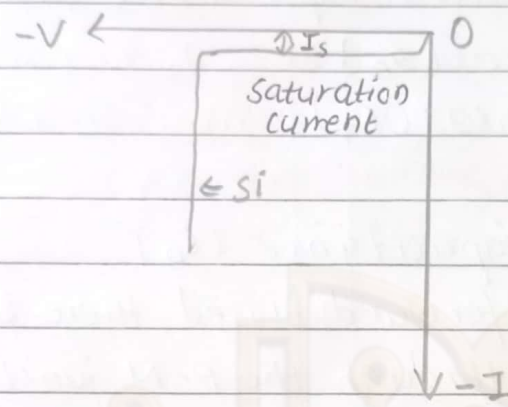
$\eta$ " Silicon " = 2

## 2. Reverse Bias

When we increase the temperature in reverse bias cond$^n$, the saturation current also increases. The reverse saturation current will double in every 10° rise in temperature.

It is given by,

$$I_{S(new)} = I_{S(old)} 2^{\frac{\Delta T}{10}} \quad \text{——} (1)$$




where,

$\Delta T$ = change in temp$^r$

For eg: Find the reverse saturation ~~current~~ current at 125°C where room temp$^r$ is 25°C the reverse saturation current at room temp$^r$ is 50nA.

Soln

Here,

$T_1 = 25°C$

$T_2 = 125°C$

$I_{S(old)} = 50nA$

$I_{S(new)} = ?$

we have,

$$I_{S(new)} = I_{S(old)} 2^{\frac{\Delta T}{10}}$$

$$= 50nA \times 2^{\left(\frac{125-25}{10}\right)}$$

$$= 50 \times 10^{-9} \times 2^{10}$$

$$= 50 \times 10^{-9} \times 1024$$

$$= 51.2\,\mu A.$$

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# Junction Capacitance

The depletion layer across the junction diode behaves as an insulating material or dielectric constant of the capacitor. Similarly, the P and N side acts as parallel plates of the capacitor. So, the junction exhibits the property of capacitance which is called junction capacitance. There are two types of junction capacitance. They are:

a. Diffusion capacitance ($C_D$)
b. Transition capacitance ($C_T$)

Imp
short note

## Diffusion capacitance ($C_D$)

When a diode is forward biased, there is a movement of electron and holes through the P-N junction. This move of charge carriers give to a capacitance effect in a dio which is called diffusion capacitance.

We know,

$$C_D = \frac{dQ}{dV} \quad \begin{matrix} \rightarrow \text{Total} \\ \rightarrow \text{P.d} \end{matrix}$$

$$\Rightarrow C_D = \frac{d}{dV_D}[ixt] \quad [\because q = ixt]$$

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$$\Rightarrow C_D = \tau \frac{d}{dV_D}[I_D]$$ where $\tau$ is the mean life tim of charge.

$$\Rightarrow C_D = \tau \frac{d}{dV_D}\left[I_S e^{\frac{V_D}{\eta V_T}}\right]$$

$$\Rightarrow C_D = \frac{\tau \times I_S}{\eta V_T}\left[e^{\frac{V_D}{\eta V_T}}\right]$$

$$\Rightarrow C_D = \frac{\tau \times I_D}{\eta V_T}$$

$$\therefore \boxed{C_D \propto I_D}$$

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## Transition capacitance ($C_T$)

When a diode is reverse biased, the depletion region increases. The depletion region consists of a layer of +ve and -ve charge ions. These ions behave as +ve and -ve plate of a parallel plate capacitor due to which capacitive effect is produced in a diode known as transition capacitance.

Thus, transition is the capacitive effect shown by the diode when it is reverse biased.

Mathematically,

$$C_T = \frac{\varepsilon A}{W} \quad \text{---(1)}$$

where, $\varepsilon$ = permitivity of semiconduct

$A$ = Area of parallel plates (or, are of P and N side)

$W$ = width of the depletion region.

The depletion width ($W$) is given by,

$$W^2 = \frac{2 V_B \varepsilon}{e}\left[\frac{1}{N_A} + \frac{1}{N_D}\right] \quad \text{---(11)}$$

where,

$V_B$ = Breakdown voltage

$e$ = electronic charge

$N_A$ = no. of acceptor atoms

$N_D$ = no. of donor atoms.

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Substituting value of $W$ in eqn (1), we get,

$$C_T = \frac{\varepsilon A}{\sqrt{\frac{2 V_B \varepsilon}{e}\left[\frac{1}{N_A} + \frac{1}{N_D}\right]}}$$

$$\Rightarrow C_T = \frac{\varepsilon A}{\sqrt{\frac{2 V_B \varepsilon}{e}\left[\frac{N_A + N_D}{N_A N_D}\right]}}$$

$$\Rightarrow \boxed{C_T \propto \frac{1}{\sqrt{V_B}}}$$

Question: $C_D$ Vs $C_T$

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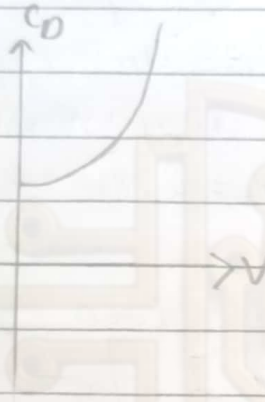
V. imp

| Diffusion capacitance ($C_D$) | Transition capacitance ($C_T$) |
|---|---|
| 1. It occurs when the diode is forward ~~diode~~ biased. | 1. It occurs when the diode is reverse ~~diode~~ biased. |
| 2. With forward biasing, the depletion ~~with~~ decreases. So, $C_D$ increases with increasing forward biasing. | 2. With reverse biasing, the depletion width increases. So, $C_T$ decreases with ~~dec~~ increasing reverse biasing. |
| 3. $C_D$ is directly proportional to forward diode current ($I_D$). ie $C_D \propto I_D$ | 3. $C_T$ is inversely proportional square root of reverse voltage. ie $C_T \propto \frac{1}{\sqrt{V_R}}$ |
| 4.   fig: $C_D$ Vs V | 4. fig: $C_T$ Vs V |

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## ~~Imp~~ Junction Breakdown

In reverse biasing, a very small current flows due to minority charge carrier. This current is called leakage saturation current ($I_S$). If reverse voltage is further increased, a large current flows suddenly at a certain reverse biased voltage.

This mechanism of producing of large reverse current is called junction breakdown. There are two types of junction breakdown available;

1. Zener Breakdown
2. Avalanche Breakdown.

1. What do you mean by junction ~~diode~~ breakdown of a diode?

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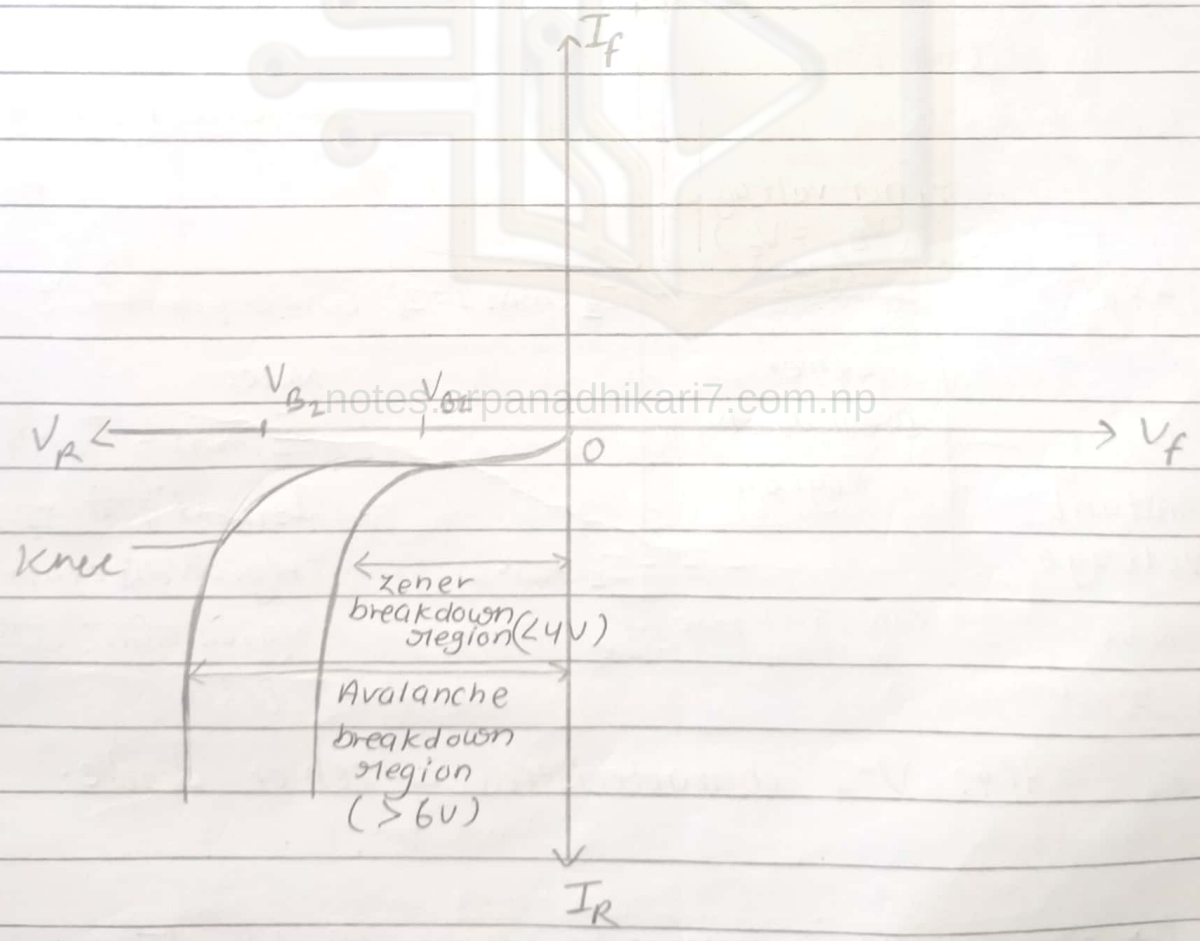
| Zener Breakdown | Avalanche Breakdown. |
|---|---|
| 1. It occurs due to high electric field intensity across the junction. | 1. It occurs due to the collision of charge carriers with other atom |
| 2. It occurs normally at low reverse voltage (< 4V). | 2. It occurs normally at high reverse voltage (> 6V) |
| 3. It occurs at high dopping level (ie heavily dopped diode). | 3. It occurs at low dopping level diode. |
| 4. The width of depletion layer is thin. | 4. The width of depletion layer is in the larger order. |
| 5. Zener Breakdown voltage is negative tempr coeff. ie. $V_B \uparrow$ T↓ | 5. Avalanche Breakdown voltage is positive tempr coff. |
| 6. V-I curve is sharper than avalanche Breakdown curve. | 6. V-I curve is flatter than zener Breakdown curve. |

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where,

$V_{B_1}$ = Zener Breakdown voltage

$V_{B_2}$ = Avalanche Breakdown voltage

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MImf

# Zener Diode

Zener diode is a special type of diode which work in a breakdown region. It acts as simple diode when forward biased. But when operated in reverse breakd they have extremely stable breakdown voltage over wid range of current. So, they are considered as the backb of voltage regulator.

Symbol:

(* clapping level high ई-ई PN junction diode भ

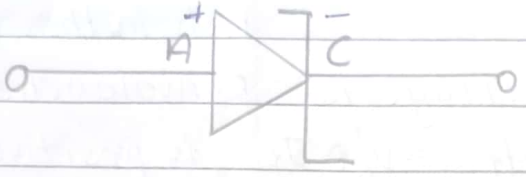
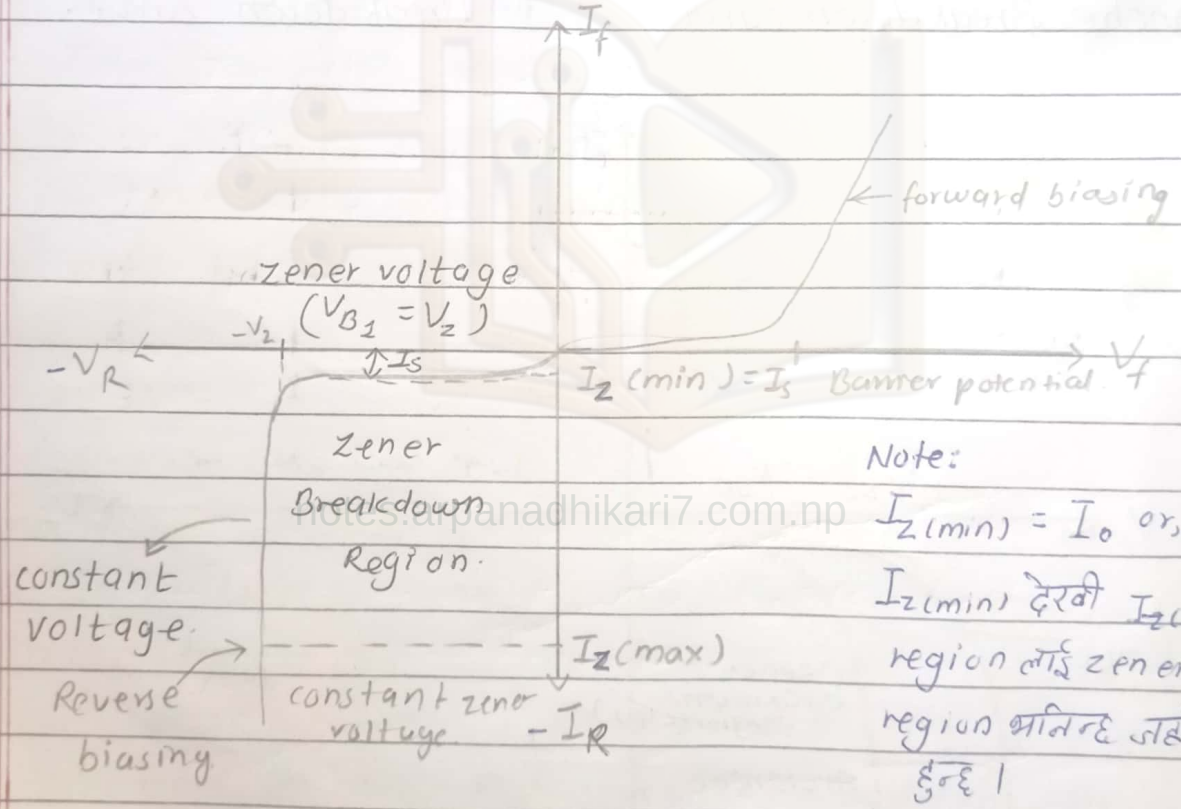



fig: circuit symbol of zener diode




Note:

$I_{z(min)} = I_0$ or, $I_s$ हुन्छ।

$I_{z(min)}$ देखी $I_{z(max)}$ सम्म region लाई zener Breakdown region भनिन्छ जहाँ $V_z$ constant हुन्छ।

fig: V-I characteristics of zener diode.

(* zener diode ले voltage regulation मा help गर्छ। input मा voltage vary गरेतापनि load मा constant voltage provide गर्छ।)

[eg shunt regulation stabilize

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## Application of zener Diode.

1. As a voltage regulator (stabilizer).
2. A clipper in wave shaping circuit.

## Imp Zener Diode as a voltage Regulator

Voltage regulator is a circuit that has an ability to maintain constant output voltage either input voltage is varying or load current is varying.

To use zener diode as voltage regulator following condition must be ~~sta~~ satisfied:-

1. Zener diode must be operated in reverse bias and connected in parallel with load.
2. Input voltage must be greater than zener voltage (i.e $V_{in} > V_z$) i.e it must be operated in a break down region.

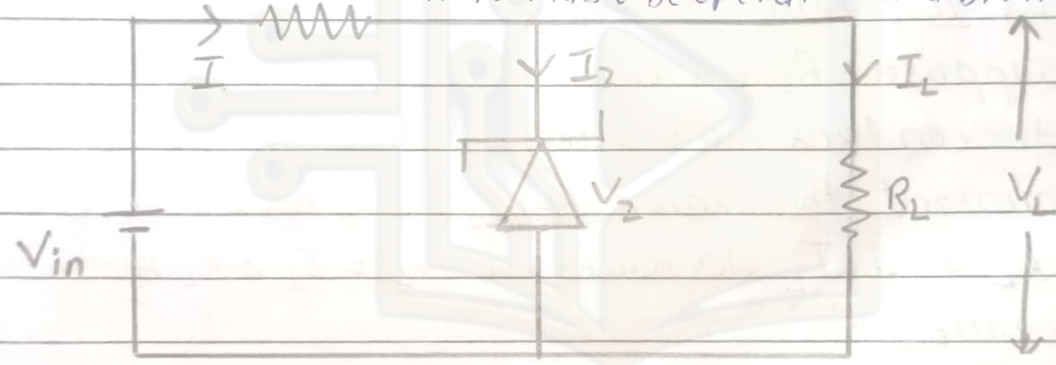


fig: zener diode as voltage regulator

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where,

$V_{in}$ = source voltage
I = Input current
$R_S$ = Series resistance
$V_z$ = zener voltage
$I_z$ = zener current
$R_L$ = load resistance
$I_L$ = ~~curre~~ load current
$V_L$ = load voltage

(Note: $R_S$ = series resistance ई। यसको काम भनेको max current flow भएर zener diode लाई damage हुनबाट बचाउनु हो हो। current limit गर्दिन्छ।)

When zener diode operates (ON), the load voltage and zener voltage remains same.

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i.e $V_L = V_Z$ —(i)

Also,

$I = I_L + I_Z$ —(ii)

and Load voltage $(V_L) = I_L \times R_L$ —(iii)

Case I : When $V_{in}$ varies

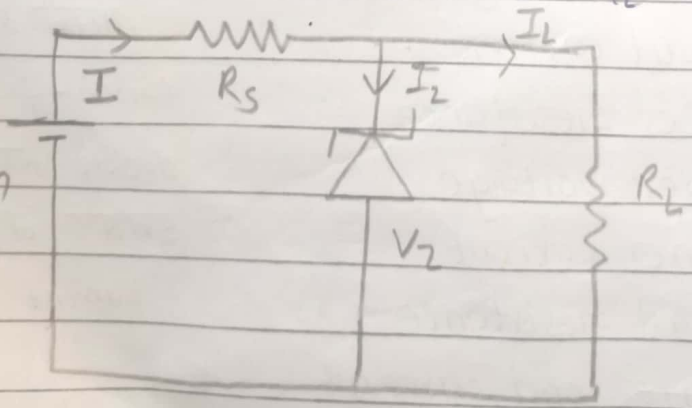
Let $V_{in}$ increases as $V_{in}$ increases the current(I) also increases. If I increases, both $I_Z$ and $I_L$ must be increased But the zener diode conducts the increased current (i.e only $I_Z$ increases) and main $I_L$ constant). Thus, $V_L = I_L \times R_L$ becomes constant even if $V_{in}$ increases.

Case II : When $R_L$ varies.

Suppose $R_L$ increases, when $R_L$ increases the zener diode conducts more current (i.e $I_Z$ increases) and minimized the value of $I_L$. So, that even increase of $R_L$ and $I_L$ decreases & $V_L = I_L \times R_L$ remains constant.

Different cases for Numerical of zener diode

* Case I: $V_{in}$ constant and $R_L$ = constant.




Replace zener diode with open source voltage 'V' and is given by;

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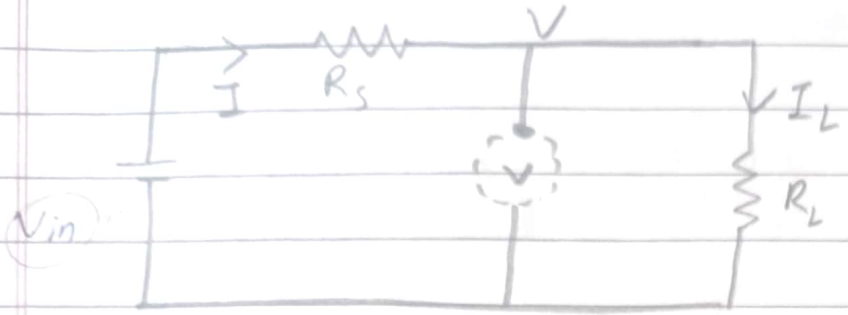
$$V = \frac{R_L}{R_S + R_L} \times V_{in} \quad \text{—— (1)}$$

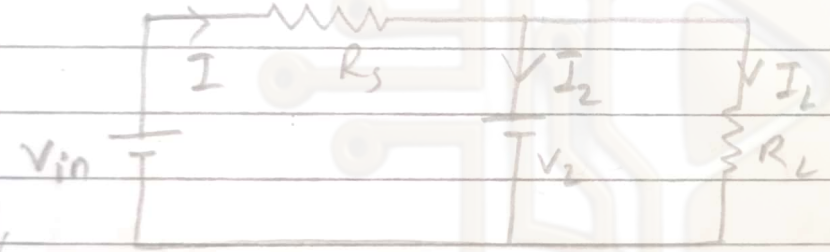



(a) If $V > V_z$ ; zener diode is ON

(b) If $V < V_z$ ; zener diode is off.

(a) If zener diode is ON i.e $V > V_z$

Replace zener diode with corresponding voltage $V_z$ and circuit becomes.




where,

(i) $V_L = V_z$

(ii) $I = \frac{V_{in} - V_z}{R_S}$

(iii) $I_L = \frac{V_L}{R_L}$

(iv) series voltage ($V_R$) = $I \times R_S$

(v) $I_z = I - I_L$

(vi) Power across diode ($P_z$) = $V_z \cdot I_z$

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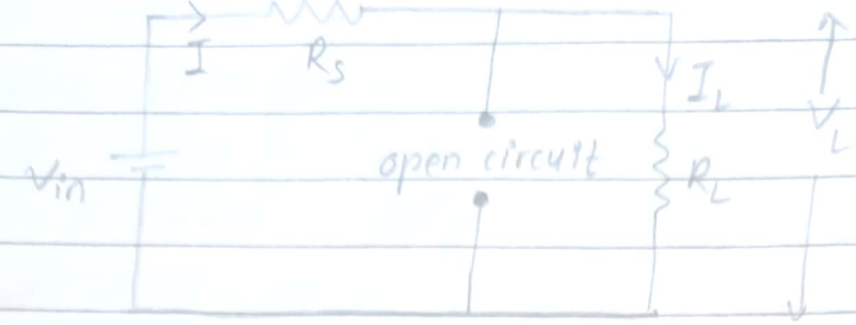
b. If zener diode is OFF:

Zener diode is replaced with open circuit.




Here,

$I_2 = 0$

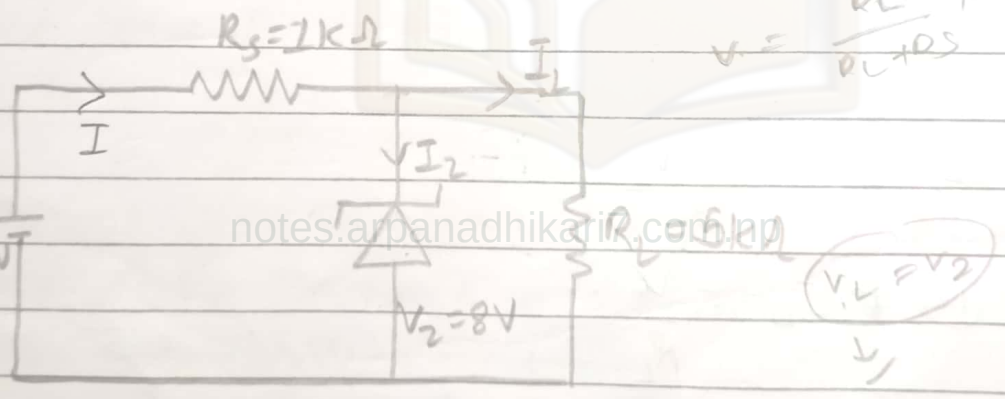
$V_L = \frac{R_L}{R_L + R_S} \times V_{in}$

$I = I_L = \frac{V_{in}}{R_L + R_S}$

$I_L = \frac{V_L}{R_L}$

$P_2 = 0 \quad [\because I_2 = 0]$

Qn.1

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Since, $V_{in} > V_2$; so the zener diode is ON.

$\therefore V_{in} = V_2 = 8V$

(i) Load voltage ($V_L$) = 8V

(ii) Zener current ($I_2$) = ?

Now, $I = \frac{V_{in} - V_2}{R_S}$

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$\Rightarrow I = \frac{18 - 8}{1\,k\Omega}$

$\Rightarrow I = 10\,mA$

Again,

$I_L = \frac{V_L}{R_L}$

$= \frac{8}{5\,k\Omega}$

$= 1.6\,mA$

Then,

zener current ($I_z$) will be $= I - I_L$

$= 10\,mA - 1.6\,mA$

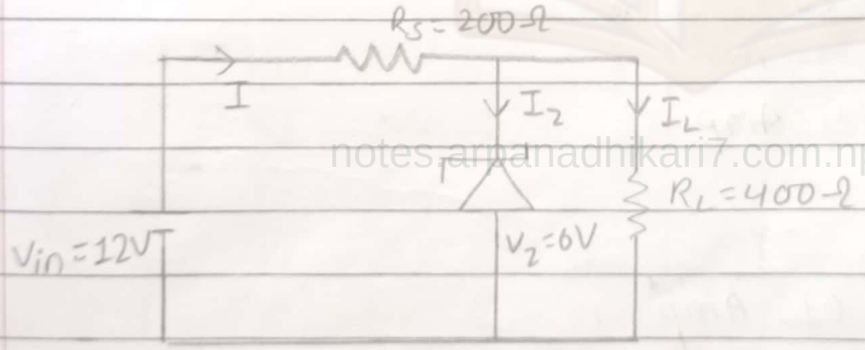
$= 8.4\,mA$

(iii) Power across diode ($P_z$) $= V_z \times I_z$

$= 8V \times 8.4\,mA$

$= 67.2\,mW$atts

Qn.2




Find;

(i) Load voltage ($V_L$)

(ii) zener current ($I_z$)

(iii) Power across diode ($P_z$) and

(iv) series voltage ($V_R$)

Is the diode operate safely when $P_z(max) = 100mW$

Solution,

Here,

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$V_{in} = 12V$

$R_S = 200\Omega$

$V_Z = 6V$

$R_L = 400\Omega$

Then;

(i) $V_L = V_Z$ [because $V > V_Z$ so; diode is ON]

$\therefore V_L = 6V$

(ii) $I_Z = I - I_L$

$= \frac{V_{in} - V_Z}{R_S} - \frac{V_L}{R_L}$

$= \frac{12 - 6}{200} - \frac{6}{400}$

$= \frac{6}{200} - \frac{6}{400}$

$= \frac{12 - 6}{400}$

$= \frac{6}{400}$

$= 0.015$ Amp

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(iii) $P_Z = V_Z I_Z$

$= 6V \times 0.015$ Amp

$= 0.09$ watt

(iv) $V_R = I \times R_S$

$= \frac{V_{in} - V_Z}{R_S} \times R_S$

$= V_{in} - V_Z$

$= 12 - 6$

$= 6V$

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If $P_z(max) = 100\ mW$

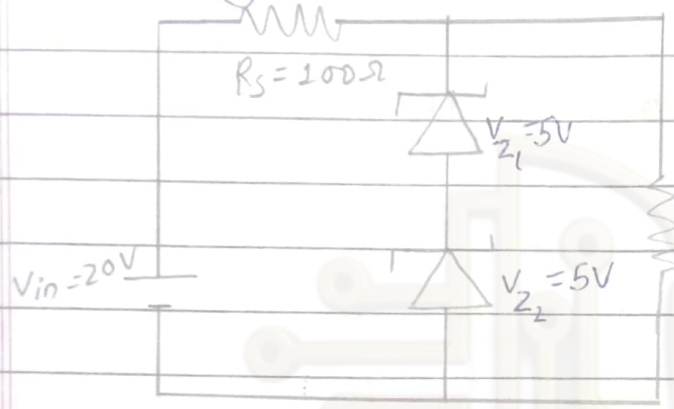
we have $P_z = 0.09\ W$

$= 90\ mW$

$P_z < P_z(max)$

So, the diode would operate safely.

Qn.1 for the given circuit find




(i) load voltage ($V_L$)

(ii) current through zener diode

(iii) voltage across series resistance

(iv) Power dissipated across zener diode.

Soln

Here,

$V_{in} = 20V$

$R_S = 100\Omega$

$V_{z_1} = 5V$

$V_{z_2} = 5V$

$R_L = 400\Omega$

Note: (concept)

*series मा छ भने

-voltage → जोडिन्छ

-current → same हुन्छ

* shunt/parallel मा छ भने

-voltage → same

-current → जोडिन्छ

Since, $V_{in} > V_z$ so, the zener diode is ON. So,

(i) $V_L = V_z = V_{z_1} + V_{z_2}$

$= 5V + 5V$

$= 10V$

(ii) $I_z = I - I_L$

$= \frac{V_{in} - V_z}{R_S} - \frac{V_L}{R_L}$ $= \frac{20-10}{100} - \frac{10}{400}$ $= \frac{16}{100} - \frac{1}{40}$

$= 0.1 - 0.025$

$= 0.075\ Amp$

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(iii) $V_R = I \times R_S$

$= \frac{V_{in} - V_L}{R_S} \times R_S$

$= 20 - 10$

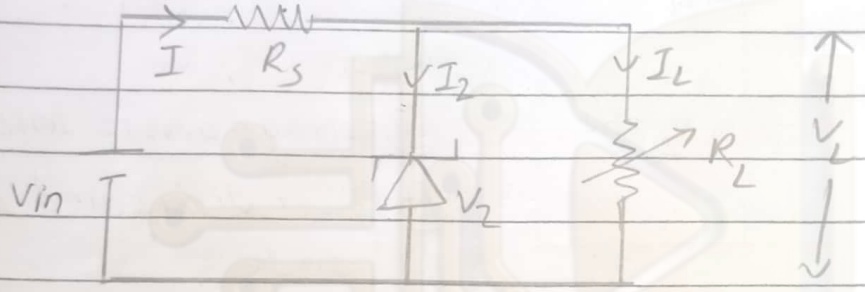
$= 10\,V$

(iv) $P_Z = V_Z I_Z$

$= 10 \times 0.075$

$= 0.75$ watt

* Case II : $R_L$ = variable and $V_{in}$ = constant




Here, we have to find the range of $R_L$ [$R_{L,min}$, $R_{L,max}$] required for safe turn 'ON' of zener diode. So, zener diode must be 'ON'.

$V_L = V_Z$ ——— (i)

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(i) for minimum $R_L$ [$R_{L,min}$]

The open source voltage (v) $= \frac{R_L}{R_L + R_S} \times V_{in}$ ——— (ii)

for minimum condition, $V = V_Z$

Now eqn (ii) becomes.

$\Rightarrow V_Z = \frac{R_L}{R_L + R_S} \times V_{in}$

or, $V_Z (R_L + R_S) = R_L \times V_{in}$

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or, $V_2 R_L + V_2 R_s = R_L \times V_{in}$

or, $R_L V_{in} - V_2 R_L = V_2 R_s$

or, $R_L [V_{in} - V_2] = V_2 R_s$

or, $R_L = \frac{V_2 R_s}{V_{in} - V_2}$

$\therefore \boxed{R_{L,min} = \frac{V_2 R_s}{V_{in} - V_2}}$ (concept: for $R_{L,min}$, $I_L$ should be max for $I_{L(max)}$, $I_z$ should be zero)

Also,

$$\boxed{I_{L,max} = \frac{V_L}{R_{L,min}}}$$

(ii) for maximum value of $R_L$ [$R_{L,max}$]

As we know,

$$R_L = \frac{V_L}{I_L}$$

$$R_{L,max} = \frac{V_L}{I_{L,min}} \quad \text{—— (i)}$$

Again,

$$I = I_L + I_z$$

or, $I_L = I - I_z$

Here,

$I_L \longrightarrow I_{L,min}$ when $I_z \longrightarrow I_{zm(max)}$

$$I_{L,min} = I - I_{zm} \quad \text{—— (ii)}$$

Now, from eqn (i) and (ii)

$$\boxed{R_{L,max} = \frac{V_L}{I - I_{zm}}}$$

Also,

$$\boxed{I = \frac{V_{in} - V_z}{R_s}}$$

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$\rightarrow R_S = 100\,\Omega$

Qn.2 A zener diode voltage regulator has 12V zener vo
with variable load resistance (200Ω – 1kΩ). Calcula
(i) maximum & minimum load current
(ii) " & " power dissipation across d
(iii) Voltage drop across series resistance.
(iv) minimum value of resistance to ensure that ze
diode is ON.

[Given: $V_{in} = 20V$]

Solution;

Given

source voltage ($V_{in}$) = 20V

$V_Z = 12V$

$R_L = 200\,\Omega$ to $1k\,\Omega$

$= 200\,\Omega$ to $1000\,\Omega$

Now; $R_S = 100\,\Omega$

(i) For maximum load current;

$$I_{L,max} = \frac{V_L}{R_{L,max}}$$

$$= \frac{V_Z}{R_{L,max}}$$

$$= \frac{12}{1000}$$

$$= 0.012 \text{ Amp}$$

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For minimum load current;

$$I_{L,min} = \frac{V_L}{R_{L,min}}$$

$$= \frac{V_Z}{R_{L,min}}$$

$$= \frac{12}{200} = 0.06 \text{ Amp.}$$

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$I = I_L + I_Z$

$= I_L \uparrow + I_Z \downarrow$

$= I_{L,min} \quad I_{Z\,max}$

(ii) For max power dissipation across diode;

$P_{Z,max} = V_Z I_{Z\,max}$

$= V_Z (I - I_{L\,min})$

$= 12 \left( \frac{V_{in} - V_Z}{R_S} - I_{L\,min} \right)$

$= 12 \left( \frac{20 - 12}{100} - 0.06 \right)$

$= 12 \left( \frac{8}{100} - 0.06 \right)$

$= 12 (0.08 - 0.06)$

$= 12 \times 0.06$ watt

$= 0.24$ watt

For min. power dissipation;

$P_{Z,min} = V_Z I_{Z\,min}$

$= V_Z (I - I_{L\,max})$

$= V_Z \left( \frac{V_{in} - V_Z}{R_S} - I_{L\,max} \right)$

$= 12 \left( \frac{8}{100} - 0.012 \right)$

$= 12 \times 0.012$ watt

$= 0.144$ watt

(iii) $V_R = I \times R_S$

$= \frac{V_{in} - V_Z}{R_S} \times R_S$

$= 20 - 12$

$= 8\ V$

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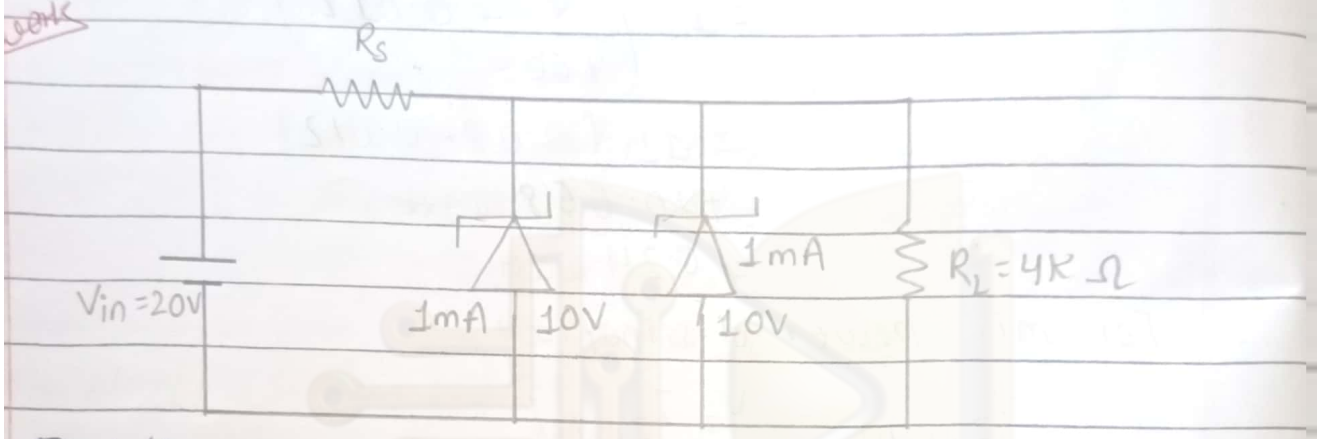
$$R_{Lmin} = \frac{V_Z R_S}{V_{in} - V_Z}$$

$$= \frac{12 \times 100\,\Omega}{20 - 12}$$

$$= \frac{1200}{8}$$

$$= 150\,\Omega$$




Find the value of series resistance ($R_s$) so that diode can operate safely.

Solution,

Given,

source voltage ($V_{in}$) = 20 V

load resistance ($R_L$) = 4 KΩ = 4000 Ω

From diagram, zener diodes are parallel to each other. So, current through diodes will be added, whereas zener voltage remains same.

zener diode current ($I_Z$) = 1mA + 1mA = 2mA = 0.002A

zener voltage ($V_Z$) = 10V

series resistance ($R_s$) = ?

Now,

$V > V_Z$ · So, zener diode is ON. Then,

$V_L = V_Z$

$\therefore V_L = 10V$

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We know that,

$$I_L = \frac{V_L}{R_L}$$

$$= \frac{10}{4}$$

$$= 2.5\,mA$$

Then,

$$I = \frac{V_{in} - V_z}{R_s}$$

$$\Rightarrow I_z + I_L = \frac{V_{in} - V_z}{R_s}$$

$$\Rightarrow 2mA + 2.5mA = \frac{20 - 10}{R_s}$$

$$\Rightarrow R_s = \frac{10}{4.5}$$

$$\Rightarrow R_s = 2.222k\Omega$$

$\therefore$ series resistance across given ckt is $2.222k\Omega$ so that diode can operate safely.

$P_z = I_z V_z$

$= 2mA \times 10$

$= 0.002A \times 10$

$= 0.02$

* Case III : $V_{in}$ = variable and $R_L$ = constant

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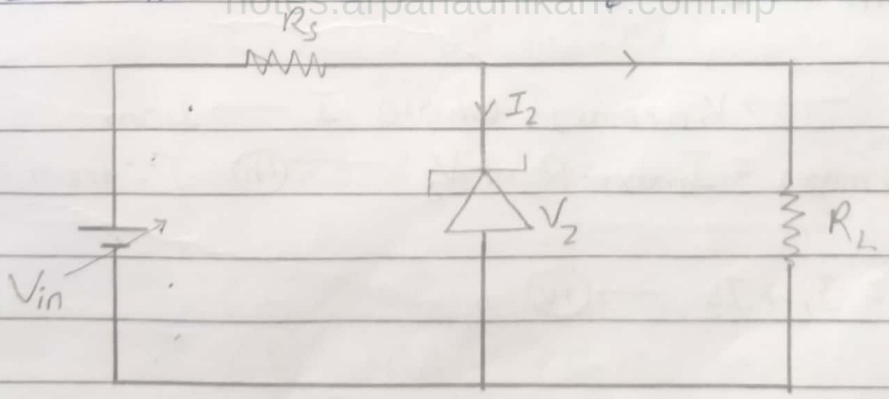


We have to find range of $V_{in}$ [i.e $V_{in,max}$ and $V_{in,min}$] required for safe turn 'ON' of zener diode. zener diode must be ON. Thus, $\boxed{V_L = V_z}$

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For minimum $V_{in}$

The open circuit voltage is given by,

$$V = \frac{R_L}{R_L + R_S} \times V_{in}$$

or, $V(R_L + R_S) = R_L V_{in}$

or, $V_{in} = \frac{V(R_L + R_S)}{R_L}$ ——— (i)

for minimum condition, $V = V_Z$

$\therefore$ eqⁿ (i) becomes;

$$V_{in(min)} = \frac{V_Z(R_L + R_S)}{R_L} \text{ and}$$

$$I_{min} = \frac{V_{in(min)} - V_Z}{R_S}$$

For maximum $V_{in}$

We know that,

$$I = \frac{V_{in} - V_Z}{R_S}$$

or, $IR_S = V_{in} - V_Z$

or, $V_{in} = V_Z + IR_S$ ——— (ii)

Here,

$V_{in} \longrightarrow V_{in(max)}$ where $I \longrightarrow I_{max}$

$V_{in(max)} = I_{(max)} \cdot R_S + V_Z$ ——— (iii) [$\because$ from eqn (ii)]

Again,

$I = I_L + I_Z$ ——— (iv)

Here,

$I \longrightarrow I_{max}$ when $I_Z \longrightarrow I_{Z(max)}$

$I_{max} = I_L + I_{Z(max)}$

Now, eqⁿ (iii) becomes.

$V_{in(max)} = (I_L + I_{Z(max)}) R_S + V_Z$

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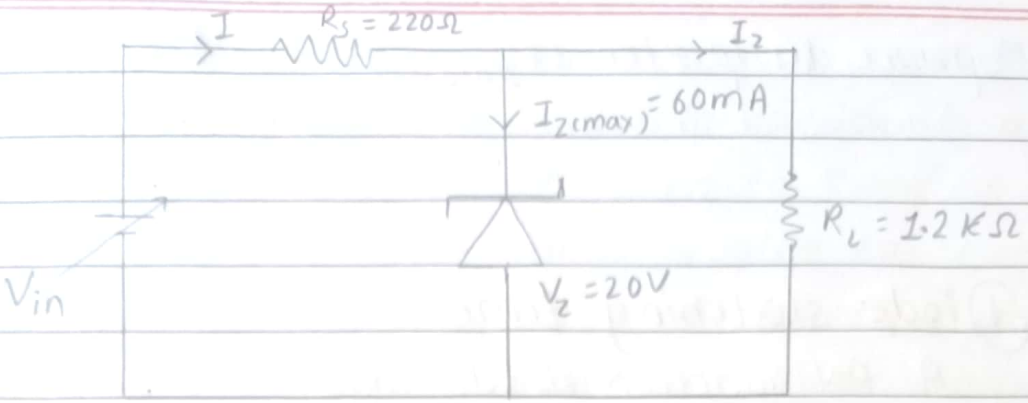
Qn.1




Calculate:

① $V_{in(min)}$

② $V_{in(max)}$

③ Max$^m$ power dissipation through diode

Solution,

Given,

Series resistance ($R_s$) = 220Ω

max$^m$ diode current ($I_{Z(max)}$) = 60mA = 0.060A

zener voltage ($V_Z$) = 20V

Load resistance ($R_L$) = 1.2kΩ = $1.2 \times 10^3 \Omega$ = 1200Ω

Now,

We know that,

① $V_{in(min)} = \dfrac{V_Z(R_L + R_s)}{R_L}$

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$= \dfrac{20(1200 + 220)}{1200}$

$= 23.666$ Volt

② $V_{in(max)} = (I_L + I_{Z(max)})R_s + V_Z$

$= \left(\dfrac{V_L}{R_L} + 0.060\right)220 + 20$

$= \left(\dfrac{20}{1200} + 0.060\right)220 + 20$

$= (0.01666 + 0.060)220 + 20$

$= 36.866$ Volt

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(iii) max$^m$ power dissipation $(P_{z(max)}) = V_z \times I_{z(max)}$
$= 20 \times 0.060$
$= 1.2$ watt

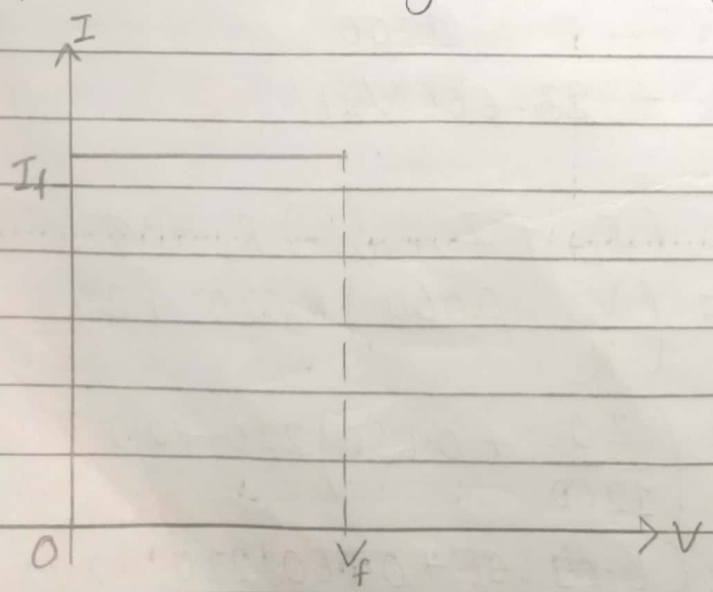
## Diode switching time

A PN junction diode may be used as an electrical switch. The electrical circuit can be made ON & OFF by forward bias or reverse bias. In both cases, diode response is accomplished by transient and an interval of time collaspe before the diode recovers its steady state. The time interval for the recovery of steady state of the diode is called recovery time. It is of two types;

1. Forward Recovery time ($t_{fr}$)
2. Reverse / Backward Recovery time ($t_{rr}$)

### 1. Forward recovery time ($t_{fr}$)

The time required to the diode to recover from reverse bias to forward bias condition, The diode takes certain interval of time called FRT. It is defined as the time interval between instant of 10% diode voltage to 90% of its final value.




[fig: Forward Recovery time curve]

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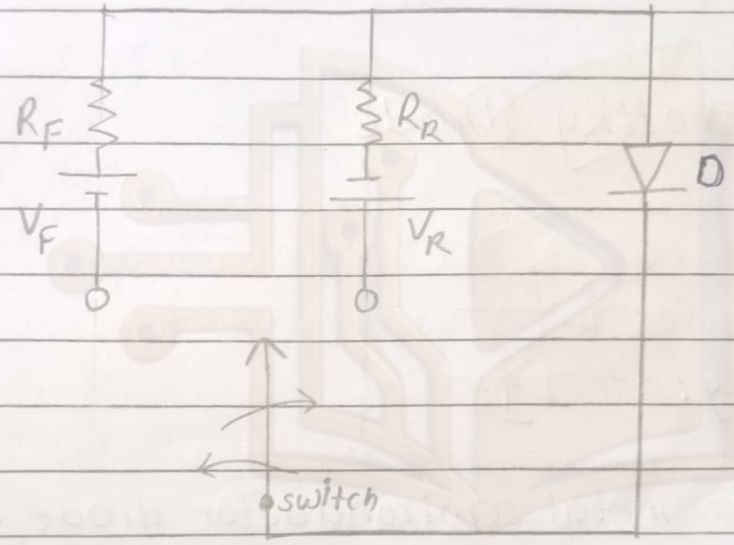
2 Reverse Recovery time ($t_{rr}$)

When PN Junction is forward biased, a large no. of electrons cross over from N region to P region and a large no. of holes cross over from P region to N region. They establish a large no. of minority carriers in each region.

If the external voltage in the diode is suddenly reversed, the diode current will not immediately fall to zero. A large current flows due to large no. of minority carriers for certain interval of time ($t_1 - t_2$) as shown in figure below;

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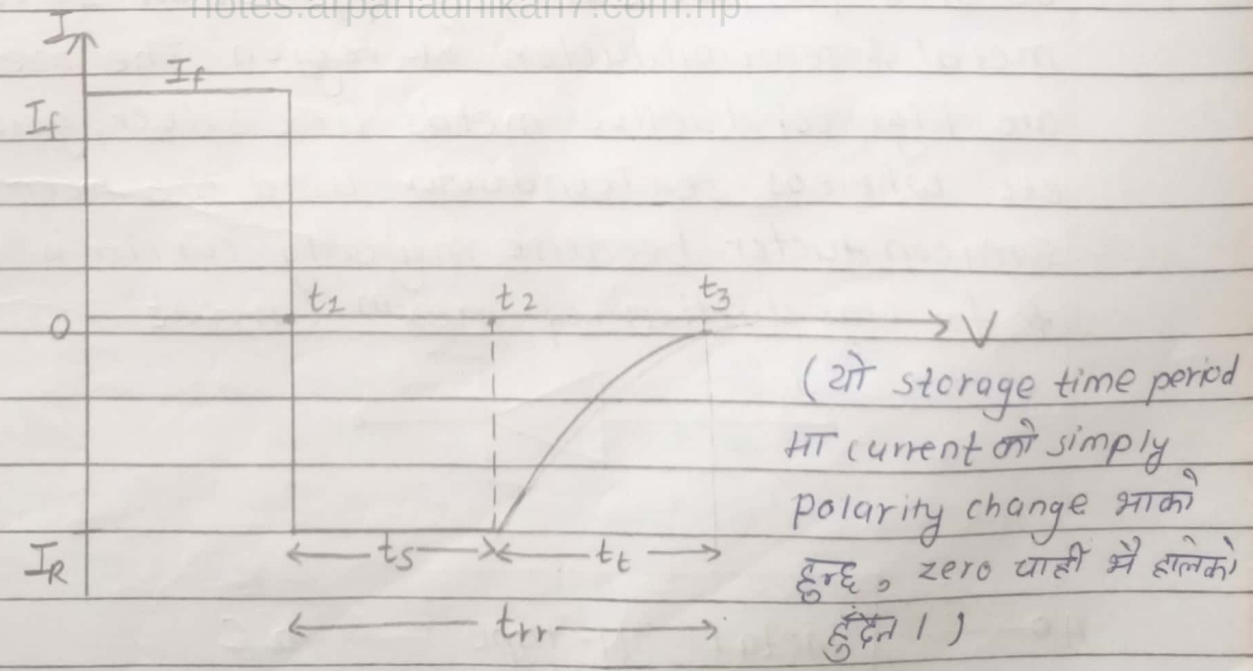


(यो storage time period मा current को simply polarity change भाको हुन्छ, zero चाही भै हालेको हुँदैन।)

[fig: Reverse recovery time]

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During time interval ($t_1$ to $t_2$), the stored minority charge carriers causes large reverse current ($I_R$). This time interval is called storage time ($t_s$). After $t_2$, diode begins to reverse and diode current begins to fall exponentially to normal steady state.

The time ellapse betⁿ $t_2$ and time instant when diode is normally recovered transition time ($t_t$). The sum of storage time & transition time is called reverse recovery time.

$$\text{i.e } t_{rr} = t_s + t_t$$

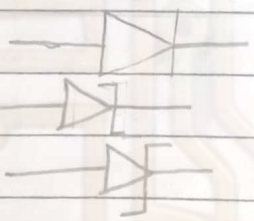
## Schottky Diode

symbol:

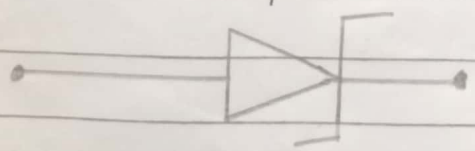
PN :

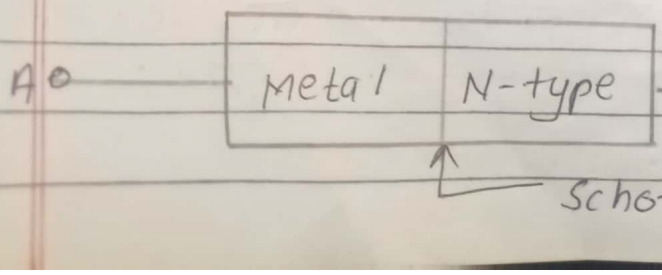
Zener:

Schottky:



It is a metal semiconductor diode designed to be used for fast switching application. It consists of metal & semiconductor N-region. The metals used are high conductive metals like silver, gold, platinum etc. whereas semiconductor used are normally N-type semiconductor because majority carrier is electron & for conduction of maxm current



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It is unipolar diode (i.e only electron conducts current). It is also called hot carrier diode because the electron must gain high energy to cross metal semiconductor junction to conduct current. The metal semiconductor junction is called schottky junction. The metal region of schottky diode is called positive terminal or anode whereas the semiconductor region is called negative terminal or cathode.

The V-I characteristics of schottky diode is shown bel

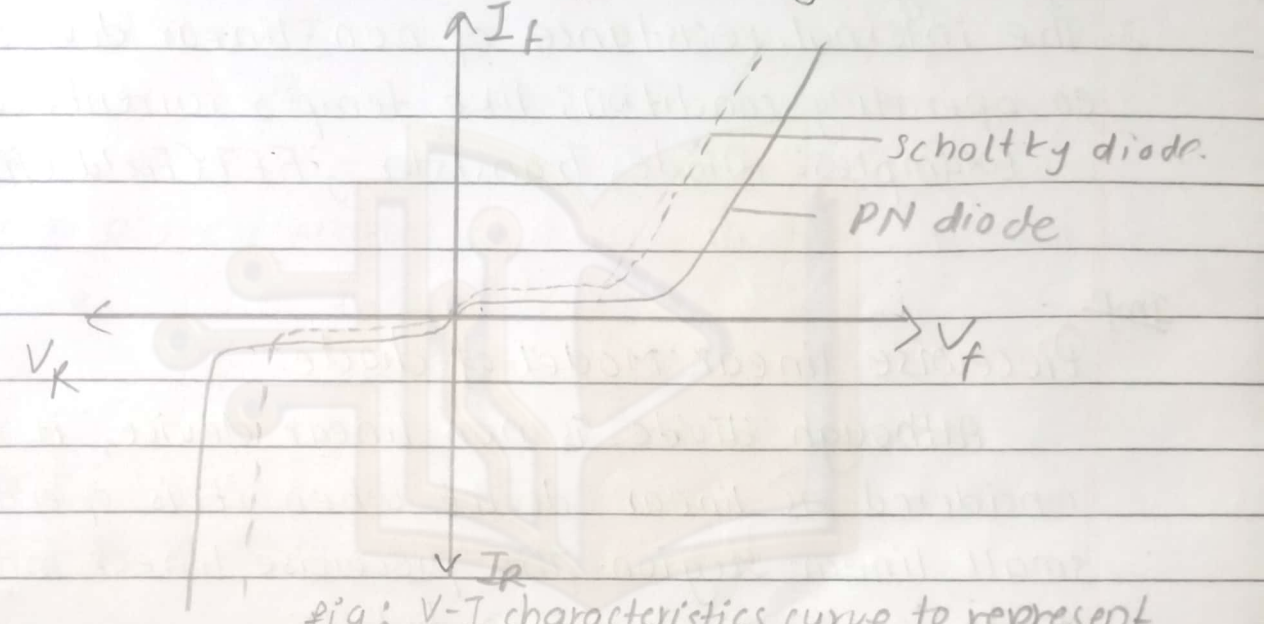


fig: V-I characteristics curve to represent schottky diode

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Applications

1. To rectify the very high frequency signals.
2. As a switching device in digital computer
3. In clipping & clamping circuits.
4. In mixing & detecting circuits used in communication.

(depletion region पातलो हुने भएकोले reverse recovery time धेरै लाग्दैन त्यही भएर fast switching device को रूपमा काम गर्दछ।)

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## Non-linear Model

The electronic device which donot follow ohm's law or whose V-I characteristic curve is not straight line are non-linear device. The properties of non-linear device are given below:

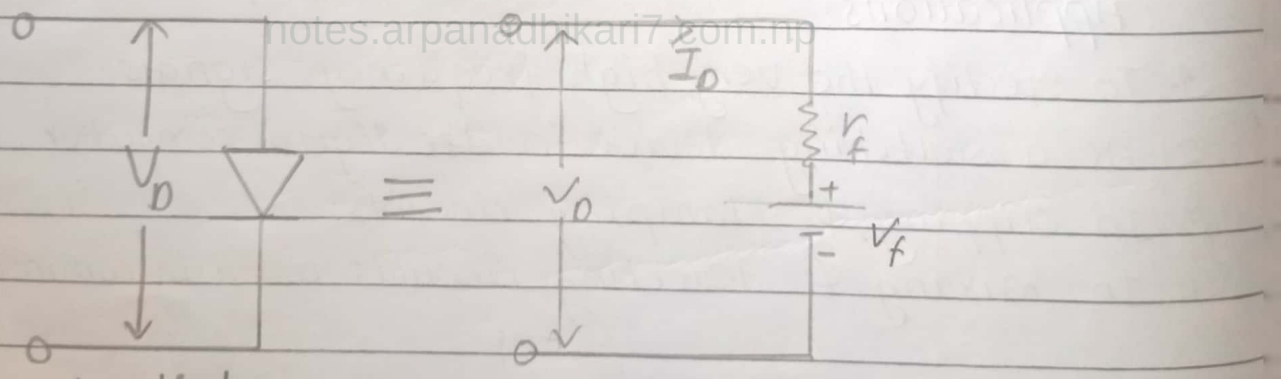
1. They donot follow network theory like Thevinin's theory, Norton's theory, superposition theory.
2. The circuit containing non-linear device can be analyse by using KCL or KVL.
3. The internal resistance of non-linear device depends on operating conditions like temp°, current, voltage, etc

Examples: Diode, Transistor, FET (Field effect Transis

Imp

## Piecewise linear Model of diode

Although diode is non linear device, it can be considered as linear device when it is operated in small linear region. The piecewise linear model of diode is shown is figure below:




(a) diode

b) Piecewise linear model of dio

We have,

Dynamic resistance $(r_f) = \frac{\Delta V_D}{\Delta I_D}$

$$r_f = \frac{\partial V}{\partial I_D} \quad \text{---} \quad (1)$$

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and,

$$I_D = I_S\left(e^{\frac{V_D}{\eta V_T}} - 1\right)$$

$$\frac{dI_D}{dV_D} = \frac{I_S e^{V_D/\eta V_T}}{\eta V_T} = \frac{I_D}{\eta V_T}$$

or, $\frac{1}{r_f} = \frac{I_D}{\eta V_T}$

or, $\boxed{r_f = \frac{\eta V_T}{I_D}}$ where, $r_f$ is diode forward resistance

$$V_T = \frac{KT}{q}$$

From above figure (b);

$$V_D = I_D r_f + V_f$$

$$V_f = V_D - I_D r_f$$

Note:

- to + जाँदा = + and viceversa.

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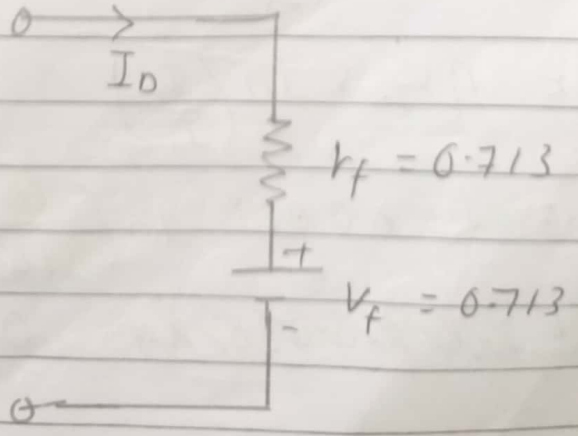
* Find the piece wise linear model of silicon diode with $\eta = 1.6$ and $I_S = 10^{-11}$ Amp in the vicinity operating point. Assume $I_D = 1$ mAmp.

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Given;

$\eta = 1.6$

$I_S = 10^{-11}$ Amp

$I_D = 1mA = 1 \times 10^{-3} A$

We know,

$$V_T = \frac{KT}{q}$$

$$= \frac{1.38 \times 10^{-23} \times 298}{1.6 \times 10^{-19}} \quad (\text{i.e } 25+273)$$

$$= 0.025\ V$$

Now,

$$r_f = \frac{\eta V_T}{I_D}$$

$$= \frac{1.6 \times 0.025}{1 \times 10^{-3}}$$

$$= 41.4\ \Omega$$

Again,

$$V_f = V_D - I_D r_f \quad \text{---} (1)$$

where,

$$I_D = I_S \left(e^{\frac{V_D}{\eta V_T}}\right)$$

$$\Rightarrow e^{\frac{V_D}{\eta V_T}} = \frac{I_D}{I_S}$$

$$\Rightarrow \frac{V_D}{\eta V_T} = \ln\left(\frac{I_D}{I_S}\right)$$

$$\Rightarrow V_D = \eta V_T \times \ln\left(\frac{I_D}{I_S}\right)$$

$$\Rightarrow V_D = 1.6 \times 0.025 \ln\left(\frac{10^{-3}}{10^{-11}}\right)$$

$$\Rightarrow V_D = 0.76\ V$$

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From eqⁿ ①

$V_f = 0.76 - 10^{-3} \times 41.4$

$V_f = 0.7186\,V$

[Note: If $I_S$ is not given take $V = 0.7\,V$ for Si and $0.3\,V$ for Ge]

# Taking appropriate example of non-linear device show that non-linear device doesnot follow principle of superposition theorem.

Proof:

Consider a non-linear device (square low device) whose characteristics eqn is given by;

$$I_S = A(V_S - V_{TR})^2 \quad \text{——} ①$$

where,

$I_S$ = current through non-linear device

$V_S$ = voltage across "

$A$ = constant = $1\,mA/V^2$

$V_{TR}$ = Threshold voltage = 0

Now,

eqⁿ ① becomes

$$I_S = V_S^2 \quad \text{——} ⑪$$

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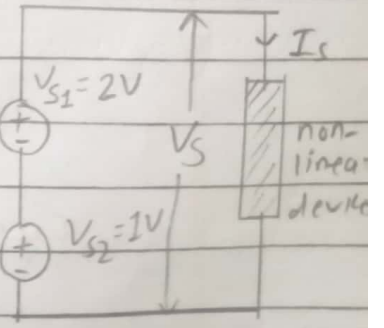
Now, let the non-linear device be connected to source.

Here,

$V_S = V_{S_1} + V_{S_2}$

$= 2V + 1V$

$= 3V$




from eqⁿ ⑪, $I_S = (3)^2 = 9\,mA$

Now, to apply superposition theorem firstly $V_{S_1} = V_S = 2$

∴ from eq ⑪, $I_S = V_S^2$

$I_{S_1} = (V_{S_1})^2 = (2)^2 = 4\,mA$

Now,

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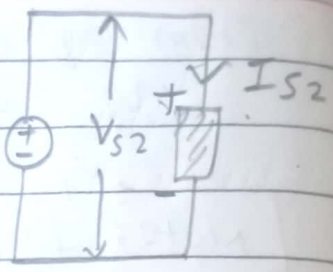
consider source $V_2$ only,

$$V_{S_2} = V_2 = 1V$$

and,

$$I_{S_2} = (V_{S_2})^2 = (1)^2 = 1\,mA$$



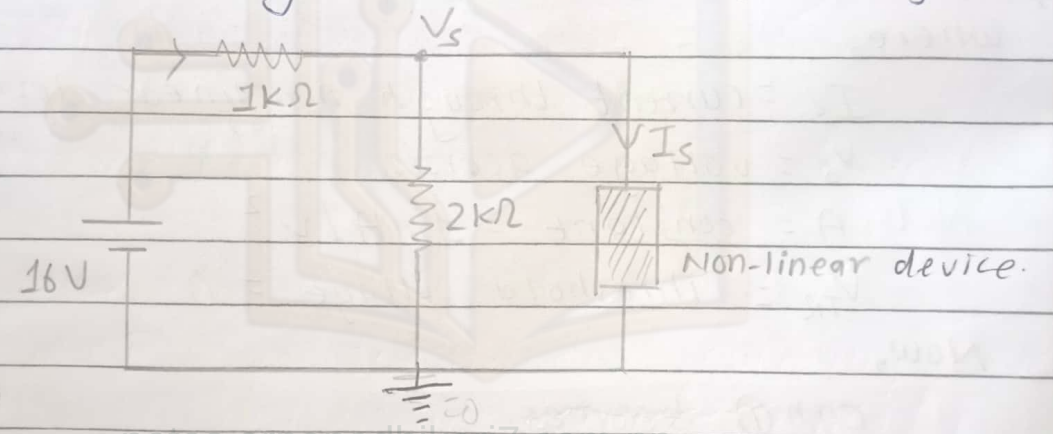

from, superposition theorem,

$$I_{S_1} + I_{S_2} = 4mA + 1mA = 5A$$

We have, $I_S = 9mA$

since $I_S \neq (I_{S_1} + I_{S_2})$, this proves that non-linear device doesn't follow the principle of superposition.

# From the circuit given below, find $I_S$ and $V_S$.




Soln

Here we have,

$$I_S = A(V_S - V_{TR})^2 \quad \text{——①}$$

where,

$$A = 1\,mA/V^2$$

$$V_{TR} = 0$$

Eqn ① becomes

$$I_S = V_S^2 \quad \text{——②}$$

Now using KCL,

Total incoming current = Total outgoing current.

$$\text{or, } \frac{16 - V_S}{1} = \frac{V_S - 0}{2} + I_S$$

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$$\text{or, } 16 - V_S = \frac{V_S}{2} + V_S^2 \quad [\because I_S = V_S^2]$$

$$\text{or, } 16 - V_S = \frac{V_S + 2V_S^2}{2}$$

$$\text{or, } 32 - 2V_S = V_S + 2V_S^2$$

$$\text{or, } 2V_S^2 + 3V_S - 32 = 0 \quad \text{— (iii)}$$

On solving (iii) we get,

$$V_{S1} = 3.31\,V$$

$$V_{S2} = -4.81\,V$$

$$\therefore I_{S1} = (3.31)^2 = 10.95\,mA$$

$$I_{S2} = (-4.81)^2 = 23.13\,mA$$

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# DC Power Supply

DC power supply is required for the operation of must of the electronic device & circuits. Nowadays almost all electronic devices include a circuit that converts AC supply into a DC supply.

The basic block diagram of DC power supply is shown below:

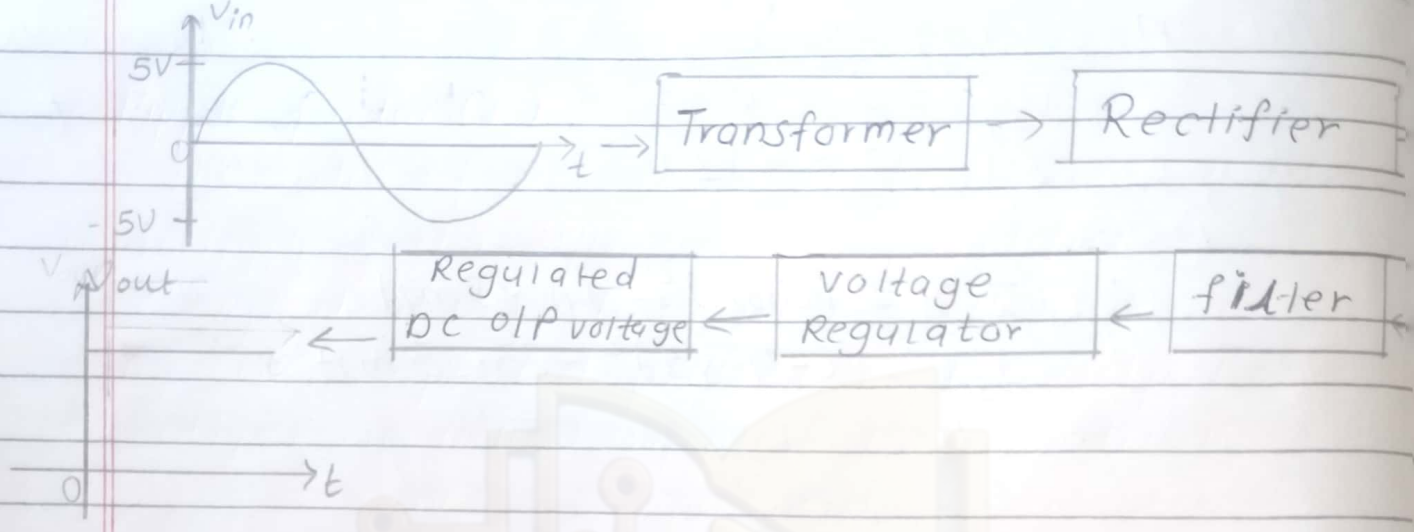


fig: Block diagram of DC power supply.

a. Transformer.

The input to the transformer is normally an AC sig that is generated by line voltage such as power fro an electric outlet. The transformer's main function i to step down (lower the amplitude) or step up (incr the amplitude). The signal to produce the desired DC level required at output of the power supply.

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b. Rectifier

The signal at the output of transformer is fade to the rectifier. The diode circuit which are used t convert input AC signal to DC signal are called rectifiers. It produces pulsating DC signal. A pulsating DC signal is the signal that does not

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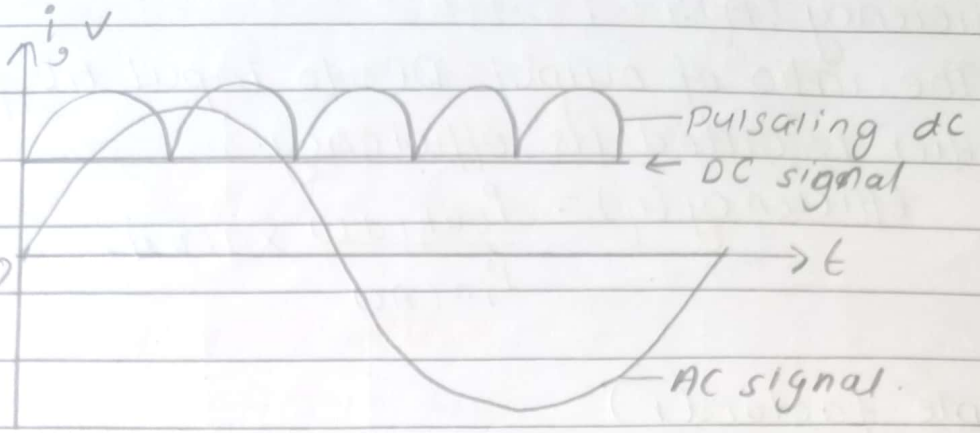
change its polarity but its magnitude is the function of time.




Rectifiers are of two types:

a. Half wave rectifier
b. Full wave rectifier

c. Filter

In order to convert pulsating DC signal into a non-pulsating DC signal, a filter is needed. Normally a simple capacitor filter suffices. The output of filter is DC voltage which usually has some ripple or small AC variation.

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d. Regulator.

The regulator has following two functions:

1. To smooth the signal from the filter producing DC signal with no ripple.
2. To produce a constant voltage ~~and~~ at the output.

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## Rectifier's parameter

### 1. Efficiency ($\eta$):

The ratio of output DC to input AC power of a rectifier is called its efficiency.

$$\text{Efficiency } (\eta) = \frac{P_{out(dc)}}{P_{in(AC)}} \times 100\%$$

### 2. Ripple factor (r)

The ratio of AC component to DC component is known as ripple factor. The output of rectifier is not pure DC signal. It consists of DC as well as AC component. This AC component present at the output of a rectifier are called ripples. Mathematically,

$$r = \frac{\text{ac components}}{\text{dc components}}$$

In terms of current,

$$r = \frac{I_{ac}}{I_{dc}}$$

We know,

$$I_{rms}^2 = I_{ac}^2 + I_{dc}^2$$

$$I_{ac} = \sqrt{I_{rms}^2 - I_{dc}^2}$$

Now,

$$r = \frac{\sqrt{I_{rms}^2 - I_{dc}^2}}{I_{dc}}$$

$$r = \sqrt{\frac{I_{rms}^2}{I_{dc}^2} - 1}$$

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3 Peak inverse voltage (PIV)

During operation of rectifier, diode is either forward biased or reverse ~~diod~~ biased.
The maximum voltage across reverse bias diode of a rectifier is called peak inverse voltage.

## Types of Rectifier.

### 1 Half wave Rectifier (HWR)

The rectifier circuit which converts any of half cycle either +ve half cycle or -ve half cycle of input AC signal to pulsating dc signal is called HWR. It consists of a diode & step down transformer connected to a load ($R_L$) as shown in figure.

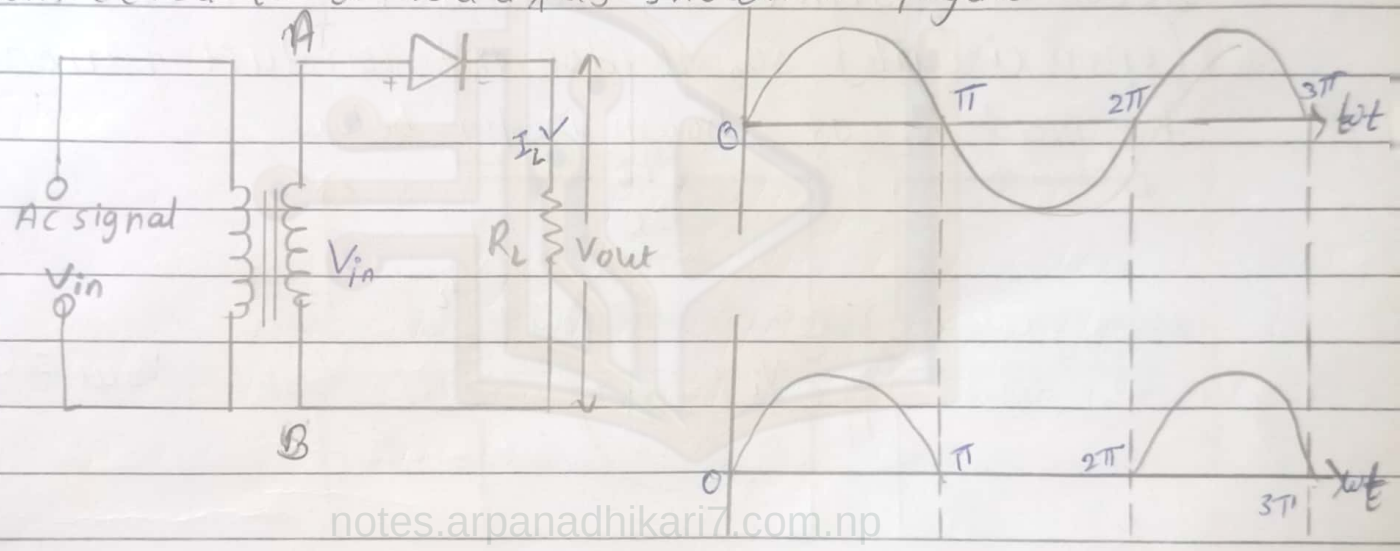


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fig: I/o waveform of Half wave rectifier.

Operation:

(i) for +ve cycle of input AC.
(ii) for -ve cycle of input AC.

(i) For +ve cycle of Input AC.

During this cycle, the end A of secondary winding of transformer become positive where end B become negative. So, the diode is forward biased

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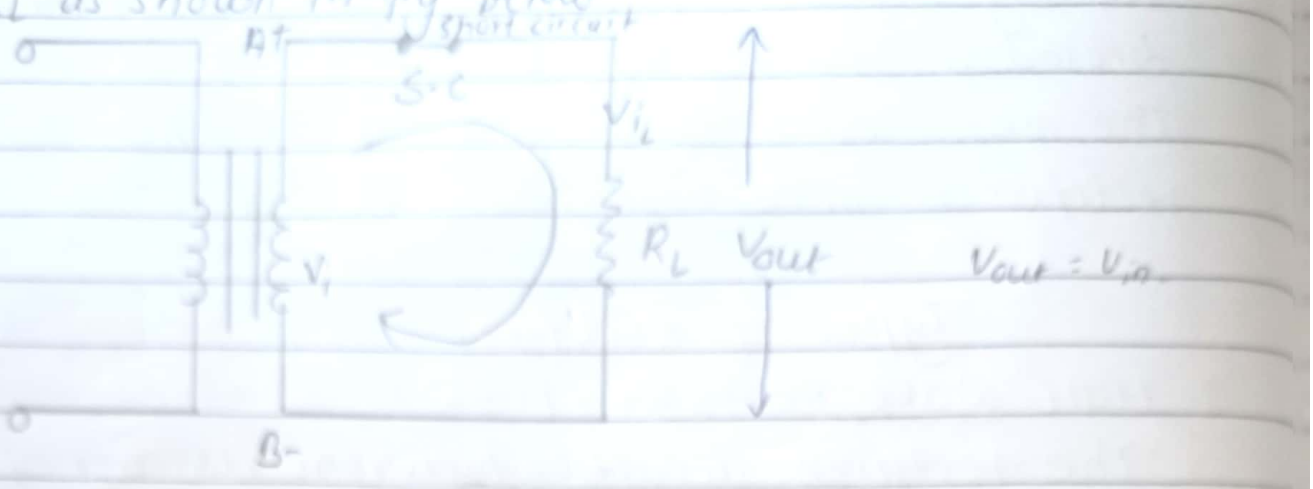
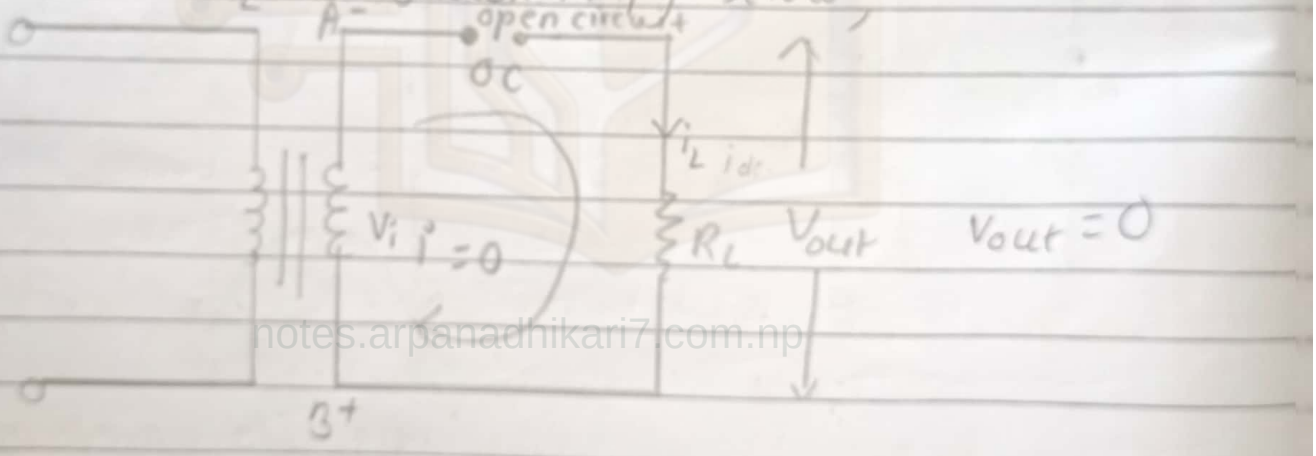
(short circuit) so, it conducts current through the load $R_L$ as shown in fig below.




(ii) For -ve cycle of Input AC

During this cycle, the end A of secondary winding of transformer becomes negative whereas end B becomes positive. So, the diode is reverse biased (open circuit). So, it doesn't conduct current through the load $R_L$ as shown in fig below:

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$$V_m = \sqrt{2}\, V_{in}$$

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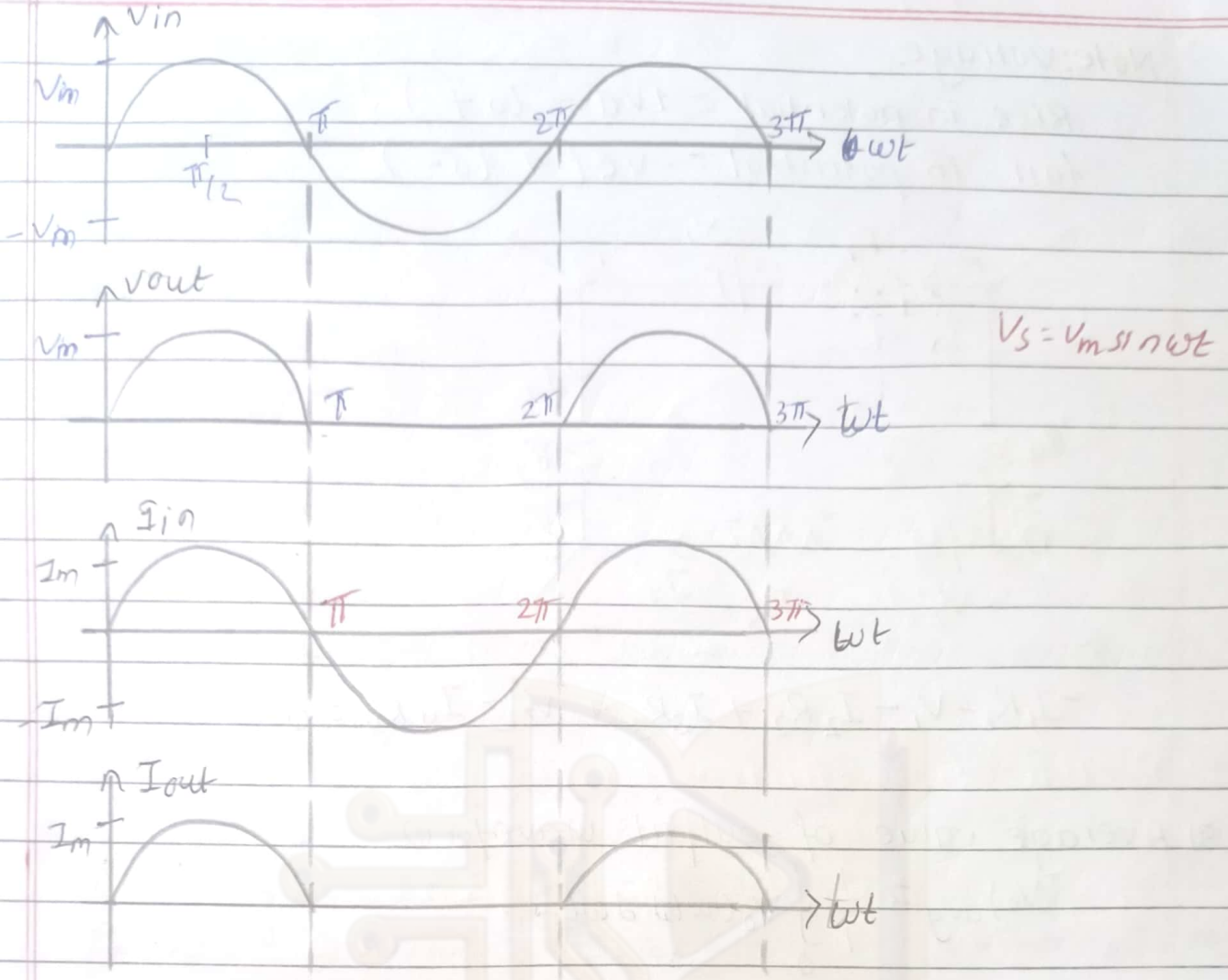


$V_s = V_m \sin \omega t$

Fig: I/O curve of HWR for current and voltage

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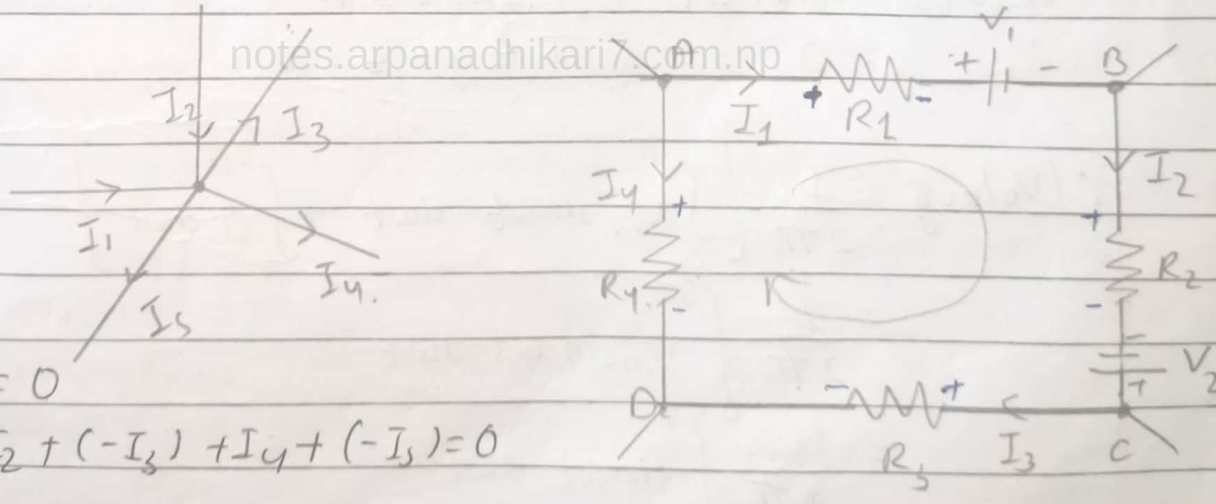


$\Sigma I = 0$

$I_1 + I_2 + (-I_3) + I_4 + (-I_5) = 0$

$-I_1R_1 - V_1 - I_2R_2 + V_2 - I_3R_3 + I_4R_4 = 0$

Note: Resistor: Movement of direction & current को direction Same भए voltage drop in resistor -ve लिने

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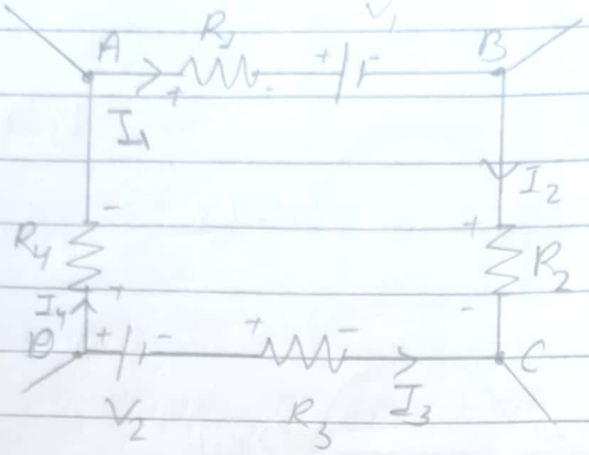
Note: Voltage.

Rise in potential = +ve (− to +)

fall in potential = −ve (+ to −)




$$-I_1R_1 - V_1 - I_2R_2 + I_3R_3 + V_2 - I_4R_4 = 0$$

9) Average value of output waveform:

$$(V_o)_{avg} = \frac{1}{T}\int_0^T V_o(\omega t)\, d\omega t$$

Time period $(T) = 2\pi - 0 = 2\pi$

$V_o(\omega t) = V_m \sin\omega t \quad 0 \le \omega t \le \pi$

$= 0 \quad \pi \le \omega t \le 2\pi$

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$$\therefore (V_o)_{avg} = \frac{1}{2\pi}\left[\int_0^{\pi} V_m \sin\omega t \cdot d\omega t + \int_{\pi}^{2\pi} 0 \cdot d\omega t\right]$$

$$= \frac{1}{2\pi}\int_0^{\pi} V_m \sin\omega t \cdot d\omega t$$

$$= \frac{V_m}{2\pi}\int_0^{\pi} \sin\omega t \cdot d\omega t$$

$$= \frac{V_m}{2\pi}\{-\cos\omega t\}_0^{\pi} = \frac{V_m}{2\pi}\{-\cos\pi - (-\cos 0)\}$$

$$= \frac{V_m}{2\pi}\{-\cos\pi + \cos 0\}$$

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$$(V_o)_{avg} = \frac{V_m}{2\pi}\{-(-1)+1\}$$

$$= \frac{2V_m}{2\pi}$$

$$\therefore (V_o)_{avg} = \frac{V_m}{\pi}$$

Similarly $(i_L)_{avg} = \frac{I_m}{\pi}$

b. Root mean square (RMS) value of output waveform

$$(V_o)_{rms} = \sqrt{\frac{1}{T}\int_0^T V_o^2(\omega t)\cdot d\omega t}$$

$\cos 2x = 1 - 2\sin^2 x$

$\sin^2 x = \frac{1-\cos 2x}{2}$

$\sin^2 \omega t = \frac{1-\cos 2\omega t}{2}$

$\int dx = x$

$\int d\omega t = \omega t$

$\int \cos ax = \frac{\sin ax}{a}$

Time period $(T) = 2\pi - 0 = 2\pi$

$V_o(\omega t) = V_m \sin \omega t \quad 0 \le \omega t \le \pi$

$= 0 \quad \pi \le \omega t \le 2\pi$

$$(V_o)_{rms} = \sqrt{\frac{1}{T}\int_0^T V_o^2(\omega t)\, d\omega t}$$

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$$= \sqrt{\frac{1}{2\pi}\left\{\int_0^{\pi} V_m^2 \sin^2 \omega t \cdot d\omega t + \int_{\pi}^{2\pi} 0 \cdot d\omega t\right\}}$$

$$= \sqrt{\frac{V_m^2}{2\pi}\int_0^{\pi} \sin^2 \omega t \, d\omega t}$$

$$= \sqrt{\frac{V_m^2}{2\pi}\int_0^{\pi} \left(\frac{1-\cos 2\omega t}{2}\right) d\omega t}$$

$$= \sqrt{\frac{V_m^2}{4\pi}\int_0^{\pi} (1-\cos 2\omega t)\, d\omega t}$$

$$= \sqrt{\frac{V_m^2}{4\pi}\left\{\omega t - \frac{\sin 2\omega t}{2}\right\}_0^{\pi}}$$

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$$(V_o)_{rms} = \sqrt{\frac{V_m^2}{4\pi}\left\{\pi - \frac{\sin 2\pi}{2} - 0 + \frac{\sin 0}{2}\right\}}$$

$$= \sqrt{\frac{V_m^2}{4\pi} \times \pi}$$

$$= \sqrt{\frac{V_m^2}{4}}$$

$$\therefore (V_o)_{rms} = \frac{V_m}{2}$$

Sampurna Ghimire

Similarly,

$$(i_L)_{rms} = \frac{I_m}{2}$$

c. Rectification efficiency ($\eta$):

It is the ratio of dc power output to the ac power input. (ie efficiency ($\eta$) = $\frac{\text{dc power output}}{\text{ac power input}} \times 100$ = power output)

$$\Rightarrow \eta = \frac{I_{dc}^2 R_L}{I_{rms}^2 (r_f + R_L)} \times 100$$

where,

$r_f$ = forward resistance of diode.

$R_L$ = load resistance

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$$\eta = \frac{\left(\frac{I_m}{\pi}\right)^2 R_L}{\left(\frac{I_m}{2}\right)^2 (r_f + R_L)} \times 100$$

$$= \frac{4}{\pi^2} \frac{R_L}{(r_f + R_L)} \times 100$$

Since, $r_f + R_L \approx R_L$

$$\therefore \eta = \frac{4}{\pi^2} \times \frac{R_L}{R_L} \times 100$$

$$= 40.56\% \quad = 40.6\%$$

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Therefore, maximum possible efficiency of half wave rectifier is 40.6%.

d. Frequency of output waveform:
(no. of cycle per second)

$$\therefore f_o = f_{in}$$

e. Ripple factor (rf)

$$r.f = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$$

$$= \sqrt{\left(\frac{\frac{I_m}{2}}{\frac{I_m}{\pi}}\right)^2 - 1}$$

$$= 1.21$$

f. Transformer Utilization factor (TUF)

It is defined as the ratio of power delivered to the load to the AC rating of transformer secondary.

$$TUF = \frac{\text{Power delivered to load}}{\text{Secondary ac rating of Transformer}}$$

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$$= \frac{I_{dc}^2 R_L}{V_{rms} I_{rms}}$$

$$= \frac{\left(\frac{I_m}{\pi}\right)^2 R_L}{\frac{V_m}{\sqrt{2}} \times \frac{I_m}{2}}$$

$$= \frac{\frac{I_m^2}{\pi^2} \times R_L}{\frac{I_m (r_d + R_L)}{\sqrt{2}} \times \frac{I_m}{2}}$$

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since $r_d + R_L \approx R_L$

$$\therefore TUF = \frac{\left(\frac{I_m}{\pi}\right)^2 R_L}{\frac{I_m R_L}{\sqrt{2}} \times \frac{I_m}{2}}$$

$$= \frac{2\sqrt{2}\, I_m^2 R_L}{\pi^2 I_m^2 R_L}$$

$$= 0.286$$

## Advantages & Disadvantages of Half-wave rectifier

### Advantages

Circuit is simple.

cost is low.

### Disadvantages

Its rectification efficiency is low i.e 40.6

Ripple factor is high i.e 1.21

Transformer utilization factor is quite low i.e 0.28

Average voltage or current is also low.

Therefore, this type of rectifier is not satisfact for DC power supply.

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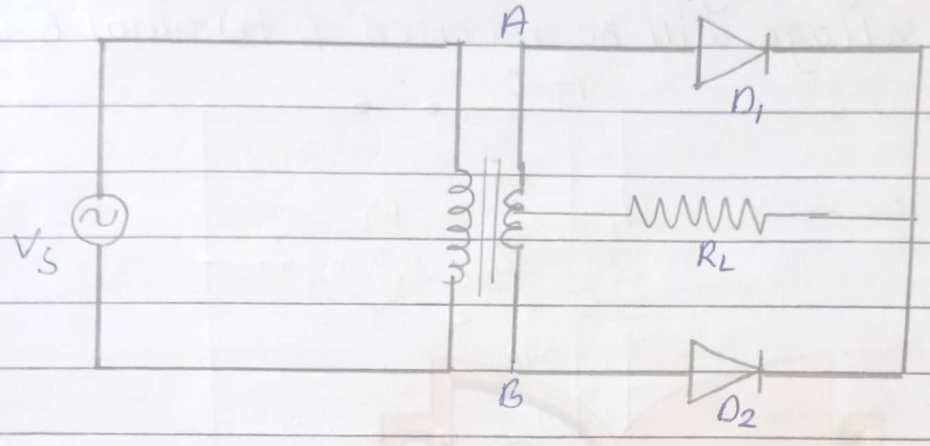
2. Full wave rectifier

It has two types.

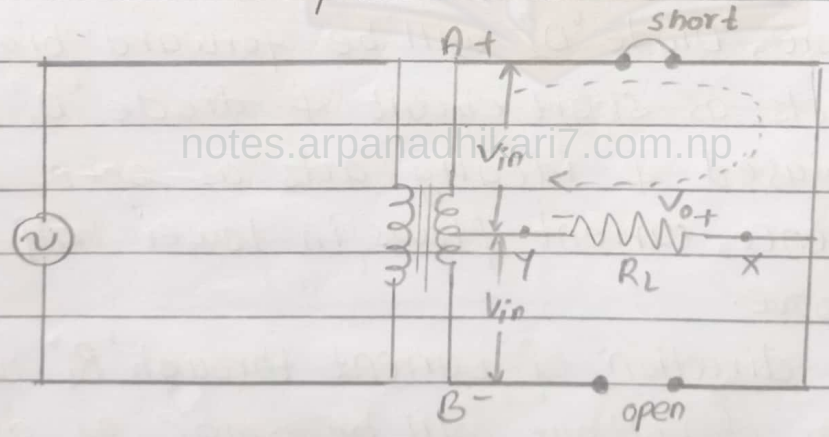
1. Center tapped full wave rectifier
2. Full wave bridge rectifier

1. Center tapped full wave rectifier




Full wave rectifier consists of center tapped transformer and two diode as shown in figure.

During positive half of the input, let terminal A is positive with respect to B.

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Diode '$D_1$' will be forward biased and ideally acts as short circuit and diode '$D_2$' will be reverse biased and ideally acts as open circuit. So, current will circulate on upper half as shown in figure above. The direction of current through $R_L$ will be X to Y and the polarity of voltage '$V_o$' will be as shown in fig above.

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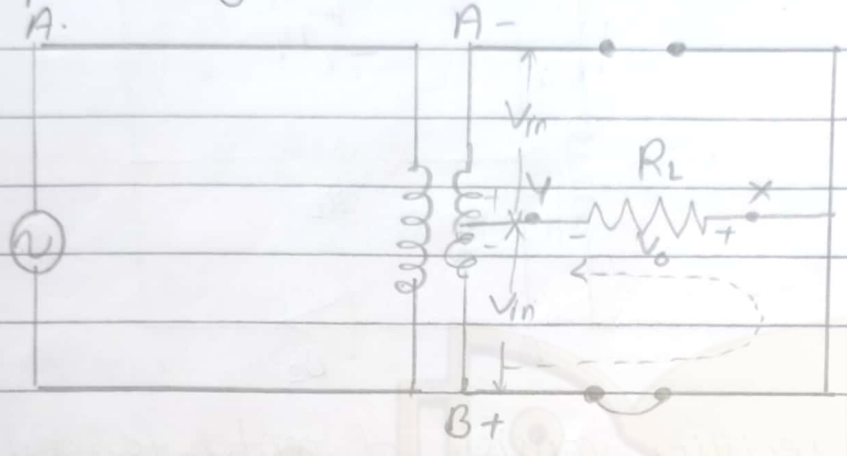
~~$V_{in} = V_m \sin \omega t$~~

Applying KVL

$V_{in} - V_o = 0$

$\therefore \boxed{V_o = V_{in}}$ —①

During negative half of the input, the polarity of input voltage will be reversed & terminal B will be positive wrt A.




Applying KVL

$V_{in} - V_o = 0$

$V_o = V_{in}$

Therefore, Diode '$D_2$' will be forward biased and ideally acts as short circuit & diode '$D_1$' will be reverse biased & ideally acts as open circuit.

Hence, current flows in lower half as shown in figure above.

The direction of current through '$R_L$' will be X to Y & polarity of voltage will be same as during positive half of the input.

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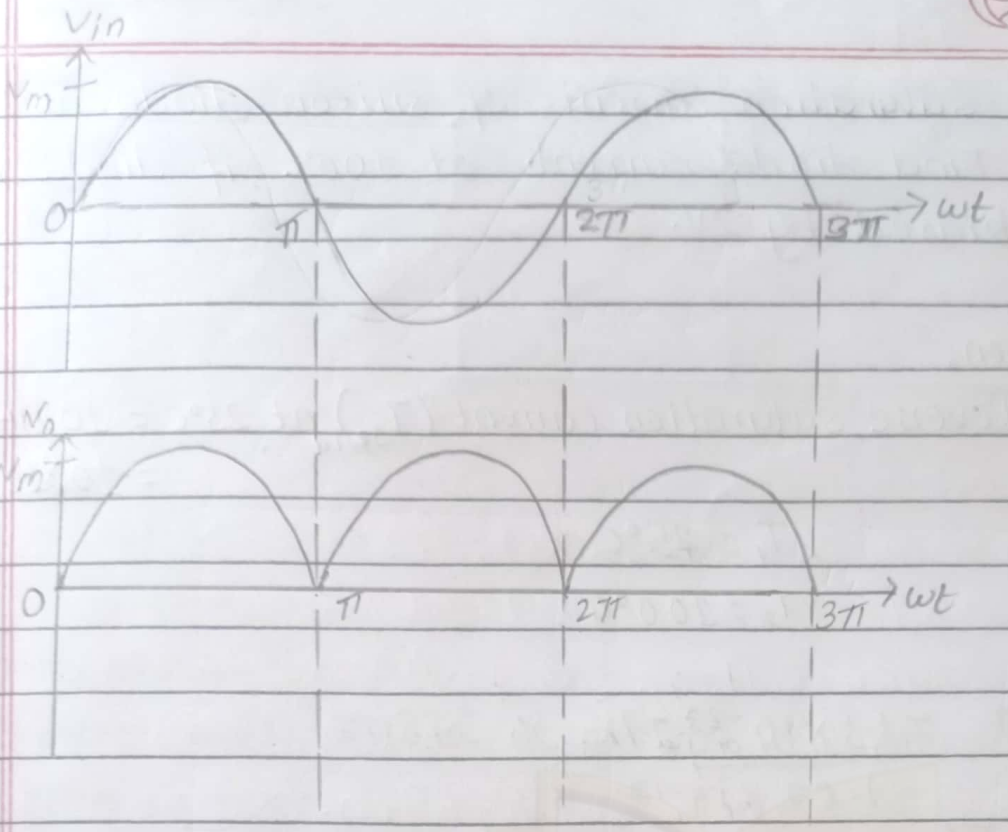


A. Average value of output waveform:

$$(V_o)_{avg} = \frac{1}{T}\int_0^T V_o(\omega t)\, d\omega t$$

$T = \pi$

$$\therefore (V_o)_{avg} = \frac{1}{\pi}\int_0^{\pi} V_m \sin\omega t \, d\omega t$$

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$$= \frac{V_m}{\pi}\int_0^{\pi} \sin\omega t \cdot d\omega t$$

$$= \frac{V_m}{\pi}\left[-\cos\omega t\right]_0^{\pi}$$

$$= \frac{V_m}{\pi}\left[-\cos\pi - (-\cos 0)\right]$$

$$= \frac{V_m}{\pi}\left[-(-1) - (-1)\right]$$

$$\therefore \boxed{(V_o)_{avg} = \frac{2V_m}{\pi}}$$ similarly

$$I_{dc} = \frac{2I_m}{\pi}$$

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swork

Date
Page

1. If reverse saturation current of silicon diode at 25°C is 10 μA. Find diode current at 100°C if diode is forward biased by 2V.

solution,

Here Given,

Reverse saturation current ($I_{sat}$) at 25°C = 10 μA

$= 10 \times 10^{-6}$ Amp

$T_1 = 25°C$

$T_2 = 100°C$

$V_T = \frac{kT}{q} = \frac{1.3 \times 10^{-23} J/k \times 298 \cdot 15}{1.67 \times 10^{-19}}$

$= 231.97 \times 10^{4} = 2.319 \times 10^{6}$ ✓

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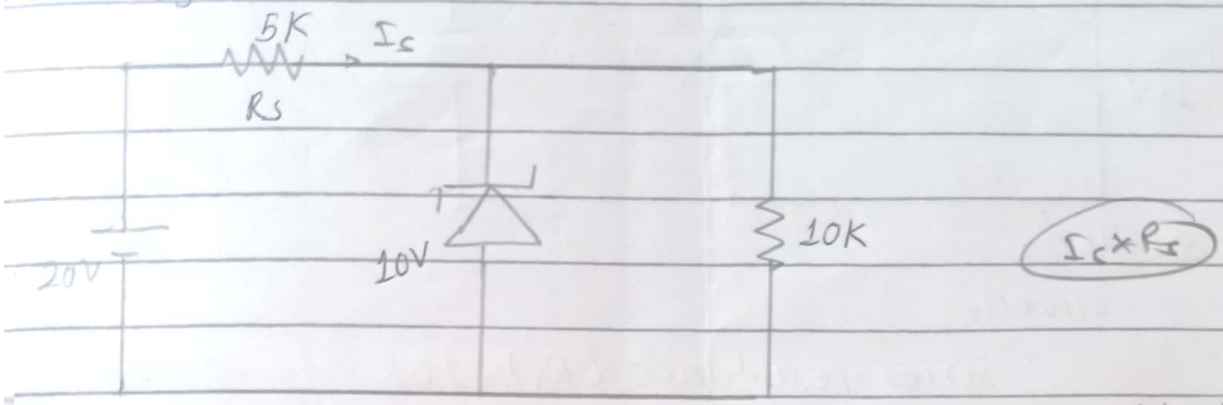
Find load voltage, current through zener diode and voltage across $R_s$.




solution, Here,

series resistance ($R_s$) = 5K

load resistance ($R_L$) = 10K

zener voltage ($V_z$) = 10V

source " ($V_{in}$) = 20V

load voltage ($V_L$) = ?

current through diode ($I_z$) = ?

voltage across $R_s$ ($V_R$) = ?

We know,

If $V_{in} > V_z$, zener diode is ON so,

$V_L = V_z = 10V$

Then,

$$I_z = I - I_L$$
$$= \frac{V_{in} - V_z}{R_s} - \frac{V_L}{R_L}$$
$$= \frac{20-10}{5} - \frac{10}{10}$$
$$= \frac{10}{5} - 1$$
$$= 1 \text{ Amp}$$

$$V_R = I \times R_s$$
$$= 2 \times 5$$
$$= 10V$$

$$I_s = \frac{V_{in} - V_z}{R_s}$$

$$I_s R_s = V_{in} - V_z$$

$$I_s = \frac{V_{in} - V_z}{R_s}$$
$$= \frac{20-10}{5000} = 0.002$$

$$I_s R_s = V_{in} - V_z$$
$$\Rightarrow 0.002 \times 5000 = 20 - 10$$
$$\Rightarrow 10 = 10$$

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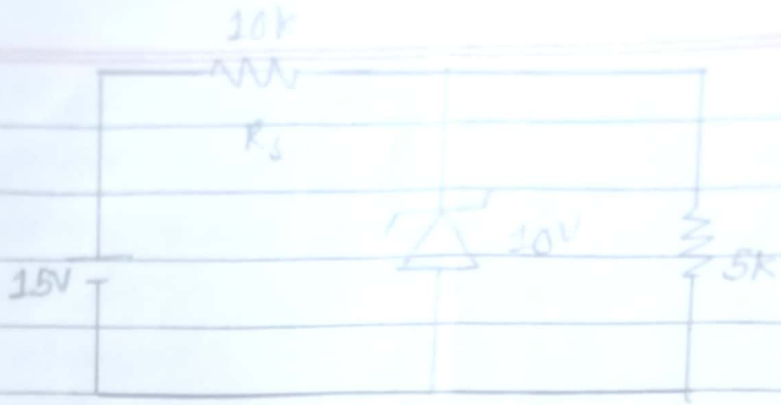


Solution,

Given,

Series resistance ($R_s$) = 10k

Load " ($R_L$) = 5k

Source voltage ($V_{in}$) = 15V

Zener voltage ($V_z$) = 10V

since, $V_{in} > V_z$ zener diode is on. so,

$V_z = V_L = 10V$

$\therefore$ Load voltage ($V_L$) = 10V

current through diode ($I_z$) : $I - I_L$

$$= \frac{V_{in} - V_z}{R_s} - \frac{V_L}{R_L}$$

$$= \frac{15-10}{10000} - \frac{10}{5000}$$

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$$= \frac{5}{10000} - \frac{2}{1000}$$

$$= 0.0005 - 0.002$$

$$= -$$

series voltage ($V_s$) = $I \times R_s$

$$= \frac{5}{10000} \times 10$$

$$= 0.005\ V$$

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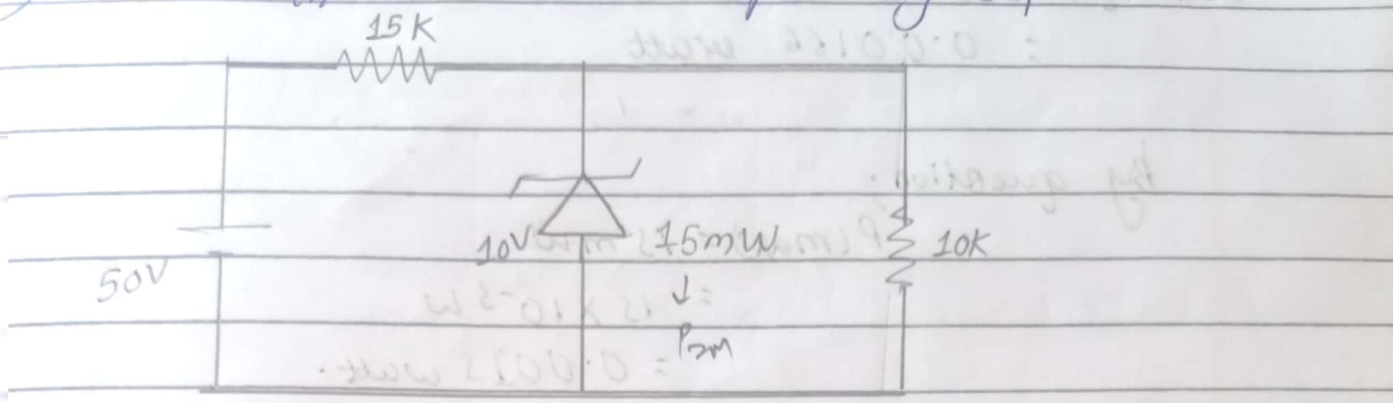
Determine either diode is operating safe or not.




Solution,

Given,

$V_{in} = 50V$

$R_S = 15K = 15000\ \Omega$

$R_L = 10K = 10000\ \Omega$

$V_Z = 10V$

Power across diode ($P_Z$) = 15 mW = $15 \times 10^{-3}$ W

Now,

$$V_Z = \frac{R_L}{R_L + R_S} \times V_{in}$$

$$= \frac{10000}{10000 + 15000} \times 50$$

$$= \frac{10000}{25000} \times 50$$

$$= 20V.$$

Here, $V_{in} > V_Z$ so, zener diode is ON.

calculated

if $P_{ZM} > I_{ZM} V_Z$ Safe

if $P_{ZM} < I_{ZM} V_Z$ not Safe

$$P_Z = I_{ZM} V_Z$$

$$= \frac{V_{in} - V_Z}{R_S} - \frac{V_Z}{R_L}$$

$$= \frac{50 - 10}{15000} - \frac{10}{10000}$$

$$= \frac{4}{1500} - \frac{1}{1000}$$

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$P_2 = 0.00266 - 0.001$

$= 0.00166$ watt

By question;

$P(max) = 15\ mW$

$= 15 \times 10^{-3}\ W$

$= 0.0015$ watt.

Here,

$P(max) < P_2$

Therefore, diode cannot operate safely.

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Rms value of output waveform

$$(V_o)_{rms} = \sqrt{\frac{1}{T}\int_0^T V_o^2(\omega t)\, d\omega t}$$

$$= \sqrt{\frac{1}{\pi}\int_0^{\pi} V_m \sin^2\omega t \cdot d\omega t}$$

$$= \sqrt{\frac{1}{\pi}\int_0^{\pi} V_m \left\{\frac{1-\cos 2\omega t}{2}\right\} d\omega t} \qquad \left[\because \cos 2x = 1 - 2\sin^2 x\right]$$

$$= \sqrt{\frac{V_m^2}{2\pi}\int_0^{\pi} (1-\cos 2\omega t)\, d\omega t}$$

$$= \sqrt{\frac{V_m^2}{2\pi}\int_0^{\pi} 1\, d\omega t - \int_0^{\pi} \cos 2\omega t \cdot d\omega t}$$

$$= \sqrt{\frac{V_m^2}{2\pi}\left[\omega t - \frac{\sin 2\omega t}{2}\right]_0^{\pi}}$$

$$= \sqrt{\frac{V_m^2}{2\pi}\left\{\pi - \frac{\sin 2\pi}{2} - 0 + \frac{\sin 0}{2}\right\}}$$

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$$= \sqrt{\frac{V_m^2}{2\pi}\left\{\pi - 0 - 0 + 0\right\}}$$

$$= \frac{V_m}{\sqrt{2}}$$

$$\therefore (V_o)_{rms} = \frac{V_m}{\sqrt{2}} \qquad \text{——— (1)}$$

Similary

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

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C Rectification efficiency:

$$\eta = \frac{\text{output power (dc)}}{\text{Input power (ac)}} \times 100$$

$$= \frac{I_{dc}^2 R_L}{I_{rms}^2 (r_f + R_L)} \times 100$$

$$= \frac{\left(\frac{2I_m}{\pi}\right)^2 R_L}{\left(\frac{I_m}{\sqrt{2}}\right)^2 (r_f + R_L)} \times 100$$

If $(r_f + R_L) \simeq R_L$

$$\therefore \eta = \frac{\left(\frac{2I_m}{\pi}\right)^2 R_L}{\frac{I_m^2}{2} R_L} \times 100$$

$$= \frac{8}{\pi^2} \times 100$$

$$= 81.05\%$$

D. Riffle factor $= \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$

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$$= \sqrt{\left(\frac{\frac{I_m}{\sqrt{2}}}{\frac{2I_m}{\pi}}\right)^2 - 1}$$

$$= \sqrt{\frac{I_m^2}{2} \times \frac{\pi^2}{4I_m^2} - 1}$$

$$= \sqrt{\frac{\pi^2}{8} - 1}$$

$$= 0.48$$

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E. Transformer utilization factor (TUF):

$$TUF = \frac{\text{Power delivered to load}}{\text{Secondary rating of transformer}}$$

$$= \frac{I_{dc}^2 R_L}{(V_{in})_{rms} \times (I_s)_{rms}}$$

$$= \frac{\left(\frac{2I_m}{\pi}\right)^2 R_L}{\frac{V_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}}}$$

$$= \frac{\left(\frac{2I_m}{\pi}\right)^2 R_L}{\frac{I_m (r_f + R_L) I_m}{2}}$$

If $(r_f + R_L) \approx R_L$

$$TUF = \frac{\frac{4I_m^2}{\pi^2} R_L}{\frac{I_m^2 R_L}{2}}$$

$$TUF = \frac{4I_m^2 R_L}{2\pi^2} \times \frac{2}{I_m^2 R_L}$$

$$= \frac{8}{\pi^2}$$

$$= 0.810$$

In average, overall

Overall TUF $= \frac{2 \times 0.786 + 0.81}{2}$

$= 0.691$.

F. Frequency of output waveform.

$$f_o = 2 f_{in}$$

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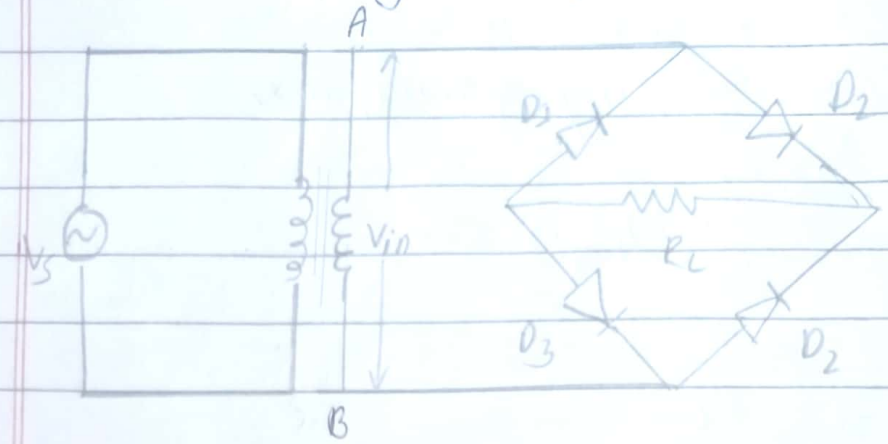
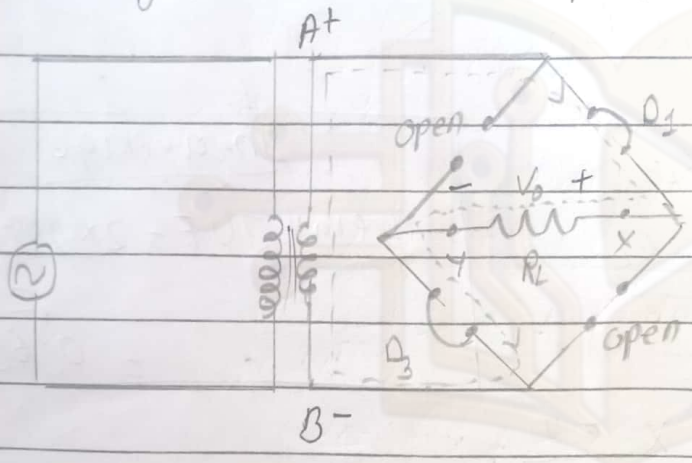
11. Full wave bridge rectifier



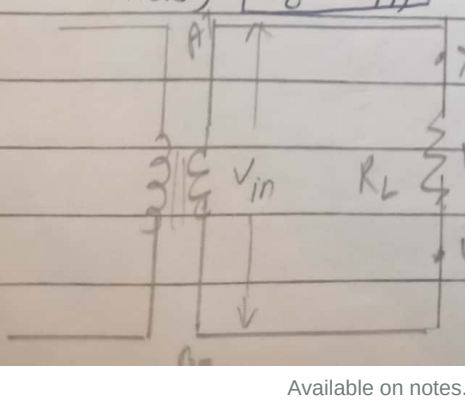

A full wave bridge rectifier consists of four PN junction diode as shown in figure above.

During positive half of input;




During positive half of input, let terminal A of sec. winding of transformer be positive wrt terminal B. In this condition, diode $D_1$ & $D_3$ will be forward biased and ideally acts as short circuit and diode $D_2$ and $D_4$ will be reverse diode & ideally acts as open. Thus, current flows through $D_1$, $R_L$ and $D_3$ as shown in fig above.

Thus; $V_o = V_{in}$

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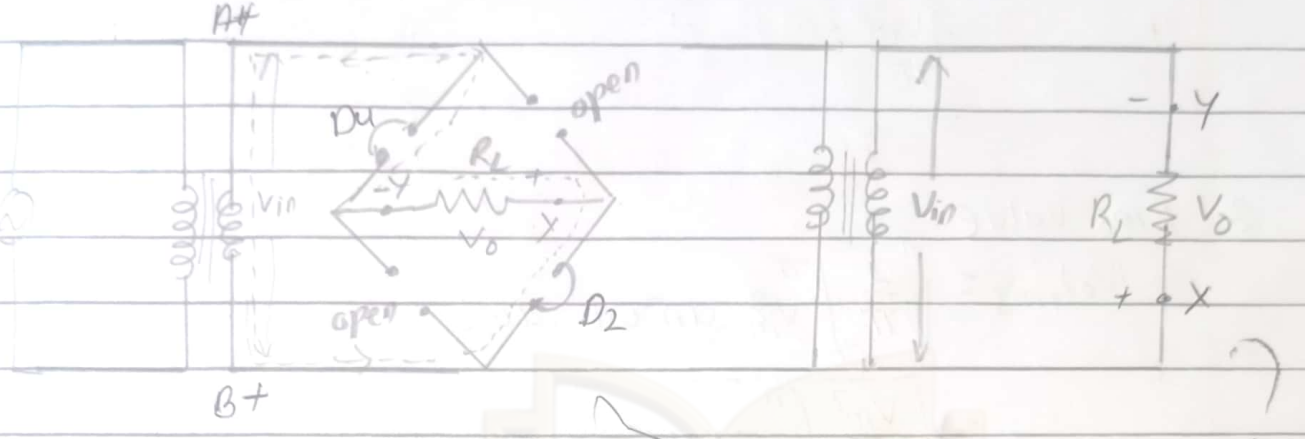
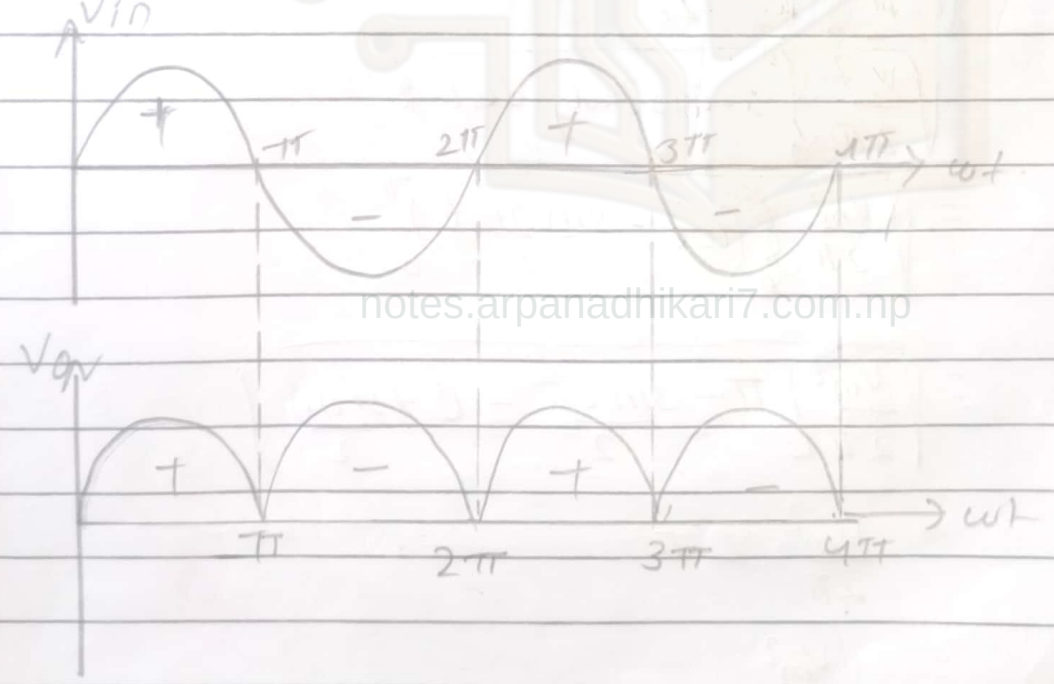
During negative half of the input, the polarity of terminal A will be reversed i.e terminal B will be positive wrt A. Thus, diode $D_2$ & $D_4$ will be forward biased and ideally acts as short circuit and diode $D_1$ and $D_3$ will be reverse diode & ideally acts as open circuit. Thus, current flows from terminal B to diode $D_2$, $R_L$ and diode $D_4$.




Here, the direction of current and hence the polarity of voltage across $R_L$ is same as that during +ve half of input

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1. Average value of output waveform:

$$(V_o)_{avg} = \frac{1}{\pi}\int_0^{\pi} V_m \sin\omega t \cdot d\omega t$$

Time period of $V_o$ =

Different $R_f$ = 2f

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$$(V_o)_{avg} = \frac{V_m}{\pi}\{-\cos\omega t\}_0^{\pi}$$

$$= \frac{V_m}{\pi}(-\cos\pi + \cos 0)$$

$$= \frac{V_m}{\pi}(-(-1) + 1)$$

$$= \frac{2V_m}{\pi}$$

Sampurna Ghimire

Rms Value

$$(V_o)_{rms} = \sqrt{\frac{1}{\pi}\int_0^{\pi} V_m^2 \sin^2\omega t \, d\omega t}$$

$$= \sqrt{\frac{V_m^2}{\pi}\int_0^{\pi} \sin^2\omega t \cdot d\omega t}$$

$$= \sqrt{\frac{V_m^2}{\pi}\int_0^{\pi} \left(\frac{1-\cos 2\omega t}{2}\right) d\omega t}$$

$$= \sqrt{\frac{V_m^2}{2\pi}\int_0^{\pi} (1-\cos 2\omega t)\, d\omega t}$$

$$= \sqrt{\frac{V_m^2}{2\pi}\left(\omega t - \frac{\sin 2\omega t}{2}\right)_0^{\pi}}$$

$$= \sqrt{\frac{V_m^2}{2\pi}\left[\pi - \frac{\sin 2\pi}{2} - 0 + \frac{\sin 0}{2}\right]}$$

$$= \frac{V_m^2}{2\pi} \times \pi$$

$$= \sqrt{\frac{V_m^2}{2}}$$

$$\therefore (V_o)_{rms} = \frac{V_m}{\sqrt{2}}$$

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## Rectification Efficiency ($\eta$):

$$\eta = \frac{(P_o)_{dc}}{(P_{in})_{ac}} \times 100$$

$$= \frac{(I_{dc})^2 R_L}{(I_s)^2_{rms}(2r_d + R_L)} \times 100$$

$$= \frac{\left(\frac{2I_m}{\pi}\right)^2 R_L}{\left(\frac{I_m}{\sqrt{2}}\right)^2 \times (2r_d + R_L)}$$

since, $2r_d + R_L \approx R_L$

$$\therefore \eta = \frac{\left(\frac{2I_m}{\pi}\right)^2 R_L}{\left(\frac{I_m}{\sqrt{2}}\right)^2 R_L} \times 100$$

$$= \frac{4I_m^2}{\pi^2} \times \frac{2}{I_m^2} \times 100$$

$$= \frac{8}{\pi^2} \times 100$$

$$= 81.056\%$$

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## Riffle factor (R.f)

$$R\text{-}f = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$$

$$= \sqrt{\left(\frac{\frac{V_m}{\sqrt{2}}}{\frac{2V_m}{\pi}}\right)^2 - 1}$$

$$= \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1}$$

$$= 0.488$$

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Transformer utilization factor:

$$TUF = \frac{\text{Power delivered to load}}{\text{Sec. rating of T/F}} \times 100$$

$$= \frac{I_{dc}^2 R_L}{(I_s)_{rms} \times (V_{in})_{rms}}$$

$$= \frac{\left(\frac{2I_m}{\pi}\right)^2 R_L}{\frac{I_m}{\sqrt{2}} (I_s)_{rms} \times (I_s)_{rms} (2r_d + R_L)}$$

$$= \frac{\frac{4I_m^2}{\pi^2} R_L}{\left(\frac{I_m}{\sqrt{2}}\right)^2 R_L}$$

$$= \frac{4I_m^2}{\pi^2} \times \frac{2}{I_m^2}$$

$$= \frac{8}{\pi^2}$$

$$= 0.810$$

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For center tap;

$$TUF = \frac{2 \times 0.286 + 0.81}{2}$$

$$= 0.693$$

Frequency.

$$f_0 = 2f_{in}$$

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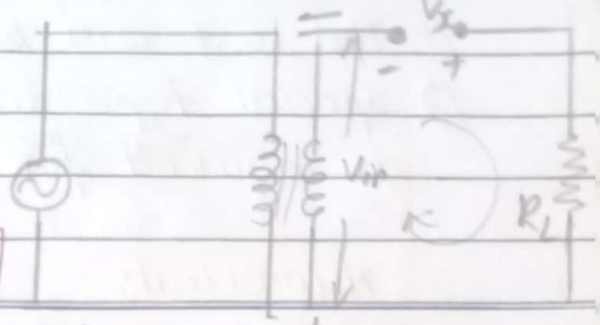
$V_{in} = V_m \sin \omega t$

## Peak inverse voltage (PIV):

It is the maximum voltage across reverse bias diode.

For half wave rectifier,



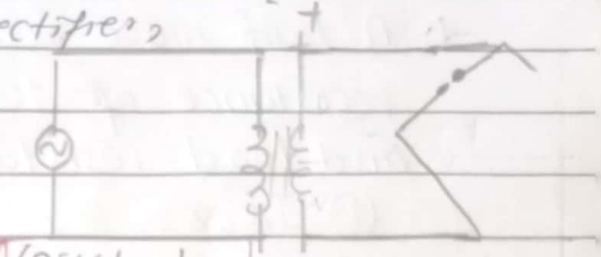

Using KVL

$-V_{in} + V_x = 0$

$V_x = V_{in}$

For half wave, $\boxed{PIV = V_m}$

For full wave bridge rectifier,



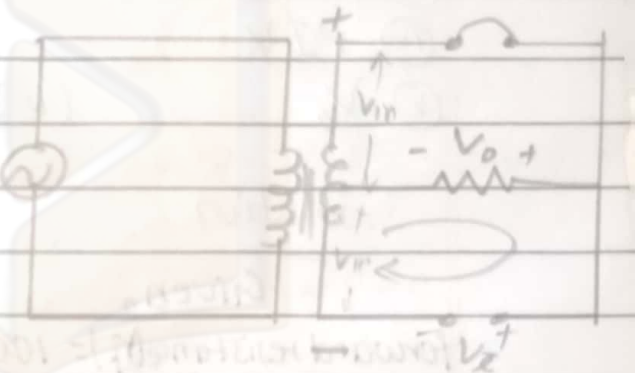
Using KVL,

$V_{in} - V_2 = 0$

$\therefore V_{in} = V_2 = V_o$

For full wave bridge rectifier $\boxed{(PLV) = V_m}$

Center tapped full wave rectifier.




$V_{in} + V_o - V_x = 0$

$\Rightarrow V_o + V_{in} = V_x$

$\Rightarrow V_x = V_{in} + V_{in}$

$\therefore V_x = 2V_{in}$

$\therefore$ For center tap transformer full wave rectifier,

$\boxed{PIV = 2V_m}$

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## Advantages of Full wave rectifier over half wave:

Average & Rms value of output waveform is higher.

Rectification Efficiency is higher in full wave rectifier.

For half wave $\eta = 40.6\%$ whereas in full wave $\eta = 81.05\%$

Ripple factor in full wave rectifier (i.e 0.48) is less than in half wave rectifier (i.e 1.21)

Transformer utilization factor is higher in full-wave bridge (i.e 0.810) as compared to half-wave rectifier (0.286) whereas for center tapped full wave TUF = 0.691.

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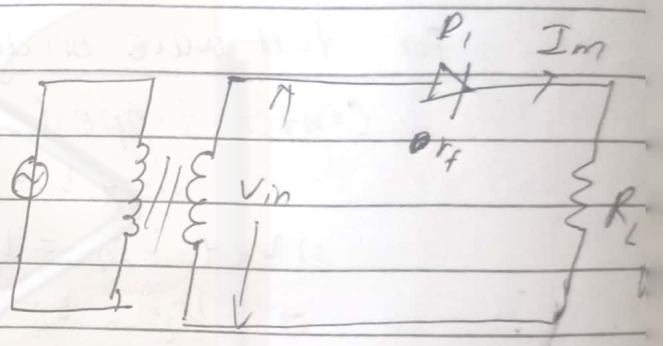
## Disadvantages of full wave rectifier:

1. Circuit is complex as compared to half wave.
2. The cost will be higher.
3. Special type transformer i.e Center tapped transformer is required for center tap full wave rectifier.

## Numericals.

1. A half wave rectifier uses a diode with a forward resistance of 100Ω. If the input voltage is 200V and load resistance is 2kΩ Determine;
   (i) $I_{max}$
   (ii) $I_{dc}$
   (iii) $I_{rms}$
   (iv) PIV
   (v) $V_o$
   (vi) $P_{dc}$
   (vii) $\gamma$-f
   (viii) TUF
   (ix) η




Solution

Given,

forward resistance($r_f$) = 100Ω

Input voltage($V_{in}$) = 200V

load resistance($R_L$) = 2kΩ = 2000Ω

Then,

(i) $I_{max}$ = ?

$$I_m = \frac{V_m}{r_f + R_L}$$

$$= \frac{282.84}{100 + 2000}$$

$$= 0.1346 \text{ Amp}$$

$$= 0.1346 \times 1000$$

$$= 134.6 \text{ mA}$$

$V_{in} = V_{out} = 200V$

$V_m = V_{in}\sqrt{2}$

$= 200 \times \sqrt{2}$

$= 282.84V$

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(ii) $I_{dc} = \frac{I_m}{\pi}$

$= \frac{0.09523}{3.14}$ $\quad \frac{0.1346}{3.14}$

$= 0.03033$ Amp. $\quad = 0.04289$ Amp

$= 42.87$ mA

(iii) $I_{rms} = \frac{I_m}{2}$

$= \frac{0.09523}{2}$ $\quad \frac{0.1346}{2}$

$= 0.04761$ Amp. $\quad = 0.0673$ Amp

$= 67.3$ mA

(iv) PIV = $V_m$ = 282.84V

(v) $V_o = V_{dc}$ $\quad V_{dc} = I_{dc} R_L$

$= 42.827 \times 10^{-3} \times 2000 = 85.65V$

(vi) $P_{dc} = I_{dc}^2 R_L$

$= (0.03033)^2 \times 2000$ | $(0.04289)^2 \times 2000$

$= 9.1990 \times 2000$ | $= 3.6791$ watt

$=$

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(vii) $r\text{-}f = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$

$= \sqrt{\left(\frac{0.0673}{0.0428}\right)^2 - 1}$ $\quad = 1.209020$

(viii) $TUF = \frac{I_{dc}^2 R_L}{V_{rms} \times I_{rms}}$

$= \frac{(0.04269)^2 \times 2000}{\frac{V_m}{\sqrt{2}} \times \frac{I_m}{2}}$ $\quad = \frac{3.64487}{\frac{282.84}{\sqrt{2}} \times 134.6}$

$= 0.1353$

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iv) $\eta = \frac{I_{dc}^2 R_L}{I_{rms}^2 (r_f + R_L)}$

$= \frac{(0.04289)^2 \times 2000}{(0.0673.)^2 (100+2000)} \times 100 = \frac{3.67791042}{9.51150}$

$= 0.3868 \times 100$

$= 38.68\%$

$(P_{in})_{ac} = I_{rms}^2 (r_f + R_L)$

$= (0.0673)^2 (100 + 2000)$

$= 9.5115$ watt

$\eta = \frac{P_{dc}}{(P_{in})_{ac}} \times 100$

$= \frac{3.68}{9.512} \times 100$

$= 38.68\%$

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Qn.2 A 220V, 50Hz ac voltage is applied to the prima of 4:1 step down transformer which is used in a bridge rectifier having a load resistance of 1kΩ Assuming ideal diode determine

(i) $V_{dc}$
(ii) $P_{dc}$
(iii) PIV
(iv) $f_o$
(v) $\eta$
(vi) $\gamma$-f

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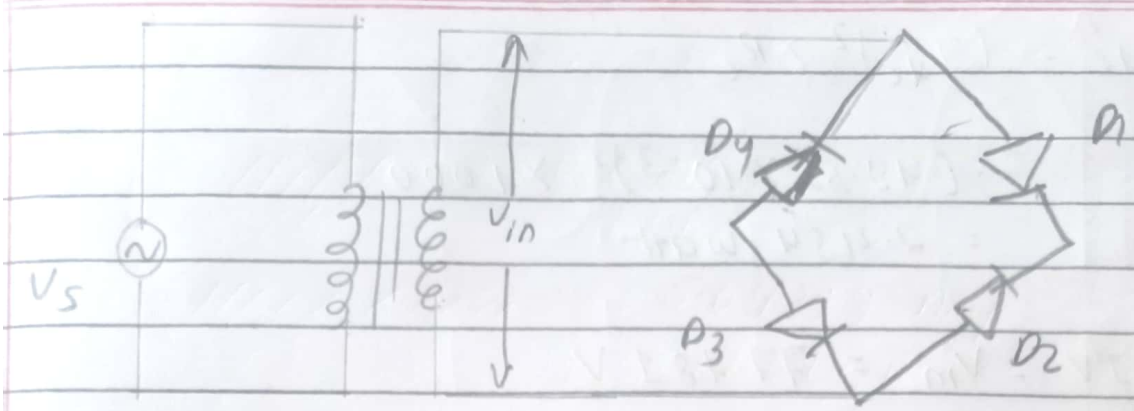
4:1




Solution,

Given,

$V_s = 220\,V$ $\qquad R_L = 1\,k\Omega = 1000\,\Omega$

$f_{in} = 50\,Hz$ $\qquad r_f = 0$ [since by question diode is idealdiode]

| Primary | secondary |
|---|---|
| 4 | 1 |
| 1 | 1/4 |
| 220 | $1/4 \times 220 = 55\,V$ |

$\therefore V_{in} = 55\,V$

$V_m = \sqrt{2}\, V_{in}$

$= \sqrt{2} \times 55$

$= 77.781\,V$

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$I_m = \frac{V_m}{(2r_f + R_L)} = \frac{77.78}{0 + 1000} = 77.781 \times 10^{-3}\,Amp$

$= 77.781\,mA.$

$I_{dc} = \frac{2I_m}{\pi} = \frac{2 \times 77.781}{3.14} = 49.542\,mA$

① $V_{dc} = I_{dc} \times R_L$

$= 49.542 \times 10^{-3} \times 1000$

$= 49.542\,V$

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(iii) $P_{dc} = (I_{dc})^2 \times R_L$

$= (49.52 \times 10^{-3})^2 \times 1000$

$= 2.454$ watt

(iii) $PIV = V_m = 77.781$ V

(iv) $f_o = 2f_{in}$

$= 2 \times 50$

$= 100$ Hz

(v) $(P_{in})_{ac} = I_{rms}^2 (2R_f + R_L)$

$= I_{rms}^2 R_L$

$= \left(\frac{I_m}{\sqrt{2}}\right)^2 \times R_L$

$= \left(\frac{77.781 \times 10^{-3}}{\sqrt{2}}\right)^2 \times 1000$

$= 3.0249$ watt.

$\eta = \frac{(P_o)_{dc}}{(P_{in})_{ac}} \times 100$

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$= \frac{2.45}{3.0249} \times 100$

$= 81.126\%$

(vi) $r\text{-}f = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$

$= \sqrt{\left(\frac{\frac{I_m}{\sqrt{2}}}{\frac{2I_m}{\pi}}\right)^2 - 1}$

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$$r_f = \sqrt{\left(\frac{0.054999}{0.04952}\right)^2 - 1} = \sqrt{1.23354 - 1}$$

$$= \sqrt{0.23354} = 0.4832$$

## Clipper Circuits

Clippers are the network that employ diodes to clip a way, a portion of an input signal without distorting remaining part of applied waveform. It removes a portion of applied wave without distorting other part. Some clipper circuits are ~~positive~~

1. Positive series clipper
2. Positive shunt clipper
3. Negative series clipper
4. Negative shunt clipper
5. Biased clipper.

### Positive series clipper.

The circuit diagram of positive series clipper is shown in figure:

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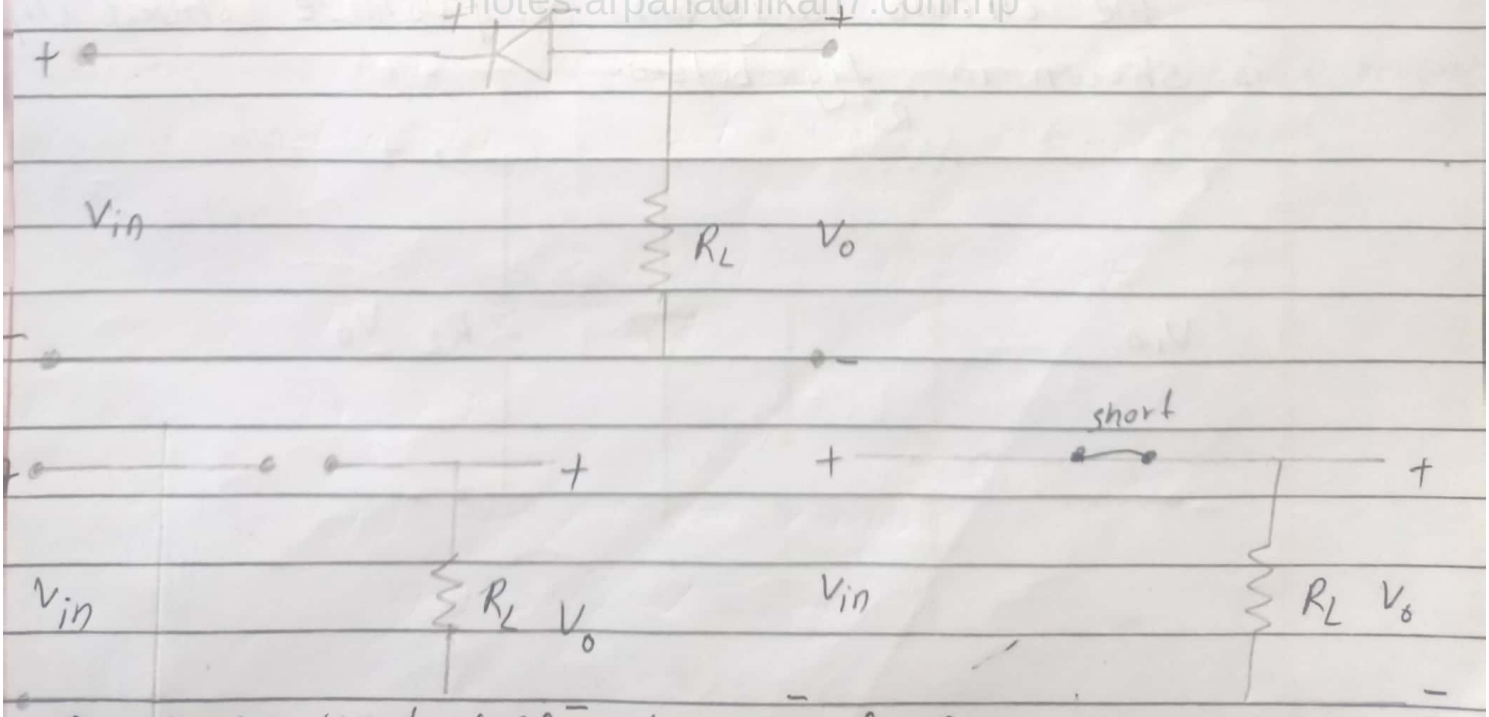


fig: During +ve half of input $V_o = 0$

fig: During -ve half of input $V_o = V_{in}$

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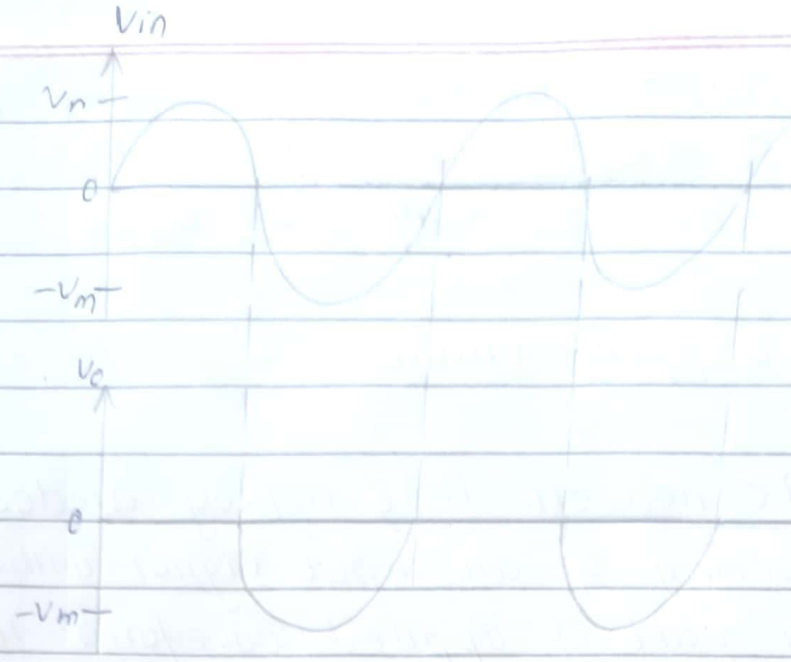


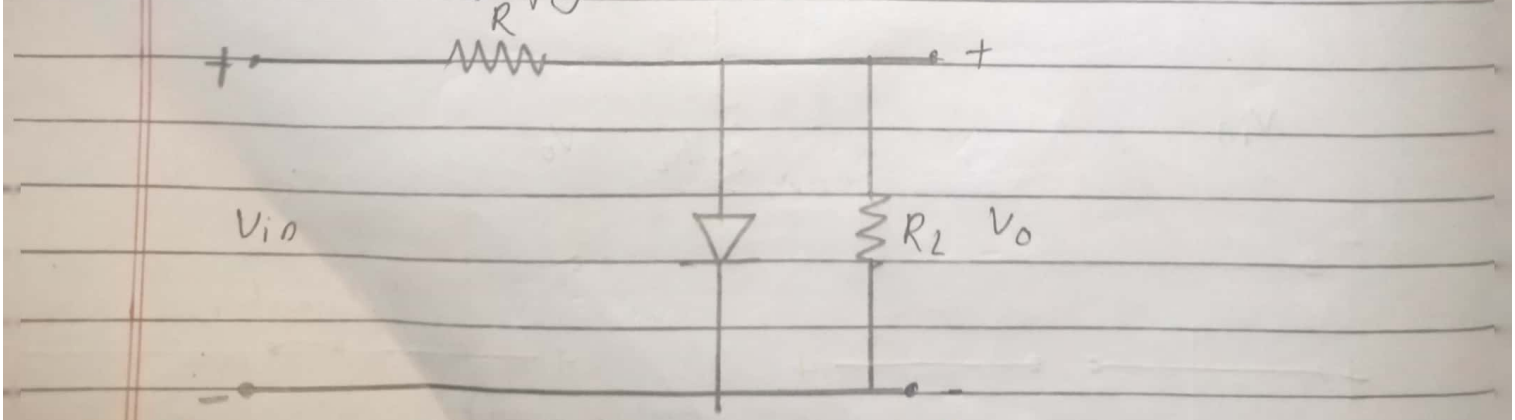

During +ve half of the input, the diode will be reverse biased & ideally acts as open. So, output voltage $V_o =$

During -ve half of the input the diode will forward biased & ideally acts as short ckt. So, output voltage $V_o = V_{in}$

Here, the +ve half of the input is clipped and only -ve half appears across node $R_L$ as shown in fig.

2. Positive shunt clipper

The circuit diagram of positive shunt clipper is shown in fig below.

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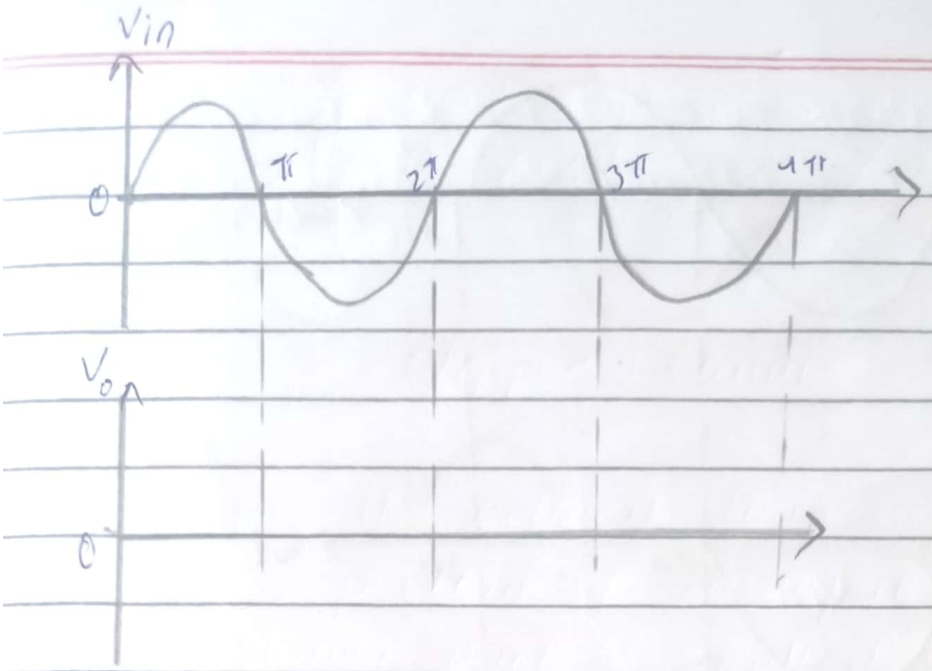
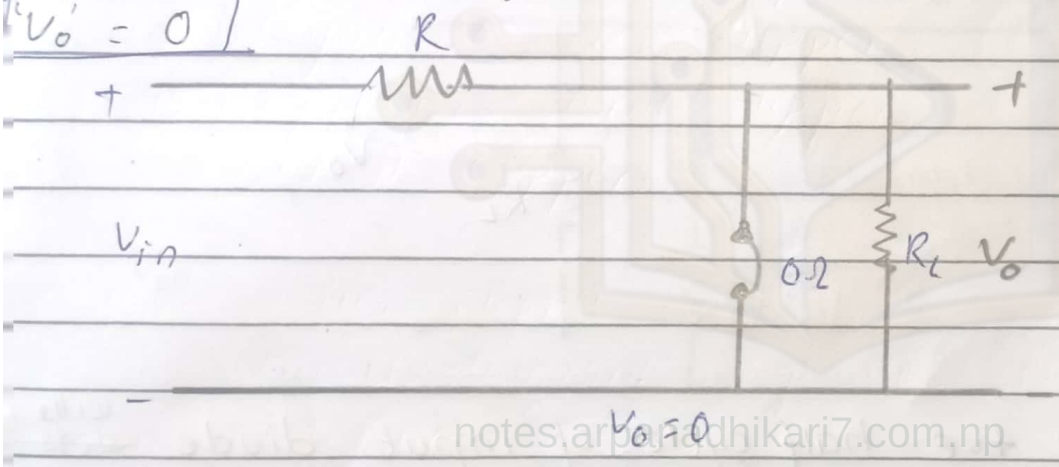



fig: Graph representation of +ve half of input

During +ve half of the input, diode 'D' will be forward biased and ideally acts as short. Therefore, voltage

$V_o = 0$

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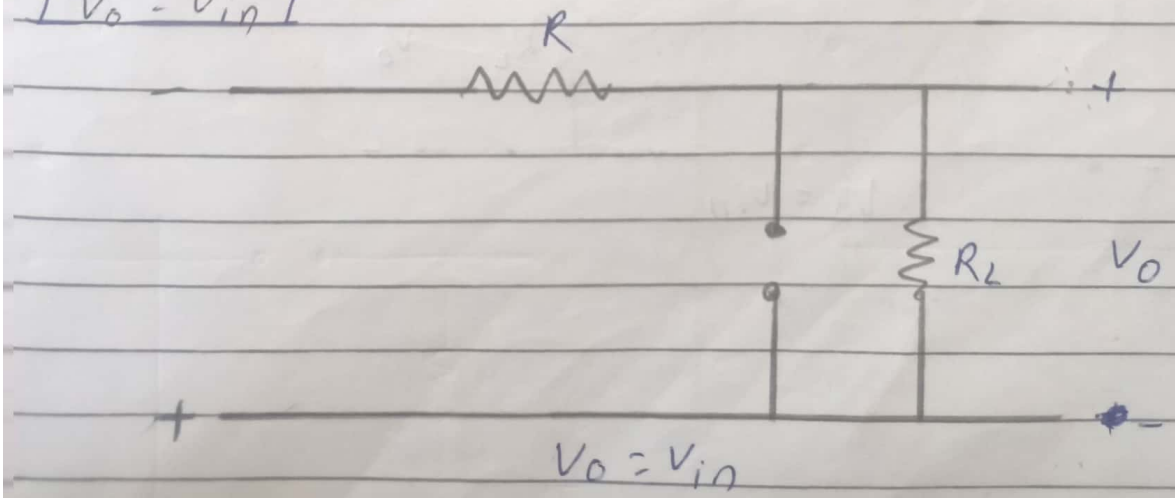
During -ve half of the input, the diode 'D' will be reverse biased and ideally acts as open ckt. Therefore,

$V_o = V_{in}$

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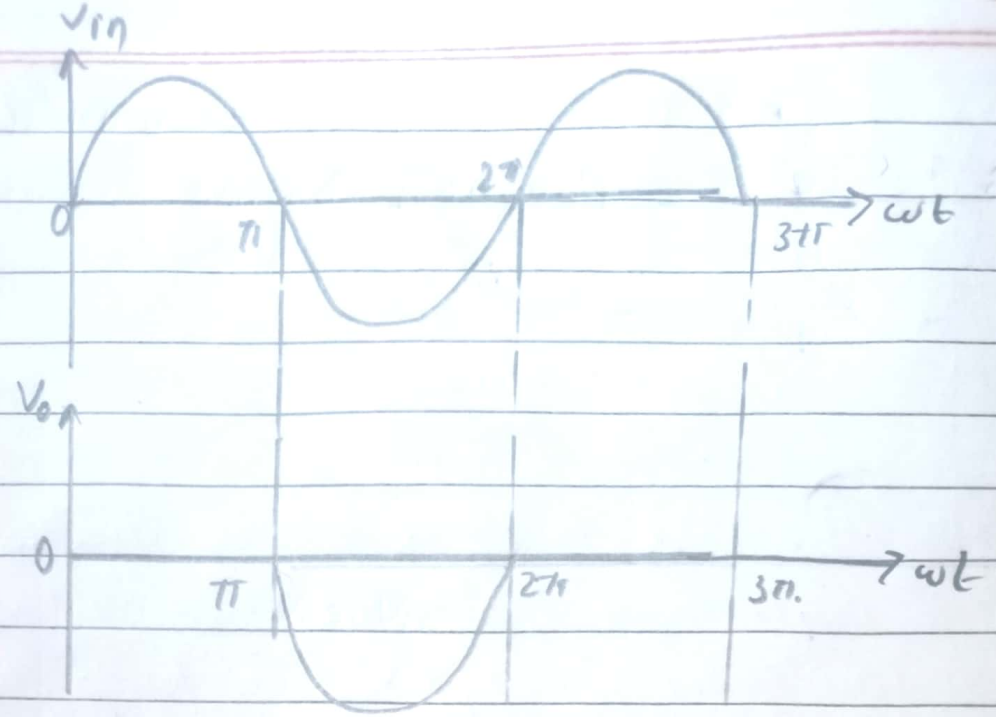
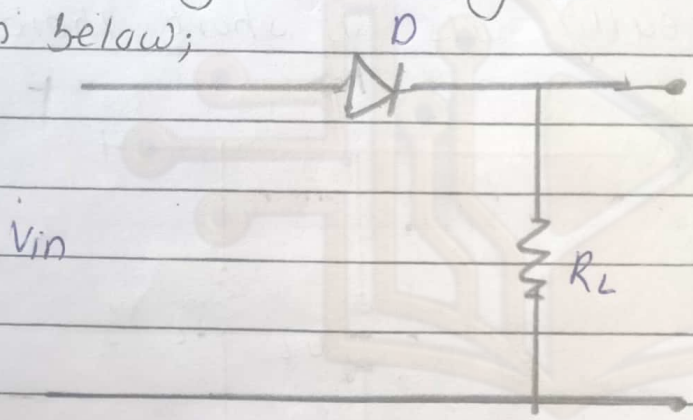



3. Negative series clipper:

The ckt diagram of negative series clipper is shown below;

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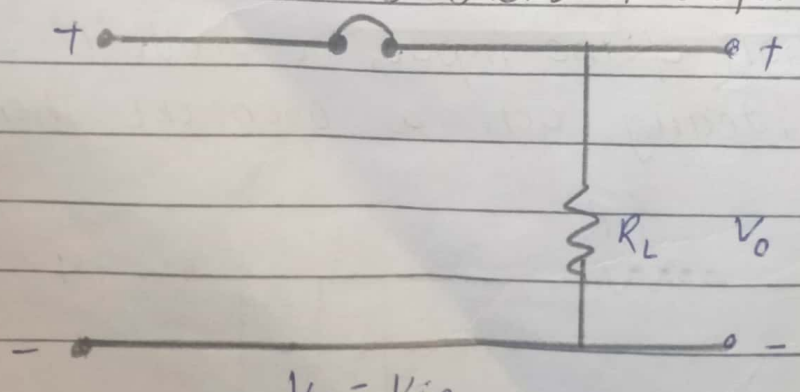
During +ve half cycle, of input diode ~~acts as~~ will be forward biased and acts as short. Therefore, $V_o = V_{in}$.

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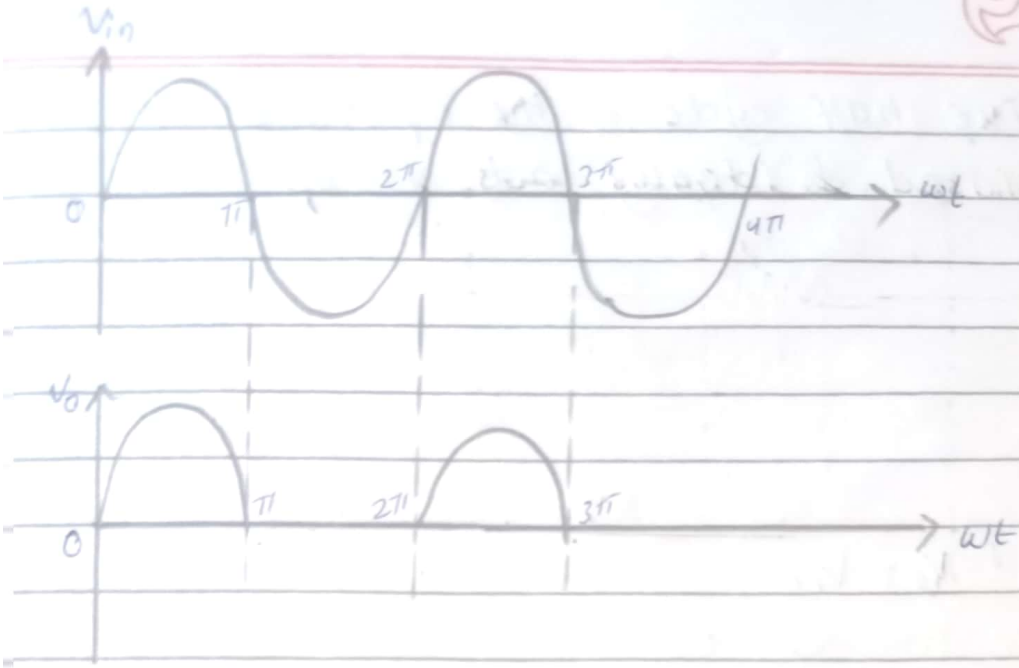


During negative half of the input, diode D will be reverse biased and ideally acts as open ckt. Therefore $\boxed{V_o = 0}$

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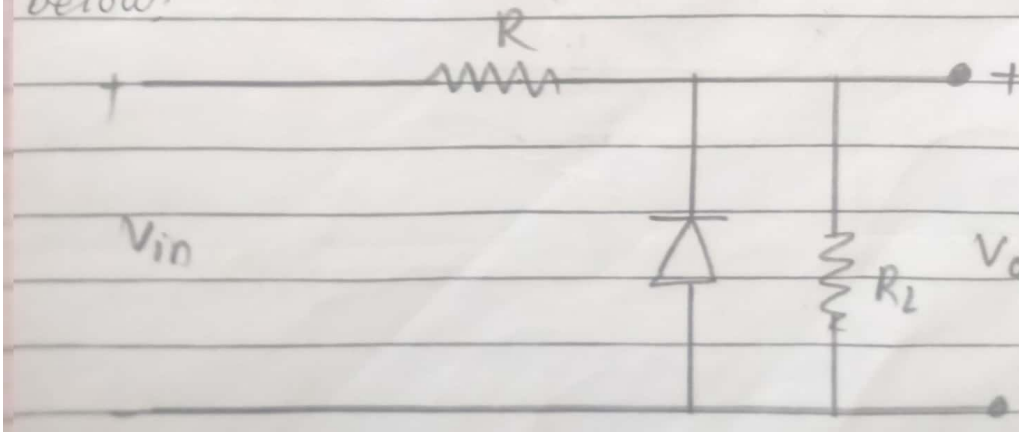
## Negative shunt clipper

The ckt diagram of negative shunt clipper is shown below.

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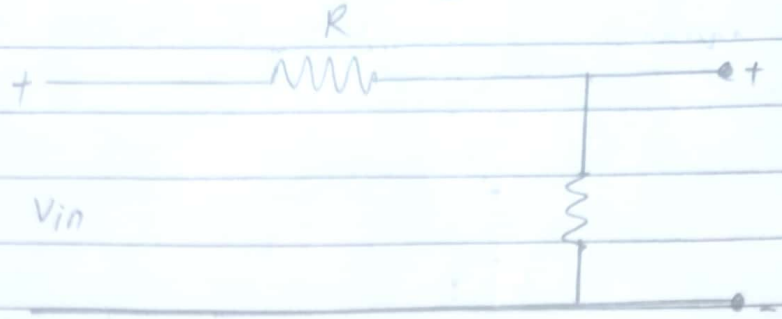
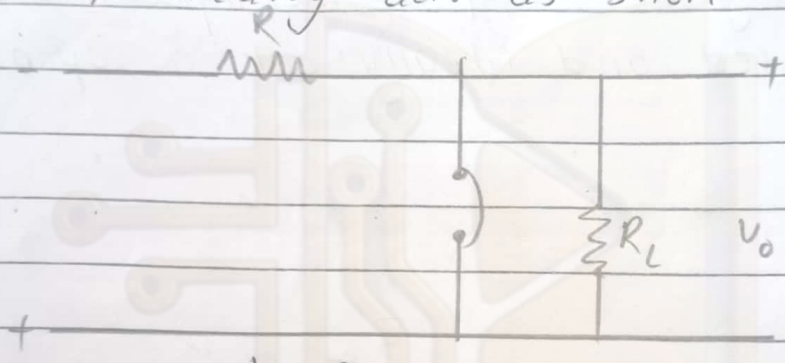
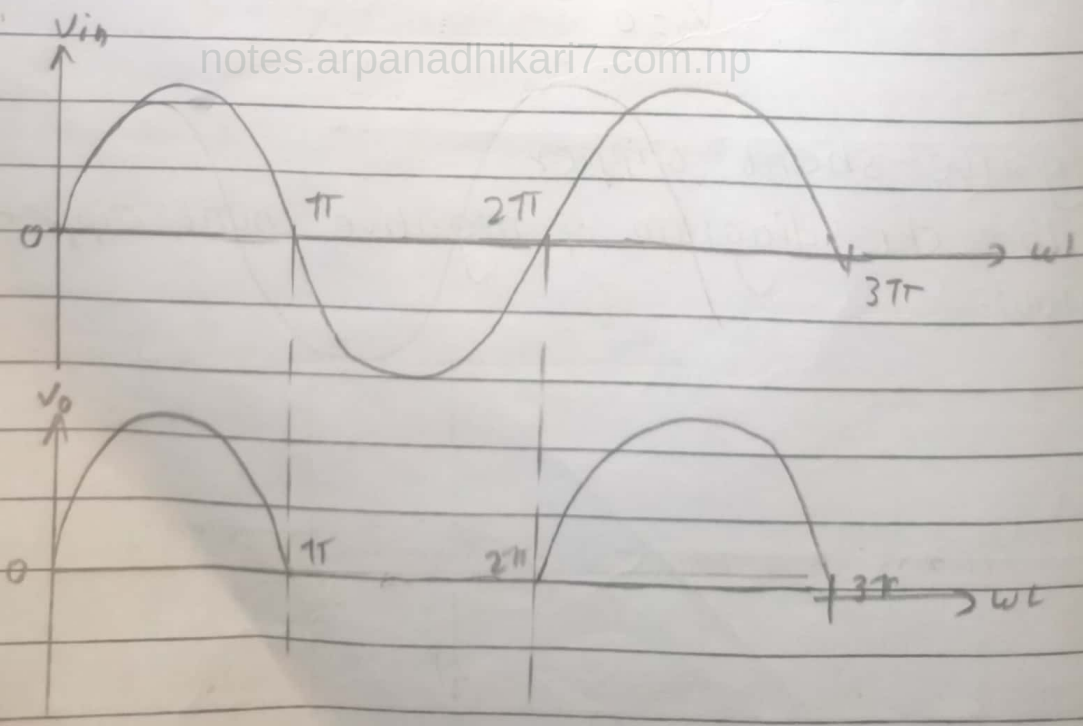
During +ve half cycle of the input, diode D acts as reverse biased & ideally acts as open ckt. Therefore $V_o = V_{in}$.




During -ve half of the input, diode D will be forward biased & ideally acts as short. Therefore, $V_o = 0$

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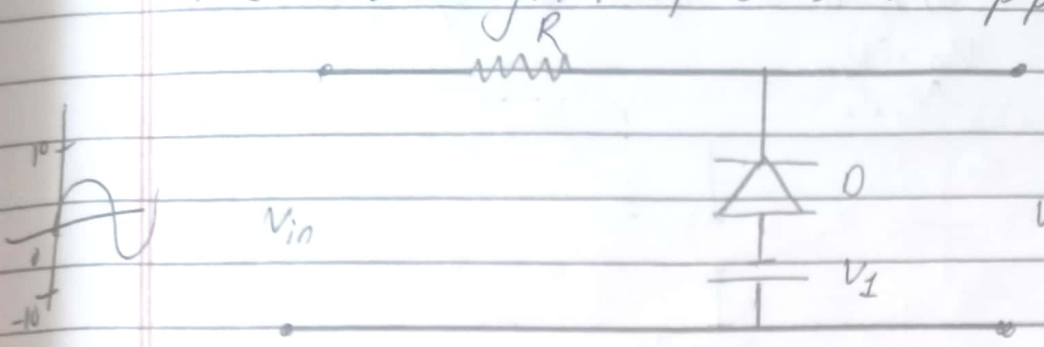
5 Biased clipper

The ckt diagram of biased clipper is shown below:

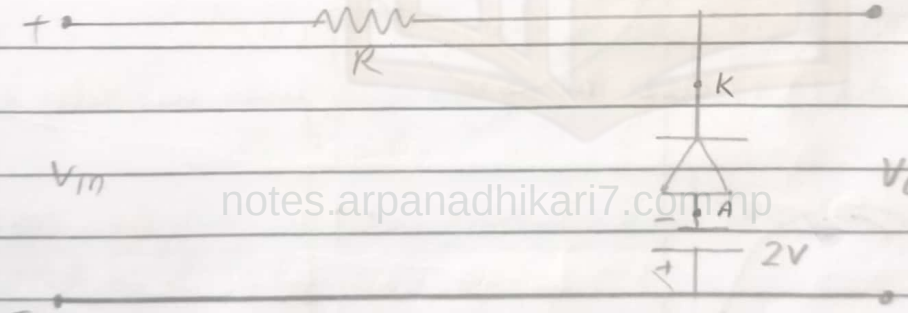



A DC voltage source is present in biased clipper as shown in figure

Note: Points to be considered while solving clipper ckt.

The diode will be on when voltage at anode is greater that voltage at cathode. When diode is on, it acts as short ckt.

Let us consider an example;

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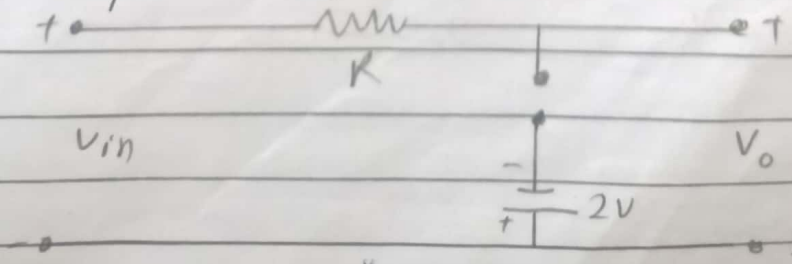
Diode will be ON when voltage at cathode is less that -2V.

During +ve half of input, diode will be OFF and acts as open ckt.




$\therefore V_o = V_{in}$

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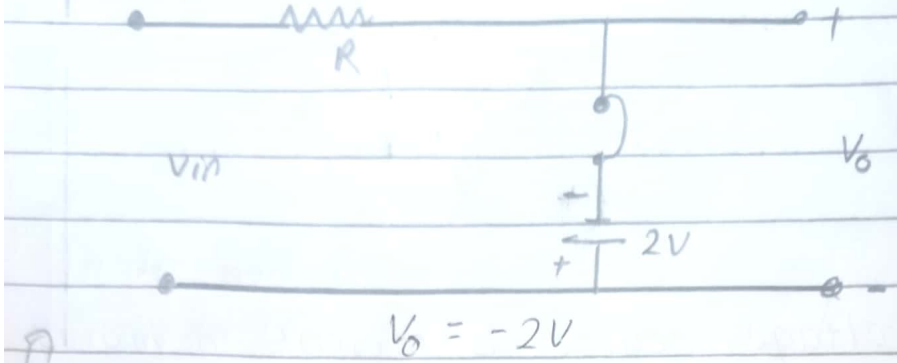
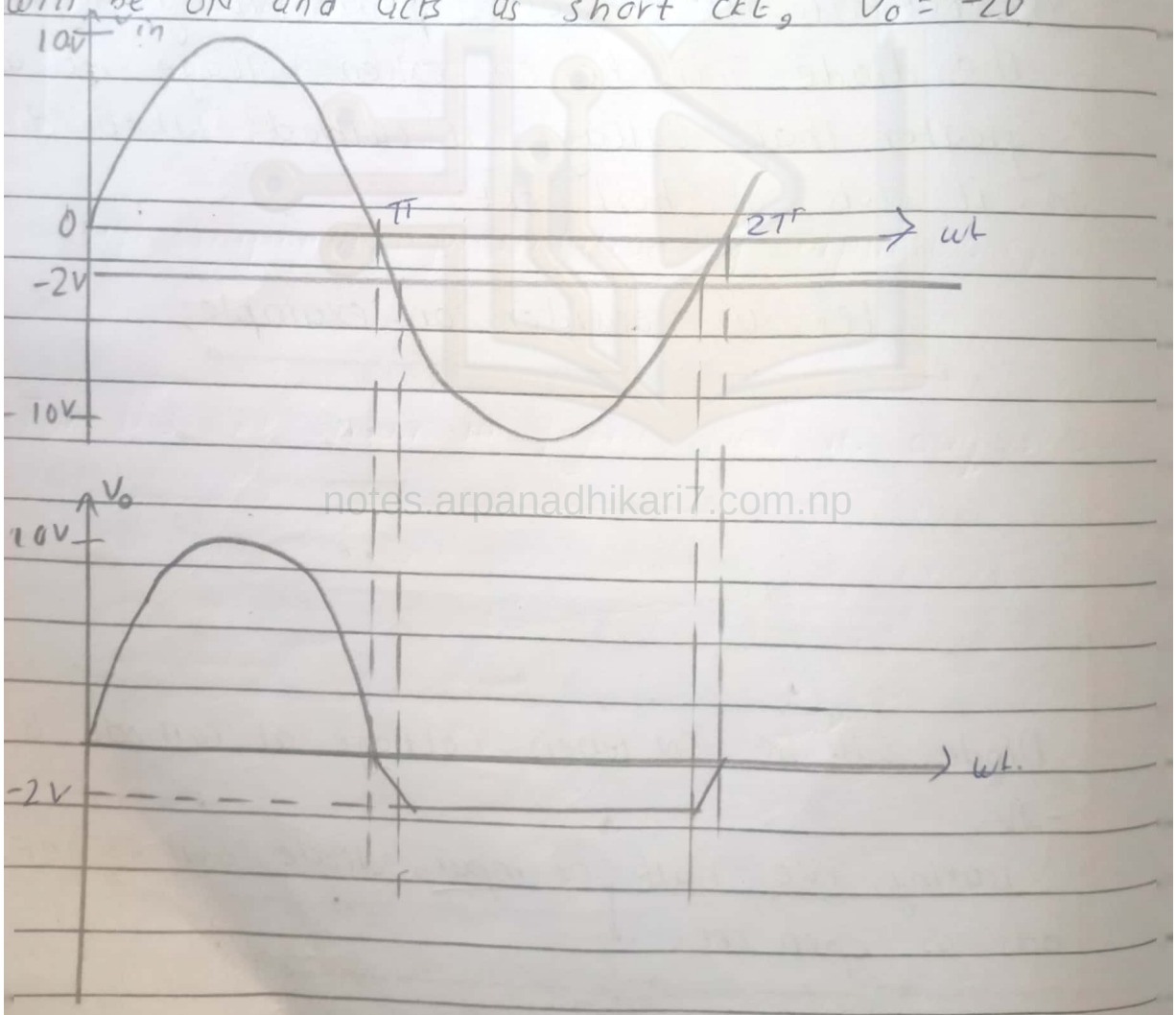
During -ve half, when input varies from 0V, Diode will be OFF and acts as open ckt. So,

$V_o = V_{in}$




$V_o = -2V$

When input varies from -2V to -10V, Diode will be ON and acts as short ckt, $V_o = -2V$

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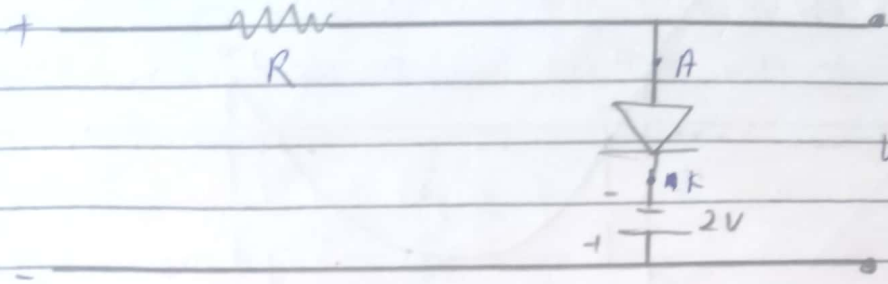
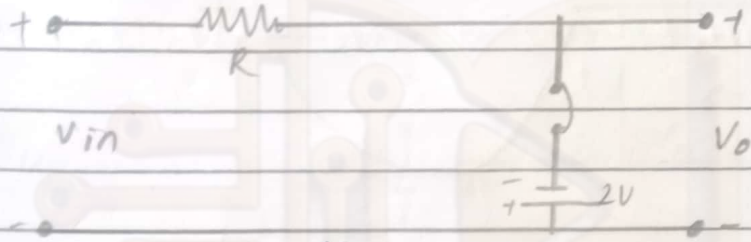
Q:1 For the given clipper ckt, find its ~~oupt~~ output waveform




Diode will be ON, when voltage at anode is greater than -2V.

During +ve half of input, diode will be ON & acts as short ckt.




$\therefore V_o = -2V$

During -ve half of input, when input varies from 0V to -2V, Diode will be ON and acts as short ckt,

$V_o = -2V$

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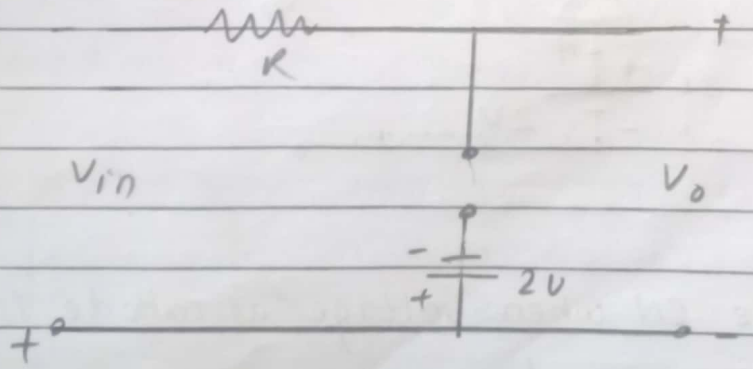
When input varies from -2V to -10V, Diode will be OFF and acts as open ckt.




$V_o = V_{in}$

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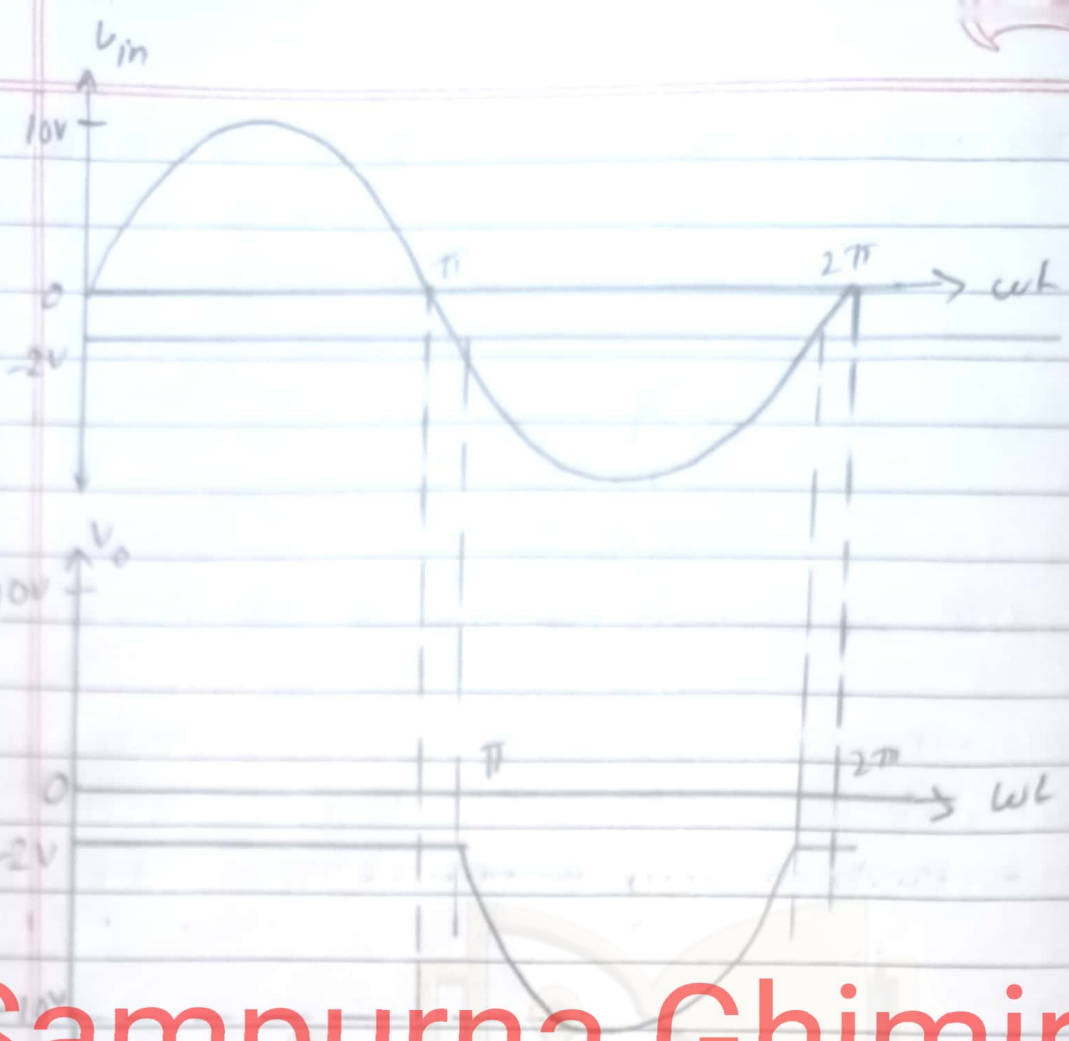


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fig: Graphical representation of output waveform

Ans

Qn2 For the ckt shown in figure, find its output wave

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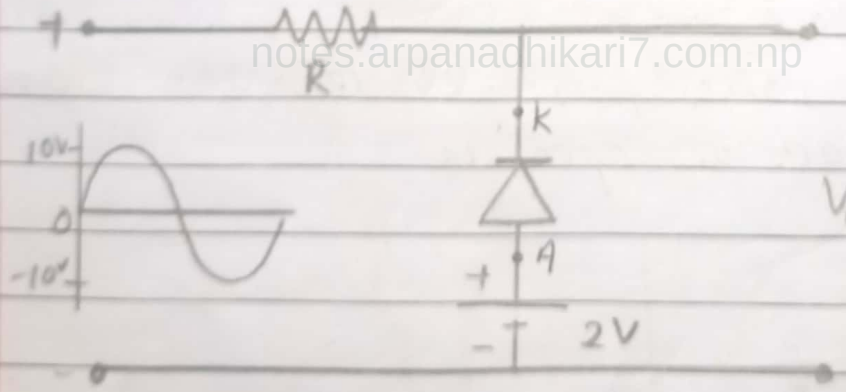


Solution,

Diode will be ON when voltage at cathode is greater than 2V.

During positive half of input, diode will be reverse biased and acts as open ckt.

$\therefore V_o = V_{in}$

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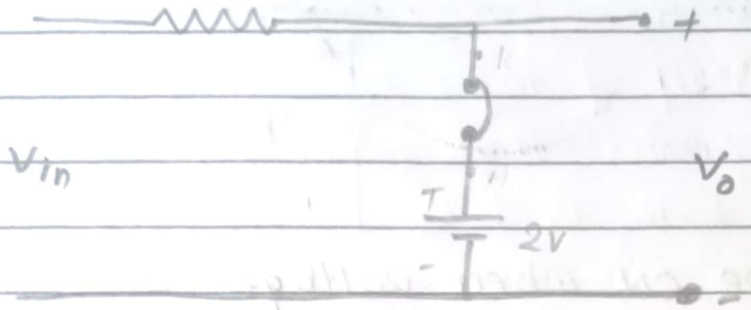
(A) During ~~negative~~ Positive half of input;

(i) When $V_{in}$ is less than 2V ($V_{in} \leq 2V$)

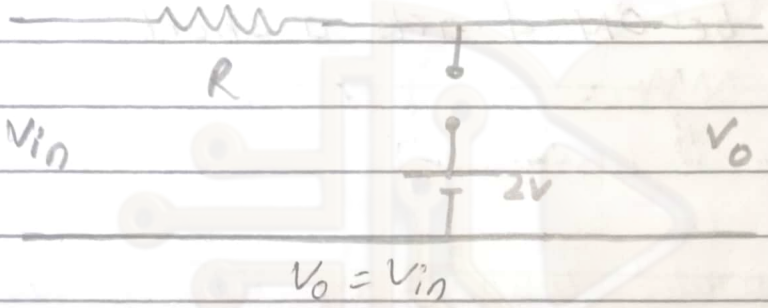
Diode will be ON & acts as short.




$V_o = 2V$

(ii) When $V_{in}$ is greater than 2V ($V_{in} \geq 2V$)

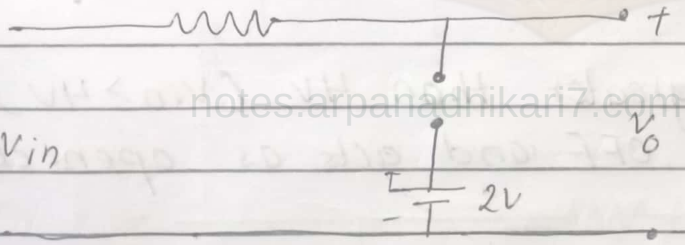
Diode with be OFF and acts as open ckt.




$V_o = V_{in}$

(B) During -ve half of input.

Diode will be ON and acts as short ckt

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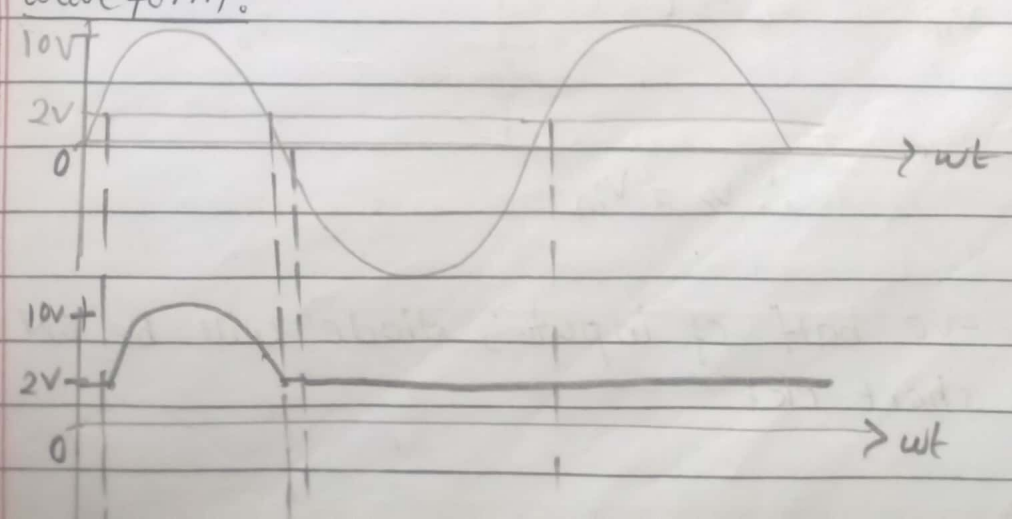
$\therefore V_o = 2V$

Wave form:

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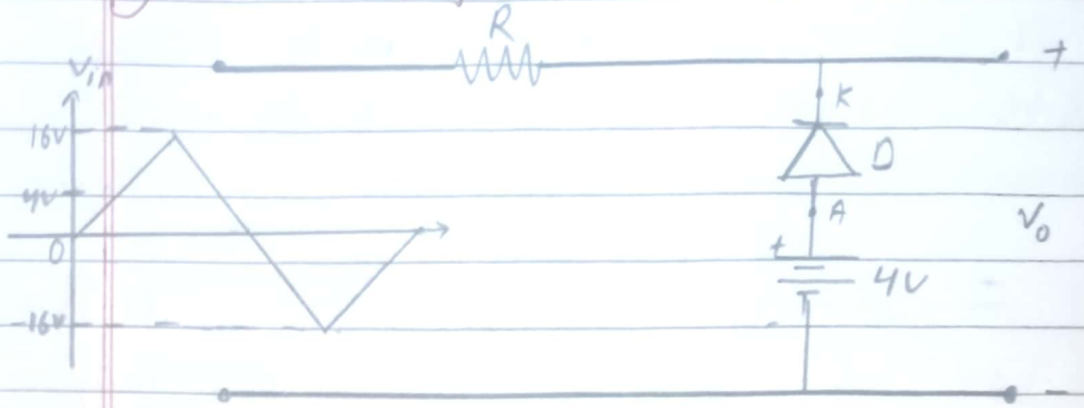
Q.1 Determine $V_o$ for the network shown below;

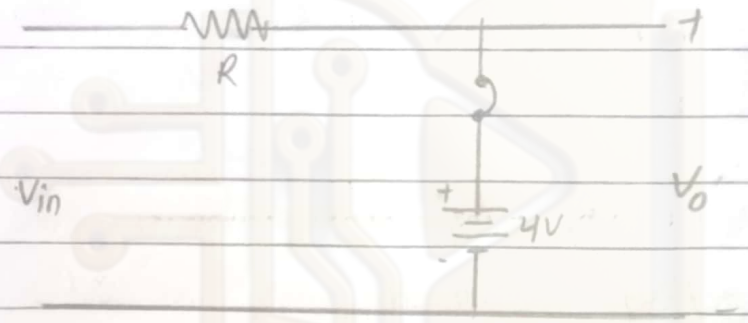



Diode will be ON when voltage at cathode is less than 4V.

During +ve half of input;

(i) When $V_{in}$ is less than 4V ($V_{in} \leq 4V$)

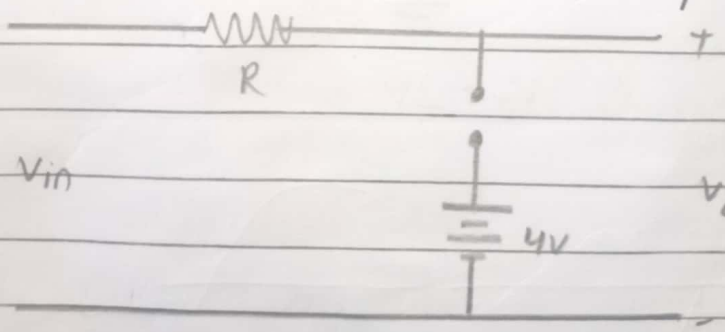
Diode will be ON & acts a short.




$\therefore V_o = 4V$

(ii) When $V_{in}$ is greater than 4V ($V_{in} \geq 4V$)

Diode will be OFF and acts as open ckt.




$\therefore V_o = V_{in}$

During -ve half of input; diode will be ON and acts as short ckt.

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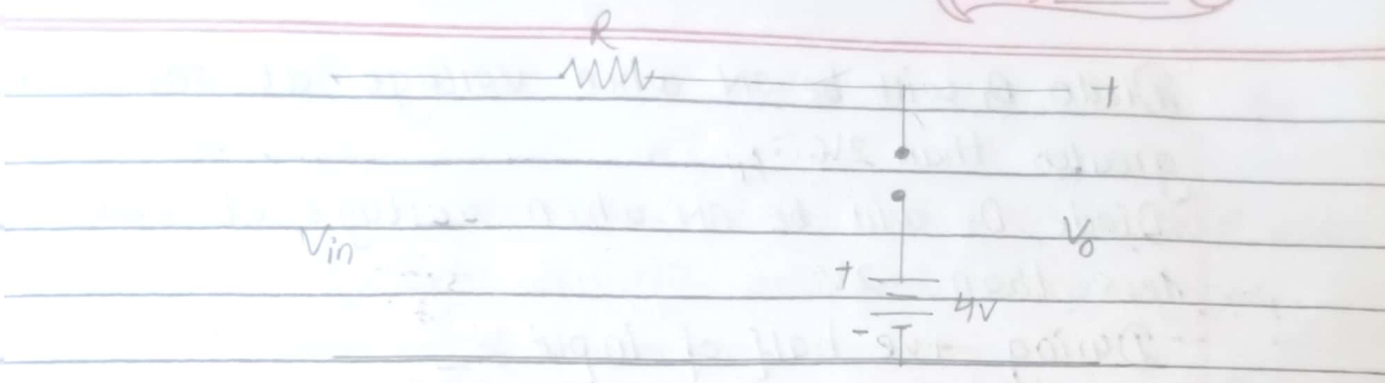
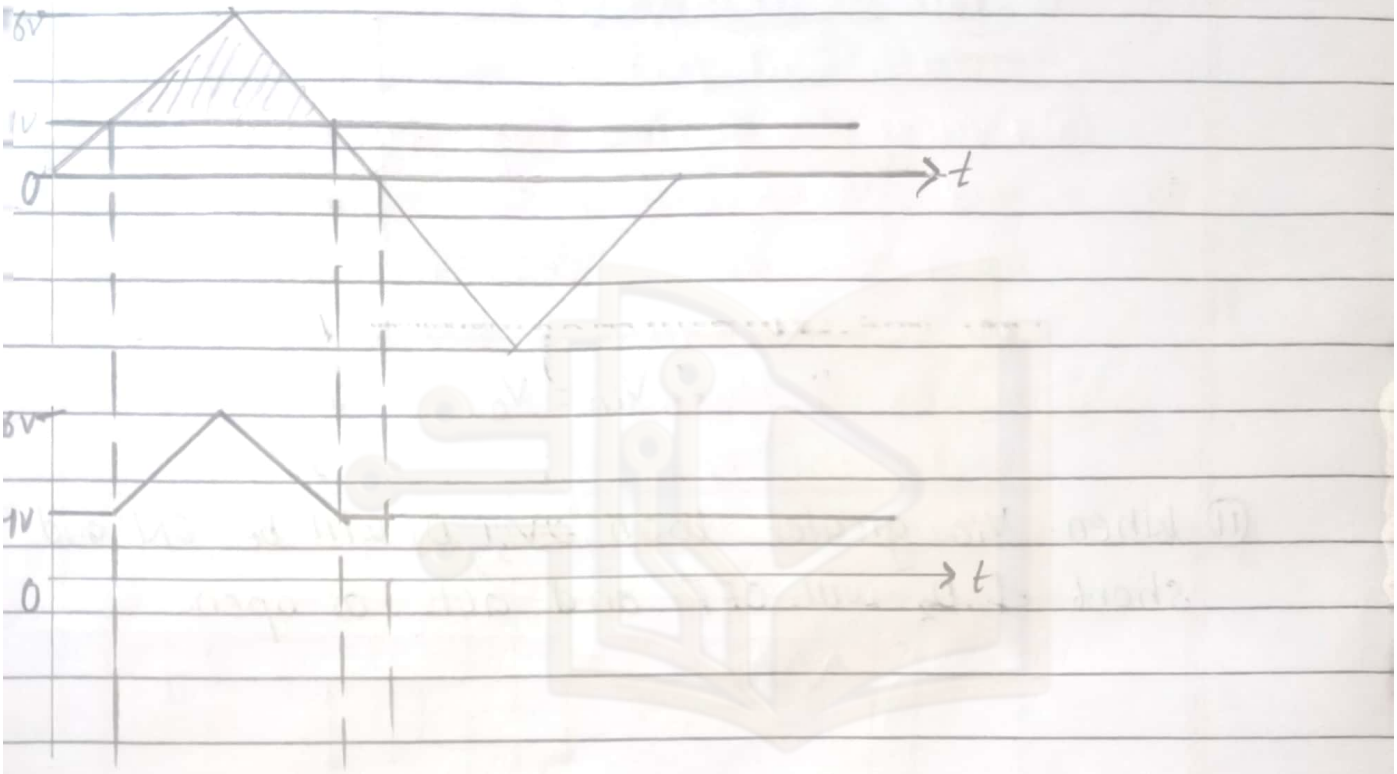



$\therefore V_o = 4V$

waveform:

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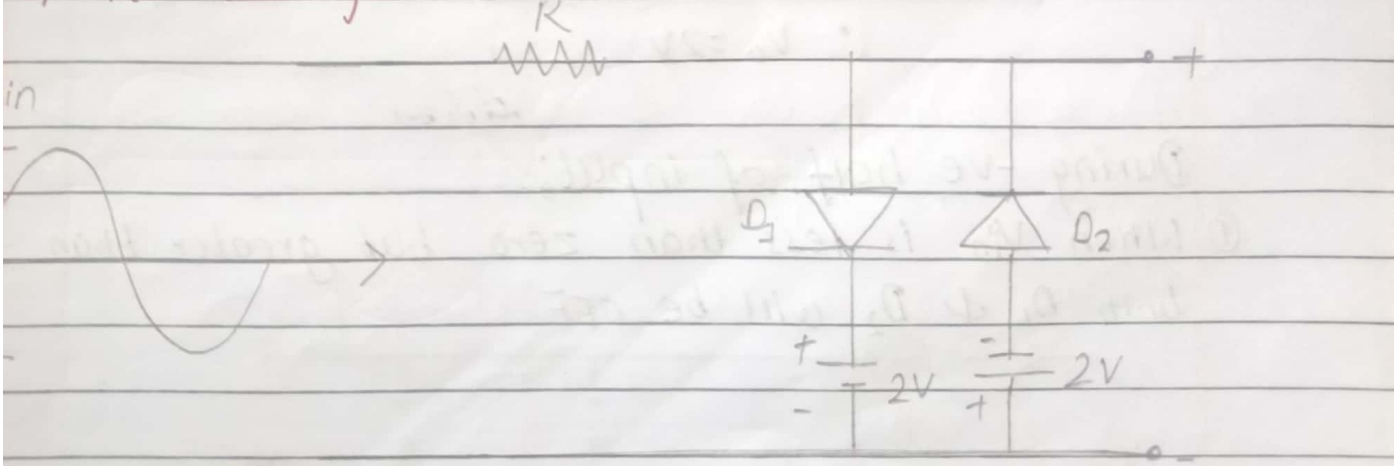
For the given clipper circuit, find output voltage ($V_o$) & its waveform:

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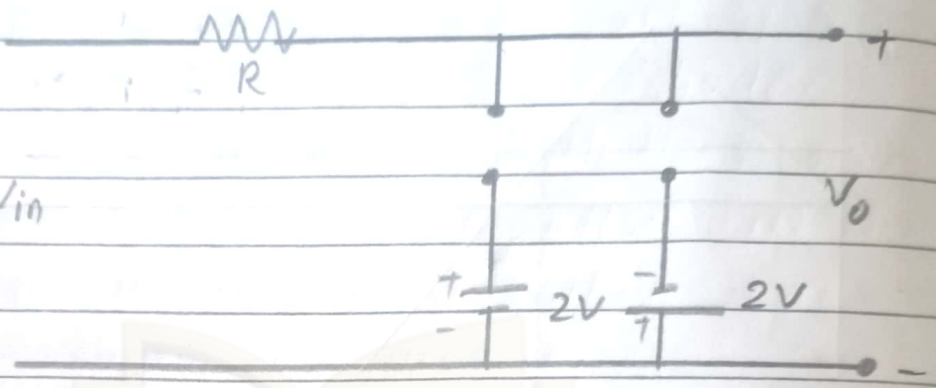
Diode $D_1$ will be ON when voltage at anode is greater than 2V.

Diode $D_2$ will be ON when voltage at cathode is less than -2V.

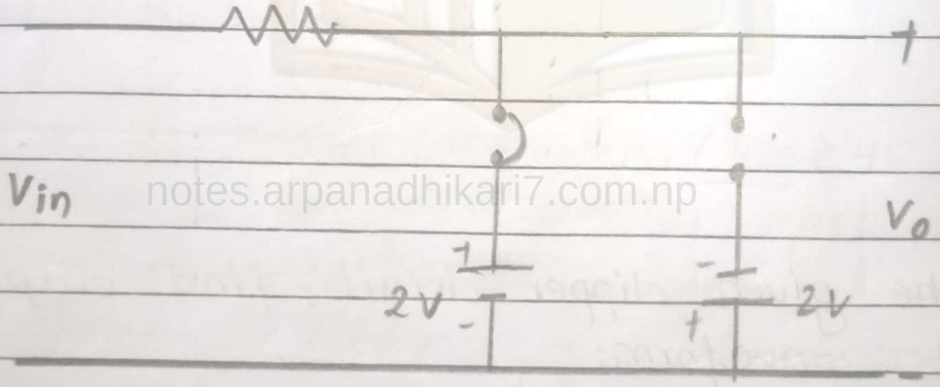
During +ve half of input;

(i) When $V_{in}$ is greater than zero but less than 2V ($0 < V_{in} < 2V$); both $D_1$ & $D_2$ will be OFF & acts as op




$\therefore V_{in} = V_o$

(ii) When $V_{in}$ greater than 2V; $D_1$ will be ON and acts short & $D_2$ will OFF and acts as open.

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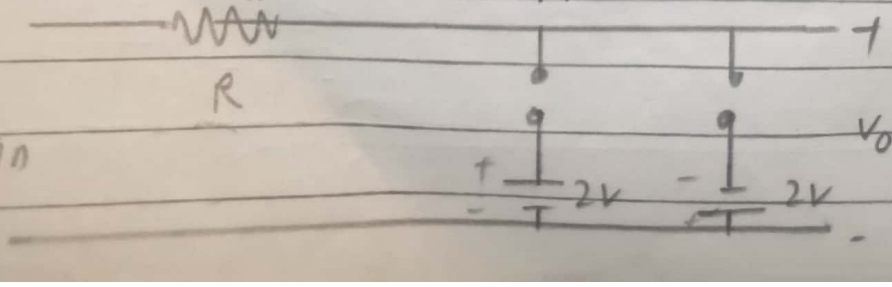
$\therefore V_o = 2V$

During -ve half of input;

(i) When $V_{in}$ is less than zero but greater than -2V; both $D_1$ & $D_2$ will be OFF

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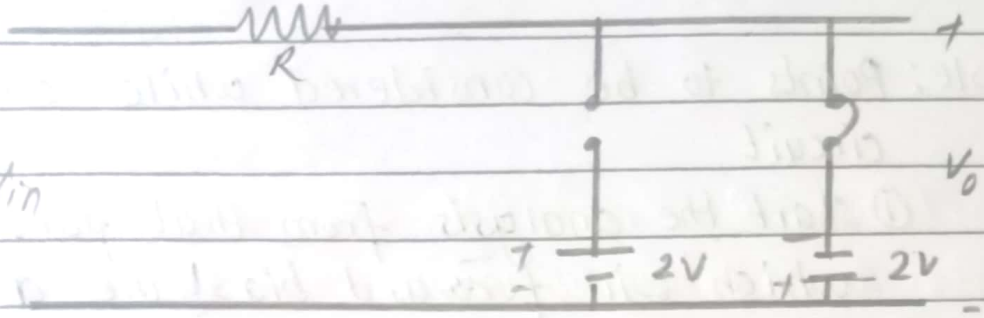
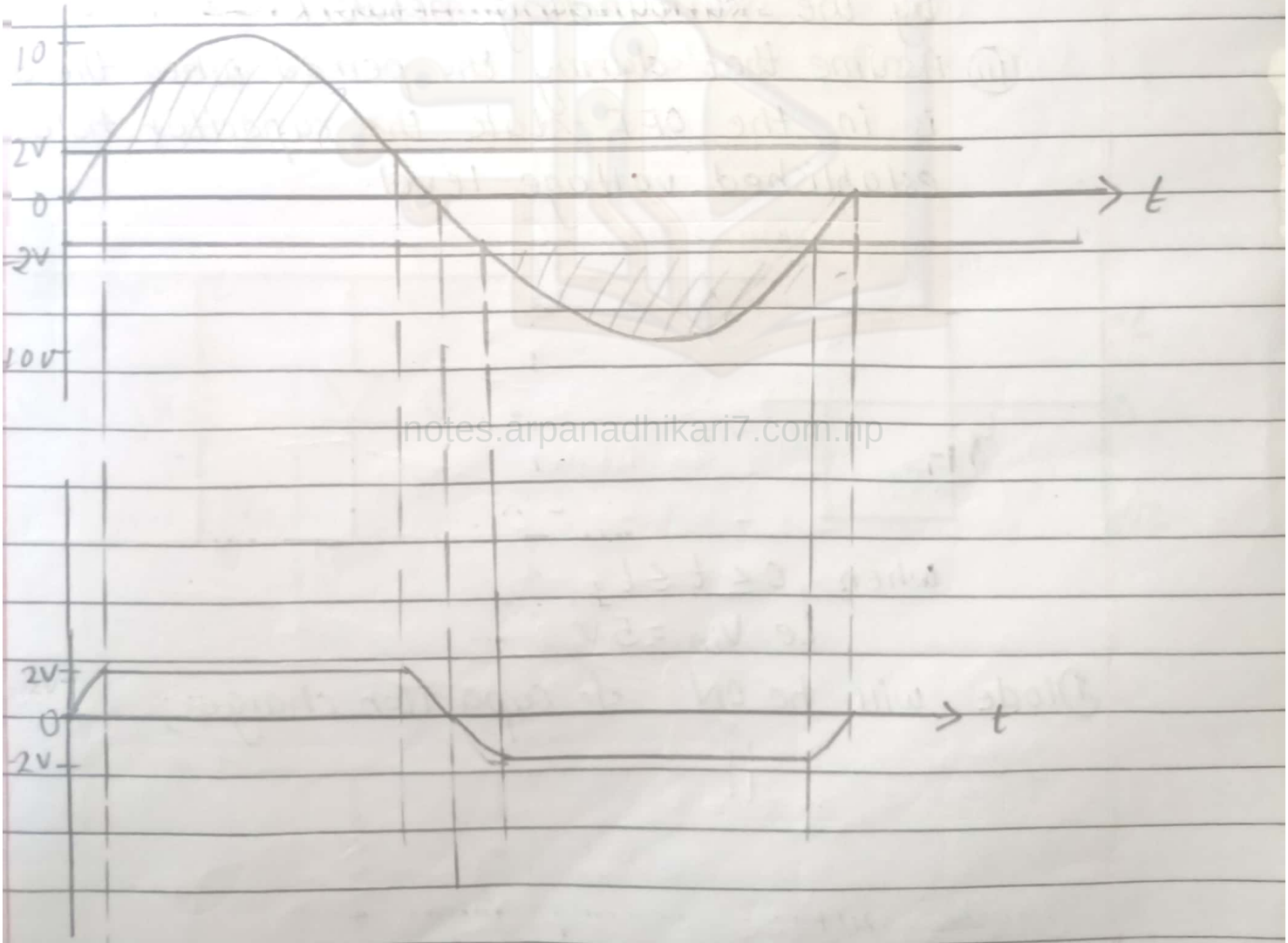
$\therefore V_o = V_{in}$

When $V_{in}$ is less than $-2V$, $D_2$ will be ON and acts as short & $D_1$ will be OFF and acts as open ckt.




$\therefore V_o = -2V$

## waveform

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# Clamper circuit

A clamper is a network constructed of a diode resistor and a capacitor that ships a waveform to a different DC level without changing the appearance of the applied signal.

Note: Points to be considered while solving clamper circuit;

(i) Start the analysis from that part of the input which will forward bias the diode.

(ii) During the period, that the diode is in the ON state, assume that the capacitor will charge up instantaneously to a voltage level determine by the surrounding network.

(iii) Assume that during the period when the diode is in the OFF state the capacitor holds its established voltage level.

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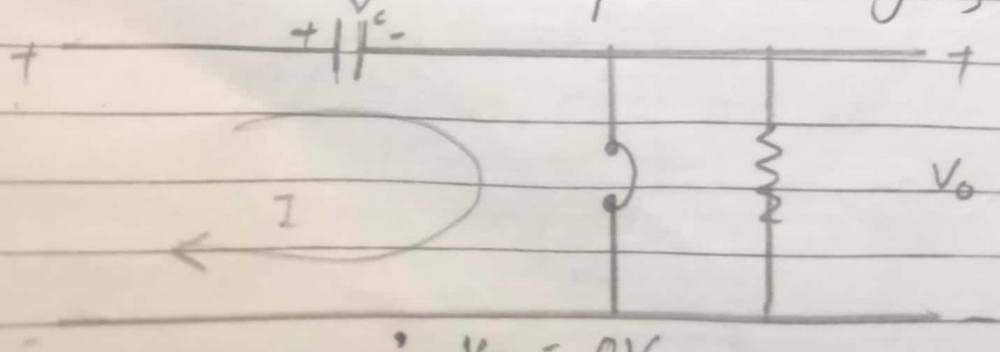
when $0 \leq t \leq t_1$

i.e $V_{in} = 5V$

Diode will be ON & capacitor charges;




$\therefore V_o = 0V$

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Applying KVL,

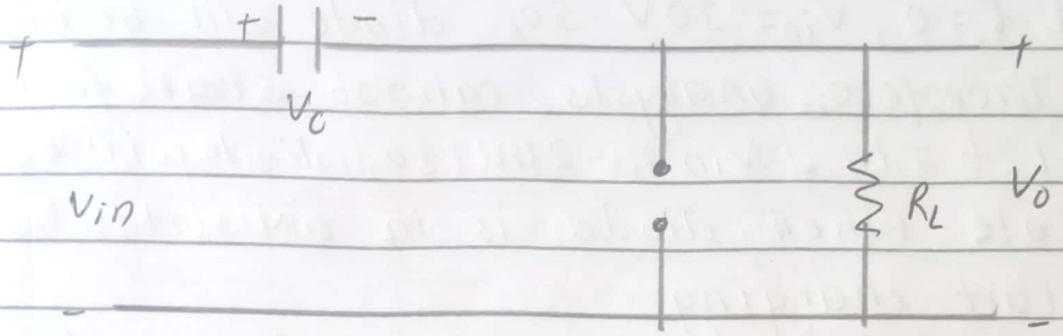
$$V_{in} - V_C = 0$$

$$\text{or } V_C = V_{in} = 5V$$

when $t_1 \leq t \leq t_2$

i.e $V_{in} = -5V$

Diode will be OFF




Applying KVL at outer loop,

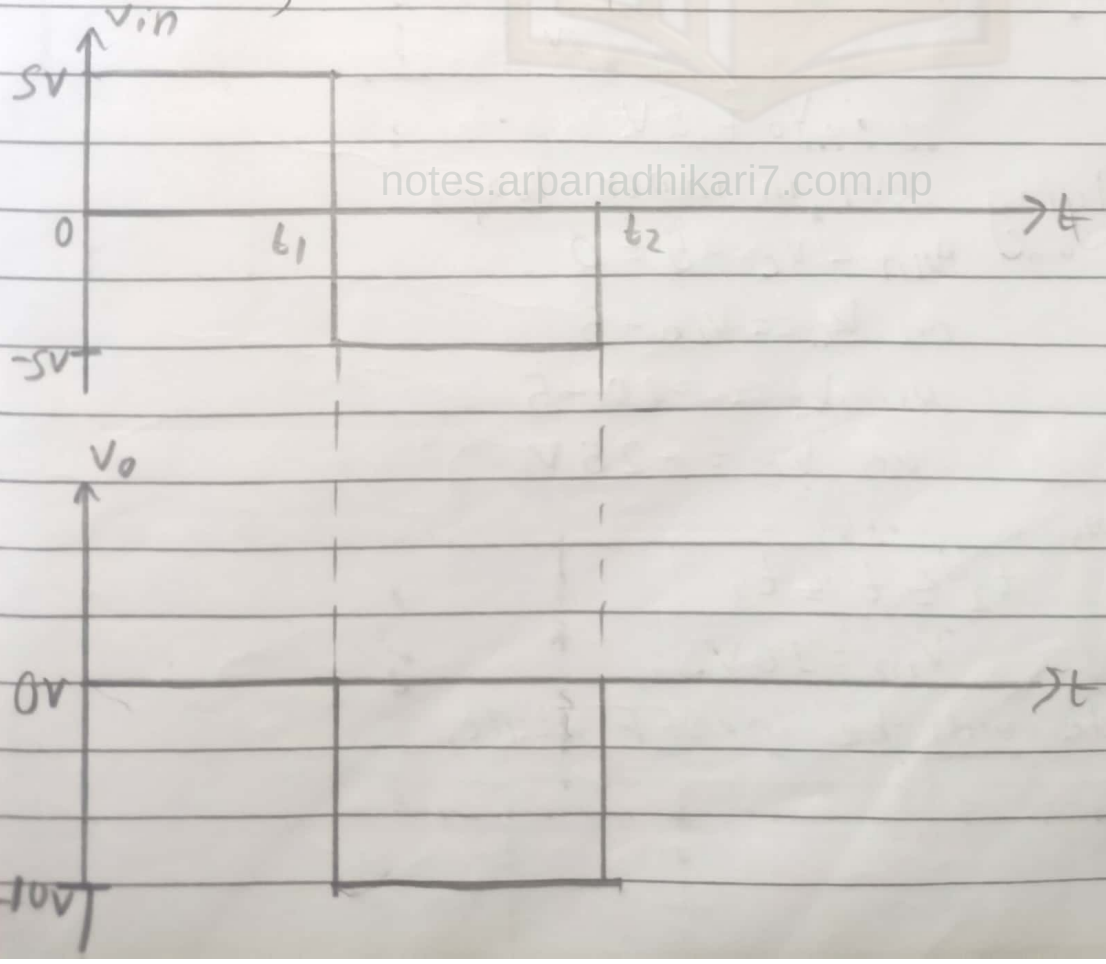
$$V_{in} - V_C - V_o = 0$$

$$\text{or } V_o = V_{in} - V_C$$

$$= -5 - 5$$

$$= -10V$$

waveform;

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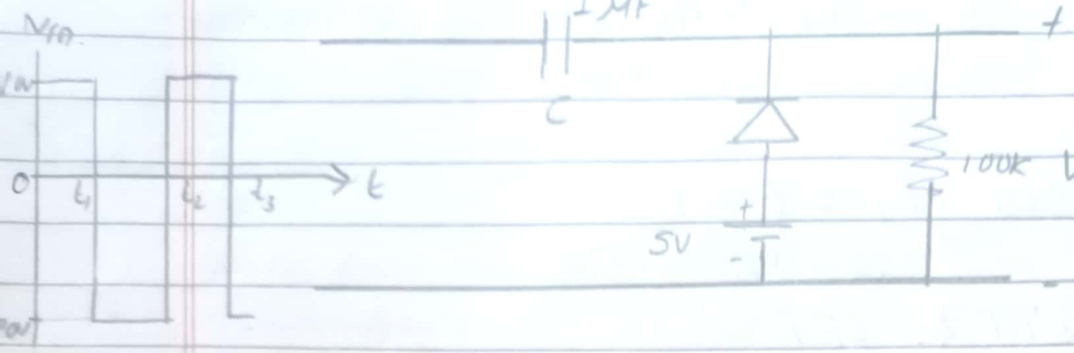
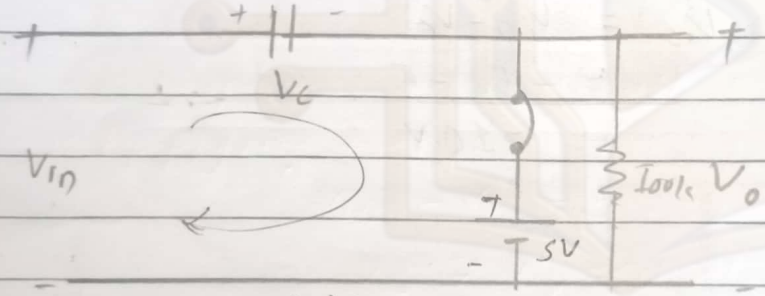
Qn.1 Determine $V_o$ for the network shown in figure




At $t = 0$, $V_{in} = 10V$ so, diode will be in OFF state. Therefore, analysis cannot start from $t=0$.

At $t = t_1$, $V_{in} = -20V$ so, diode will be on ON state. When diode is in ON state, the capacitor will start charging.

Let us start the analysis from $t = t_1$ when $V_{in} = -20V$, diode will be in ON state.




$\therefore V_o = 5V$

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Applying KVL, in inner loop,

$V_{in} - V_C - 5 = 0$

or, $V_C = V_{in} - 5$

or, $V_C = -20 - 5$

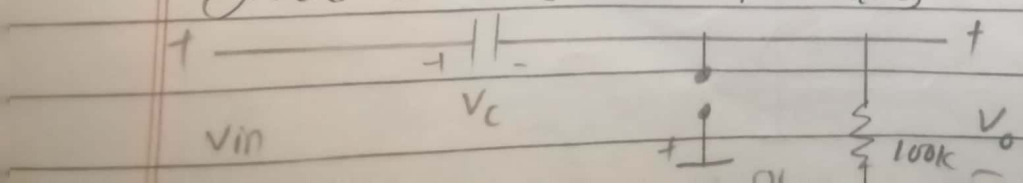
or, $V_C = -25V$

Now,

$t_2 \leq t \leq t_3$

$V_{in} = 10V$

Diode will be in OFF state,

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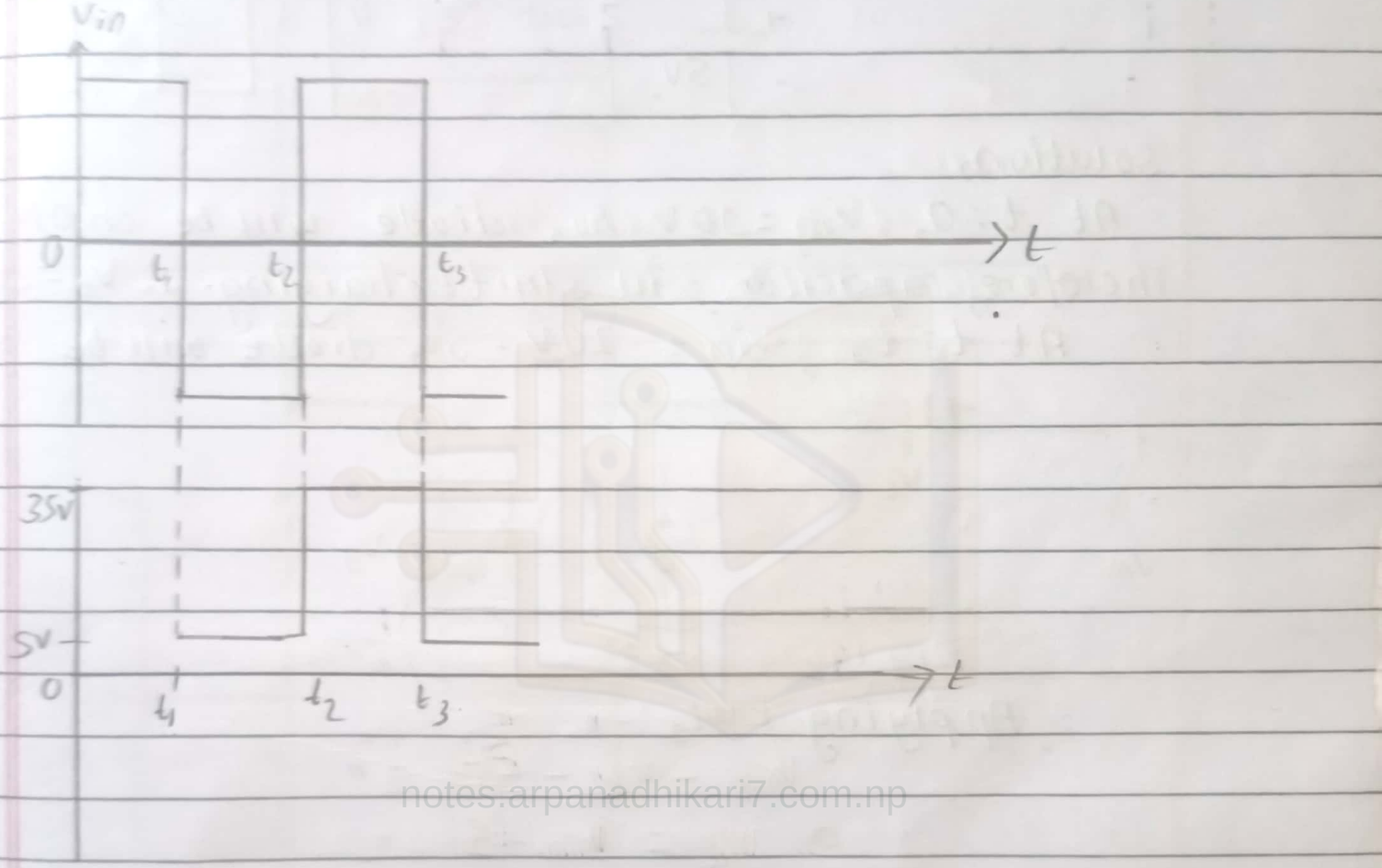
Applying KVL in outer loop,

$$V_{in} - V_C - V_o = 0$$

or, $V_o = V_{in} - V_C$

or, $V_o = 10 - (-25)$

$\therefore V_o = 35V$

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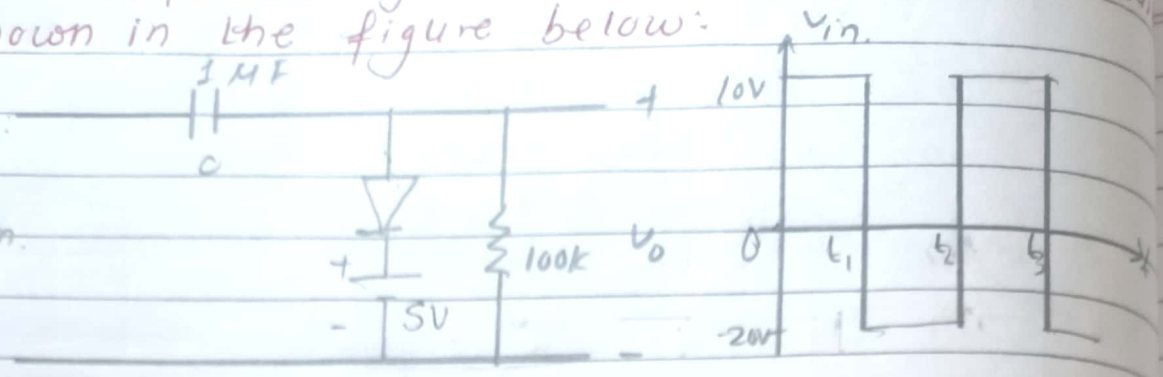
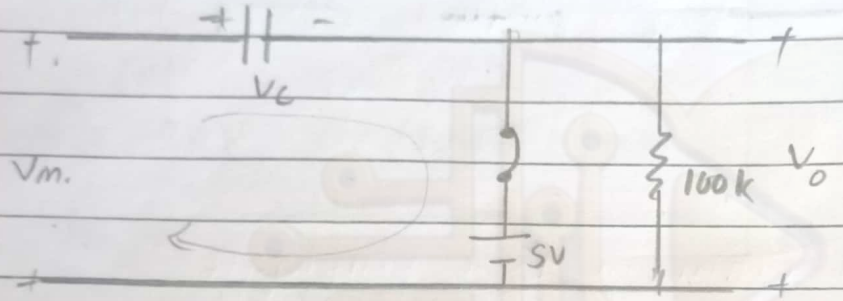
Qn.2. Find output voltage & its waveform for the circuit shown in the figure below:




Solution,

At $t=0$, $V_{in} = 10V$. So, diode will be on ON state. Therefore, capacitor will start charging. $\therefore V_o = 5V$

~~At $t=t_1$, $V_{in} = -20V$. So, diode will be~~




Applying KVL,

$$V_{in} - V_C - 5 = 0$$

$$\text{or, } V_C = V_{in} - 5$$

$$\text{or, } V_C = 10 - 5$$

$$\text{or, } V_C = 5V$$

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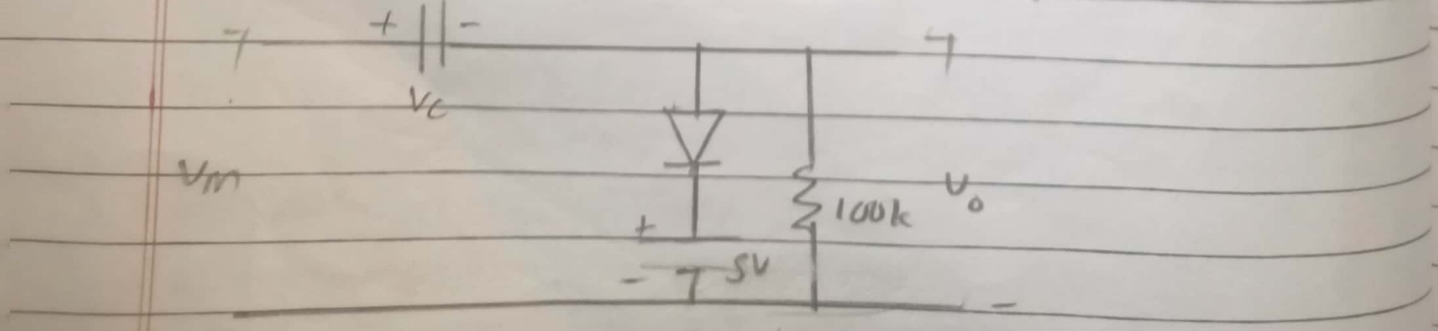
Now,

$$t_1 \leq t \leq t_2$$

$$V_{in} = -20V$$

Diode will be in OFF state.

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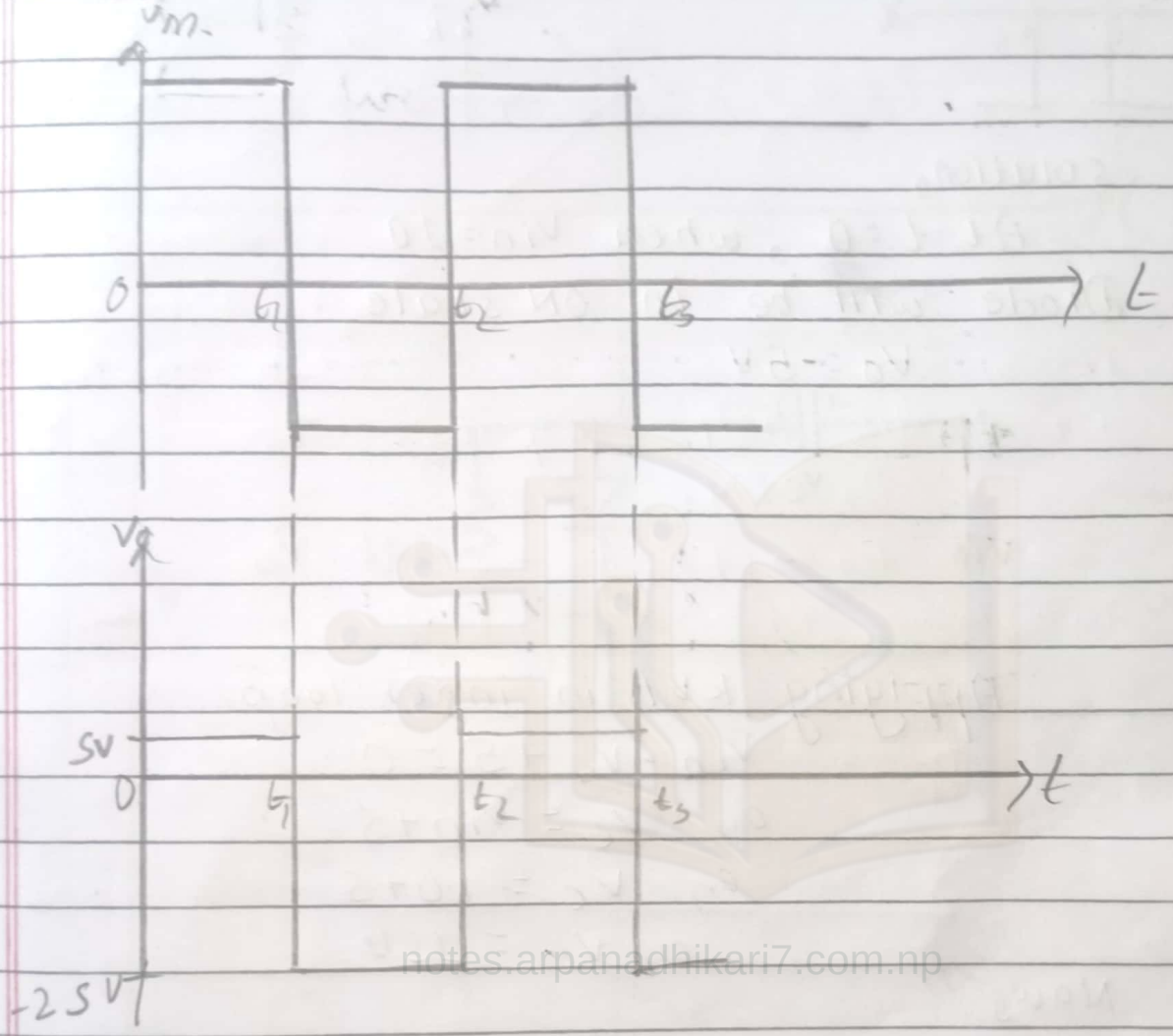
Applying KVL in outer loop;

$$V_{in} - V_C - V_o = 0$$

$$\therefore V_o = V_{in} - V_C$$

$$= -20 - 5$$

$$= -25\,V$$

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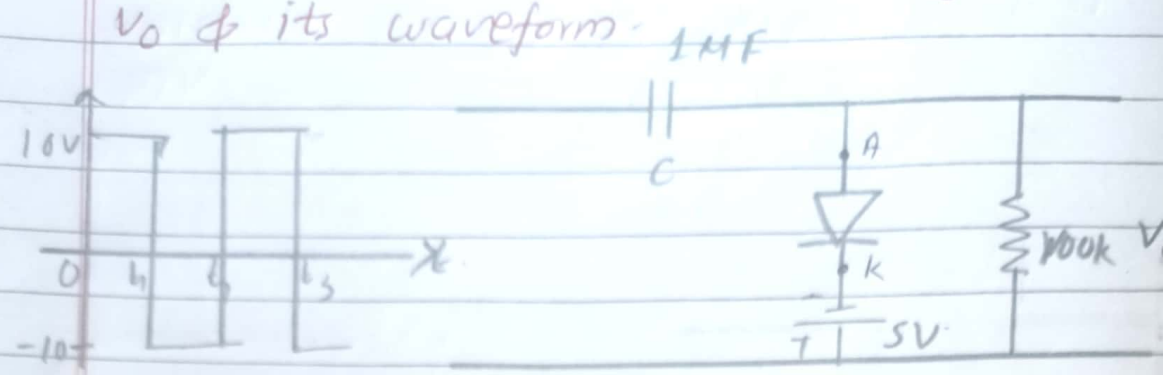
Q.n.3 For the circuit shown in the figure below, find $V_0$ & its waveform.

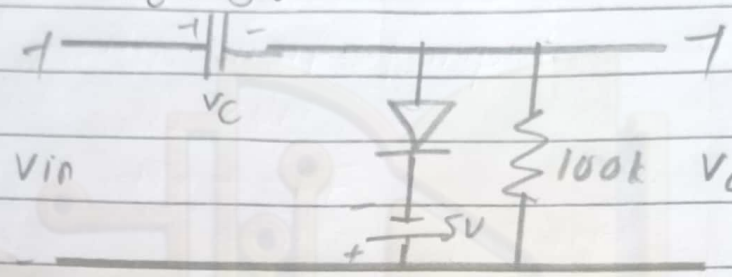



Solution,

At $t = 0$, when $V_{in} = 10$

Diode will be in ON state

$V_0 = -5V$




Applying KVL in inner loop.

$$V_{in} - V_C + 5 = 0$$

$$\text{or, } V_C = V_{in} + 5$$

$$\text{or, } V_C = 10 + 5$$

$$\text{or, } V_C = 15V$$

Now,

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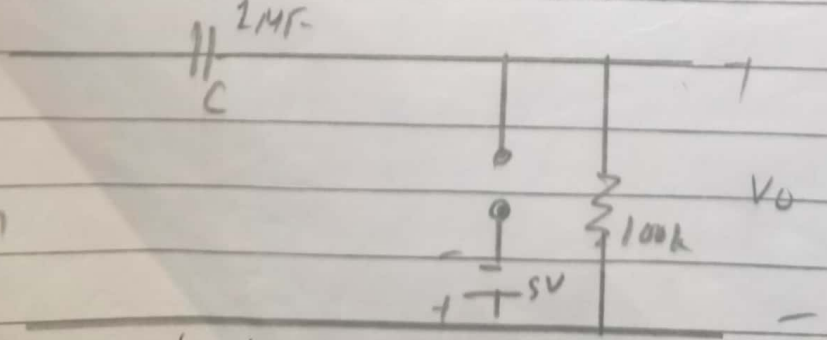
$$t_1 \leq t \leq t_2$$

$$V_{in} = -10V$$

Diode will be in OFF state.




Applying KVL in outer loop,

$$V_{in} - V_C - V_0 = 0$$

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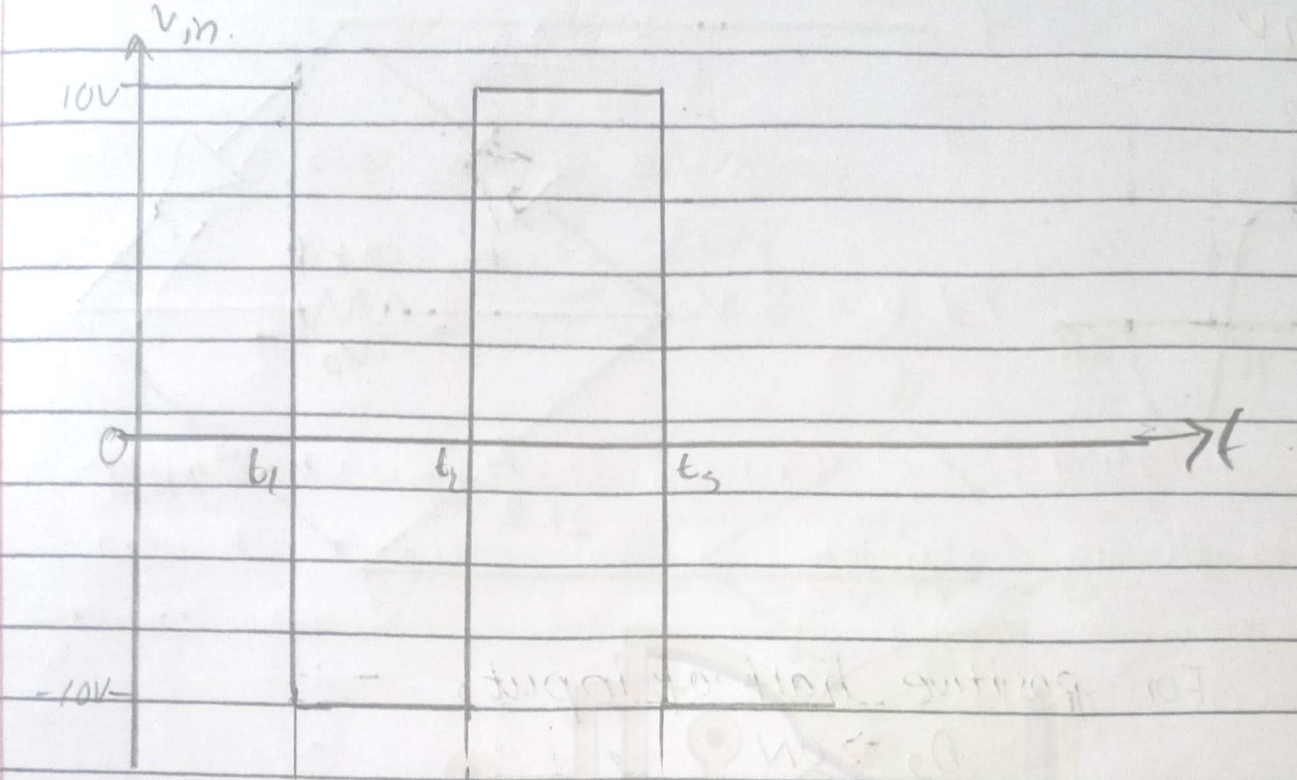
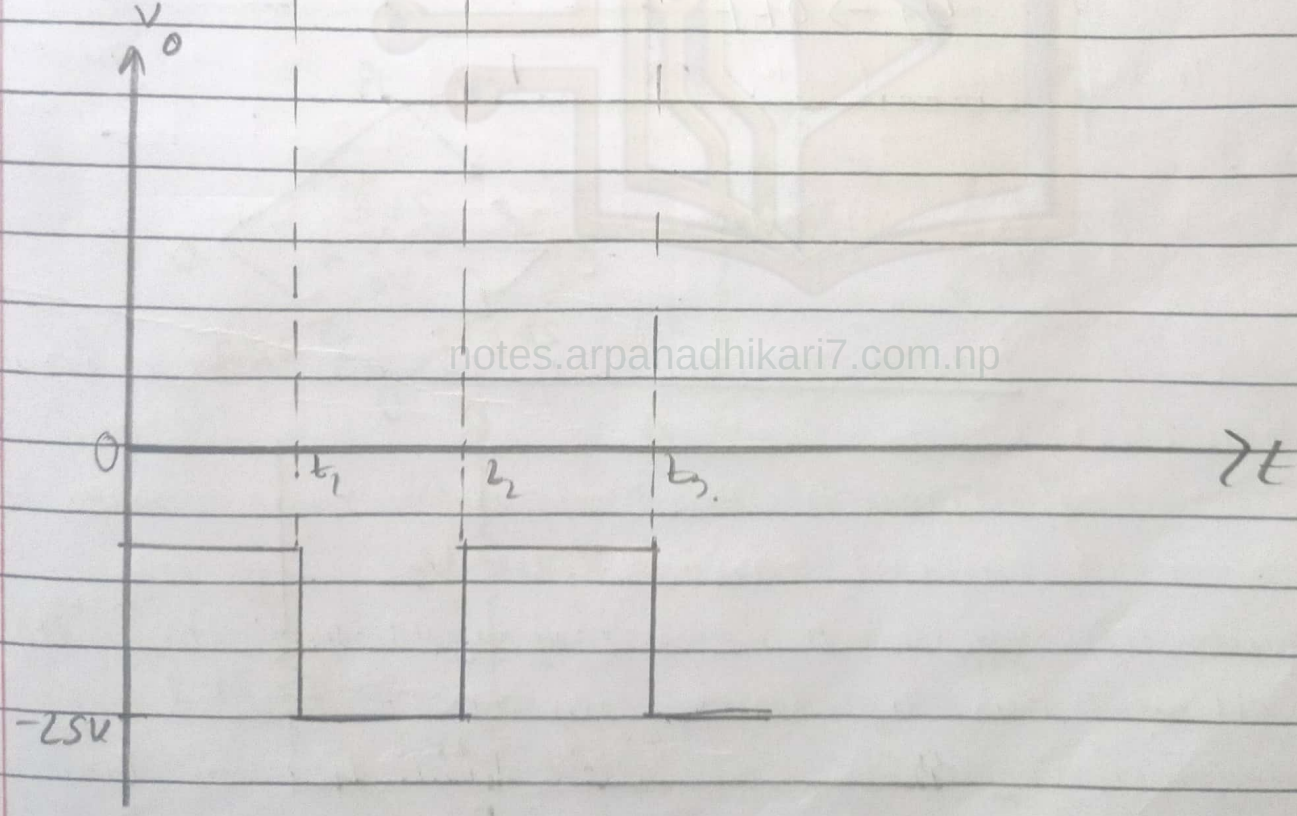
On $V_0 = V_{in} - V_C$

$\therefore V_0 = -10 - 15 = -25\ V$

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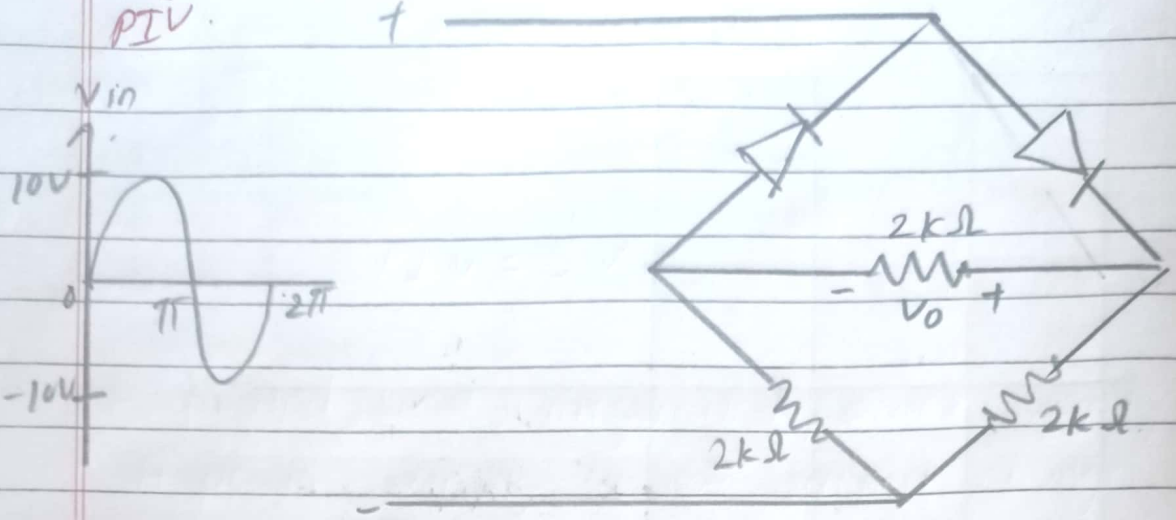
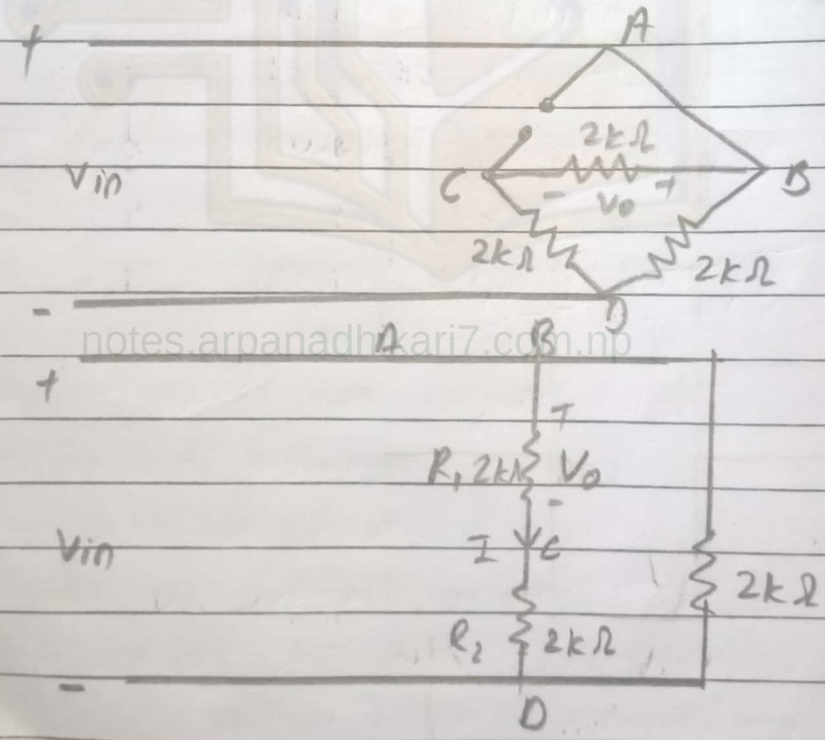
Ans 4: Determine the output waveform for the network & calculate the ~~o/p~~ $V_o$ output dc level & required PIV.




For positive half of input,

$D_2 \rightarrow ON$

$D_1 \rightarrow OFF$




$$I = \frac{V_{in}}{R_1 + R_2}$$

$$V_o = I \times R_1$$

$$= \frac{V_{in}}{R_1 + R_2} \times R_1$$

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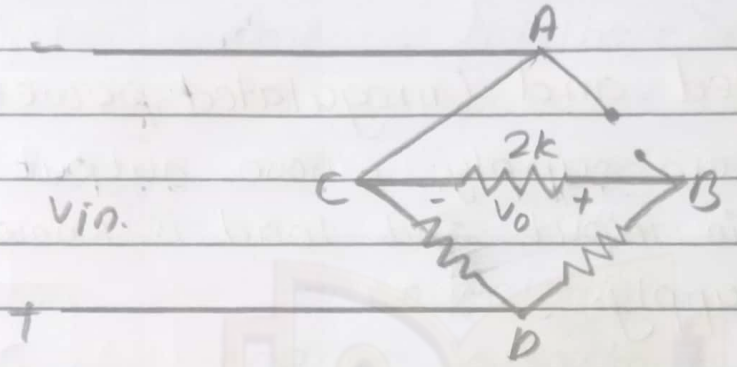
$$V_0 = V_{in} \times \frac{2}{2+2}$$

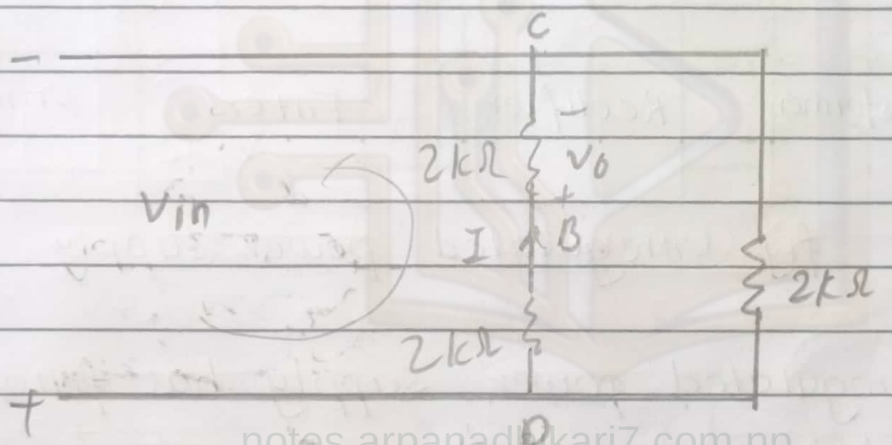
$$= 0.5\, V_{in}$$

During -ve half of input

$D_1 \rightarrow ON$

$D_2 \rightarrow OF$

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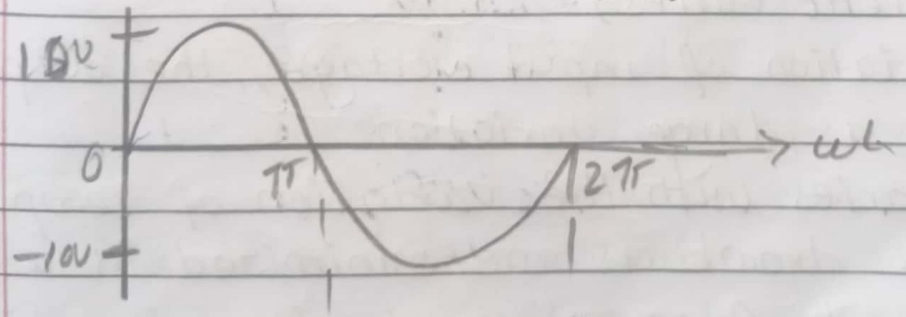
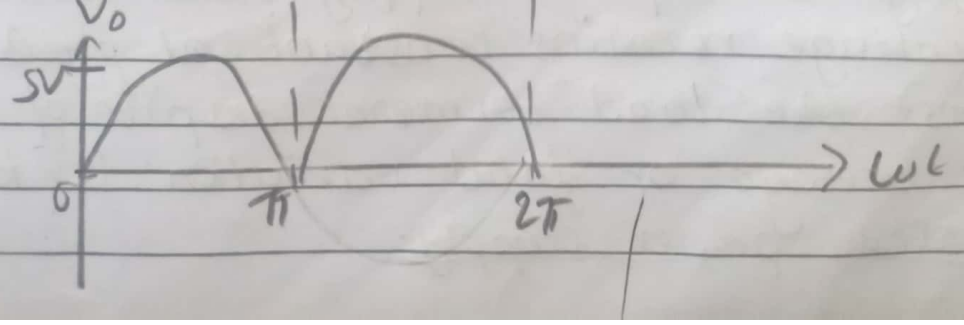
$$V_0 = \frac{V_{in} \times 2}{2+2} = 0.5\, V_{in}$$

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Output dc voltage $= \frac{2V_m}{\pi}$

$= \frac{2 \times 5}{\pi}$

$= 3.18$

$PIV = 5V$

## Regulated and Unregulated power supply

A power supply whose output changes with change in input and load is known as unregulated power supply.

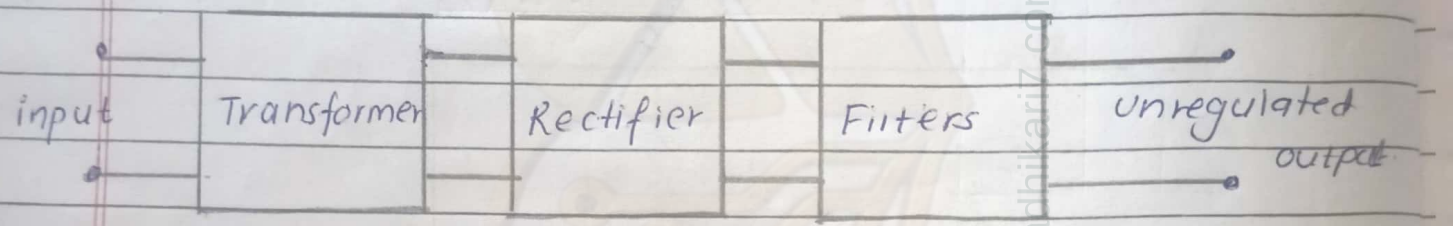


fig: Unregulated power supply

The unregulated power supply has following drawbacks;

1. Poor Regulation (i.e the output voltage varies when the load current varies).
2. With the variation of input voltage, the output voltage has a large variation.
3. The output varies with the variation of temperature.

These drawbacks are minimized in a regulated power supply. A power supply ~~flows~~ in which the output voltage remains constant at pre-determined value across the load terminal despite of change in input voltage or load variation is known as regulated power supply.

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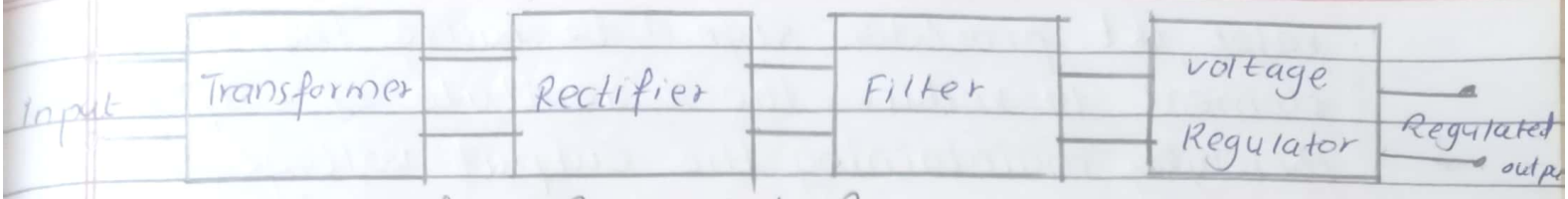


fig: Regulated Power supply.

## * Voltage Regulator

Voltage regulator circuit consists of a comparator circuit, control element, feedback network and reference voltage. The comparator ckt compares the reference voltage with the feedback voltage and converts the difference into error signal. This error signal controls the control element in order to compensate the change in output voltage. There are mainly two types of voltage regulators. They are

(i) Series Voltage Regulator
(ii) Shunt " "

### (i) Series Voltage Regulator

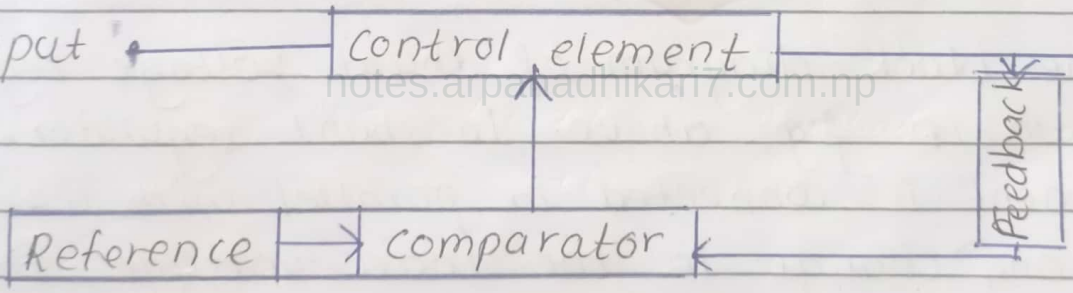


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fig: Series voltage regulator

The block diagram of series V.R shown in figure above. The series element controls the magnitude of the input voltage that gets to the output. The output voltage is sampled by a ckt that provides a feedback voltage to be compared to a reference voltage. If the output voltage increases, the compa

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rator ckt provides signal to cause the series con element to reduce the magnitude of output voltag thereby maintaining the output voltage.

on other hand, if output voltage falls, the comparator ckt provides signal to cause the series control element to increase the magnitude of outp voltage thereby maintaining the constant output voltage.

(ii) Shunt voltage Regulator.

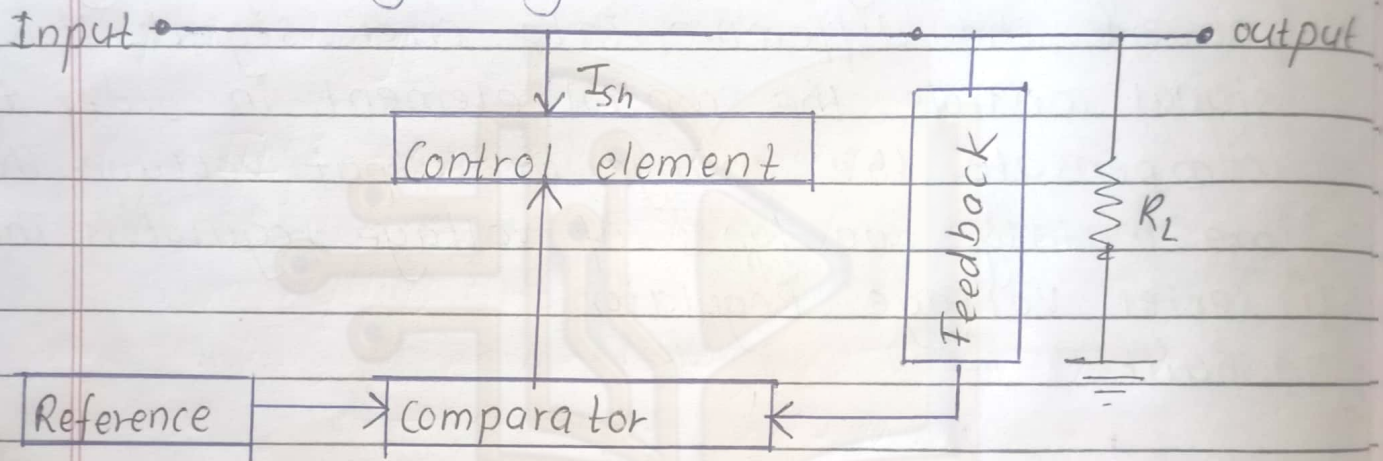


fig: Shunt voltage Regulator.

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The block diagram of shunt voltage regulator is shown in fig above. In shunt regulator, the cont element is connected in parallel with load. If ther is any change in the load voltage, the sampli or feedback ckt provides a feedback signal to the comparator ckt. The comparator ckt compa the feedback signal with the reference signal ~~sign~~ ~~generates~~ ~~the~~ a control signal which dec the amount of current ($I_{sh}$) to keep the load voltage constant.

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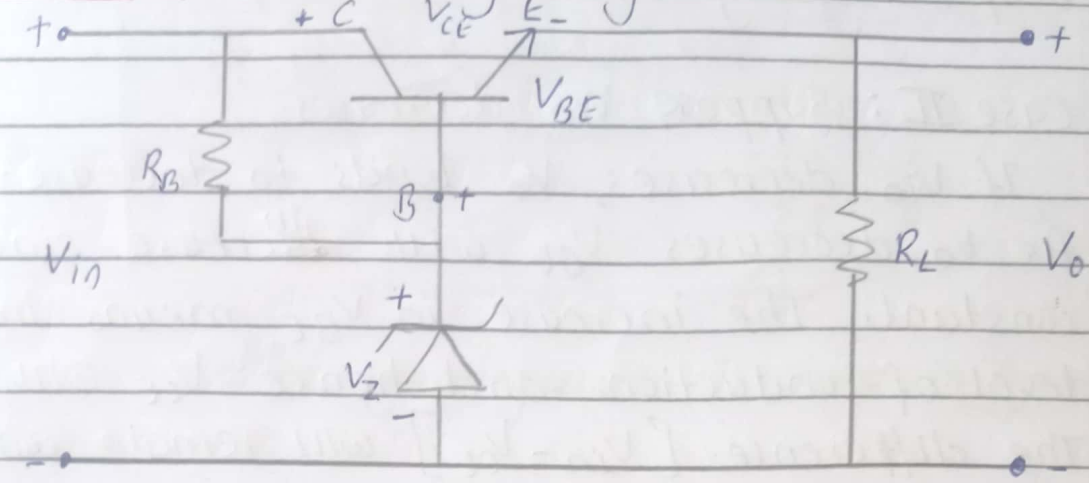
## Transistor Series voltage regulator




Applying KVL, $V_z - V_{BE} - V_o = 0$

or, $V_o = V_z - V_{BE}$ —— (I)

Applying KVL at outer loop,

$V_{in} - V_{CE} - V_o = 0$

or, $V_o = V_{in} - V_{CE}$ —— (II)

**Note:** $V_{BE}\uparrow$ : level of conduction ↑ but $V_{CE}\downarrow$

**Case I: Suppose $V_{in}$ increases**

If $V_{in}$ increases, $V_o$ tends to increase.

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From eqⁿ (I),

Since, $V_z$ is constant, $V_o$ increases but $V_{BE}$ decreases.

Here, $V_{BE}$ = level of conduction ↓

but $V_{CE}$ ↑

$\therefore V_o = V_{in} - V_{CE}$ will remains constant.

Prem

When $V_{in}$ increases, $V_o$ tends to increase. From eqⁿ (I) As $V_o$ increases $V_{BE}$ will decrease since $V_z$ remains constan The decrease in $V_{BE}$ means decrease in level of conducti on & hence $V_{CE}$ will increases. The difference [$V_{in} - V_{CE}$] will remain constant. Hence, the output

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voltage $V_o$ remains constant in respective of incre in input voltage.

Case II : Suppose $V_{in}$ decreases

If $V_{in}$ decreases, $V_o$ tends to decrease. From eqn (i) As $V_o$ decreases $V_{BE}$ will increase since $V_z$ rema constant. The increase in $V_{BE}$ means increase in level of conduction and hence $V_{CE}$ will decrease. The difference $[V_{in} - V_{CE}]$ will remain constant. He $V_o$ remains constant in respective of decrease in input voltage.

Case III : Suppose $R_L$ decreases

If $R_L$ decreases, $V_o$ tends to decrease:

From eqn (i),

Since, $V_z$ is constant, $V_o \downarrow$, $V_{BE} \uparrow$

$V_{BE} \uparrow \rightarrow$ level of conduction $\uparrow$

$I_E \uparrow$

$I_L \uparrow$

$V_o = I_L R_L$

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$\therefore V_o = I_L R_L$ remains constant.

Case IV : Suppose $R_L$ increases:

If $R_L$ increases, $V_o$ tends to increase.

From eqn (i),

Since, $V_z$ is constant, $V_o \uparrow$, $V_{BE} \downarrow$

$V_{BE} \downarrow \rightarrow$ level of conduction $\downarrow$

$I_E \downarrow$

$I_L \downarrow$ $\quad V_o = I_L R_L$

$\therefore V_o = I_L R_L$ remains constant.

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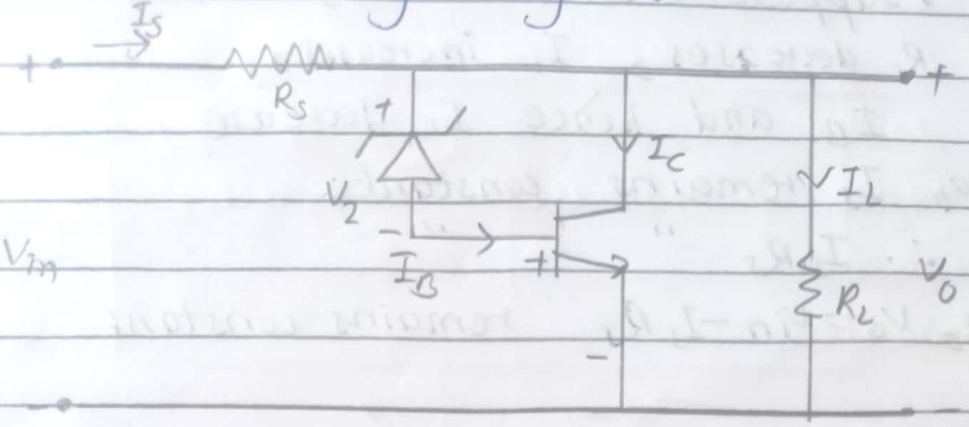
# Transistor Shunt voltage regulator




Transistor shunt voltage regulator is shown in figure above.

Now, applying KVL;

$$V_{BE} + V_Z - V_0 = 0$$

$$\text{or, } V_0 = V_Z + V_{BE} \quad \text{—— (I)}$$

($V_{BE}$ → constant)

Applying KVL at outer loop,

$$V_{in} - I_S R_S - V_0 = 0$$

$$V_0 = V_{in} - I_S R_S$$

Also,

$$I_S = I_B + I_C + I_L \quad \text{—— (III)}$$

**Case I** : Suppose $V_{in}$ increases;

If $V_{in}$ increases, $V_0$ tends to increase from eqⁿ (I);

$V_0 \uparrow$, $V_{BE} \uparrow$, level of conduction $\uparrow$, $I_B \uparrow$, $I_C \uparrow$

$\therefore I_S \uparrow$, $I_S R_S \uparrow$

$\therefore V_0 = V_{in} - I_S R_S$ remains constant.

**Case II** : Suppose $V_{in}$ decreases:

If $V_{in}$ decreases, $V_0$ tends to decrease,

From eqⁿ (I);

$V_0 \downarrow$, $V_{BE} \downarrow$, level of conduction $\downarrow$, $I_B \downarrow$, $I_C \downarrow$

$\therefore I_S \downarrow$, $I_S R_S \downarrow$

$\therefore V_0 = V_{in} - I_S R_S$ remains constant

↓

$10\,V - 5\,V = 5\,V$

$8\,V - 3\,V = 5\,V$

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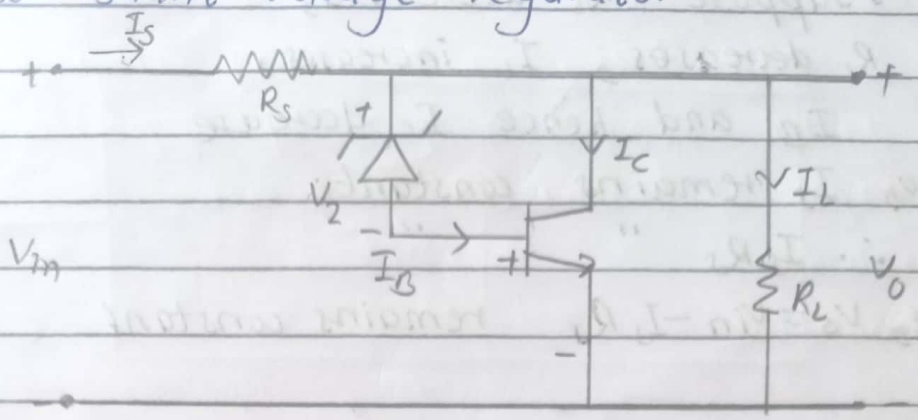
# Transistor shunt voltage regulator:




Transistor shunt voltage regulator is shown in figure above.

Now, applying KVL;

$$V_{BE} + V_2 - V_0 = 0$$

$$\text{or, } V_0 = V_2 + V_{BE} \quad \text{—— (i)}$$

↳ constant

Applying KVL at outer loop,

$$V_{in} - I_S R_S - V_0 = 0$$

$$V_0 = V_{in} - I_S R_S$$

Also,

$$I_S = I_B + I_C + I_L \quad \text{—— (iii)}$$

**Case I** : Suppose $V_{in}$ increases;

If $V_{in}$ increases, $V_0$ tends to increase from eqⁿ (i);

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$V_0 \uparrow$, $V_{BE} \uparrow$, level of conduction $\uparrow$, $I_B \uparrow$, $I_C \uparrow$

$\therefore I_S \uparrow$, $I_S R_S \uparrow$

$\therefore V_0 = V_{in} - I_S R_S$ remains constant.

**Case II** : Suppose $V_{in}$ decreases:

If $V_{in}$ decrease, $V_0$ tends to decrease,

From eqⁿ (i);

$V_0 \downarrow$, $V_{BE} \downarrow$, level of conduction $\downarrow$, $I_B \downarrow$, $I_C \downarrow$

$\therefore I_S \downarrow$, $I_S R_S \downarrow$

$\therefore V_0 = V_{in} - I_S R_S$ remains constant

↳ 10 V − 5V = 5V

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Case III : Suppose $R_L$ decreases,

If $R_L$ decreases, $I_L$ increases,

$I_B$ and hence $I_C$ decrease

So, $I_S$ remains constant.

$\therefore I_S R_S$ " "

Hence, $V_o = V_{in} - I_S R_S$ remains constant.

Case IV : Suppose $R_L$ increases

If $R_L$ increases, $I_L$ decreases.

$I_B$ and hence $I_C$ increases.

So, $I_S$ remains constant

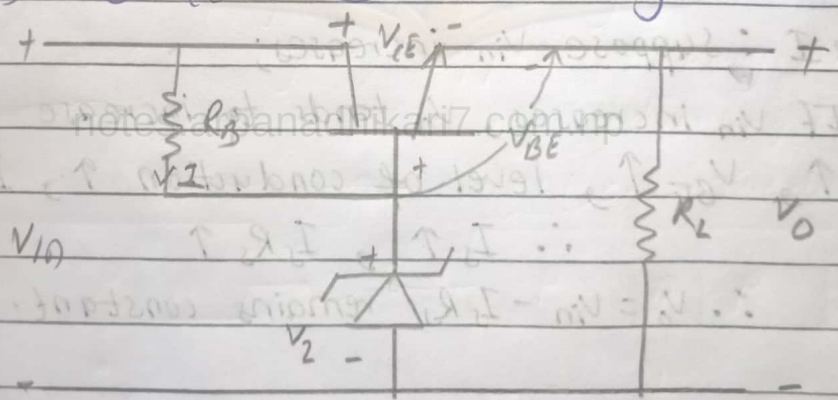
$\therefore I_S R_S$ " "

Hence, $V_o = V_{in} - I_S R_S$ remains constant.

## Numerical

① In the ckt shown in figure, $V_2 = 12V$, $V_{in} = 20V$, $R_B = 200\Omega$ and $V_{BE} = 0.65V$. Find,

(i) $V_o$, (ii) $V_{CE}$ & (iii) current through $200\Omega$




Solution,

Given,

$V_2 = 12V$

$V_{in} = 20$

$R_B = 200\Omega$

$R_{BE} = 0.65V$

Then,

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(i) Applying KVL,

$V_2 - V_{BE} - V_0 = 0$

or $V_0 = V_2 - V_{BE}$

$= 12 - 0.65$

$= 11.35\ V$

(ii) Applying KVL at outer loop,

$V_{in} - V_{CE} - V_0 = 0$

$V_{CE} = V_{in} - V_0$

$= 20 - 11.35$

$= 8.65\ V$

(iii) Again,

Applying KVL,

$V_{in} - IR_B - V_2 = 0$

$\Rightarrow 20 - 12 = I \times 200$

or $I = 0.04$ Amp.

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Chapter-2

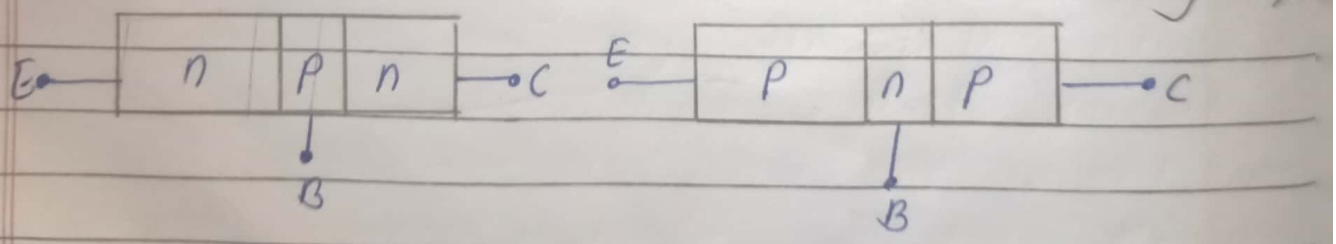
# Transistor

A transistor is a three layer, three terminal device which is formed by sandwiching either P type or N-type semiconductor between a pair of opposite type. There are 2 types of transistor. They are;

1. npn transistor
2. pnp transistor

A transistor consists of 2 P-N junctions. Therefore, a transistor may be regarded as a combination of two diodes connected back to back. There are three terminals in a transistor, one taken from each type of S/C. These three terminals are named as emitter, base and collector. The middle section of a transistor is very thin layer and a terminal is taken from it which is known as base. The section on one side of base is emitter and the section on the opposite side is the collector.

The emitter is heavily dopped so that it can inject a large no. of charge carriers into the base. The base is lightly dopped & very thin, it passes most of the emitter injected charge carrier to the collector. The collector is moderately dopped




The emitter is always forward biased with respect to base and the collector is always reverse biased with respect to base.

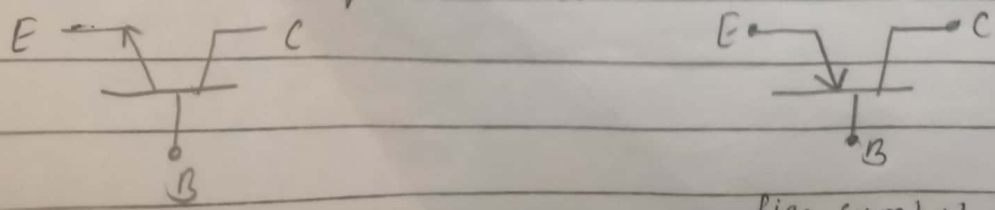



Fig: Symbol of n-p-n transistor

Fig: symbol of p-n-p transi

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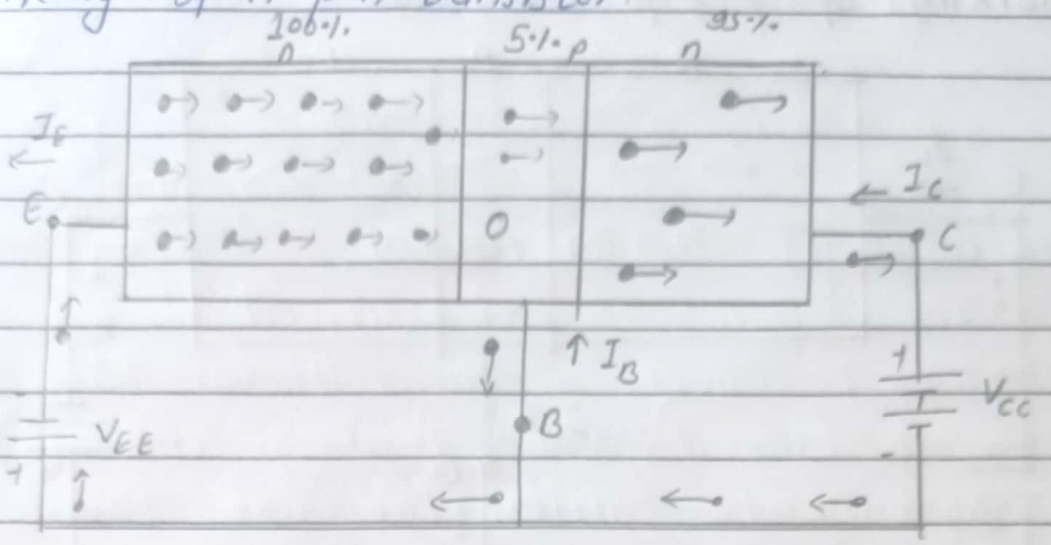
Working of n-p-n transistor:




The n-p-n transistor with forward biased to emitter base junction and reverse biased to collector base junction is shown in figure above.

The forward bias causes the electrons in N-type emitter to flow towards the base. This constitutes the emitter current ($I_E$). As these electrons flows through the P-type base, they tends to combine with holes. As the base is lightly dopped and very thin, therefore only few electrons (less than 5%) combine with holes to constitutes the base current ($I_B$). The remainder (more than 95%) crosses over into the collector region to constitute collector current '$I_C$'. In this way, almost the entire emitter current flows in the collector circuit. The emitter current is the sum of base current and collector current.

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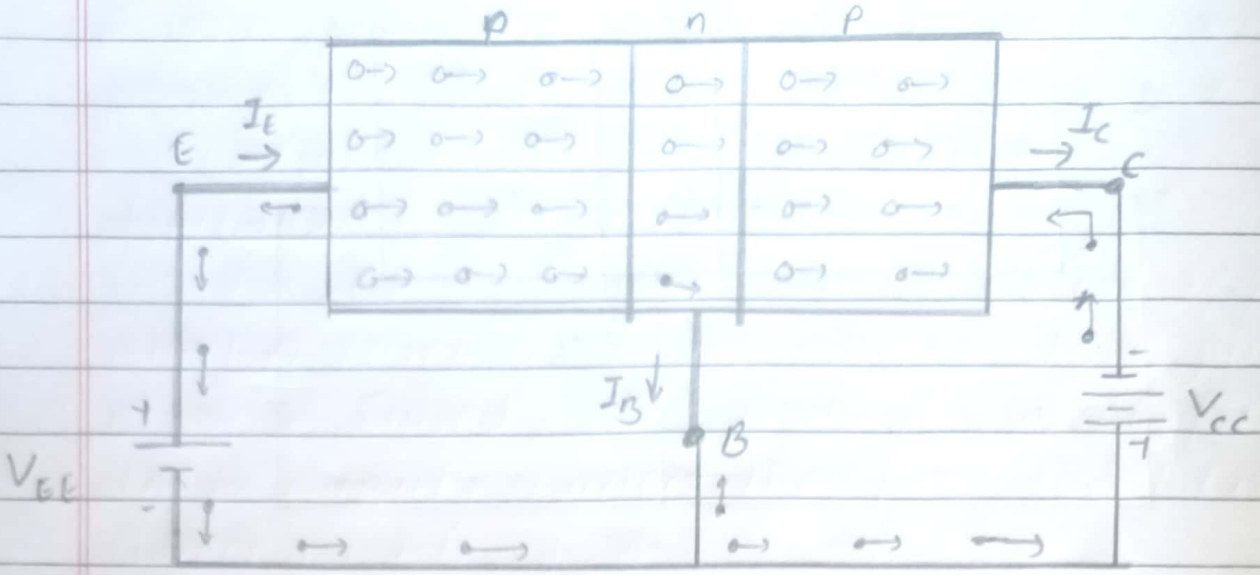
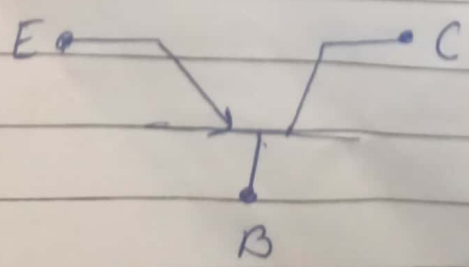
## Working of p-n-p transistor




The basic connection of p-n-p transistor is in figure above. The forward bias causes the holes in the P-type emitter to flow towards the base. This contitute the emitter current '$I_E$'. As the holes cross into N-type base, they tends to combine with electron. As the base is lightly dopped & very thin, only a few holes (i.e less than 5%) combine with the electrons. This constitutes the base current '$I_B$'.

The remainder (more than 95%) cross into the collector region to constitute collecter current ($I_C$). In this way, almost the entire emitter current flows in the collector circuit.

It may be noted that the current conduction within p-n-p transistor is by holes. However, in external connecting wires, the current conduction is by electrons.




$\therefore I_E = I_C + I_B$

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# Transistor Connection

The transistor can be connected in a circuit in the following three ways.

Common base connection

Common emitter connection

Common collector connection

## Common base connection

In common base connection, the input is applied bet$^n$ emitter and base and output is taken from collector and base. Here, base of the transistor is common to both input and o/p circuits. The common base transistor circuit is shown in figure below:

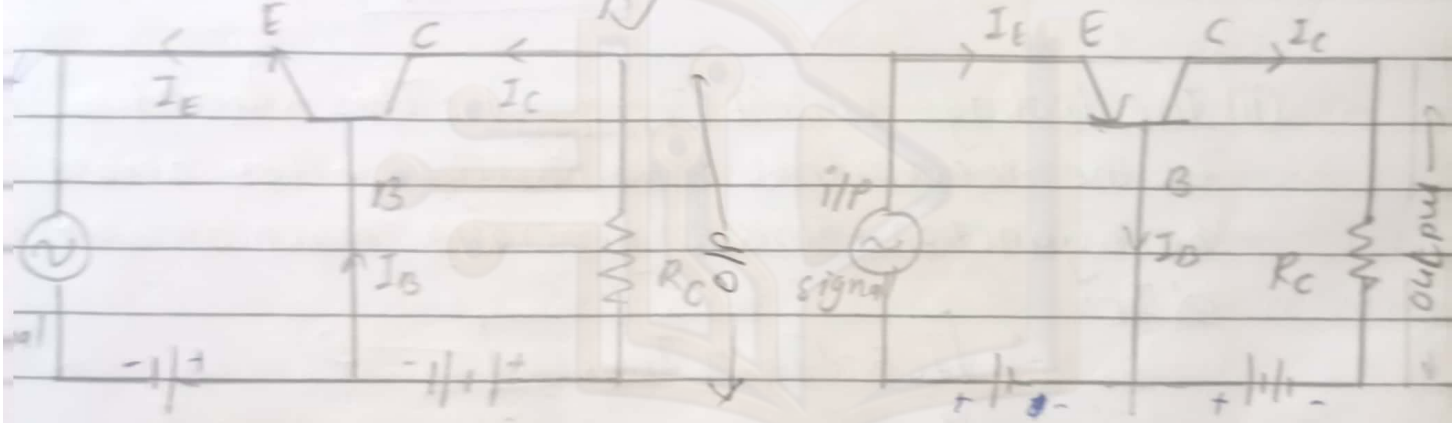


fig: CB circuit for n-p-n transistor

fig: CB circuit for p-n-p transistor

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## Current amplification factor ($\alpha$):

It is the ratio of o/p current to I/p current. In a common base connection, the input current is emitter current ($I_E$) and output current is the collector current ($I_C$).

The ratio of change in collector current to the change in emitter current at constant collector base voltage ($V_{CB}$) is known as current amplification factor.

ie $$\alpha = \frac{\Delta I_C}{\Delta I_E}$$ at constant $V_{CB}$

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If only ~~dc output~~ direct current (dc) is considered,

$$\alpha = \frac{I_C}{I_E}$$

Expression of collector current

The whole emitter current do not reach the collector. Moreover, as the collector base junction is reverse biased, some leakage current ~~follows~~ flows due to minority carriers. Therefore, the total collector current consists of

(i) That part of emitter current which reaches the collector terminal i.e $\boxed{I_C = \alpha I_E}$.

(ii) The leakage current ($I_{leakage}$): This current is due to the movement of minority carries across collector base junction. This is generally much smaller than $\alpha I_E$.

$$\therefore \boxed{I_C = \alpha I_E + I_{leakage}} \quad \text{—— (i)}$$

If $I_E = 0$, $I_{leakage}$ is abbreviated as $I_{CBO}$

where, $I_{CBO}$ = current to collector base junction when emitter is open.

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$$I_C = \alpha I_E + I_{CBO}$$
$$\Rightarrow I_C = \alpha (I_B + I_C) + I_{CBO}$$
$$\Rightarrow I_C = \alpha I_B + \alpha I_C + I_{CBO}$$
$$\Rightarrow I_C - \alpha I_C = \alpha I_B + I_{CBO}$$
$$\Rightarrow I_C (1 - \alpha) = \alpha I_B + I_{CBO}$$

$$\therefore \boxed{I_C = \frac{\alpha}{(1-\alpha)} I_B + \frac{I_{CBO}}{1-\alpha}} \quad \text{—— (ii)}$$

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## Input characteristics

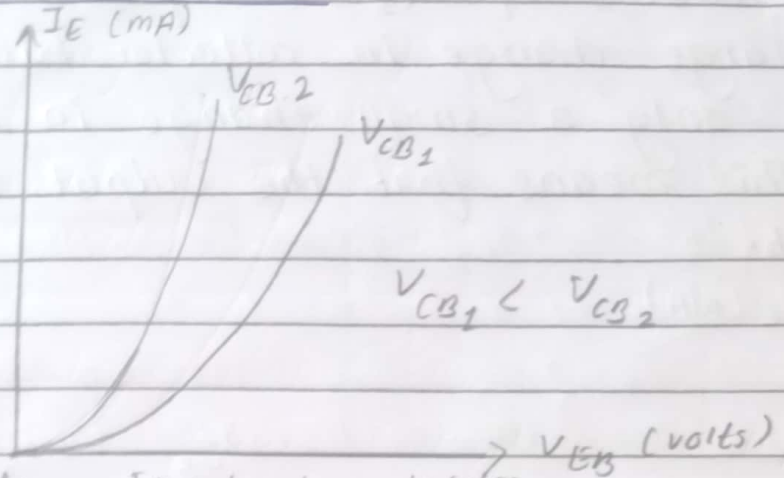


fig ; Input characteristic

It is the curve bet$^n$ emitter current ($I_E$) & emitter base voltage ($V_{EB}$) at constant collector Base voltage ($V_{CB}$). The emitter current is generally taken along Y-axis and emitter base voltage is taken along X-axis.

The emitter current ($I_E$) increases rapidly with small increase in emitter base voltage ($V_{EB}$). It means that the input resistance is very small. The emitter current is almost independent of collector base voltage ($V_{CB}$).

$$\text{Input resistance } (r_{in}) = \frac{\Delta V_{EB}}{\Delta I_E} \quad \text{at constant } V_{CB}.$$

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## Output characteristics.

It is the curve bet$^n$ collector current '$I_C$' & collector base voltage '$V_{CB}$' at constant '$I_E$'.

Generally, collector current is taken along y-axis and collector base voltage ($V_{CB}$) along X-axis. The collector current ($I_C$) varies with $V_{CB}$ only at very low voltage (i.e less than 1V). The transistor never operate in this region. When the value of $V_{CB}$ increases, the collector current '$I_C$' becomes constant as indicated by straight horizontal curves. It means that the current '$I_C$' is

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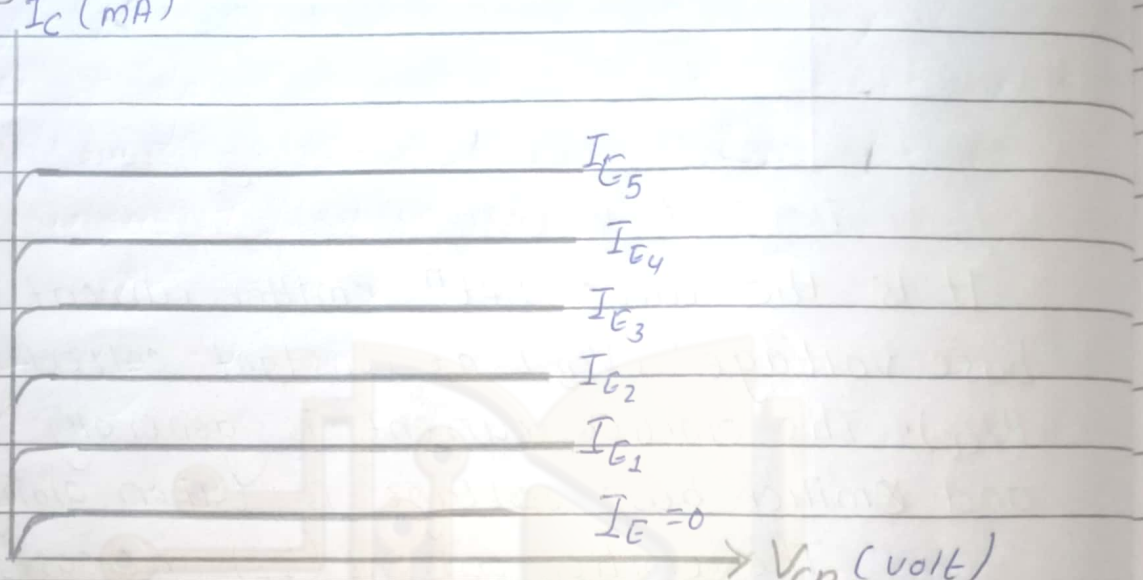
now independent of '$V_{CB}$' & depends upon '$I_E$'.
A very large change in collector base voltage
produces only a small change in collector current.

This means that the output resistance is very high.




frg: output characteristics of transistor CB connection

Here, $I_{E_1} < I_{E_2} < I_{E_3} < I_{E_4} < I_{E_5}$

$\therefore$ output resistance $(r_o) = \frac{\Delta V_{CB}}{\Delta I_C}$ at constant $I_E$

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## Common emitter Connection

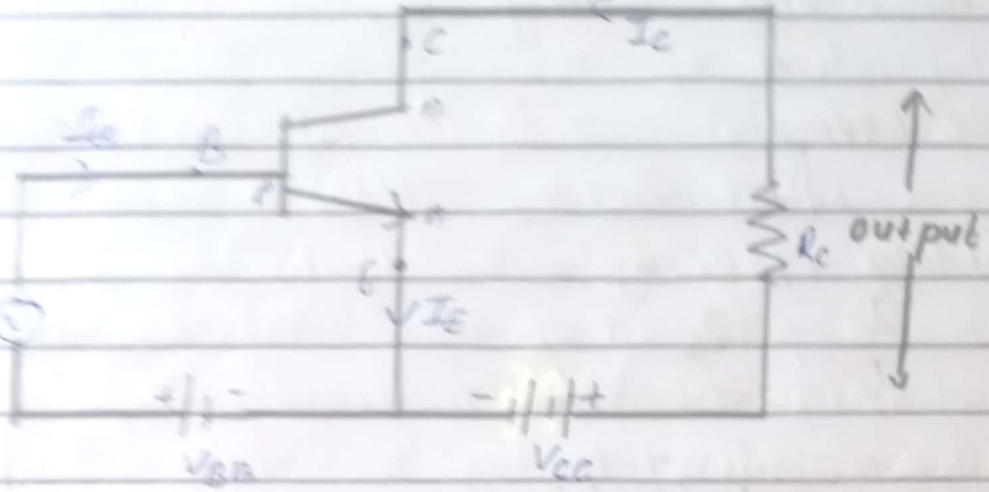


fig: CE circuit for n-p-n transistor

In common emitter connection, input is applied between base & emitter & output is taken from collector and emitter.

Here, the emitter of transistor is common to both input & output circuit. The common emitter n-p-n transistor ckt is shown in figure above.

### Current amplification factor ($\beta$)

It is the ratio of output current to input current. Here, in common emitter connection, input current is $I_B$ & output current is $I_C$.

The ratio of change in collector current to the change in base current is known as current amplification factor ($\beta$).

$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

If only dc is considered,

$$\beta = \frac{I_C}{I_B}$$

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Imp

* relationship bet$^n$ $\alpha$ and $\beta$:

As $\alpha = \frac{I_C}{I_E}$ and $\beta = \frac{I_C}{I_B}$

We know,

$$\boxed{I_E = I_B + I_C} \quad \text{---(i)}$$

Dividing eqn (i) by $I_C$:

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + \frac{I_C}{I_C}$$

or, $\frac{1}{\alpha} = \frac{1}{\beta} + 1$

or, $\frac{1}{\beta} = \frac{1}{\alpha} - 1$

or, $\frac{1}{\beta} = \frac{1-\alpha}{\alpha}$

$$\therefore \boxed{\beta = \frac{\alpha}{1-\alpha}} \quad \text{---(ii)}$$

## Expression for Collector current:

As,

$$I_C = \alpha I_E + I_{CBO}$$

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$$I_C = \alpha (I_B + I_C) + I_{CBO}$$

$$I_C = \alpha I_B + \alpha I_C + I_{CBO}$$

$$(1-\alpha) I_C = \alpha I_B + I_{CBO}$$

$$\boxed{I_C = \frac{\alpha}{1-\alpha} I_B + \frac{1}{1-\alpha} I_{CBO}} \quad \text{---(i)}$$

when, $I_B = 0$, $I_C$ is abbreviated as $I_{CEO}$;

$$\therefore I_{CEO} = \frac{1}{1-\alpha} I_{CBO} \quad \text{---(ii)}$$

$$\therefore I_C = \frac{\alpha}{1-\alpha} I_B + I_{CEO}$$

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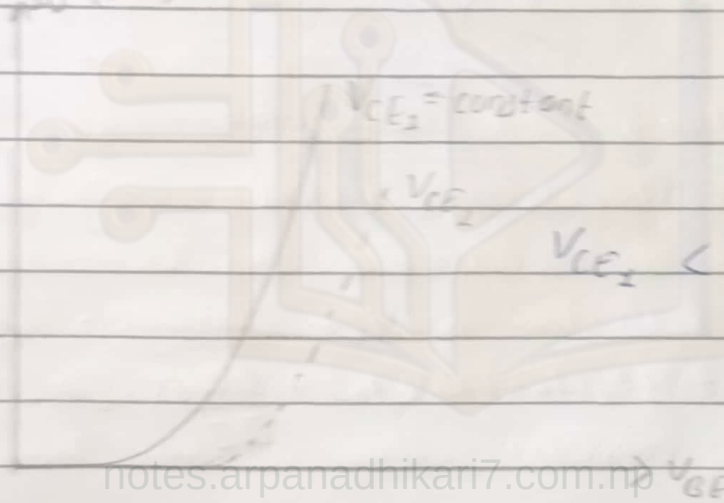
$I_C = \beta I_B + I_{CEO}$ —(iv)

Input characteristics

It is the curve bet$^n$ base current ($I_B$) & base emitter voltage ($V_{BE}$) at constant $V_{CE}$. Generally, base current $I_B$ is taken along y-axis & base emitter voltage $V_{BE}$ is taken along x-axis.

The input characteristics is similar to that of forward biased diode ~~current~~ curve. As compared to common base connection, $I_B$ increases less rapidly with $V_{BE}$. Therefore, the input resistance of common emitter ckt is higher than that of common base ckt.

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input resistance ($r_i$) = $\frac{\Delta V_{BE}}{\Delta I_B}$ at constant $V_{CE}$

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Output characteristics

It is the curve between the collector current $I_C$ & collector emitter voltage $V_{CE}$ at constant $I_B$. Generally, the collector current $I_C$ is taken along y-axis and collector emitter voltage ($V_{CE}$) along x-axis. The collector current ($I_C$) varies with $V_{CE}$ for small

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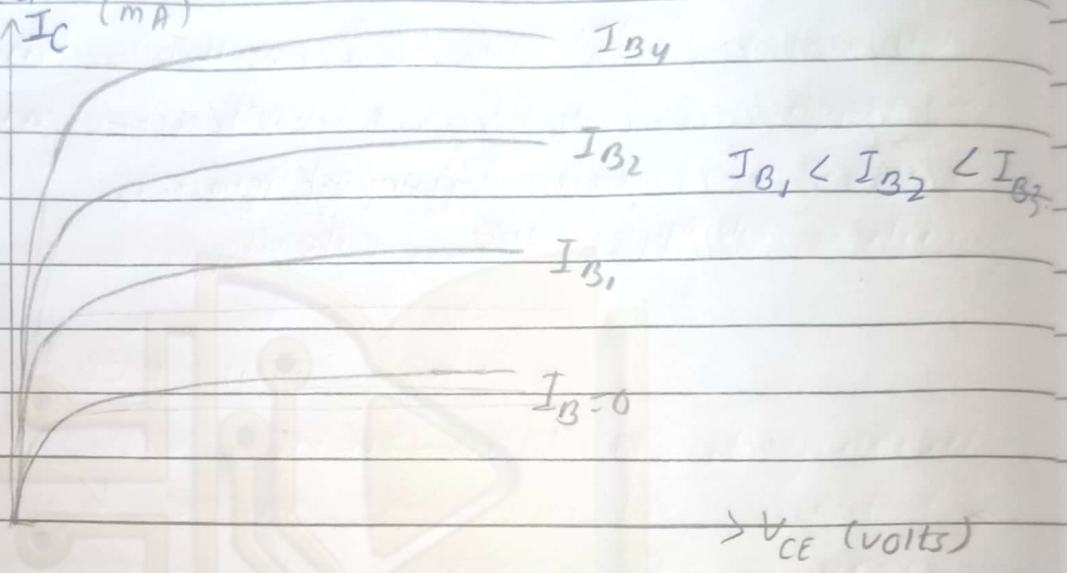
range of $V_{CE}$ (i.e betⁿ 0 and 1 volt). After this, the collector current becomes constant and independent of $V_{CE}$.

In this region, $I_C$ is almost constant. The output resistance of common emitter connection is high. It may be noted that, the output characteristics of common base ckts are horizontal but there is a ~~likeable~~ noticeable slope for the common emitter circuit.




output resistance $(r_o) = \frac{\Delta V_{CE}}{\Delta I_C}$ at constant $I_B$.

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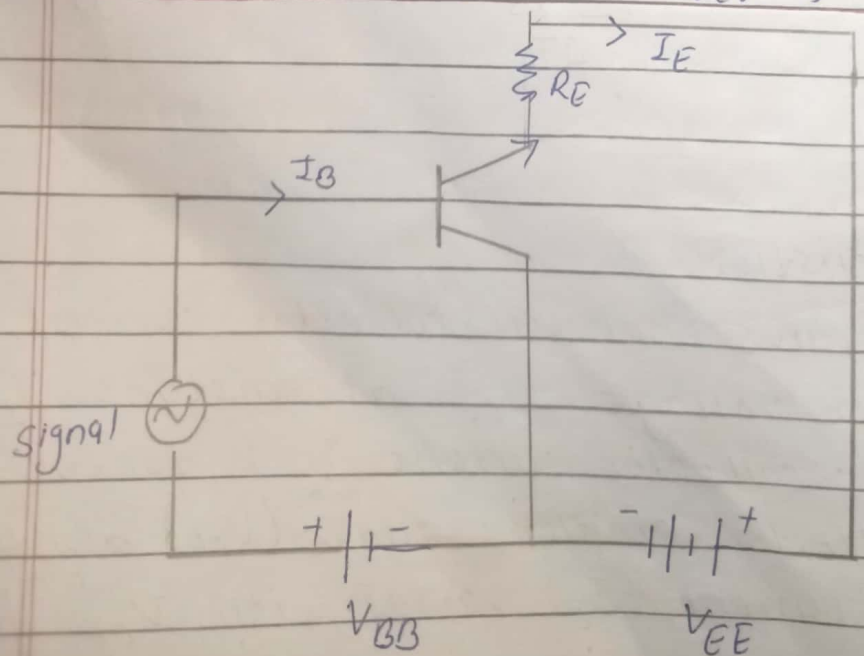
## (iii) Common collector connection

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In common collector connection, the input is applied bet$^n$ base & collector and output is taken bet$^n$ emitter and collector here, the collector of transistor is common to both input & output.

## current amplification factor ($\gamma$)

It is the ratio of change in emitter current ($I_E$) to the base current ($I_B$).

$$\gamma = \frac{\Delta I_E}{\Delta I_B}$$

if only dc is considered

$$\gamma = \frac{I_E}{I_B}$$

## Relationship bet$^n$ $\alpha$, $\beta$, and $\gamma$

Since, $\beta = \frac{I_C}{I_B}$, $\alpha = \frac{I_C}{I_E}$ and $\gamma = \frac{I_E}{I_B}$

start relationship with $\beta$

Also, $I_E = I_B + I_C$ — (1)

or, $\frac{I_E}{I_B} = \frac{I_B}{I_B} + \frac{I_C}{I_B}$

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or, $\gamma = 1 + \beta$

$$\therefore \boxed{\beta = \gamma - 1}$$

Again,

Dividing eqn (1) by $I_C$

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + \frac{I_C}{I_C}$$

or, $\frac{1}{\alpha} = \frac{1}{\beta} + 1$

or, $\frac{1}{\beta} = \frac{1}{\alpha} - 1$

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$$\therefore \boxed{\beta = \frac{\alpha}{1-\alpha}}$$

Comparision bet$^n$ common base, common ~~collected~~ emitter and common collector connection.

| Parameters | Common Base | Common Emitter | Common Collector |
|---|---|---|---|
| 1. Input resistance | low | low | Very high |
| 2. output resistance | very high | high | low |
| 3. Voltage gain | medium (about 150) | High (About 500) | less than 1 |
| 4. current gain | less than 1 | High | Appreciable |
| 5. Application | For high frequency application | For Audio frequency application | For impedance matching. |

## Transistor load line Analysis

In the transistor circuit analysis, it is generally required to determine the collector current for various collector emitter voltage. The convenient method to determine the collector current for various collector emitter voltage is load line analysis method.

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### 1. DC load line Analysis

DC load line is drawn in output characteristic curve. Let us consider a common emitter n-p-n transistor circuit as shown in figure.

It should be noted that while drawing a dc load line capacitor in the ckt acts as open circuit.

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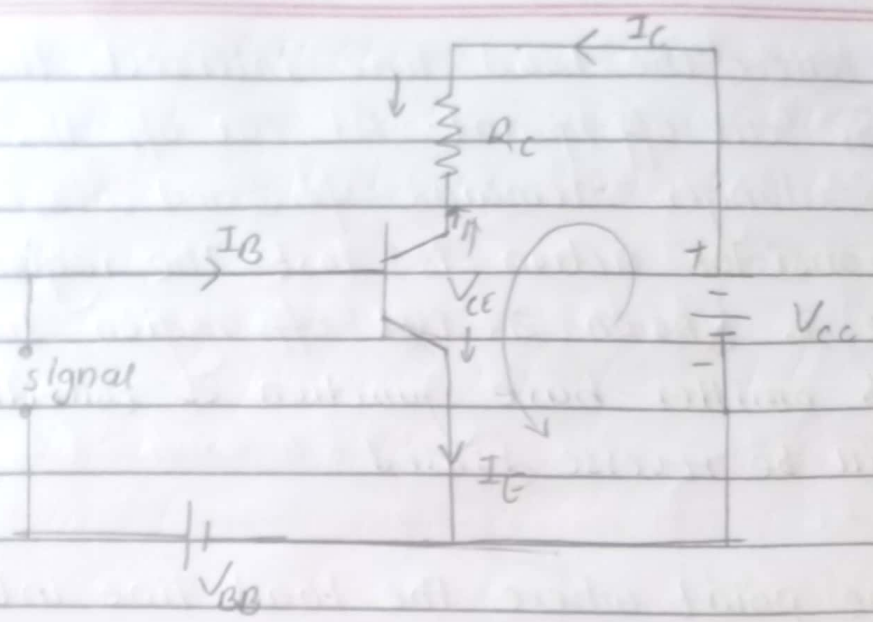



Now, Applying KVL at output,

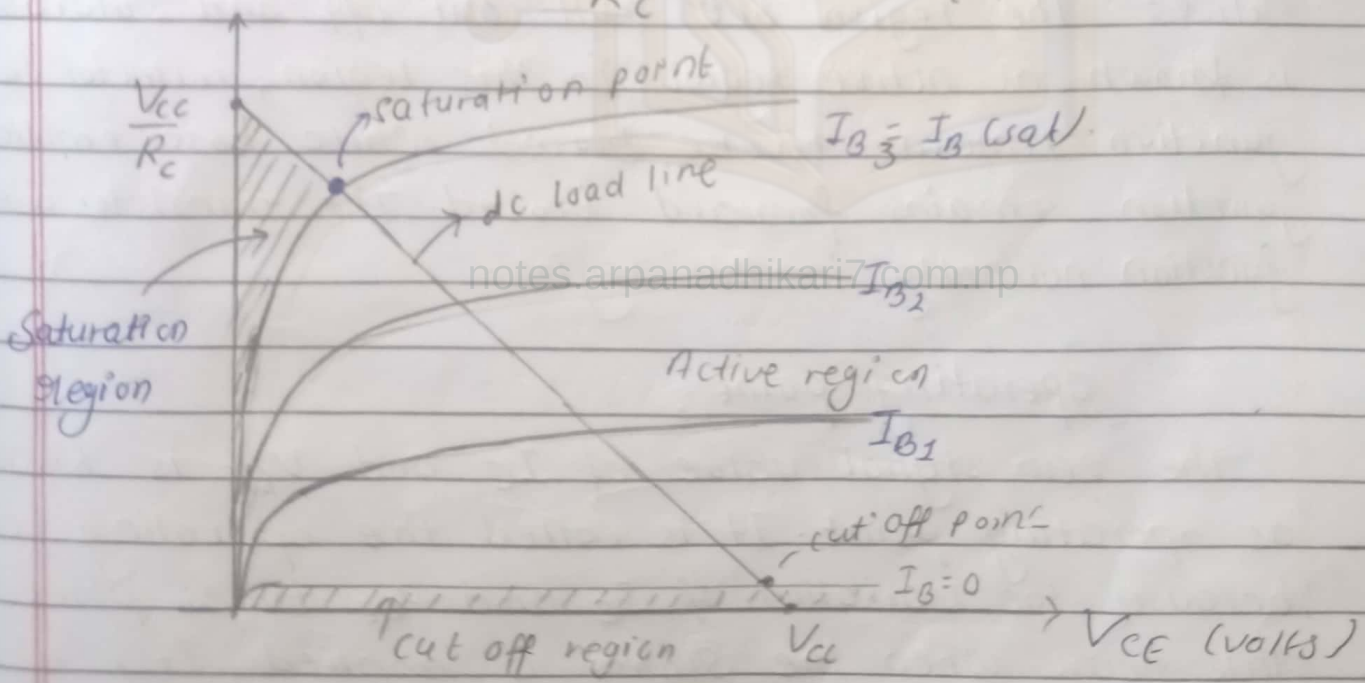
$$V_{CC} - I_C R_C - V_{CE} = 0 \quad \text{—(1)}$$

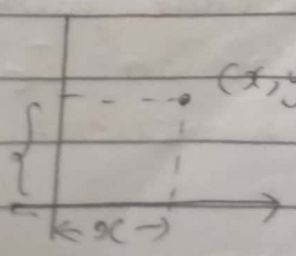
put $I_C = 0$ in eqn (1),

$$V_{CC} = V_{CE} \quad (V_{CC}, 0)$$

put $V_{CE} = 0$ in eqn (1).

$$I_C = \frac{V_{CC}}{R_C} \quad \left(0, \frac{V_{CC}}{R_C}\right)$$

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cut off:

The point where the load line intersects the $I_B = 0$ curve is known as cut off point. At cut off the base emitter junction no longer remains forward biased and normal transistor action is lost. The region below $I_B = 0$ curve is known as cut off region. In cut off region, both emitter base junction & collector base junction will be reverse biased.

saturation: The point where the load line intersects the $I_B = I_B(sat)$ is known as saturation point. At saturation collector base junction no longer remains reverse biased and normal transistor action is lost. In saturation region, both emitter base junction & collector base junction are forward biased.

Active: The region bet$^n$ cut off and saturation is known as active region. In this region, collector base junction remains reverse biased while base emitter junction remains forward biased. The transistor will function normally in this region.

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## operating point:

The zero signal value of $I_C$ and $V_{CE}$ is known as operating point. It is called the operating point because the variation of $I_C$ and $V_{CE}$ takes place about this point when the signal is applied. It is also called ~~Quotient~~ Quiescent point or 'Q-point because it is the point on $I_C - V_{CE}$ characteristics when the transistor is silent (i.e in the absence of signal.

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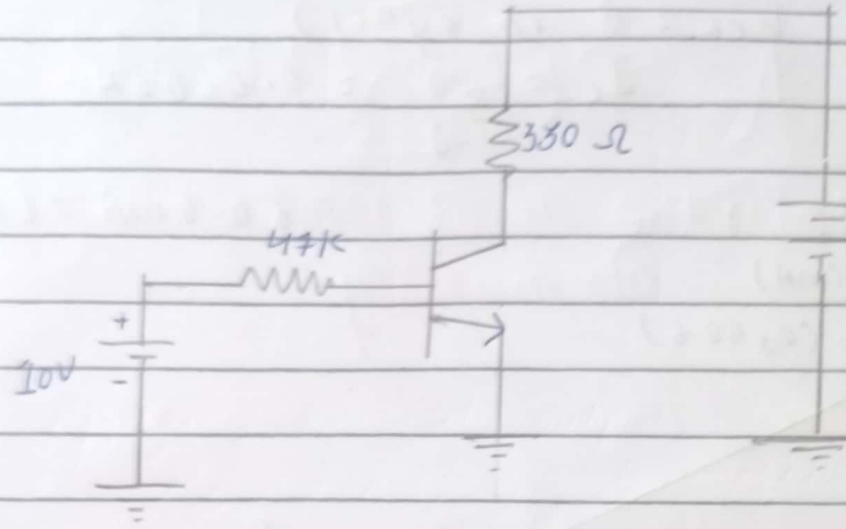
Ex: For the circuit shown in the figure, find operating point. Also draw the dc load line.

[Given: $\beta = 100$ & $V_{BE} = 0.7V$]

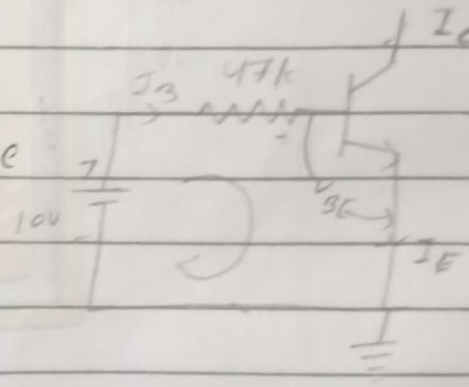



Solution,

Given,

$V_{BE} = 0.7V$

$\beta = 100$




Now, applying kvl at input side

$$10 - 47000 I_B - V_{BE} = 0$$

$$I_B = \frac{10 - V_{BE}}{47000}$$

$$= \frac{10 - 0.7}{47000}$$

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$$= 0.000197 \text{ Amp}$$

$$= 0.197 \text{ mA}$$

$$I_C = \beta I_B$$

$$= 100 \times 0.197 \text{ mA}$$

$$= 19.7 \text{ mA}$$

Now, applying kvl at output side,

$$20 - 330 \times I_C - V_{CE} = 0 \quad \text{— (1)}$$

$$\text{or, } 20 - 330 \times 19.7 \times 10^{-3} V_{CE} = 0$$

$$\text{or, } V_{CE} = 13.5V$$

$\therefore$ operating point $(V_{CE}, I_C) = (13.5V, 19.7 \text{ mA})$

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For dc load line;

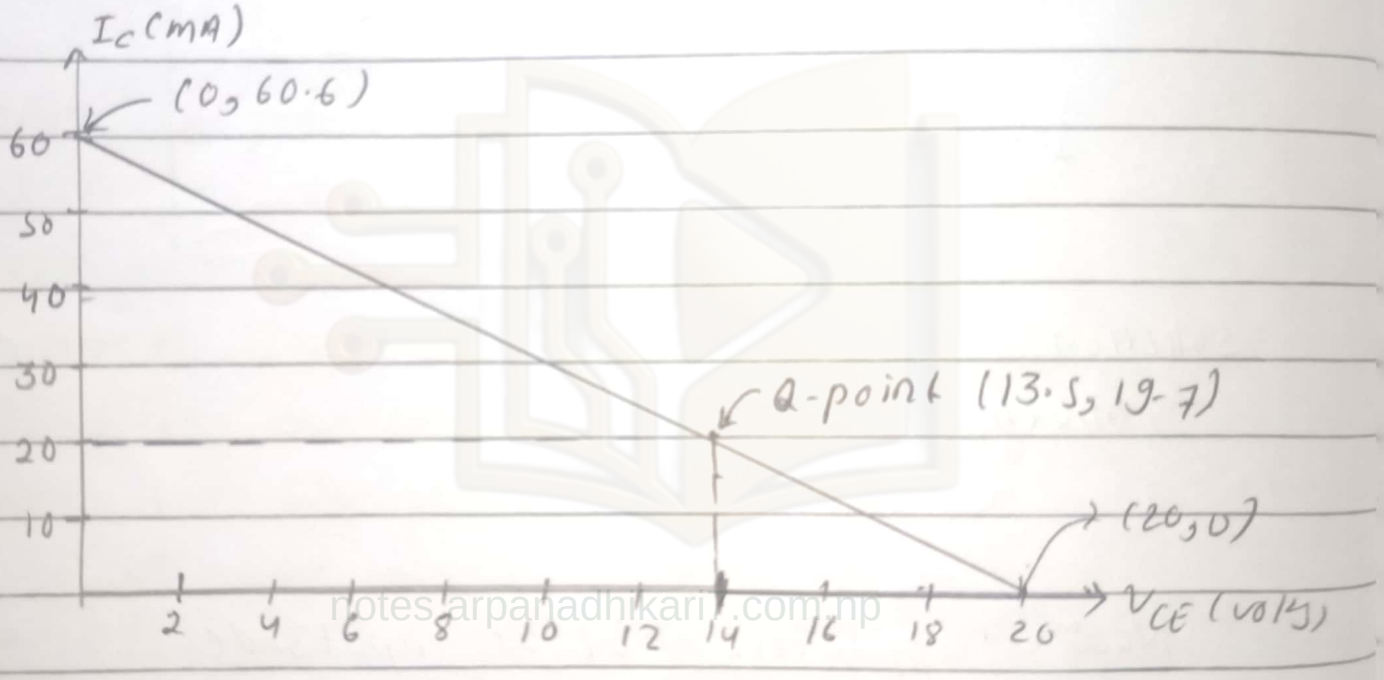
put $I_C = 0$ in eqn(1)

$V_{CE} = 20V = (20, 0)$

put $V_{CE} = 0$ in eqn(1)

$$I_C = \frac{20}{330} = 0.0606A$$

$= 60.6 mA = (0, 60.6)$

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In dc analysis => capacitor acts as open ckt

$$V_{Th} = \frac{V_{CC} \times R_2}{R_1 + R_2}$$

$$R_{Th} = \frac{R_1 \times R_2}{R_1 + R_2}$$

For ac analysis, capacitor acts as short.

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$$r_{o\ stage} = R_E \,||\, \beta \frac{r_e}{\beta}$$

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## Stabilization

The process of making operating point independent of temp change or variation in transistor parameters is known as stabilization. Once stabilization is done, the operating point is fixed. Stabilization of operating point is necessary due to following reasons:

(i) Temperature dependence of $I_C$
(ii) Individual variations
(iii) Thermal runaway

## Stability factor (S)

The rate of change of collector current $I_C$ wrt collector leakage current ($I_{CO}$). At constant $I_B$ and $\beta$ is called stability factor (S)

$$\text{ie } S = \frac{dI_C}{I_{CO}} \text{ at constant } \beta \text{ and } I_B$$

Since, we have for common emitter transistor circuits,

$$I_C = \beta I_B + (\beta+1) I_{CO}$$

Differentiating eqn (i) wrt $I_C$

$$\frac{dI_C}{dI_C} = \beta \frac{dI_B}{dI_C} + (\beta+1)\frac{dI_{CO}}{dI_C}$$

$$\text{or, } 1 = \beta \frac{dI_B}{dI_C} + (\beta+1)\frac{1}{S}$$

$$\text{or, } (\beta+1)\frac{1}{S} = 1 - \beta \frac{dI_B}{dI_C}$$

$$\text{or, } \boxed{S = \frac{\beta+1}{1-\beta\frac{dI_B}{dI_C}}}$$

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# Transistor Biasing

The proper flow of zero signal collector current and the maintenance of proper collector emitter voltage during the passage of signal is known as transistor Biasing. The basic purpose of transistor biasing is to keep the base emitter junction properly forward biased and collector base junction properly reverse biased during the application of signal.

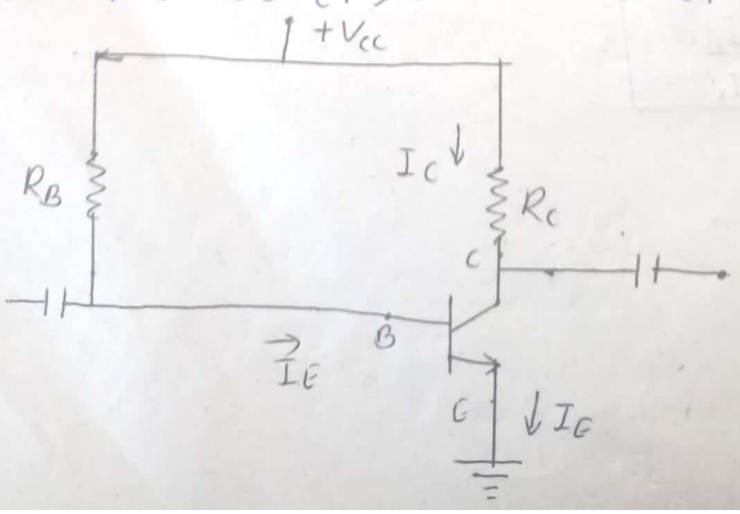
This can be achieved with a biased battery or associating a circuit with a transistor. The circuit which provides transistor biasing is known as biasing circuit. The biasing network associated with the transistor should meet the following requirements:

(i) It should ensure proper zero signal collector current.
(ii) It should ensure that emitter base junction is forward biased and collector base junction reverse biased.
(iii) It should ensure the stabilization of operating point.

## Methods of Biasing

(1) Base resistor method (fixed bias circuit)
(2) Emitter bias method
(3) Biasing with collector feedback resistor (collector to base bias circuit).
(4) Voltage divider bias

### 1. Base resistor method (fixed bias circuit)

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A high resistance $R_B$ is connected bet$^n$ the base and positive end of the supply for n-p-n transistor as shown in figure

Applying KVL at input side,

$$V_{CC} - I_B R_B - V_{BE} = 0 \quad \text{—(i)}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} \quad \text{—(ii)}$$

$$\therefore I_C = \beta I_B \quad \text{—(iii)}$$

Again, Applying KVL at output side

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C \quad \text{—(iv)}$$

Now,

diff eq$^n$ (i) w.r.t $I_C$

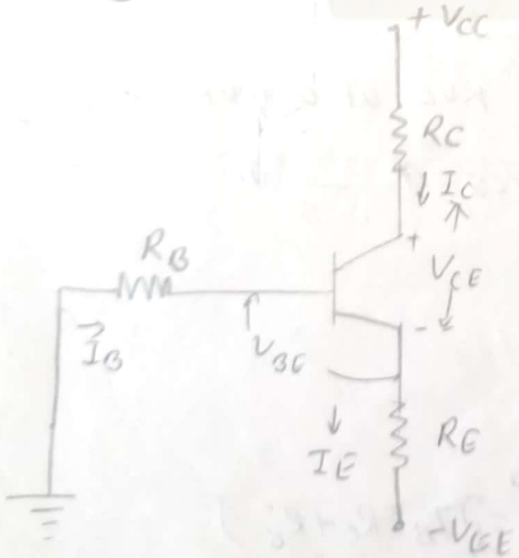
$$\frac{dV_{CC}}{dI_C} - R_B \frac{dI_B}{dI_C} - \frac{dV_{BE}}{dI_C} = 0$$

$$0 - R_B \frac{dI_B}{dI_C} - 0 = 0$$

$$\boxed{\frac{dI_B}{dI_C} = 0}$$

Hence, $\boxed{S = \beta + 1}$

2. Emitter Bias method:




Applying KVL at input side,

$$-I_B R_B - V_{BE} - I_E R_E + V_{EE} = 0$$

$$I_B R_B + (I_B + I_C) R_E = V_{EE} - V_{BE} \quad \text{—(i)}$$

Again,

Applying KVL at output side,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E + V_{EE} = 0 \quad \text{—(ii)}$$

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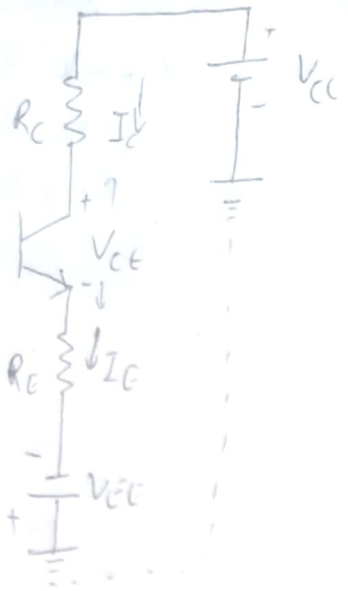



Diff. eqn (1) wrt $I_C$

$$R_B \frac{dI_B}{dI_C} + R_C \frac{dI_B}{dI_C} + R_E = -\frac{d(V_{EE} - V_{BE})}{dI_C}$$

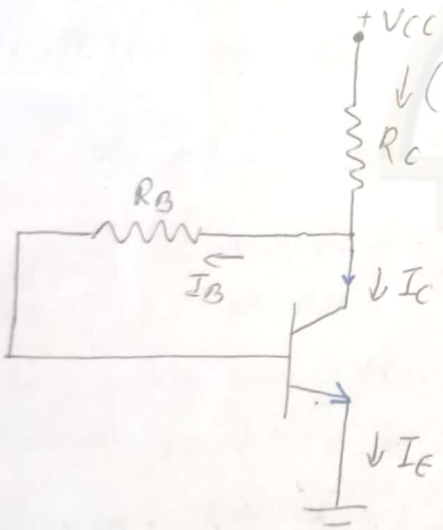
$$\frac{dI_B}{dI_C}(R_B + R_E) + R_E = 0$$

$$\therefore \frac{dI_B}{dI_C} = -\frac{R_E}{R_B + R_E}$$

Hence,

$$S = \frac{\beta + 1}{1 + \beta\left(\frac{R_E}{R_B + R_E}\right)}$$

3. Bias with collector feedback resistor.




Applying KVL at input side,

$$V_{CC} - (I_C + I_B)R_C - I_B R_B - V_{BE} = 0$$

$$\text{or, } V_{CC} - I_C R_C - I_B R_C - I_B R_B - V_{BE} = 0$$

$$\text{or, } V_{CC} - I_C R_C - I_B(R_C + R_B) - V_{BE} = 0 \quad \text{— (1)}$$

Again,

Applying KVL at output side,

$$V_{CC} - (I_C + I_B)R_C - V_{CE} = 0 \quad \text{— (1)}$$

Since,

$$S = \frac{\beta + 1}{1 - \beta \frac{dI_B}{dI_C}}$$

∴ Now, diff. eqn (1) wrt $I_C$,

$$\frac{\partial}{\partial I_C}(V_{CC} - I_C R_C - V_{BE}) = \frac{\partial I_B}{\partial I_C}(R_C + R_B)$$

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$$\text{or, } \frac{\partial V_{CC}}{\partial I_C} - R_C \frac{\partial I_C}{\partial I_C} - (R_C + R_B)\frac{\partial I_B}{\partial I_C} - \frac{\partial V_{BE}}{\partial I_C} = 0$$

$$\text{or, } 0 - R_C - (R_C + R_B)\frac{\partial I_B}{\partial I_C} - 0 = 0$$

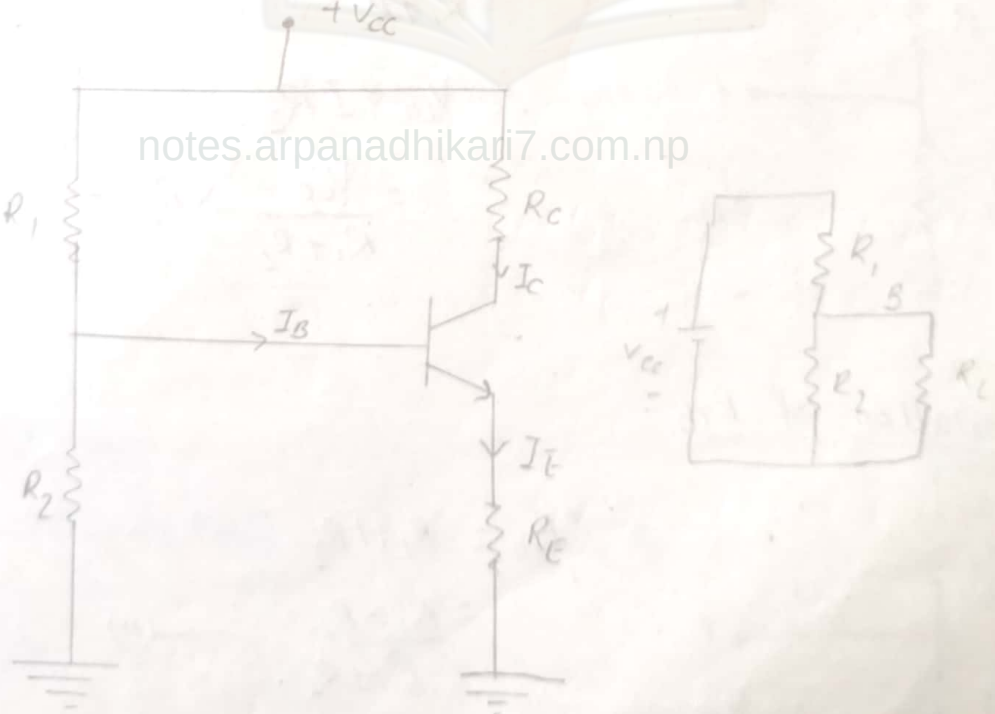
$$\text{or, } (R_C + R_B)\frac{\partial I_B}{\partial I_C} = -R_C$$

$$\text{or, } \frac{\partial I_B}{\partial I_C} = \frac{-R_C}{R_C + R_B}$$

$$\therefore S = \frac{\beta + 1}{1 + \beta \dfrac{R_C}{R_C + R_B}}$$

Since, stability factor of biased with collector feedback resister (collector to base bias) circuit is less than that of fixed bias. So, it is more stable than fixed bias.

## 4. Voltage Divider Bias (self-biased circuit)

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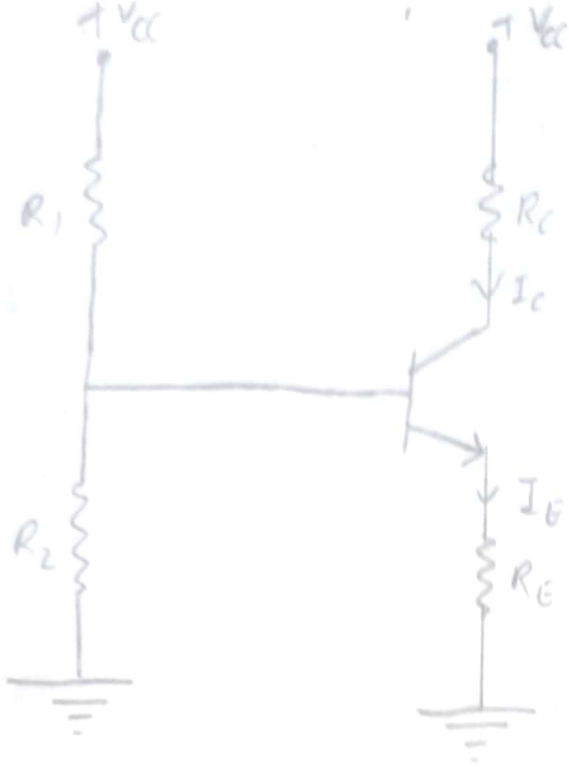


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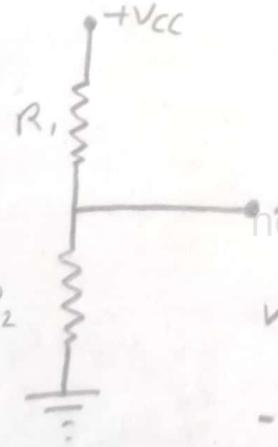
Now, replacing the input side by its equivalent Thevenins circuit.

Calculation of $V_{Th}$

$$V_{CC} - IR_1 - IR_2 = 0$$




$$I = \frac{V_{CC}}{R_1 + R_2}$$

$$\therefore V_{Th} = IR_2$$

$$= \frac{V_{CC}}{R_1 + R_2} \times R_2 \quad \text{— (i)}$$

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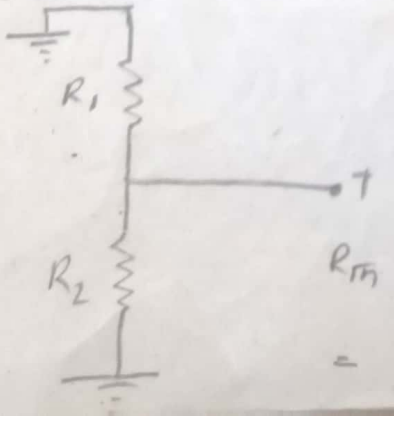
Calculation of $R_{Th}$




$$R_{Th} = R_1 \parallel R_2$$

$$= \frac{R_1 \times R_2}{R_1 + R_2} \quad \text{— (ii)}$$

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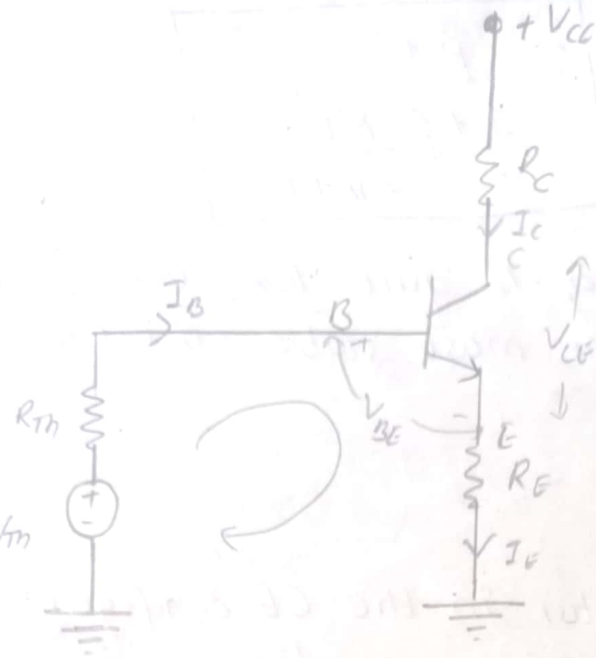


Applying KVL at input side,

$$V_{Th} - R_{Th} I_B - V_{BE} - R_E I_E = 0$$

$$\Rightarrow V_{Th} - R_{Th} I_B - V_{BE} - R_E (I_B + I_C) = 0$$

$$\Rightarrow V_{Th} - R_{Th} I_B - V_{BE} - I_B R_E - I_C R_E = 0$$

$$\Rightarrow V_{Th} - (R_{Th} + R_E) I_B - V_{BE} - I_C R_E = 0 \quad \text{——(i)}$$

Now,

Applying KVL at output side,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0 \quad \text{——(ii)}$$

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Since, $\beta = \dfrac{\beta + 1}{1 - \beta \dfrac{\partial I_B}{\partial I_C}}$

diff eqn (i) wrt $I_C$.

$$\frac{\partial V_{Th}}{\partial I_C} - (R_{Th} + R_E) \frac{\partial I_B}{\partial I_C} - \frac{\partial V_{BE}}{\partial I_C} - R_E \frac{\partial I_C}{\partial I_C} = 0$$

$$\text{or, } 0 - (R_{Th} + R_E) \frac{\partial I_B}{\partial I_C} - R_E = 0 = 0$$

$$\text{or, } \frac{\partial I_B}{\partial I_C} = -\frac{R_E}{R_{Th} + R_E} \qquad \therefore S = \frac{\beta + 1}{1 + \beta \dfrac{R_E}{R_{Th} + R_E}}$$

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~~$I = \frac{V_{cc}}{R_1 + R_2}$~~

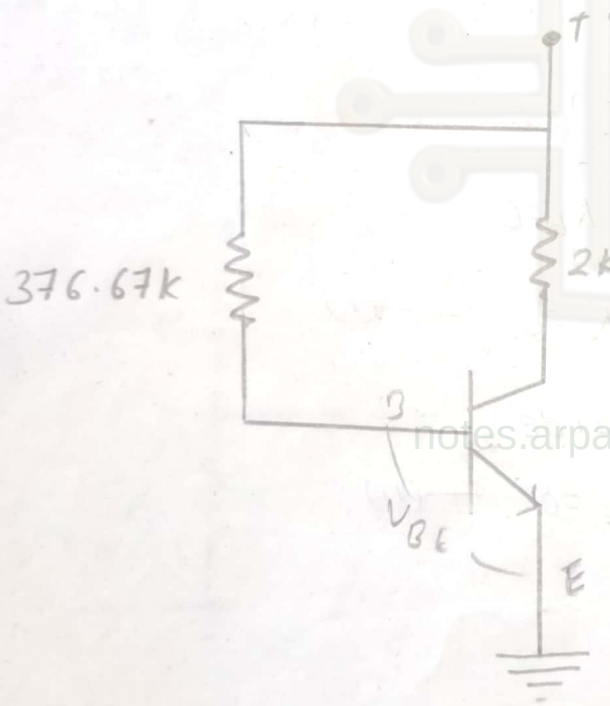
~~$V_{Th} = I R_2$~~

$$\therefore S = \frac{\beta + 1}{1 + \beta \frac{R_E}{R_{Th} + R_E}}$$

Since, the stability factor is quite low, voltage divider or self bias circuit is the most stable circuit as compared to other.

## Numerical:

Q.1 For the silicon transistor in the CE configuration as shown in figure Determine operating point (Q-point). Also draw dc load line.




Assume $\beta = 100$

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## Solution

Given,

$V_{cc} = 12\,V$

$R_C = 2k\Omega = 2000\Omega$

$R_B = 376.67k\,\Omega = 376.67 \times 1000\,\Omega$

$\beta = 100$

For silicon $V_{BE} = 0.7\,V$.

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Applying KVL at input side,

$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$= \frac{12 - 0.7}{376.67 \times 1000}$$

$$= 3 \times 10^{-5} \text{ Amp}$$

$$= 30 \times 10^{-6} \text{ Amp}$$

$$= 30 \mu A$$

$\therefore I_C = \beta I_B$

$$= 100 \times 3 \times 10^{-5}$$

$$= 3 \times 10^{-3} \text{ Amp}$$

$$= 3 \text{ mA}$$

Again, Applying KVL at output side,

$$V_{CC} - I_C R_C - V_{CE} = 0 \quad \text{——(1)}$$

$$\Rightarrow 12 - 3 \times 10^{-3} \times 2 \times 10^{3} - V_{CE} = 0$$

$$\Rightarrow V_{CE} = 12 - 6$$

$$= 6V$$

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$\therefore$ Operating point or Q-point $= (V_{CE}, I_C)$

$$= (6V, 3 \text{ mA})$$

DC load line

put $I_C = 0$ in eqn (1) we get,

$V_{CE} = V_{CC} = 12V$

$(12, 0)$

put $V_{CE} = 0$ in eqn(1).

$$I_C = \frac{V_{CC}}{R_C} = \frac{12}{2 \times 1000} = 6 \text{ mA}$$

$(0, 6 \text{ mA})$

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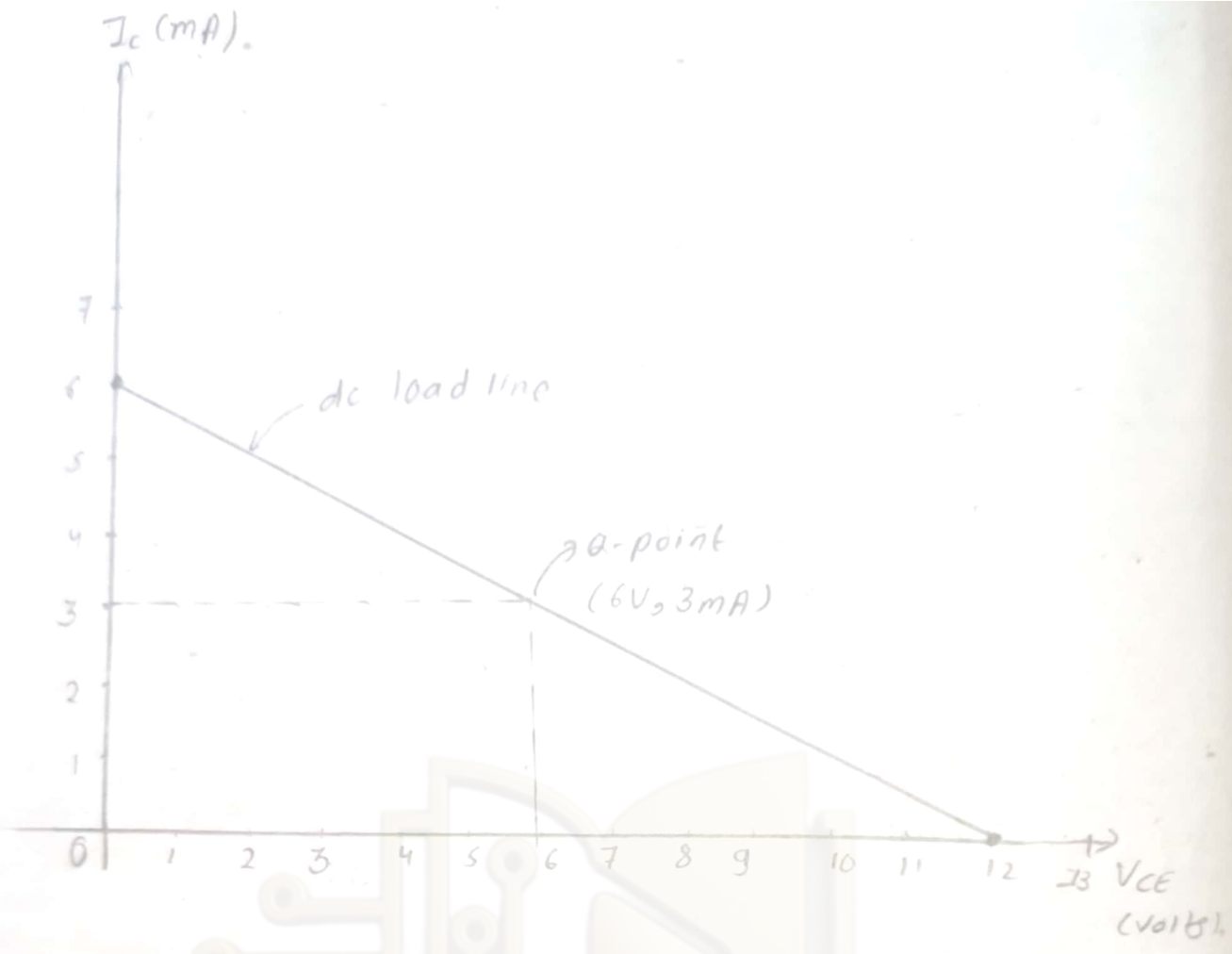


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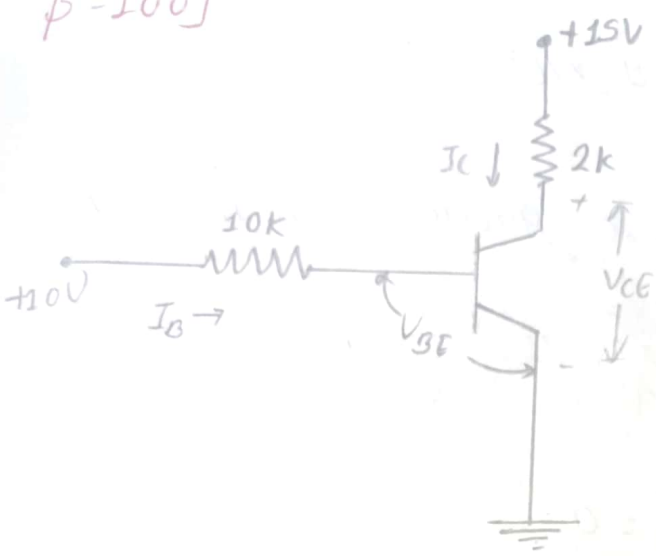
Numerical

Find the operating point & stability factor of the given circuit [Assume $\beta = 100$]




Solution

If not given $V_{BE} = 0.7$ (Assume silicon)

Applying KVL at input side,

$$10 - 10000 I_B - V_{BE} = 0 \quad \text{---(1)}$$

$$\text{or, } I_B = \frac{10 - V_{BE}}{10000}$$

$$\text{or, } I_B = \frac{10 - 0.7}{10000}$$

$$\text{or, } I_B = 9.3 \times 10^{-4} \text{ Amp.}$$

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$$I_C = \beta I_B + (\beta + 1) I_{CBO} \quad (I_{CBO} \to 0)$$

$$= 100 \times 9.3 \times 10^{-4} \text{ Amp}$$

$$= 9.3 \times 10^{-2} \text{ Amp}$$

$$= 9.3 \times \frac{10}{10} \times 10^{-2}$$

$$= 93 \times 10^{-3} \text{ Amp}$$

$$= 93 \text{ mA}$$

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again.

Applying KVL at output side,

$$15 - 2000 I_C - V_{CE} = 0$$

$$V_{CE} = 15 - 2000 \times 93 \times 10^{-5}$$

$$= -171V$$

-ve $V_{CE}$ means both emitter base junction & collector base junction are forward biased.

Now,

Diff eqn (1) wrt $I_C$.

$$\frac{\partial(10)}{\partial I_C} - 10000 \frac{\partial I_B}{\partial I_C} - \frac{\partial V_{BE}}{\partial I_C} = 0$$

$$\Rightarrow 0 - 10000 \frac{\partial I_B}{\partial I_C} - 0 = 0$$

$$\Rightarrow \frac{\partial I_B}{\partial I_C} = 0$$

$$\therefore S = \frac{\beta + 1}{1 - \beta \frac{\partial I_B}{\partial I_C}} \to 0$$

$$= \beta + 1$$

$$= 100 + 1$$

$$= 101$$

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Steps:

1. If the circuit is voltage divider biasing, replace input by its thevenin's equt circuit.
2. Apply KVL at input side, find $I_B$ & $I_C$
3. Apply KVL at output side find $V_{CE}$.
4. load line: Use eqn from KVL at output side. Find two points by taking $V_{CE} = 0$ and $I_C = 0$

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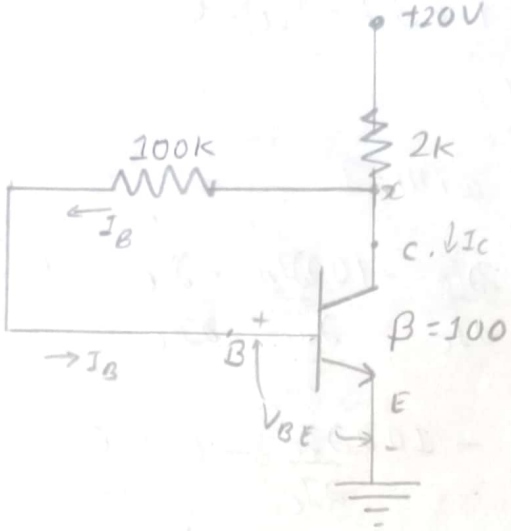
5. For stability factor diff eqn from KVL at input side wrt $I_C$.

Qn2. Find $I_B$, $I_C$, $V_{CE}$ and S for the circuit shown in figure below:




$I_E = I_B + I_C$

Note:

| V | I | R |
|---|---|---|
| volt | Amp | ohm |
| volt | mA | KΩ |

Solution

Given,

$\beta = 100$

$V_{BE} = 0.7\ V$

Applying KCL at x

$I = I_B + I_C$

Applying KVL at input side,

$$20V - 2(I_B + I_C) - 100I_B - V_{BE} = 0 \quad \text{---(1)}.$$

Or, $20 - 2(I_B + \beta I_B) - 100I_B - V_{BE} = 0$

or, $2(I_B + 100I_B) + 100I_B = 20 - V_{BE}$

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or, $302I_B = 20 - V_{BE}$

or, $I_B = \frac{20 - 0.7}{302}$

or, $I_B = 0.063$ mA

$\therefore I_C = \beta I_B$

$= 100 \times 0.063$

$= 6.3\ mA$

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Now, Applying kvl at output side,

$$20V - 2(I_B + I_C) - V_{CE} = 0$$

$$\text{or, } 20V - 2(0.063 + 6.3) - V_{CE} = 0$$

$$\text{or, } V_{CE} = 7.274V$$

Now, diff eqn (1) with $I_C$

$$\frac{\partial}{\partial I_C}(20 - 2I_B - 2I_C - 100I_B - V_{BE}) = 0$$

$$\Rightarrow 0 - 2\frac{\partial I_B}{\partial I_C} - 2 - 100\frac{\partial I_B}{\partial I_C} - 0 = 0$$

$$\Rightarrow -102\frac{\partial I_B}{\partial I_C} = 2$$

$$\Rightarrow \frac{\partial I_B}{\partial I_C} = -0.0196$$

Now,

$$S = \frac{\beta + 1}{1 - \beta\frac{\partial I_B}{\partial I_C}}$$

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$$= \frac{100 + 1}{1 - 100(-0.0196)}$$

$$= \frac{101}{1 + 1.9607}$$

$$= \frac{101}{2.9607}$$

$$= 34.113$$

Operating point $(V_{CE}, I_C) = (7.274, 6.3)$

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For dc load line

we have, $20 - 2(I_B + I_C) - V_{CE} = 0$

or, $20 - 2\left(\frac{I_B}{\beta} + I_C\right) - V_{CE} = 0$

put $I_C = 0$, $V_{CE} = 20\ V$ [20, 0]

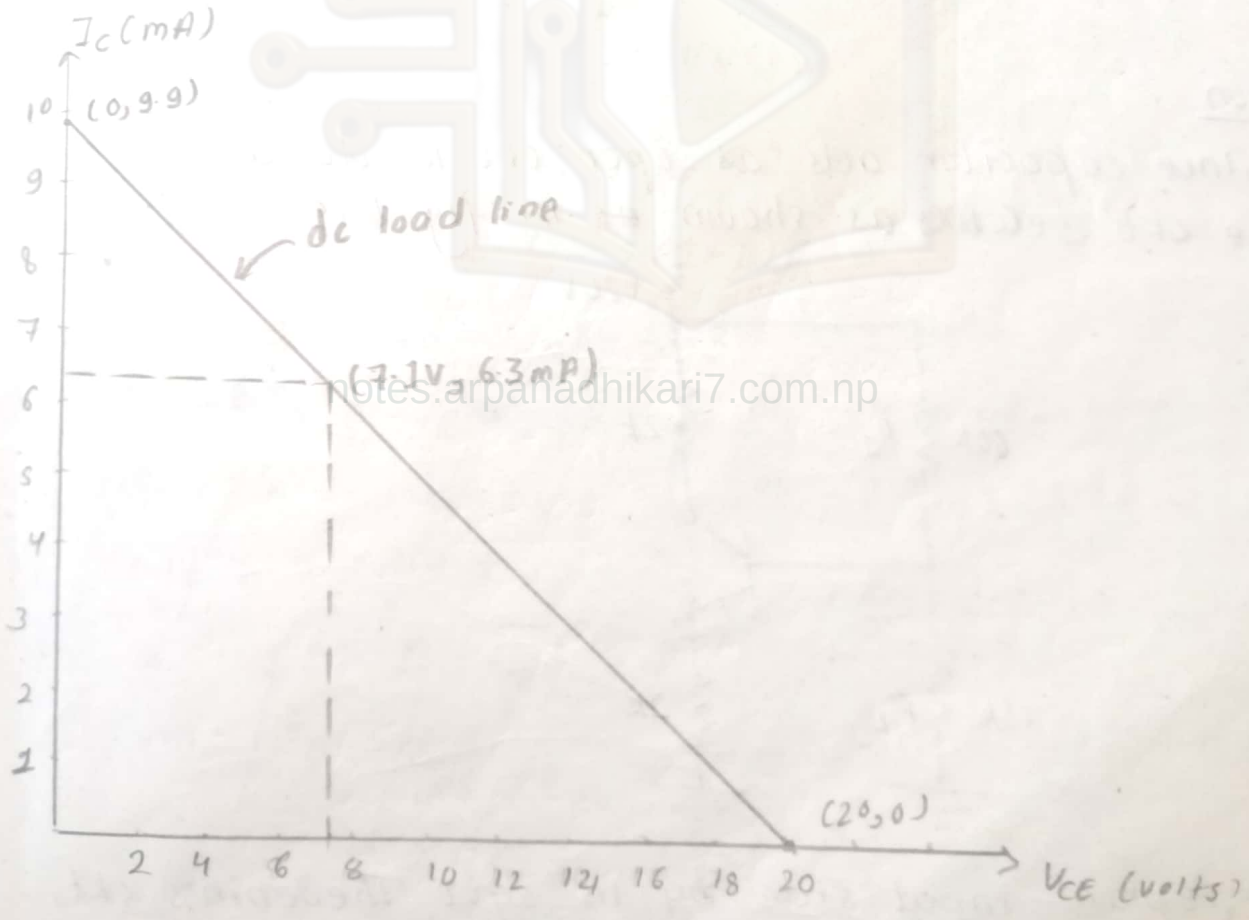
put $V_{CE} = 0$, $2\left(\frac{I_C}{\beta} + I_C\right) = 20$

$\frac{I_C}{100} + I_C = 10$

or, $101 I_C + 100 I_C = 1000$

or, $I_C = \frac{1000}{101}$

or, $I_C = 9.90\ mA$ [0, 9.9 mA]

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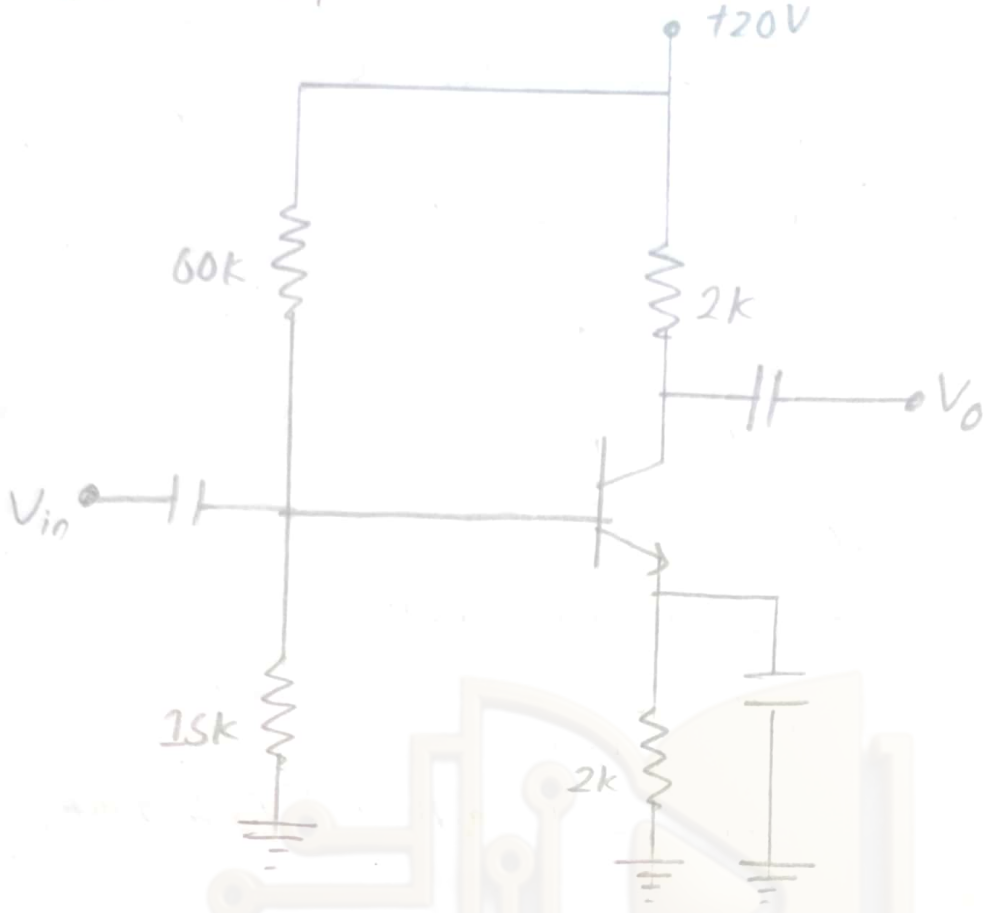
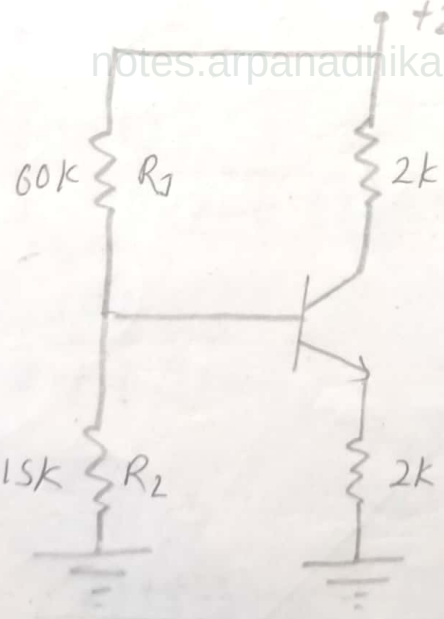
Qn.3 Find operating point (Q-point) & stability factor for the voltage divider circuit shown in figure below. [Given : $\beta = 100$]




Solution

Since capacitor acts as open ckt in dc analysis, now the ckt becomes as shown ~~as~~ in fig below

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Now, replace input side by its equt Thevenin's ckt,

$$V_{Th} = \frac{V_{cc} \times R_2}{R_1 + R_2}$$

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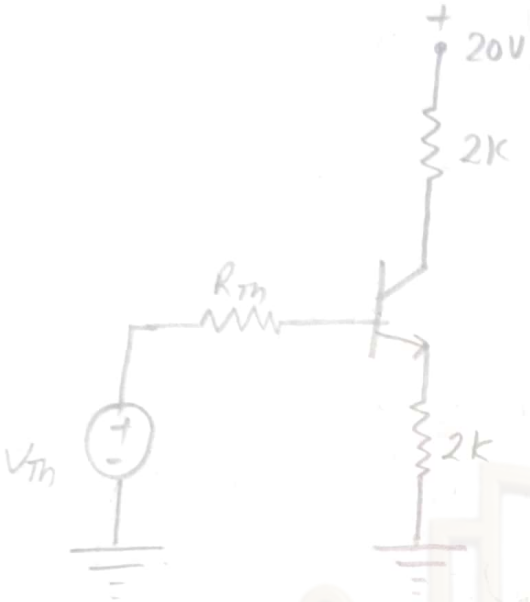
$$V_{Th} = \frac{20 \times 15}{60+15} = 4V$$

$$R_{Th} = \frac{60 \times 15}{60+15} = 12 k\Omega$$




Now, applying kvl at input side.

$$V_{Th} - R_{Th} I_B - V_{BE} - R_E I_E = 0$$

$$\text{or, } 4 - 12 \times I_B - 0.7 - 2(1+\beta) I_B = 0$$

$$\text{or, } -12 I_B - 2 I_B - 2 I_B \beta = -3.3$$

$$\text{or, } -14 I_B - 2 I_B \times 100 = -3.3$$

$$\text{or, } 14 I_B = -3.3 + 2 I_B \times 100$$

$$\text{or, } -214 I_B = -3.3$$

$$\text{or, } 14 I_B = -3.3 + 200 I_B$$

$$\text{or, } I_B = \frac{3.3}{214}$$

$$\text{or, } I_B = \frac{3.3}{186}$$

$$I_B = 0.015420 \text{ mA}$$

$$\text{or, } I_B = 0.01774 \text{ mAmp}$$

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$I_C = \beta I_B$

$= 100 \times 0.01542$

$= 1.542$ mAmp

$I_E = I_B + I_C$

$= 0.0154 + 1.542$

$= 1.5554$ mA

Applying KVL at O/P side,

$20 - 2I_C - V_{CE} - 2I_E = 0$

$\Rightarrow 20 - 2 \times 1.54 - 2 \times 1.55 = V_{CE}$

or, $V_{CE} = 13.81$ V

∴ operating point ($V_{CEQ}, I_{CQ}$)

$= (13.81\text{ V}, 1.54\text{ mA})$

For stability factor.

~~Diff. eqn (1) wrt $I_C$.~~

$V_{Th} - I_B R_{Th} - V_{BE} - 2I_E = 0$

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$\Rightarrow V_{Th} - I_B R_{Th} - V_{BE} - 2I_C - 2I_B = 0$

Diff wrt $I_C$;

$$\frac{\partial V_{Th}}{\partial I_C} - R_{Th}\frac{\partial I_B}{\partial I_C} - \frac{\partial V_{BE}}{\partial I_C} - 2\frac{\partial I_C}{\partial I_C} - 2\frac{\partial I_B}{\partial I_C} = 0$$

or, $0 - R_{Th}\frac{\partial I_B}{\partial I_C} - 0 - 2 - 2\frac{\partial I_B}{\partial I_C} = 0$

or, $\frac{\partial I_B}{\partial I_C} = \frac{-2}{(R_{Th}+2)} = \frac{-2}{(12+2)} = -0.14285$

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Now, $S = \frac{\beta+1}{1-\beta\frac{\partial I_B}{\partial I_C}}$

$= \frac{100+1}{1-100(-0.1428)}$

$= \frac{101}{1+14.28}$

$= \frac{101}{15.28}$

$= 6.6099$

Then, operating point $(V_{CE}, I_C) = (13.81, 1.54)$

For dc load line

we have,

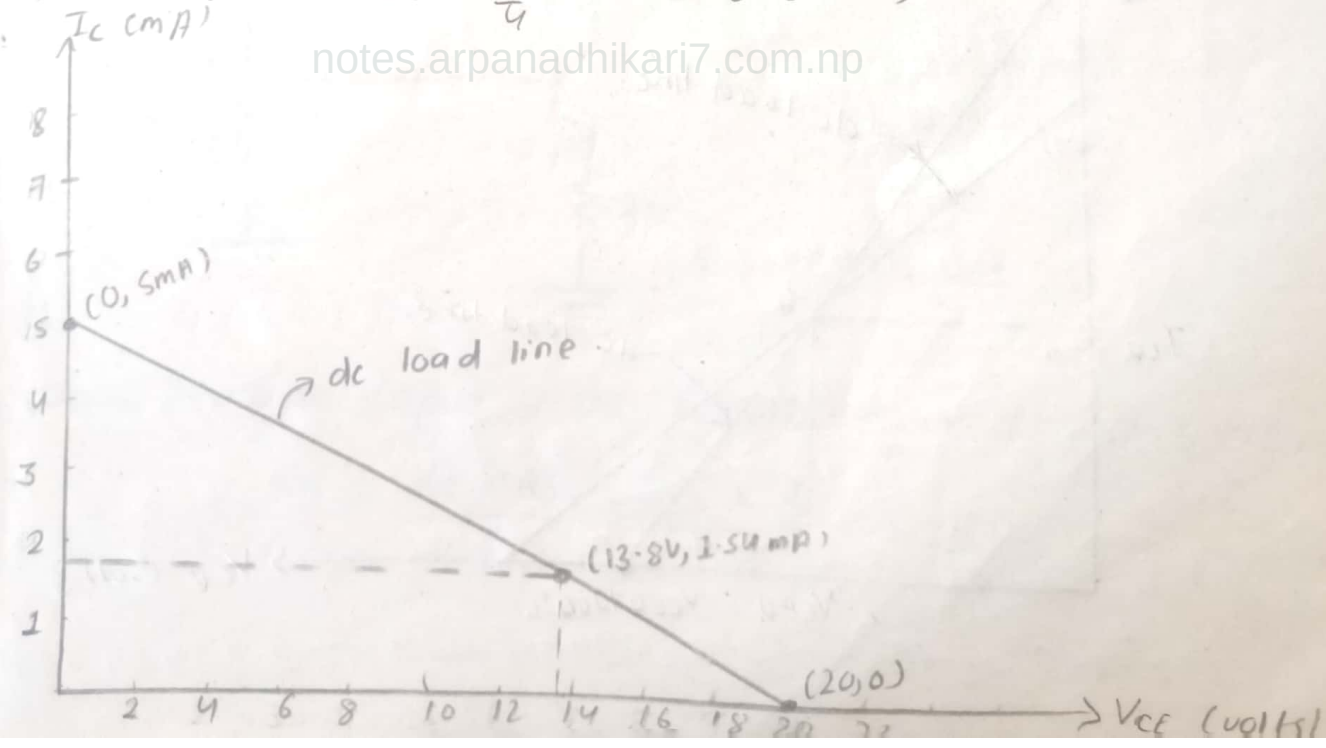
$20 - 2I_C - V_{CE} - 2I_E = 0$

or, $20 - 2I_C - V_{CE} - 2I_C = 0 \quad [\because I_E \approx I_C]$

or, $20 - 4I_C - V_{CE} = 0$ — (i)

put $I_C = 0$, $V_{CE} = 20\ V$ $(20, 0)$

put $V_{CE} = 0$, $I_C = \frac{20}{4} = 5mA$ $(0, 5mA)$

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~~An ac load line passes through Q point~~

# AC load line

An ac load line passes through Q-point and it's slope is $-\frac{1}{r_L}$ where $r_L = R_C \| R_L$. At the equation of AC load line is given by.

$$I_C - I_{CQ} = -\frac{1}{r_L}(V_{CE} - V_{CEQ}) \quad \text{——(1)}$$

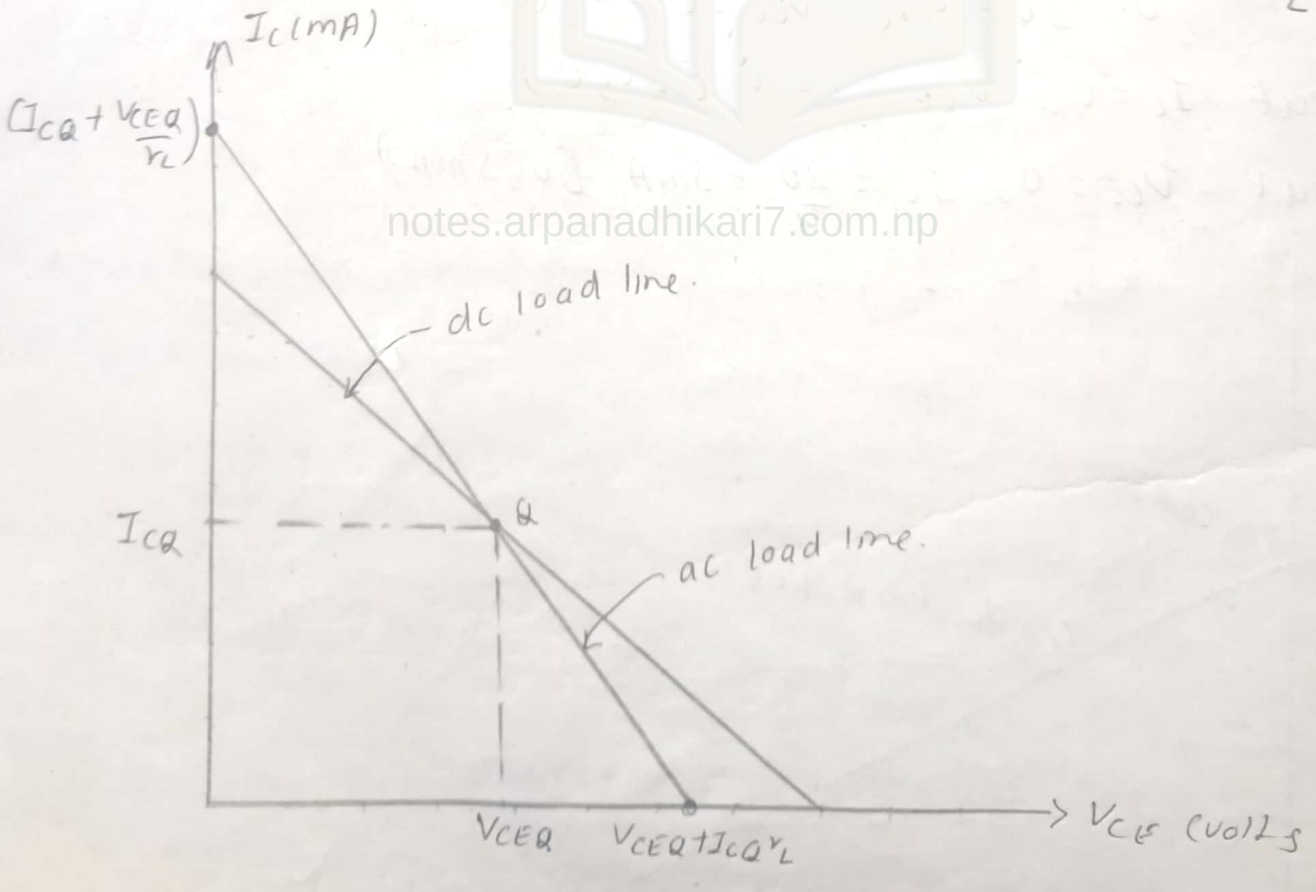
The two points for ac load line can be find as.

put $I_C = 0$ in eqn (1).

$V_{CE} = V_{CEQ} + I_{CQ} r_L$ ~~Q~~ one point is $((V_{CEQ} + I_{CQ} r_L), 0)$

put $V_{CE} = 0$ in eqn (1).

$I_C = I_{CQ} + \frac{V_{CEQ}}{r_L}$ Next point is $\left(0, (I_{CQ} + \frac{V_{CEQ}}{r_L})\right)$

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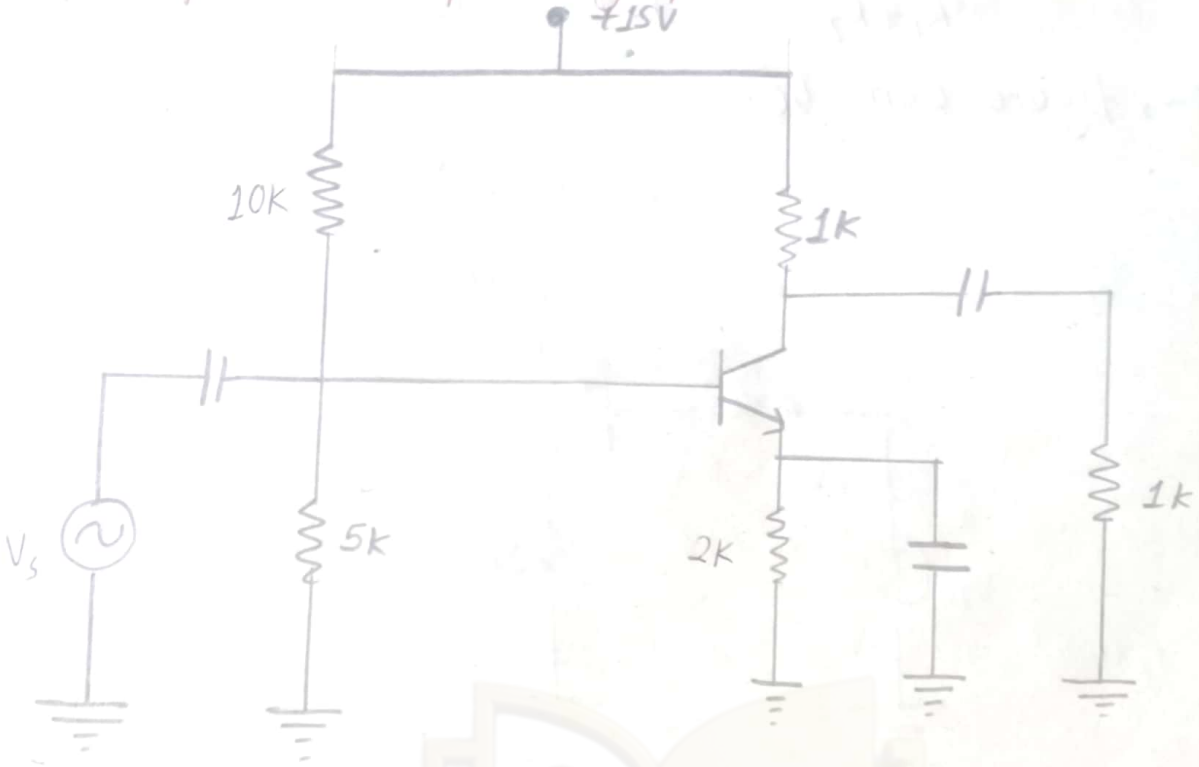
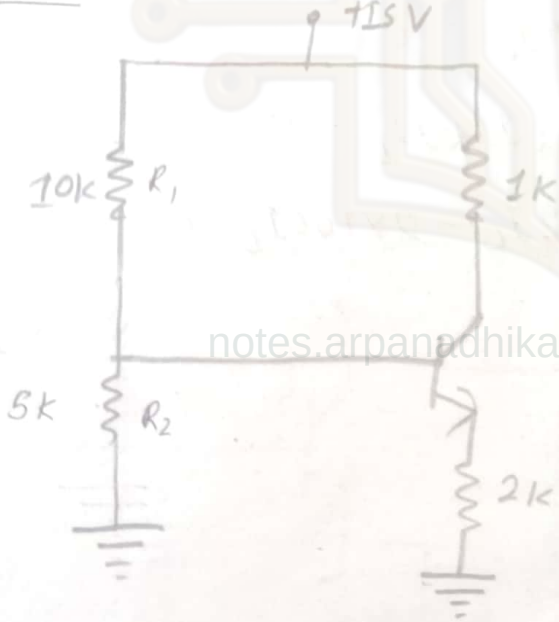
* Draw Dc load line and ac load line for the figure given below Also find the operating point.




DC Analysis

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$V_{Th}$ Now, replace input side by its equt Thevenin's ckt,

$$V_{Th} = \frac{V_{cc} \times R_2}{R_1 + R_2} = \frac{15 \times 5}{10 + 5} = 5V$$

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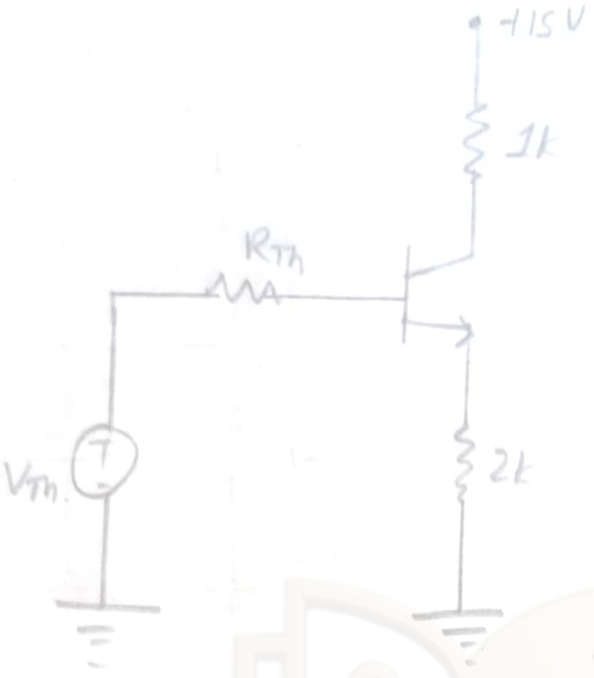
Now,

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{10 \times 5}{10 + 5} = \frac{50}{15} = 3.33 k\Omega$$

Now, figure will be.




Now, applying KVL at input side,

$$V_{Th} - I_B R_{Th} - V_{BE} - 2 I_E = 0$$

$$\text{or } V_{Th} - I_B R_{Th} - V_{BE} - 2(1+\beta) I_B = 0$$

$$\text{or, } 5 - 3.33 I_B - 0.7 - 2 I_B - 2 \times 100 I_B = 0$$

$$\text{or } -205.33 I_B = -4.3$$

$$\text{or } I_B = \frac{4.3}{205.33}$$

$$\text{or } I_B = 0.02094 \text{ mA}.$$

Then,

$$I_C = \beta I_B$$
$$= 100 \times 0.02094$$
$$= 2.094 \text{ mA}$$

$$I_E = I_B + I_C = 0.02094 + 2.094 = 2.11494 \text{ mA}$$

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Again, applying KVL at output side,

$$15 - 1I_C - V_{CE} - 2I_E = 0$$

$$\text{or } V_{CE} = 15 - 2.094 - 2 \times 2.11494$$

$$= 8.676 \text{ volts} = V_{CEQ}$$

For dc load line

We have,

$$15 - 1I_C - V_{CE} - 2I_E = 0$$

$$\text{or } 15 - I_C - V_{CE} - 2I_C = 0 \quad [\because I_C \approx I_E]$$

$$\text{or } 15 - 3I_C - V_{CE} = 0$$

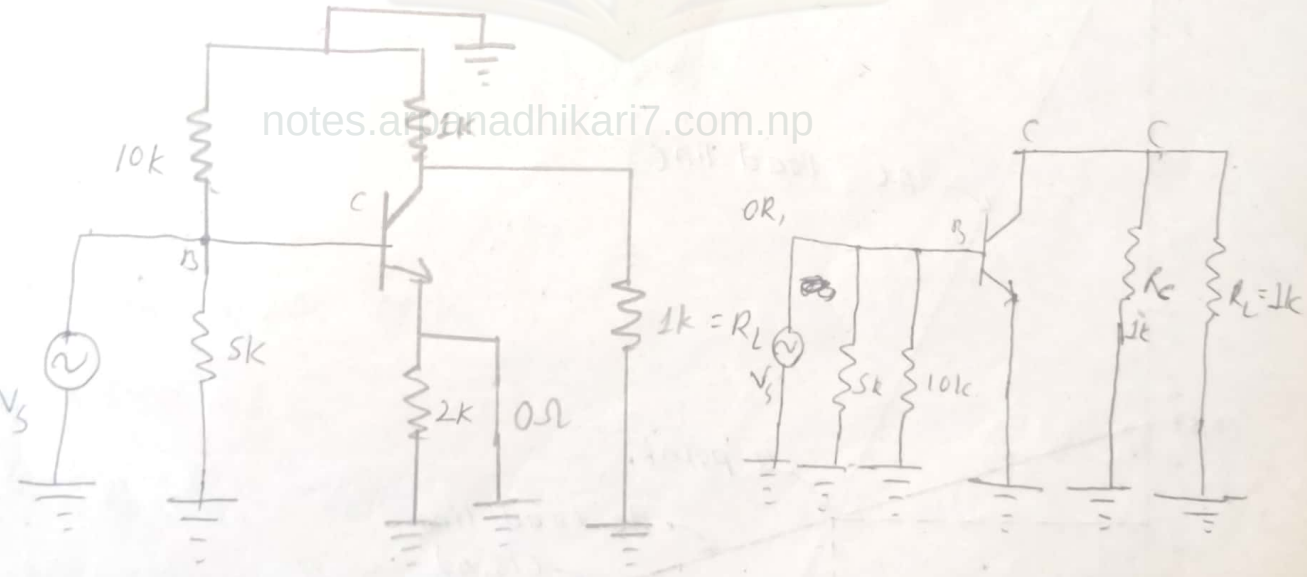
Put $I_C = 0$, $V_{CE} = 15$ $[15, 0)$

Put $V_{CE} = 0$, $I_C = 5\text{mA}$ $[0, 5]$

## AC analysis

capacitor → short

dc source → zero (ground)

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For Ac load line:

$$I_C - I_{CQ} = -\frac{1}{R_L}(V_{CE} - V_{CEQ})$$

where,

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$$r_L = R_C \| R_L = \frac{1 \times 1}{1+1} = 0.5k$$

When $I_C = 0$, $V_{CE} = V_{CEQ} + I_{CQ} r_L$

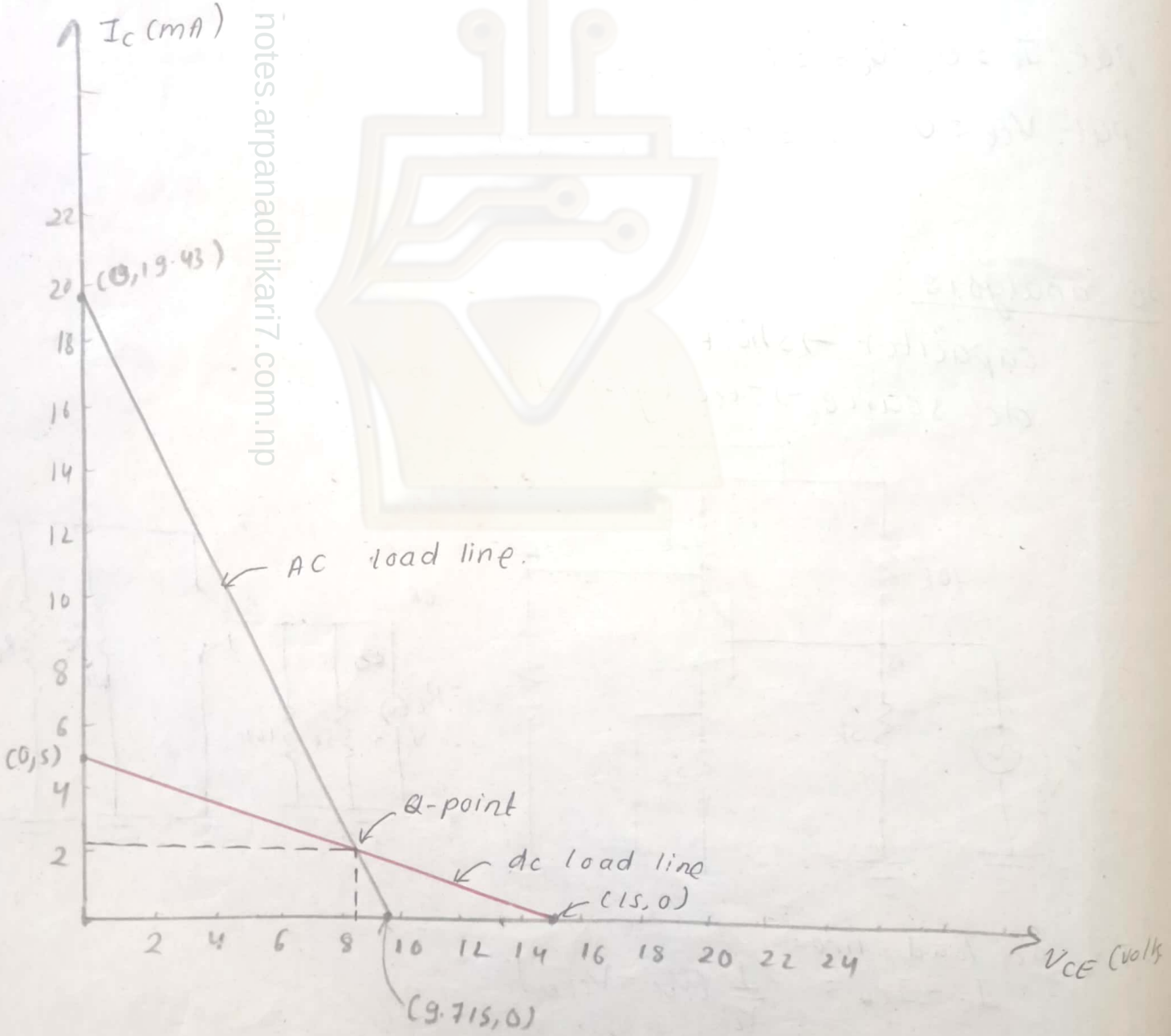
$= 8.67 + 2.09 \times 0.5$

$= 9.7211$ mA $[9.715V, 0]$

When $V_{CE} = 0$, $I_C = I_{CQ} + \frac{V_{CEQ}}{r_L}$

$= 2.09 + \frac{8.67}{0.5}$

$= 19.43$ mA $[0, 19.43]$

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## Design of transistor biasing circuit:

The following guidelines ~~may be~~ followed for designing the transistor biasing circuit.

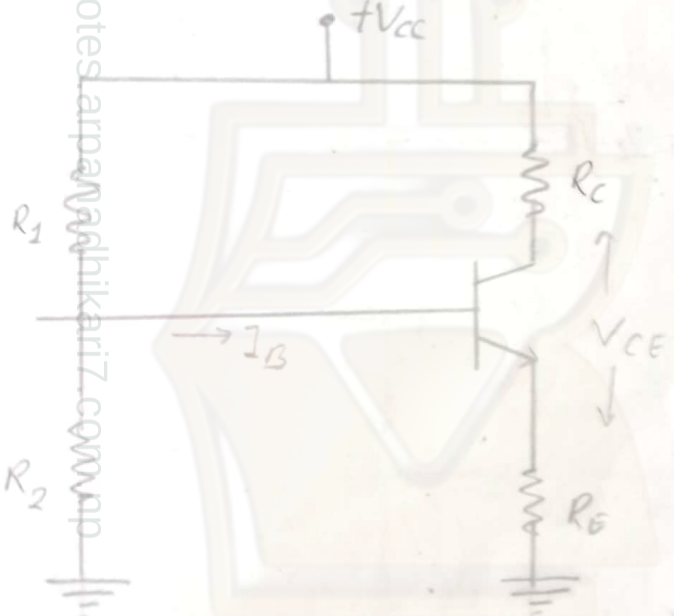
① In common practice, voltage drop across $R_E$ is taken as (10-20)% of $V_{CC}$.

② $I_C$ is chosen so that operating point lies at the middle of dc load line. It is not advisible to operate transistor below $I_C = 1mA$

③ The voltage across collector emitter terminal i.e $V_{CE}$ should be 50% of $V_{CC}$.

④ The voltage drop across $R_C$ is taken as ~~40%~~ 30% to 40% of $V_{CC}$

⑤ The value of $R_1$ and $R_2$ are selected so as to give $I_1 \geq 10 I_B$.

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$I_E R_E$ = 10% to 20% of $V_{CC}$

$V_{CE}$ = 50% of $V_{CC}$

$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$.

$V_{CC} = I_C R_C + V_{CE} + I_E R_E$

$I_C R_C$ → 30-40% of $V_{CC}$

$V_{CE}$ → 50% of $V_{CC}$

$I_E R_E$ → 10% - 20% of $V_{CC}$

$I_C = \beta I_B \Rightarrow I_B = \frac{I_C}{\beta}$

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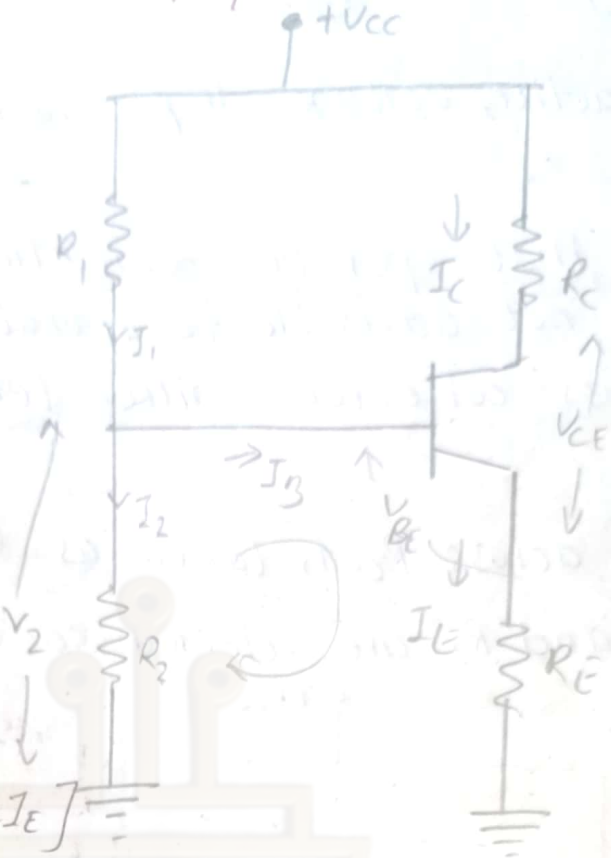
Q.n. (1) Design a common emitter self biased circuit for $I_C = 2mA$. Assume $V_{CC} = 20V$ & $\beta = 100$.




Solution

Given,

$I_C = 2mA$

$\beta = 100$

$V_{CC} = 20V$

let $I_E R_E$ = 10% of $V_{CC}$

or, $I_E R_E = 20 \times \frac{10}{100}$

$\Rightarrow R_E = \frac{2}{I_E}$

$\therefore R_E = \frac{2}{2mA}$ $[\because I_C \approx I_E]$

$= 1k\Omega$

$V_{CE}$ = 50% of $V_{CC}$

$= 20 \times \frac{50}{100}$

$= 10V$

$\therefore I_C R_C$ = 40% of $V_{CC}$

$I_C R_C = 20 \times \frac{40}{100}$

$I_C R_C = 8$

$R_C = \frac{8}{I_C}$

$= \frac{8}{2mA}$

$= 4k\Omega$

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$I_1 = 10 I_B$

$\Rightarrow I_1 = 10 \times \frac{I_C}{\beta}$

$= \frac{10 \times 2}{100}$

$= 0.2\ mA$

$I_2 = I_1 - I_B$

$= 0.2 - \frac{2}{100}$

$= 0.18\ mA$

Applying KVL,

$V_2 - V_{BE} - I_E R_E = 0$

$V_2 = V_{BE} + I_E R_E$

$= 0.7 + 2$

$= 2.7\ V$

$V_2 = I_2 R_2$

$R_2 = \frac{V_2}{I_2}$

$= \frac{2.7}{0.18\ mA}$

$= 15\ k\Omega$

Applying KVL

$V_{CC} - I_1 R_1 - I_2 R_2 = 0$

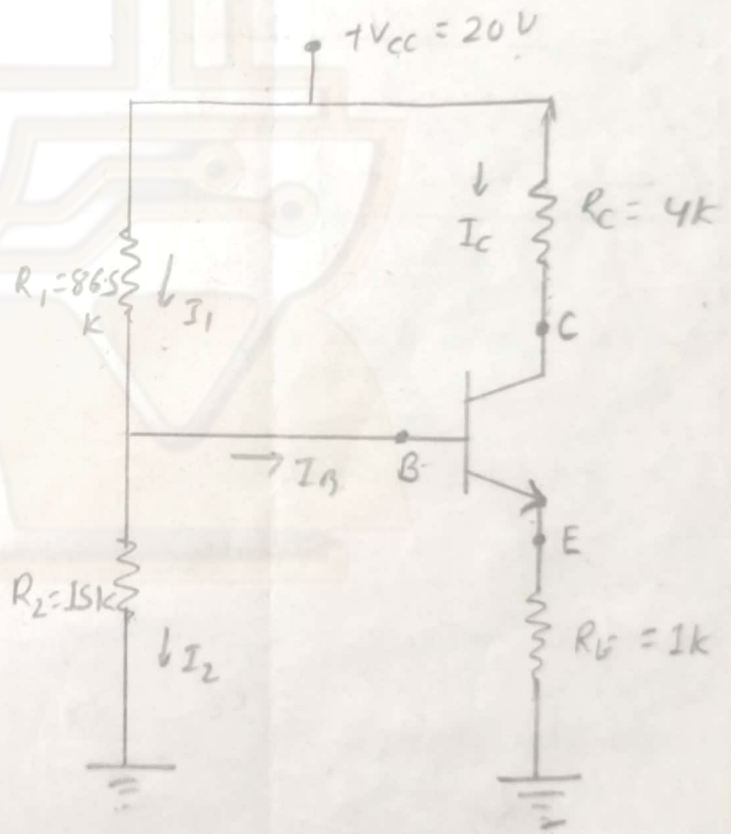
$\Rightarrow V_{CC} - I_1 R_1 - V_2 = 0$

$\Rightarrow I_1 R_1 = V_{CC} - V_2$

$\Rightarrow R_1 = \frac{V_{CC} - V_2}{I_1}$

$= \frac{20 - 2.7}{0.2\ mA}$

$= 86.5\ k\Omega$

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$I_1 = 10 I_B$

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Applying KVL,

$V_2 - V_{BE} - I_E R_E = 0$

$V_2 = V_{BE} + I_E R_E$

$= 0.7 + 2$

$= 2.7V$

$V_2 = I_2 R_2$

$R_2 = \frac{V_2}{I_2}$

$= \frac{2.7}{0.18\ mA}$

$= 15\ k\Omega$

Applying KVL

$V_{CC} - I_1 R_1 - I_2 R_2 = 0$

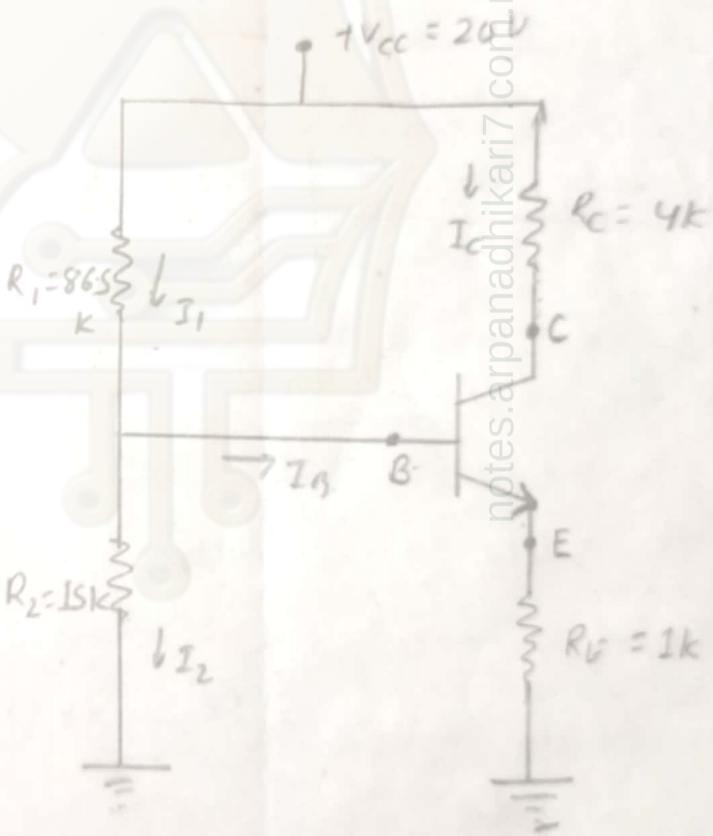
$\Rightarrow V_{CC} - I_1 R_1 - V_2 = 0$

$\Rightarrow I_1 R_1 = V_{CC} - V_2$

$\Rightarrow R_1 = \frac{V_{CC} - V_2}{I_1}$

$= \frac{20 - 2.7}{0.2\ mA}$

$= 86.5\ k\Omega$

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BJT switching time

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## Transistor ON time

(1) Delay time ($t_d$):

It is the time elapses during the delay plus the time required for the collector current to rise to 10% of its maximum (saturation) value $I_{cs}$.

(2) Rise time ($t_r$):

It is the time required for collector current to rise from 10% to 90% of $I_{cs}$.

∴ Turn ON time = $t_d + t_r$ —(i)

## Transistor OFF time:

(1) Storage time ($t_s$):

It is the time interval which elapses between the transition of input and the time when $I_c$ falls to 90% of $I_{cs}$.

(2) Fall time ($t_f$):

It is the time required for $I_c$ to fall from 90% to 10% of $I_{cs}$.

∴ Turn off time = $t_s + t_f$ —(ii)

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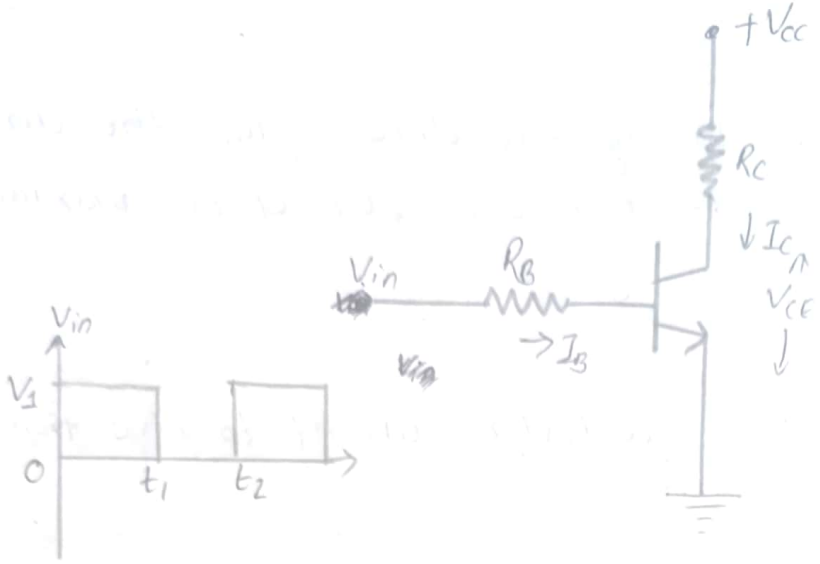
# Transistor as a Switch




At $t=0$, when $V_{in} = V_1$

Emitter-base junction will be forward biased and base current and hence collector current starts flowing in the circuit.

$$V_{in} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{in} - V_{BE}}{R_B} \quad \text{——(I)}$$

$$I_C = \beta I_B \quad \text{——(II)}$$

Also, Applying KVL at output;

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$\text{or, } V_{CE} = V_{CC} - I_C R_C \quad \text{——(III)}$$

The value of $R_B$ and $R_C$ are so chosen that,

$$I_C R_C = V_{CC}$$

$$\therefore V_{CE} = 0$$

Hence, transistor operates in saturation region and considered as switch ON.

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At $t = t_1$, when $V_{in} = 0$,

Emitter base junction will be reverse bias, and hence no base current and collector current flows in the ckt.

i.e $I_B = I_C = 0$

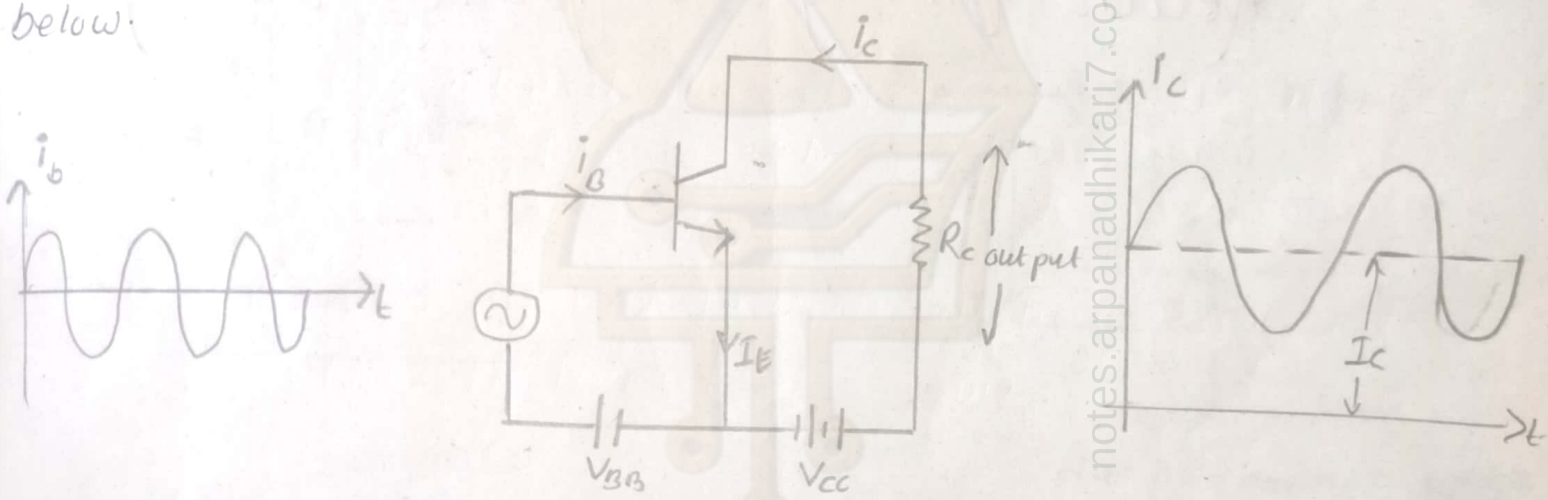
From eqn (iii)

$V_{CE} = V_{CC}$

Hence, transistor operates in cut off region and considered as switch off.

## Transistor as an Amplifier

The common emitter n-p-n transistor amplifier is shown in fig below.




Here, a biased voltage applied to keep emitter base junction forward biased regardless of the polarity of input signal. During positive half of input signal, the forward biased across emitter base junction is increased. Therefore, more electron flows from emitter to collector. This causes an increase in collector current. The increase in collector current produces a greater voltage drop across the collector load resistance $R_C$.

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However, during the -ve half of the input signal, the forward biased across emitter base junction is decreased. Therefore, the collector current decreases. This results in the decreased output voltage. Hence, an amplified output is obtained across the load as shown in fig above.

The output current (i.e collector current) consists of two components namely (i)

(i) DC collector current due to biased battery ($V_{BB}$).

(ii) The ac collecter " " " signal.

Reach through or Punch through

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Chapter-4

# Junction Field Effect Transistor (JFET)

Junction field effect transistor is a 3 terminal semiconductor device in which current conduction is by one type of carrier (i.e electrons / holes).

## Construction:

JFET consists of P type or N type silicon bar containing two PN junctions at the sides as shown in figure.

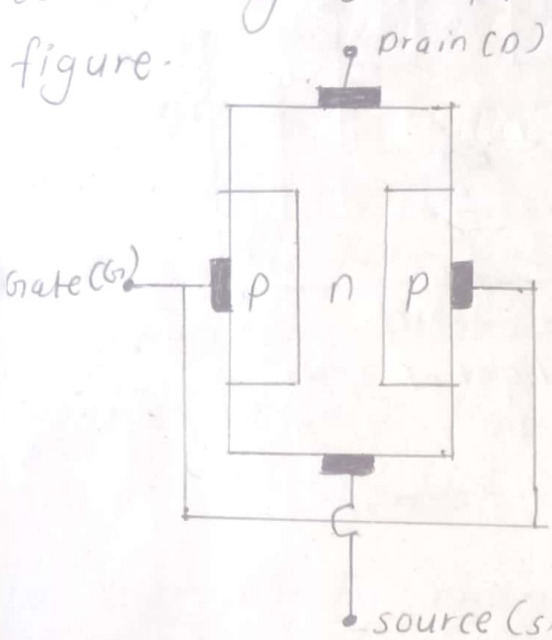


fig: n-channel JFET

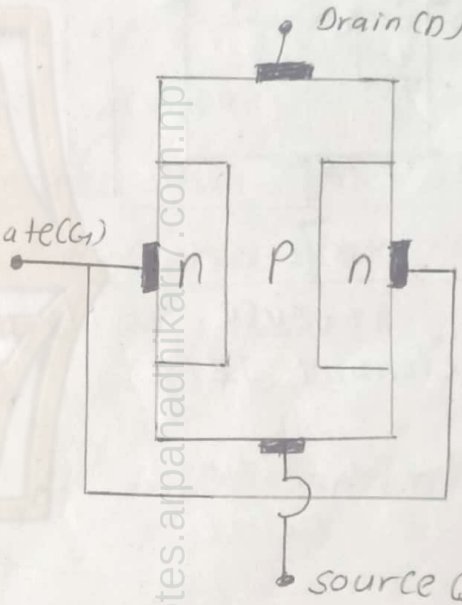


fig: p-channel JFET

The bar forms the conducting channel for the charge carriers. If the bar is of N-type, it is called n-channel JFET. And if the bar is of P-type it is called p-channel JFET. The two P-n junction forming diodes are connected internally and a common terminal called gate (G) is taken out. The other terminals are drain and source are taken out from the bar as shown in fig above.

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The input circuit (that is gate to source) is reversed biased. this means that the device has high input impedance. In ~~JFET~~ JFET, the source current is equal to the drain ~~that~~ current.

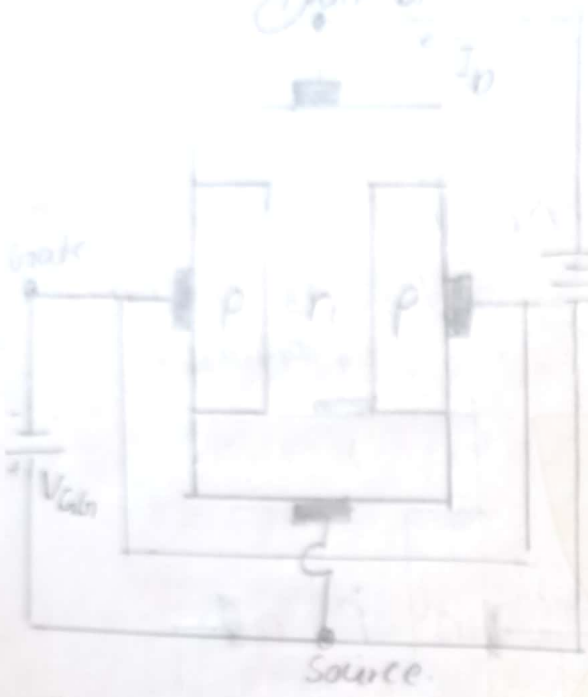


fig: n-channel JEFT

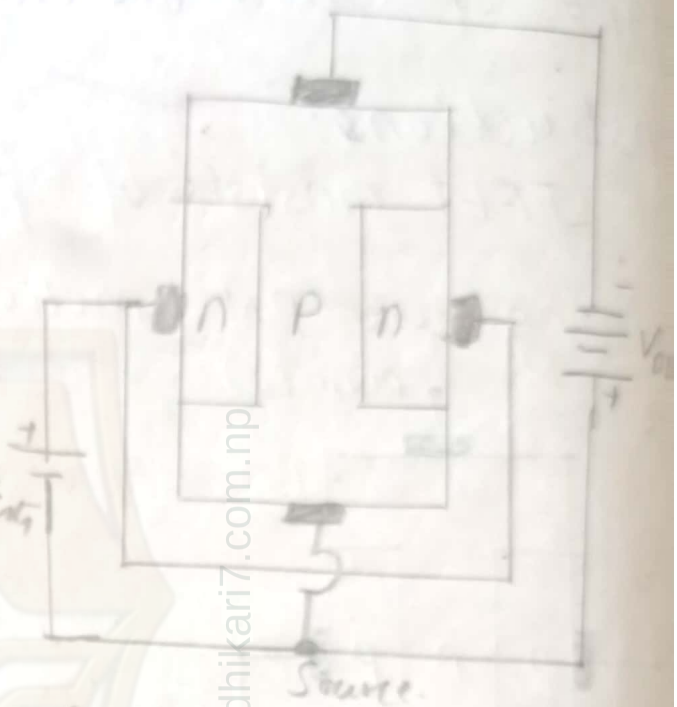


Fig: P-channel JEFT

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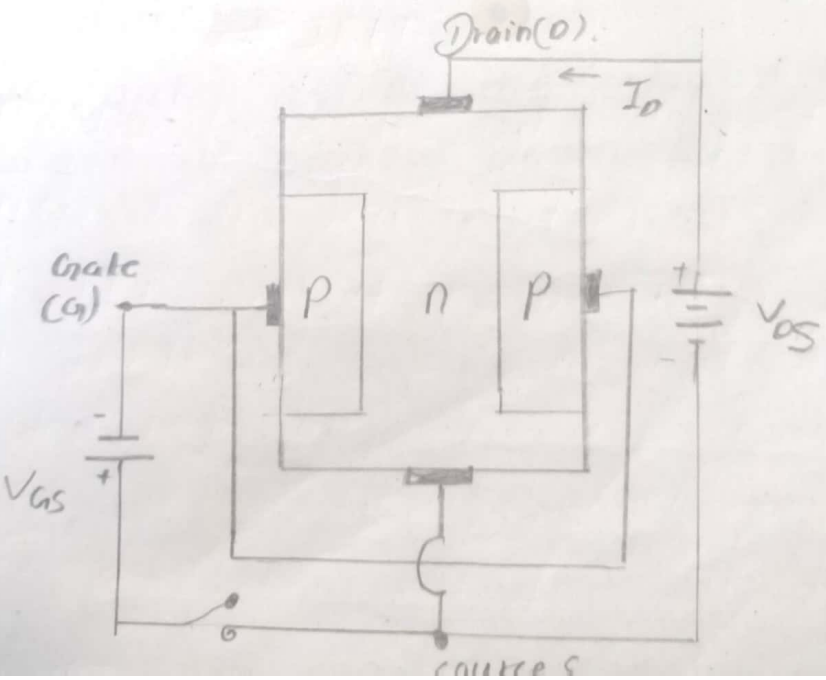
# Principle & working of JFET:

## Principle:

The two p-n junctions at the sides forms two depletion layer. The current conduction by charge carriers is through the channel bet$^n$ two depletion layers. The width and the hence the resistance of this channel can be controlled by changing the output voltage ($V_{GG}$). Greater the reverse voltage $V_{GG}$, the wider will be the depletion layer & narrow will be the conducting channel. The narrower channel means greater resistance and hence source to drain current decreases. Reverse will happen when $V_{GG}$ decreases. Thus, JFET operates on the principle that the width and hence the resistance of conducing channel can be varied by changing the reverse voltage $V_{GG}$.

Therefore, the magnitude of drain current can be changed by changing $V_{GG}$.

## Working:

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Case 1: When $V_{DS}$ is applied bet$^n$ drain & source and voltage at gate is 0, the two pn junctions at the sides of the bar stablish depletion layer. The electrons will flow from source to drain through a channel bet$^n$ the depletion layer. The size of these layers determines the width of channel and hence the current conduction through the bars.

Case 2: When reverse voltage $V_{GS}$ is applied between Gate & source, the width of depletion layer is ~~im~~ increased. This reduces the width of conducting channel thereby increasing the resistance of N-type bar. Consequently, the current from source to drain is decreased. On other hand, If reverse voltage on the gate decreased, the width of the depletion layer also decrease. This increases the width of conducting channel and hence source to drain current.

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## JFET Symbols:

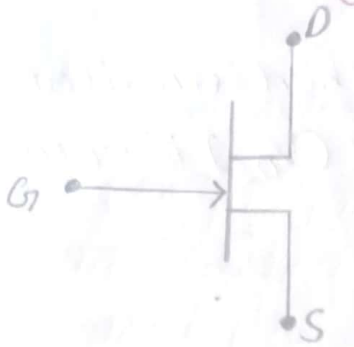


fig: n-channel JFET

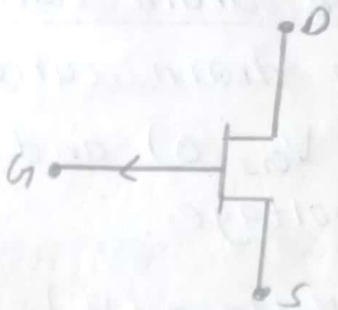


fig: p-channel JFET

## Salient features of JFET:

Following are the some of the salient features of JFET:

i) It is a three terminal voltage control semiconductor device
ii) It is always operated with gate source junction reverse biased.
iii) The gate current is 0 (ie $I_G = 0$).
iv) Drain current is equals to source current.
v) It most be operated betⁿ $V_{GS}=0$ and $V_{GS} = V_{GS(off)}$
vi) The drain current is controlled by changing the channel width.
vii) It is not subjected to thermal runaway when temperature of device increases.
viii) Since, two gates are at the same potential, both depletion layer widen or narrow down by an equal amount.

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Some Important terms:

(1) Shorted gate drain current ($I_{DSS}$)

It is the drain current with gate short circuited to source (i.e $V_{GS}=0$) and drain voltage ($V_{DS}$) is equal to pinch off voltage.

(2) Pinch off Voltage ($V_p$)

It is the minimum drain source voltage at which the drain current essentially becomes constant.

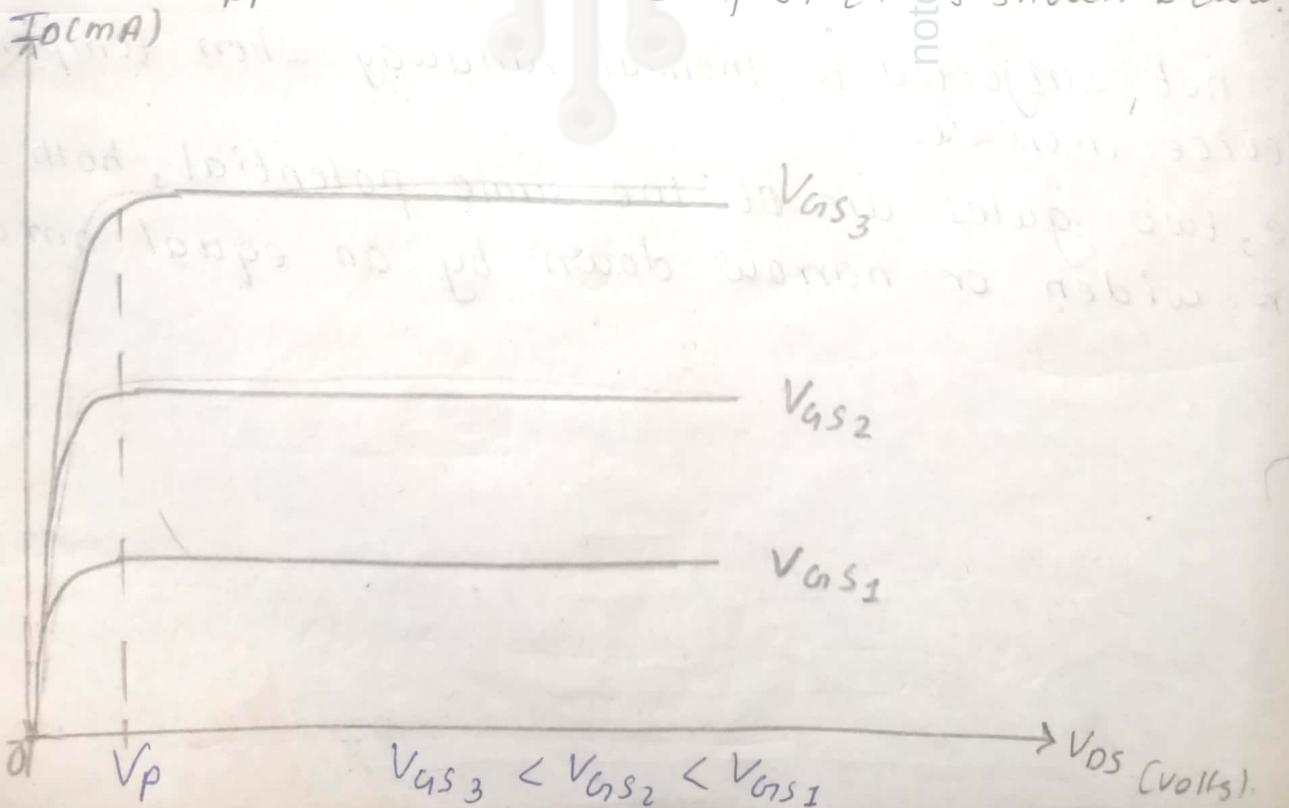
(3) Gate-source cut off voltage ($V_{GS(off)}$)

It is the gate source voltage where the channel is completely cut off and the drain current becomes zero.

## Output characteristics of JFET

The curve between drain current ($I_D$) and drain source voltage ($V_{DS}$) of JEFT at constant gate-source voltage ($V_{GS}$) is known as output characteristics of JFET.

The output characteristics of JFET is shown below:

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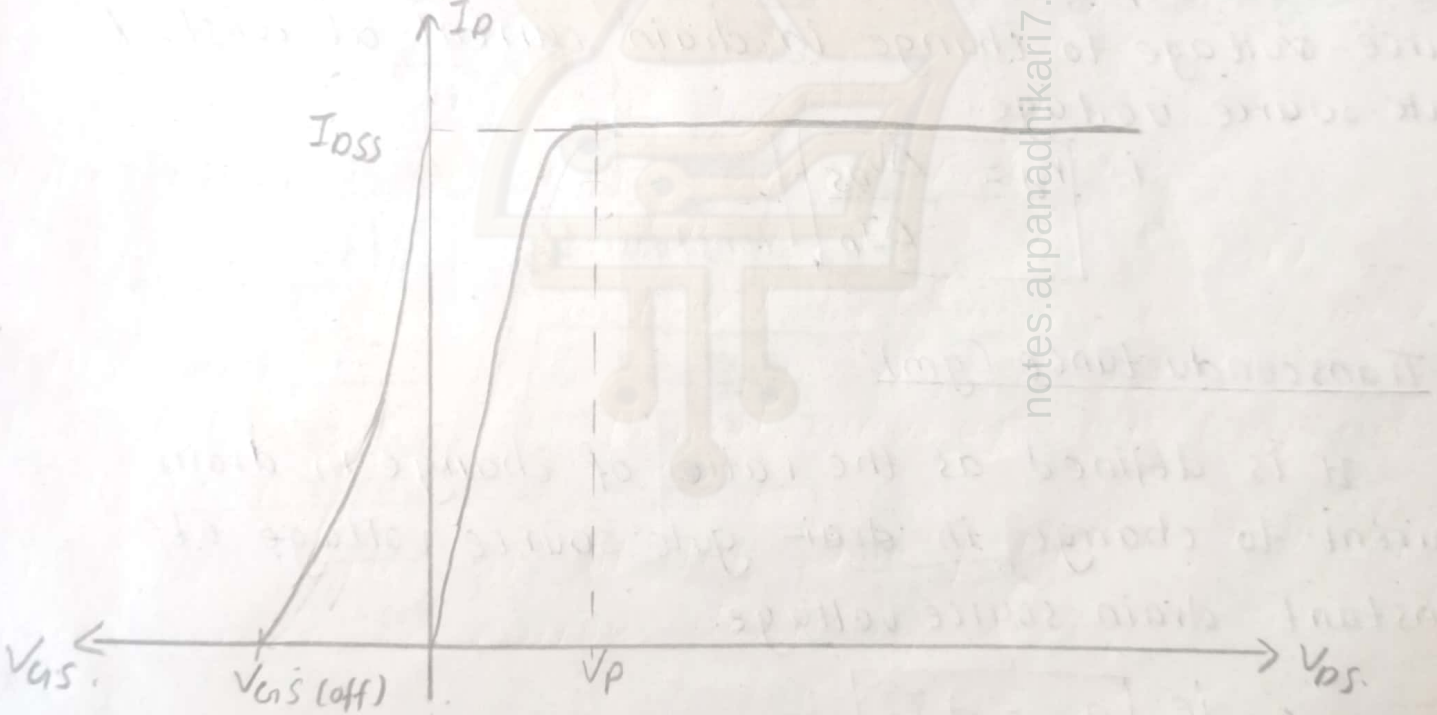
At first, the drain current $I_D$ rises rapidly with drain source voltage $V_{DS}$ and becomes constant. The drain source voltage above which drain current become constant is known as pinch of voltage. After pinch of voltage, the channel width becomes so narrow that depletion layer almost touch each other. The drain current passes through the small passage ~~with~~ bet? these two layers. Therefore, increase in drain current is very small with $V_{DS}$ above pinch of voltage. Consequently, the drain current remains constant.

## Expression for drain current:




The relation between $I_{DSS}$ and $V_P$ is shown in figure above. The gate source ~~volt~~ cut off voltage (ie $V_{GS(off)}$) on transfer characteristics is equals to pinch off voltage on output characteristics. ~~Like~~

i.e $V_P = |V_{GS(off)}|$ → magnitude

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From the transfer characteristics shown in figure, the expression for drain current is given by

Imp $$I_D = I_{DSS}\left[1 - \frac{V_{GS}}{V_{GS(off)}}\right]^2 \quad \text{---(1)}$$

where,

$I_D$ = Drain current

$I_{DSS}$ = shorted gate drain current

$V_{GS}$ = gate-source voltage

$V_{GS(off)}$ = gate-source cut off voltage

## JFET Parameters

### 1. AC drain resistance ($r_d$):

It is defined as the ratio of change in drain source voltage to change in drain current at constant gate-source voltage.

i.e $$r_d = \left.\frac{\Delta V_{DS}}{\Delta I_D}\right|_{\text{constant } V_{GS}}$$

### 2. Transconductance ($g_m$):

$g = \frac{1}{r}$

It is defined as the ratio of change in drain current to change in ~~drai~~ gate source voltage at constant drain source voltage.

i.e $$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{V_{DS} = \text{constant}}$$

Since, $$I_D = I_{DSS}\left[1 - \frac{V_{GS}}{V_{GS(off)}}\right]^2 \quad \text{---(1)}$$

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Diff eqn (i) wrt $V_{GS}$;

$$\frac{\partial I_D}{\partial V_{GS}} = \frac{\partial \left\{ I_{DSS} \left[ 1 - \frac{V_{GS}}{V_{GS(off)}} \right]^2 \right\}}{\partial V_{GS}}$$

$$= I_{DSS} \frac{\partial \left( 1 - \frac{V_{GS}}{V_{GS(off)}} \right)^2}{\partial V_{GS}}$$

$$= I_{DSS} \times \left[ \frac{-2}{V_{GS(off)}} \times \left( 1 - \frac{V_{GS}}{V_{GS(off)}} \right) \right]$$

$$= -\frac{2 I_{DSS}}{V_{GS(off)}} \times \left[ 1 - \frac{V_{GS}}{V_{GS(off)}} \right]$$

$$= \frac{2 I_{DSS}}{-V_{GS(off)}} \times \sqrt{\frac{I_D}{I_{DSS}}} \quad [\because \text{from eqn (i) equation}]$$

$$\therefore \boxed{g_m = \frac{2 I_{DSS}}{|V_{GS(off)}|} \sqrt{\frac{I_D}{I_{DSS}}}} \quad \text{——— (ii)}$$

$$g_m = \frac{2}{|V_{GS(off)}|} \sqrt{\frac{I_D \times I_{DSS} \times I_{DSS}}{I_{DSS}}}$$

$$\boxed{g_m = \frac{2}{|V_{GS(off)}|} \sqrt{I_D * I_{DSS}}} \quad \text{——— (iii)}$$

Also,

$$g_m = g_{mo} \sqrt{\frac{I_D}{I_{DSS}}}$$

where, $g_{mo} = \frac{2 I_{DSS}}{|V_{GS(off)}|}$

($g_{mo}$ → (max transconductance))

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3. Amplification factor ($\mu$).

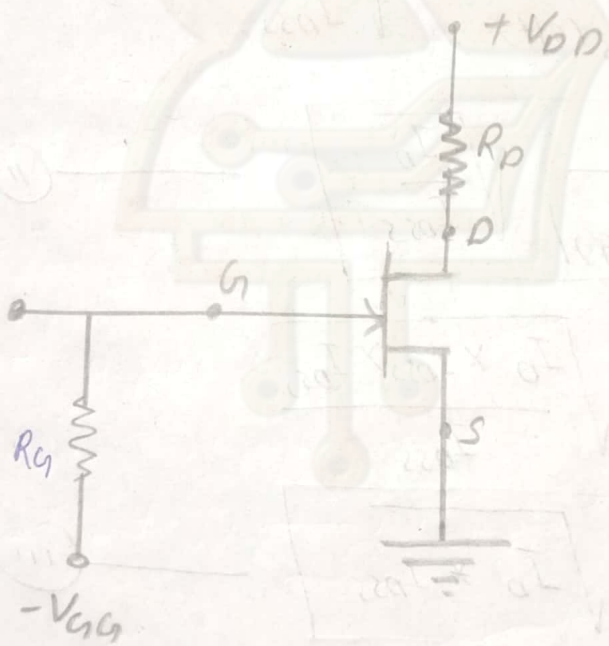
It is defined as the ratio of change in drain voltage to change in gate source voltage at constant drain current. i.e $\boxed{\mu = \frac{\Delta V_{DS}}{\Delta V_{GS}}\Big|_{I_D = \text{constant}}}$

$$\mu = \frac{\Delta V_{DS}}{\Delta I_D} \times \frac{\Delta I_D}{\Delta V_{GS}}$$

$$\boxed{\mu = r_d \times g_m} \quad \text{——} (1)$$

## JFET Biasing

### 1. JFET biasing by bias battery




Applying KVL at input;

$$-V_{GG} - I_G R_G - V_{GS} = 0$$

since $I_G = 0$ [$\because$ reverse biasing]

$$\therefore \boxed{V_{GS} = -V_{GG}} \quad \text{——} (i)$$

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We have,

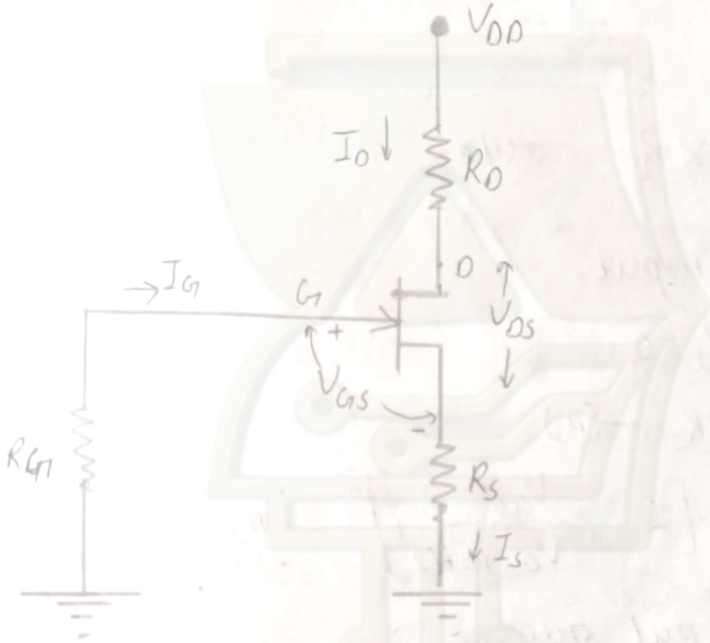
$$I_D = I_{DSS}\left[1 - \frac{V_{GS}}{V_{GS(off)}}\right]^2 \quad \text{——(II)}$$

Applying KVL at output side;

$$V_{DD} - I_D R_D - V_{DS} = 0$$

$$V_{DS} = V_{DD} - I_D R_D \quad \text{——(III)}$$

2. Self bias for ~~JFET~~ JFET :




Applying KVL at input,

$$-I_G R_G - V_{GS} - I_S R_S = 0$$

Since, $I_G = 0$

$$V_{GS} = -I_S R_S \quad \text{——(I)}$$

We have, $I_D = I_{DSS}\left[1 - \frac{V_{GS}}{V_{GS(off)}}\right]^2 \quad \text{——(II)}$

Applying KVL at output side,

$$V_{DD} - I_D R_D - V_{DS} - I_S R_S = 0$$

$$V_{DS} = V_{DD} - I_D R_D - I_D R_S \quad [I_D = I_S]$$

$$V_{DS} = V_{DD} - I_D (R_D + R_S) \quad \text{——(IV)}$$

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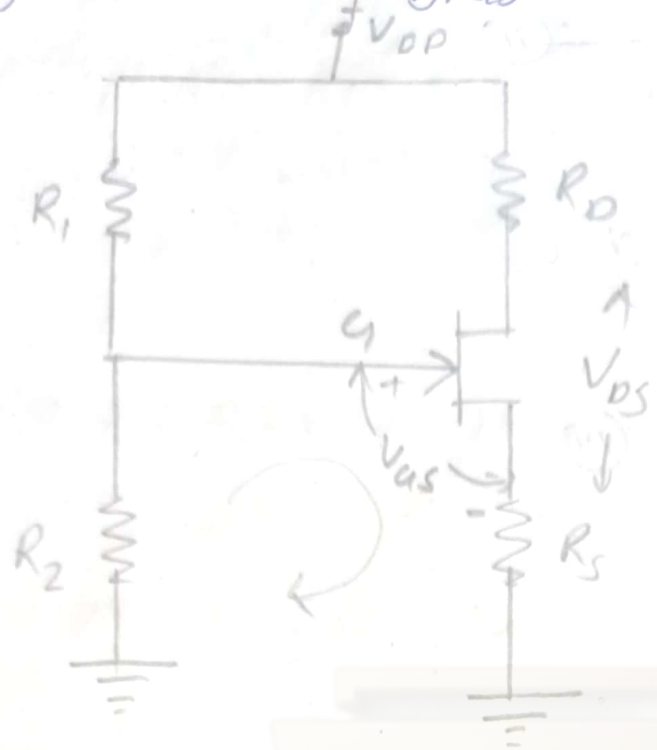
3. Voltage divider bias.




Since, $V_2 = \frac{V_{DD}}{R_1 + R_2} \times R_2$ —(i)

Applying KVL at input,

$V_2 - V_{GS} - I_S R_S = 0$

$V_{GS} = V_2 - I_D R_S$ —(ii)

We know, $I_D = I_{DSS}\left[1 - \frac{V_{GS}}{V_{GS(OFF)}}\right]^2$ —(iii)

Applying KVL at output side,

$V_{DD} - I_D R_D - V_{DS} - I_S R_S = 0$

$\Rightarrow V_{DS} = V_{DD} - I_D (R_D + R_S)$ —(iv)

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## Numericals

steps:

(i) Apply KVL at input side & find $V_{GS}$

(ii) For voltage divider bias find $V_2$ ~~bet n~~ before applying KVL at input - where, $V_2 = \frac{V_{DD}}{R_1 + R_2} \times R_2$ —(1)

(iii) Find $I_D$ by using eqn $I_D = I_{DSS}\left[1 - \frac{V_{GS}}{V_{GS(off)}}\right]^2$

(iv) Apply KVL at output side & find $V_{DS}$.

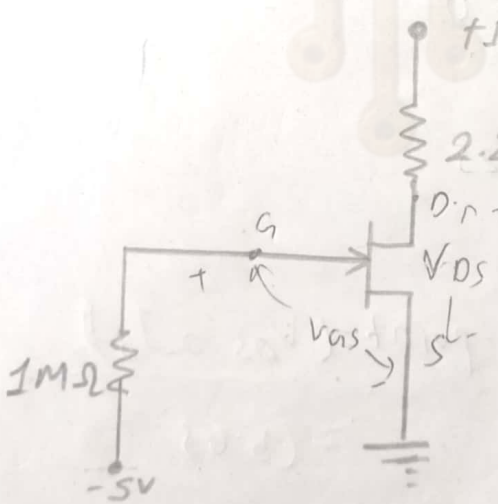
(v) Put $I_G = 0$ & $I_D = I_S$

(vi) Value of $V_{DS}$ & $I_D$ is the operating point.

(vii) Draw dc load line from the eqn obtained by applying KVL at output side.

(viii) The process is similar to that of transistor circuit.

Qn. A JFET shown in figure has $V_{GS(off)} = -8V$ and $I_{DSS} = 16mA$. Determine the value of $V_{DS}$, $I_D$ & $V_{GS}$.




Solution,

Given,

$V_{GS(off)} = -8V$

$I_{DSS} = 16\ mA = 16 \times 10^{-3} A$

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$V_{GG} = -5V$

$R_G = 1M\Omega = \cancel{1000\Omega}\ 10^6\Omega$

$R_D = 2.2\,k\Omega = 2.2 \times 10^3\Omega$

$V_{DD} = 10V$

Now,

Applying kvl at input side,

$$-V_{GG} - I_G R_G - V_{GS} = 0$$

or, $-5 - 0 \times 1000 - V_{GS} = 0$

or, $V_{GS} = -5V$

As, $I_D = I_{DSS}\left[1 - \frac{V_{GS}}{V_{GS(OFF)}}\right]^2$

$= 16\left[1 - \left(\frac{-5}{-8}\right)\right]^2$

$= 2.25\,mA$

Applying kvl at output side,

$$10 - 2.2 I_D - V_{DS} = 0$$

$V_{DS} = 10 - 2.2 I_D$

$= 10 - 2.2 \times 2.25$

$= 5.05V$

Q-point $= (V_{DS}, I_D)$

$= (5, 0)$

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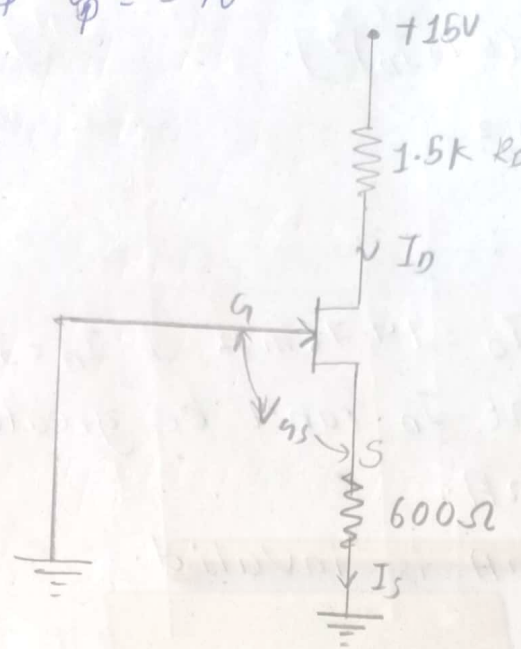
Q.2. For the circuit shown in figure Determine $I_D$ & $V_{DS}$. Assume $I_{DSS} = 10\,mA$ & $V_p = -4V$




Solution

Given,

$V_p = -4V$

we know,

$V_p = |V_{GS(off)}|$

$\therefore V_{GS(off)} = -4V$

$I_{DSS} = 10\,mA$

$R_D = 1.5k$

Now, Applying kvl at input side,

$$-V_{GS} - \frac{600}{1000} I_S = 0 \quad \text{(Ω to kΩ)}$$

$$V_{GS} = -\frac{600}{1000} I_D \quad [\because I_D = I_S]$$

$$V_{GS} = -0.6 I_D \quad \text{—(1)}$$

As, $I_D = I_{DSS}\left[1 - \frac{V_{GS}}{V_{GS(off)}}\right]^2$

$$= 10\left[1 - \left(\frac{-0.6 I_D}{-4}\right)\right]^2$$

$$= 10\left[1 - \frac{0.6 I_D}{4}\right]^2$$

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$I_D = 10[1 - 0.15 I_D]^2$

$= 10[1 - 2 \times 0.15 I_D + (0.15 I_D)^2]$

$I_D = 10 - 3 I_D + 0.225 I_D^2$

or, $0.225 I_D^2 - 4 I_D + 10 = 0$

The value of $I_D$ is $I_D = 14.76\,mA$ & $I_D = 3\,mA$.

Since, the drain current $I_D$ can't be greater than $I_{DSS}$. Therefore, $I_D = 3\,mA$.

Here, $I_D = 14.76\,mA$ is invalid.

Applying KVL at output side.

$15 - I_D \times 1.5 \times 10^3 - V_{DS} - I_S \times \frac{600}{1000} = 0$

or, ~~$15 - 3 \times 1.5 \times 10^3 - 0.6 \times 3 \times 600 = 0$~~ $V_{DS}$ [$\because I_D = I_S$]

or, ~~$V_{DS} = 6.9\,Volts$~~ $V_{DS} = 15 - 1.5 I_D - 0.6 I_D$

$= 15 - 2.1 \times 3$

$V_{DS} = 8.7\,V$

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# Meta oxide semiconductor field effect transistor (MOSFET)

Types:

1. Depletion type MOSFET (D-MOSFET)
2. Enhancement type MOSFET (E-MOSFET)

## (1) Depletion type MOSFET (D-MOSFET)

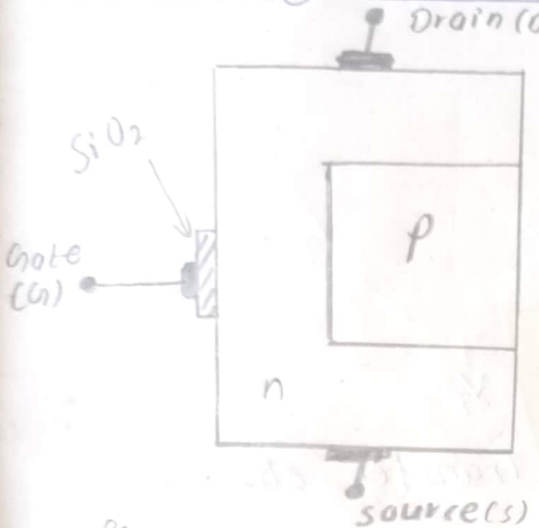


fig: n-channel D-MOSFET

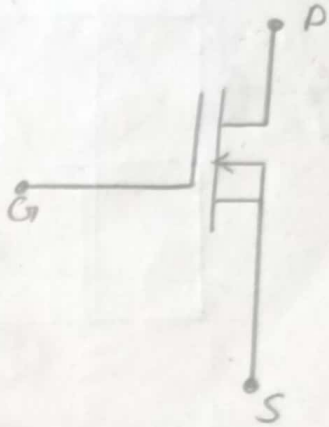


fig: symbol of DMOSFET

DMOSFET can be operated in both depletion mode and enhancement mode.

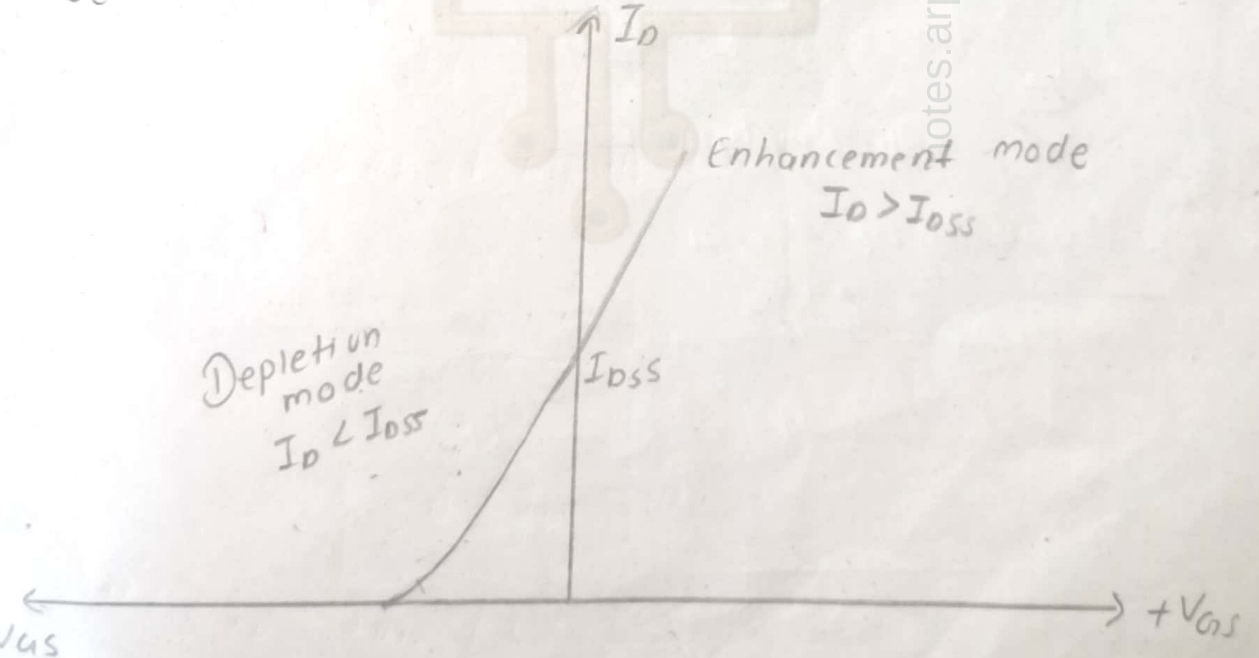


fig: Transfer characteristics of MOSFET

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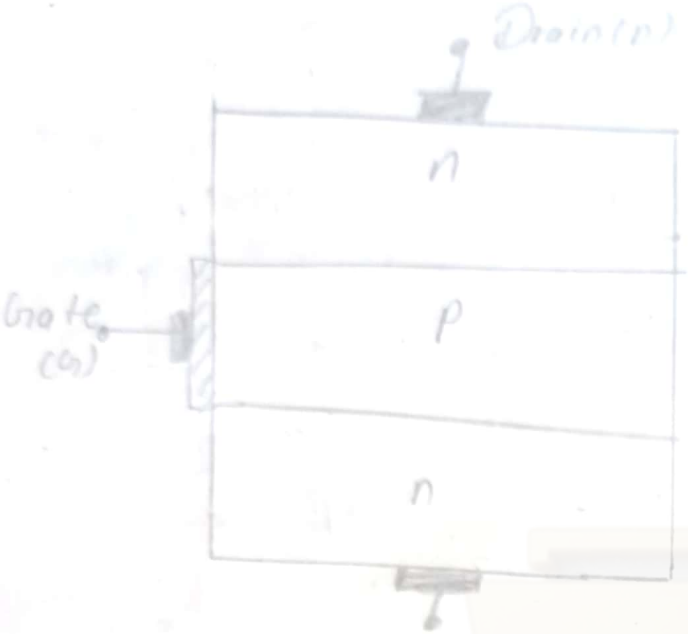
2. Enhancement type MOSFET (EMOSFET)

Operate in enhancement mode only.



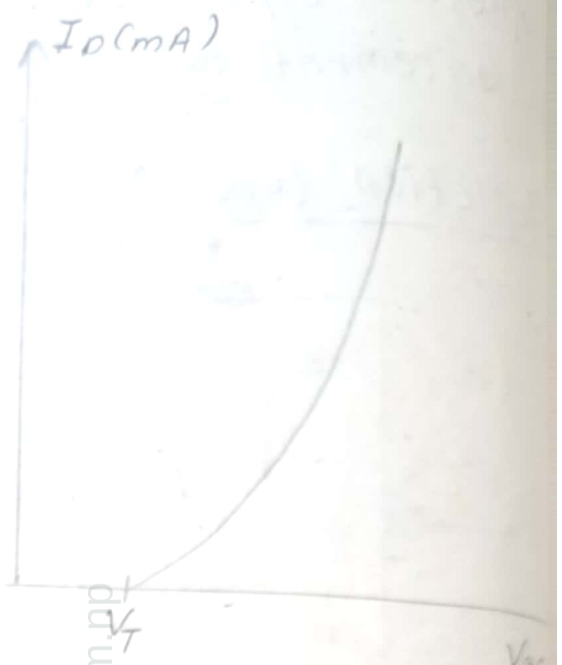




fig: Transfer characteristic

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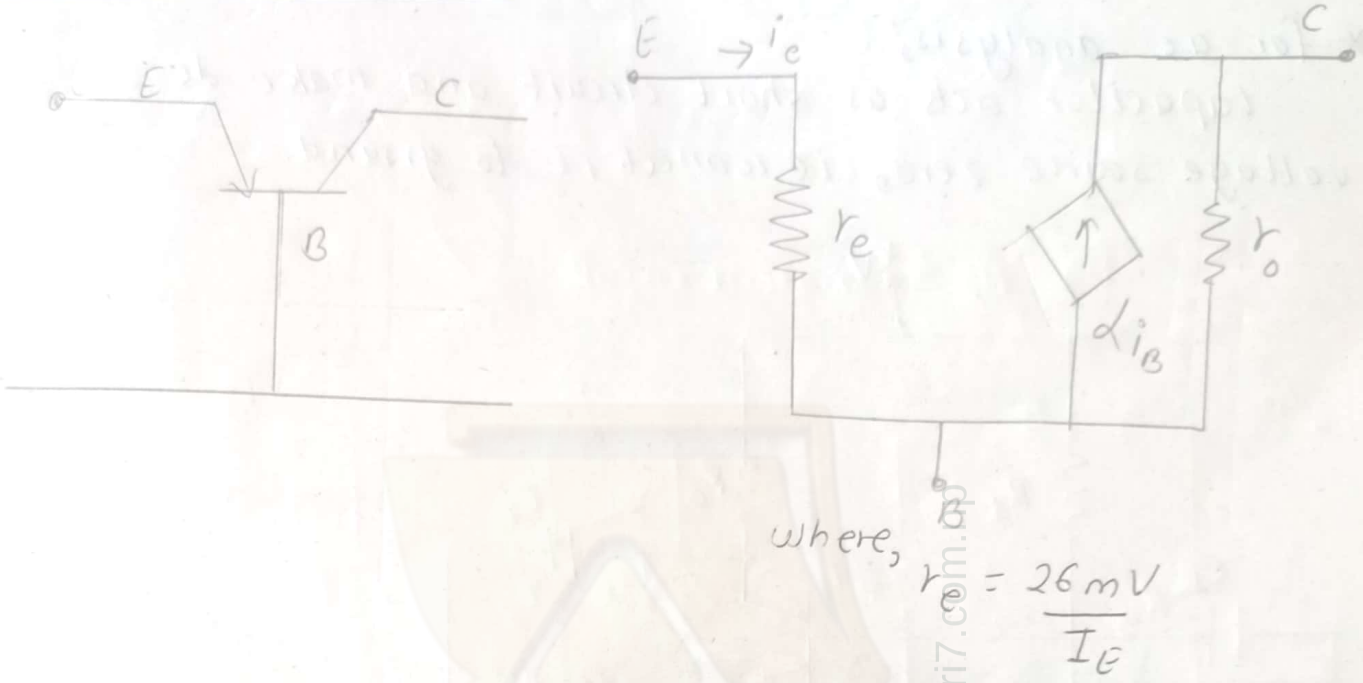
# Small signal Analysis

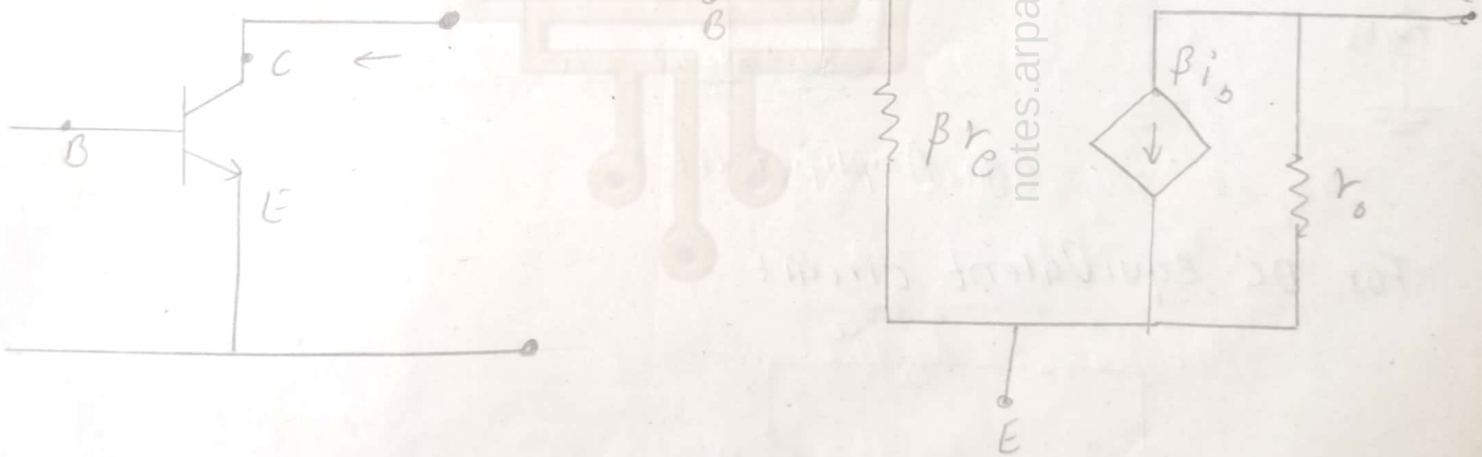
## $r_e$ - model

### 1. Common base connection




where, $r_e = \frac{26mV}{I_E}$

### 2. Common Emitter connection

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# DC and AC equivalent circuits

* for dc analysis,
  capacitor acts as open circuits.

* for ac analysis,
  capacitor acts as short circuit and make dc voltage source zero, i.e connect it to ground.

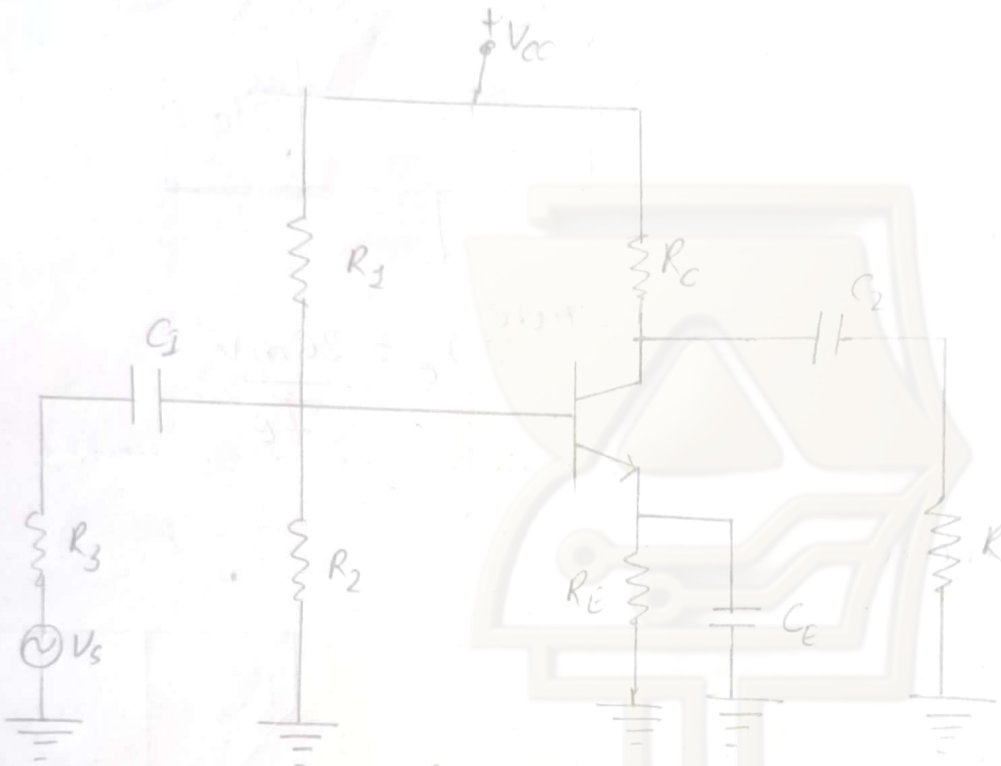
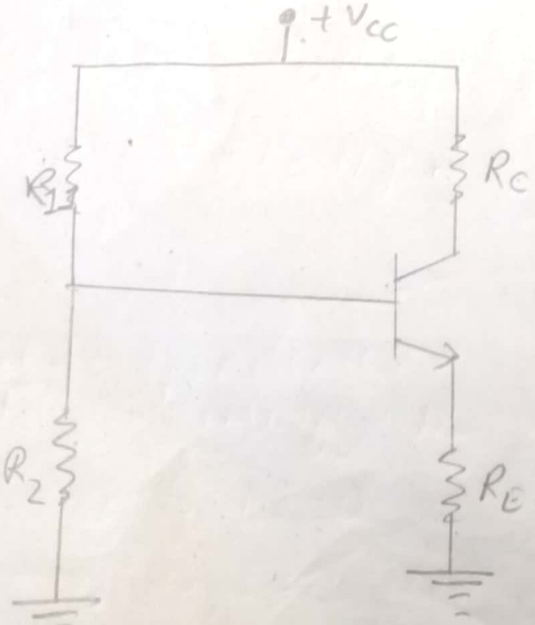



fig: Amplifier ckt

For DC equivalent circuit

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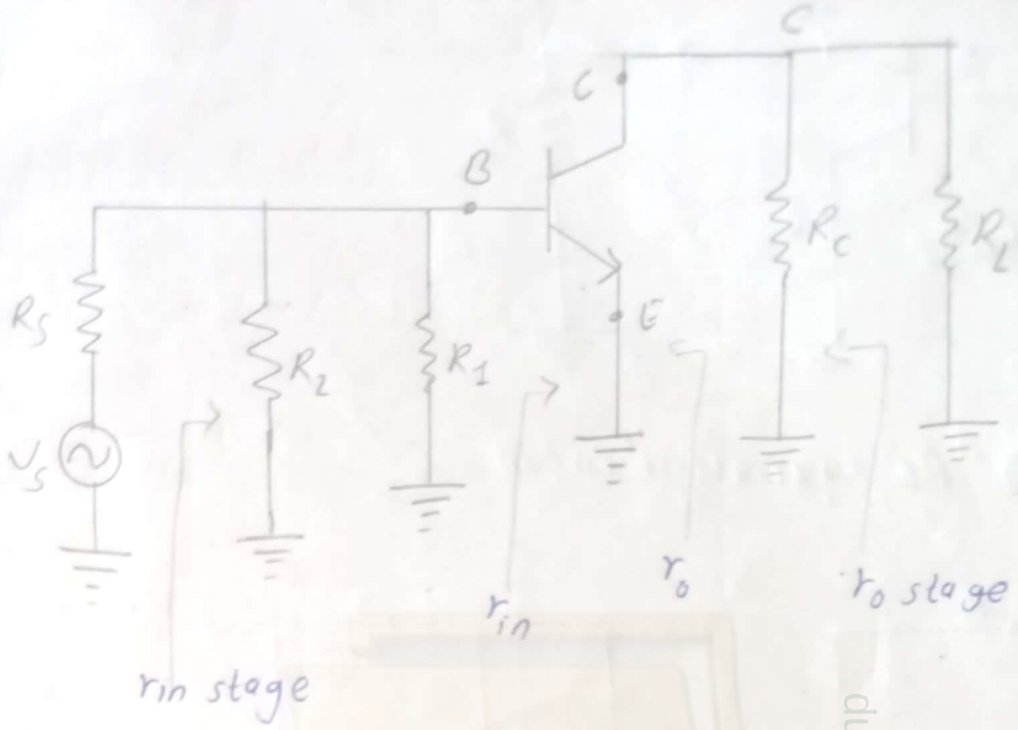
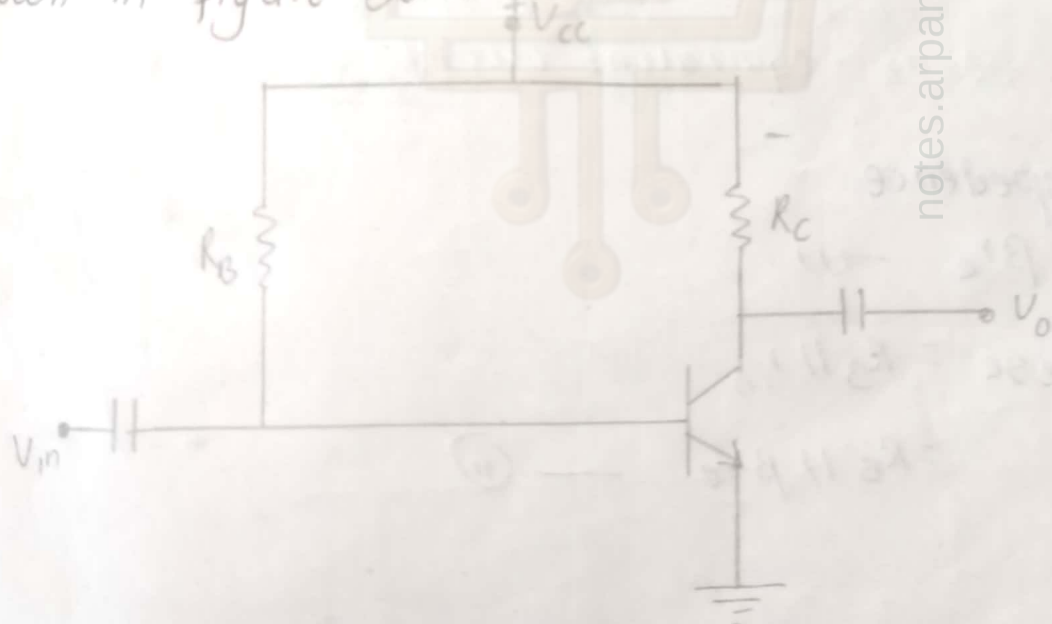
For AC equivalent circuit:




## 1. Small ~~angle~~ signal Analysis of common Emitter circuit

let us consider a common emitter fixed biased circuit as shown in figure below.




Now, for ac analysis capacitor acts as short and make $V_{cc} = 0$ by connecting it in the ground.

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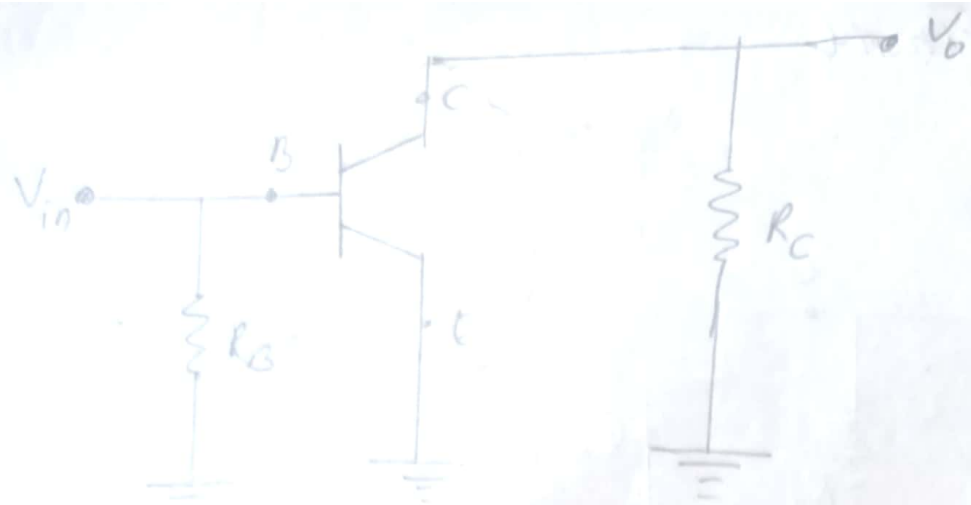



fig: AC equivalent circuit

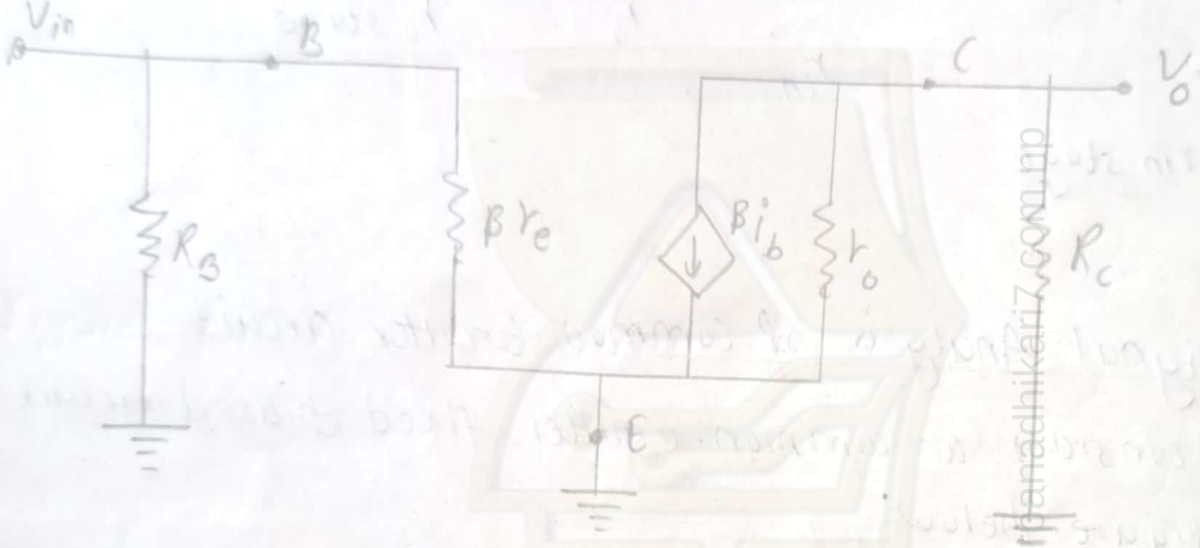


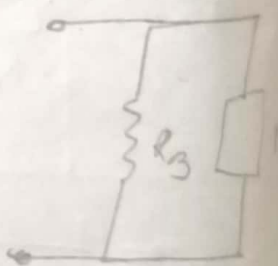

fig: $r_e$ - equivalent circuit

(a) Input impedence

$$r_{in} = \beta r_e \quad \text{—(i)}$$

$$r_{in\ stage} = R_B \,//\, r_{in}$$

$$= R_B \,//\, \beta r_e \quad \text{— (ii)}$$




(b) Output impendence.

While determining the o/p impendence, the input voltage is set to be 0.

$$r_{o\ stage} = r_o \,//\, R_C \quad \text{— (iii)}$$

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(c) Voltage gain ($A_v$) = $\frac{V_o}{V_{in}}$

Applying KCL at output

$\beta i_b + I_1 = 0$

$\Rightarrow \frac{\beta i_b + V_o}{(r_o \| R_c)} = 0$

$\Rightarrow \beta i_b = -\frac{V_o}{(r_o \| R_c)}$

$\Rightarrow \beta \frac{V_{in}}{\beta r_e} = \frac{-V_o}{(r_o \| R_c)}$

$\Rightarrow \frac{V_{in}}{r_e} = \frac{-V_o}{(r_o \| R_c)}$

$\Rightarrow \frac{V_o}{V_{in}} = -\frac{(r_o \| R_c)}{r_e}$

$\therefore A_v = -\frac{(r_o \| R_c)}{r_e}$ ——(iv)




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Voltage divider circuit with emitter

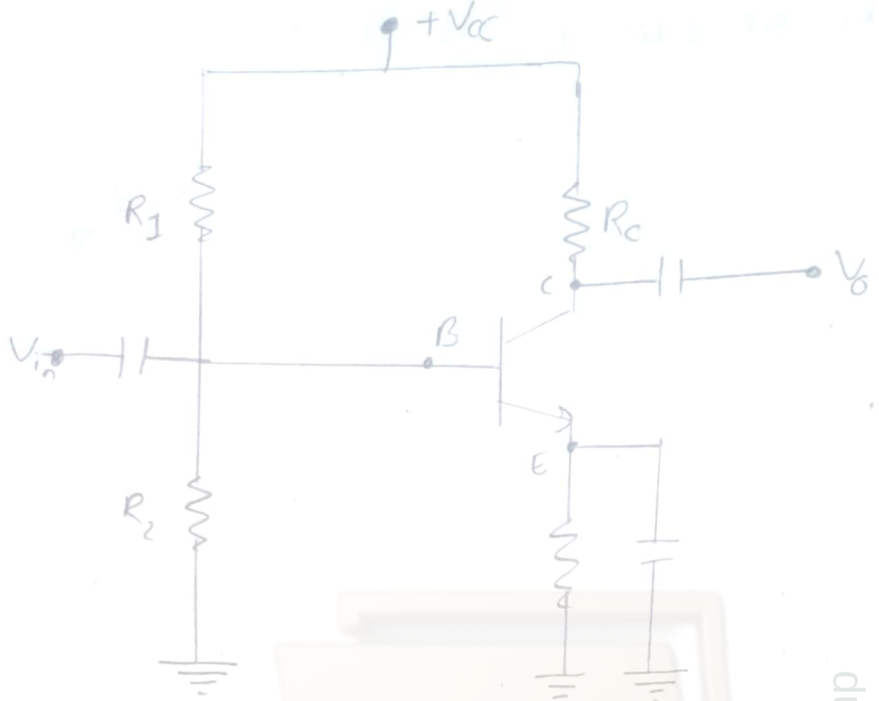
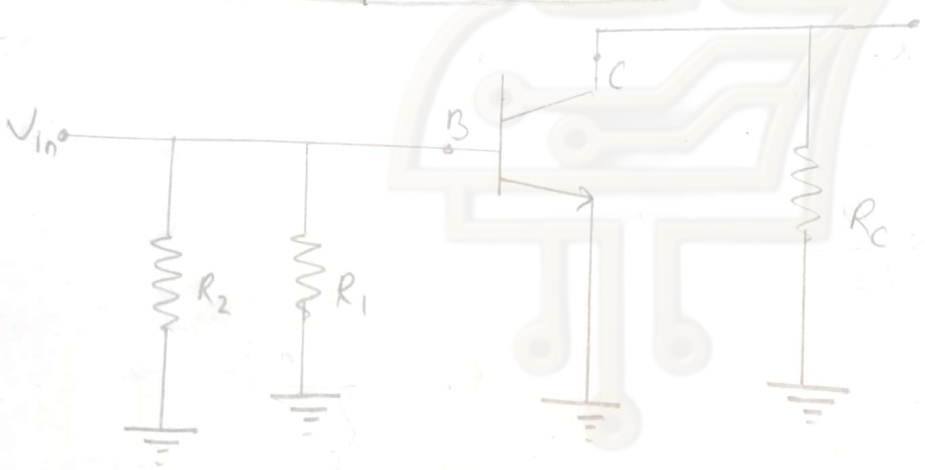
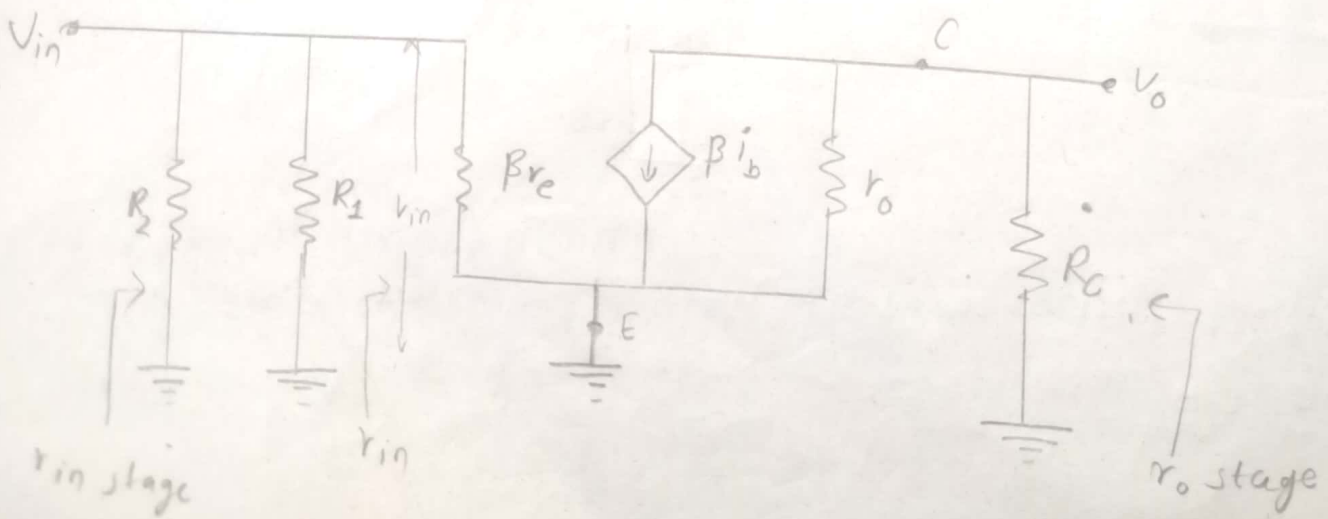



Figure for Ac equivalent circuit.

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(a) Input impedence.

Applying KVL at input side,

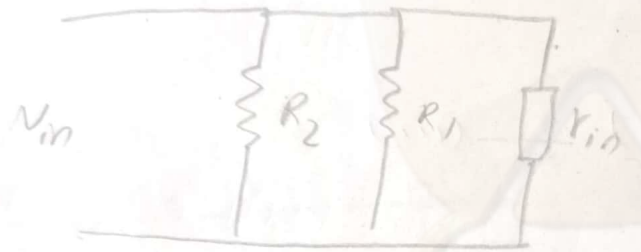
$$V_{in} - i_b \beta r_e = 0$$

$$V_{in} = i_b \beta r_e$$

$$\frac{V_{in}}{i_b} = \beta r_e$$

$$r_{in} = \beta r_e \quad \text{——(I)}$$

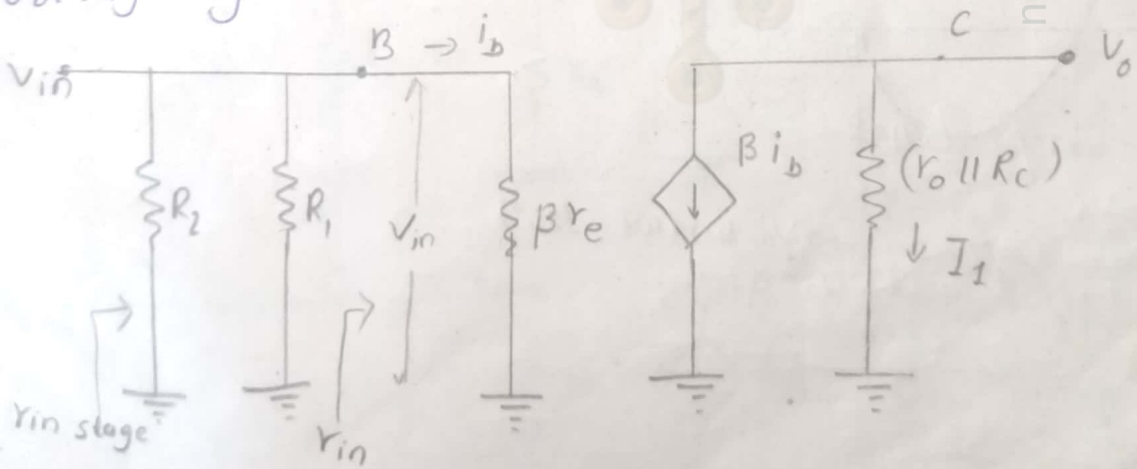
$$r_{in\ stage} = (R_1 \| R_2 \| r_{in}) \quad \text{——(II)}$$




(b) output impedence

$$r_o \text{ stage} = (r_o \| R_c)$$

(c) Voltage gain ($A_V$)




(d) $$A_V = \frac{V_o}{V_{in}}$$

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Applying KCL at output side,

$$\beta i_b + I_1 = 0$$

$$\beta i_b + \frac{V_o}{(r_o \parallel R_c)} = 0 \quad \left[ I_1 = \frac{V_o - 0}{r_o \parallel R_c} \right] \text{ nodal analysis}$$

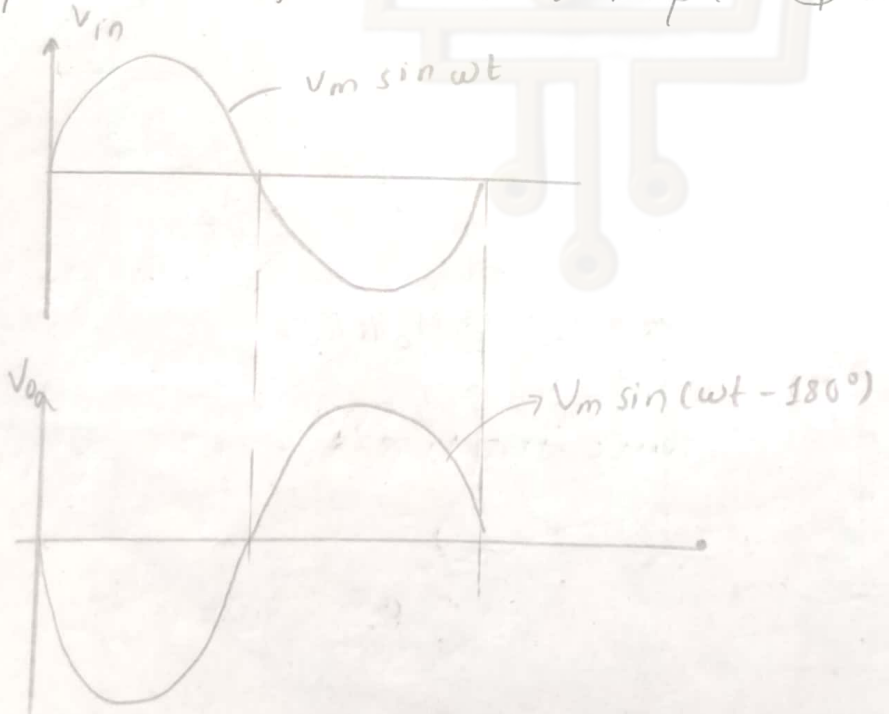
$$\beta i_b = \frac{-V_o}{(r_o \parallel R_c)}$$

$$\beta i_b + \frac{V_{in}}{\beta r_e} = \frac{-V_o}{(r_o \parallel R_c)}$$

$$\therefore \frac{V_o}{V_{in}} = -\frac{(r_o \parallel R_c)}{r_e}$$

$$A_v = -\frac{(r_o \parallel R_c)}{r_e} \quad \text{---(iii)}$$

The negative sign in ~~powe~~ voltage gain represents 180° phase shift between input & output.

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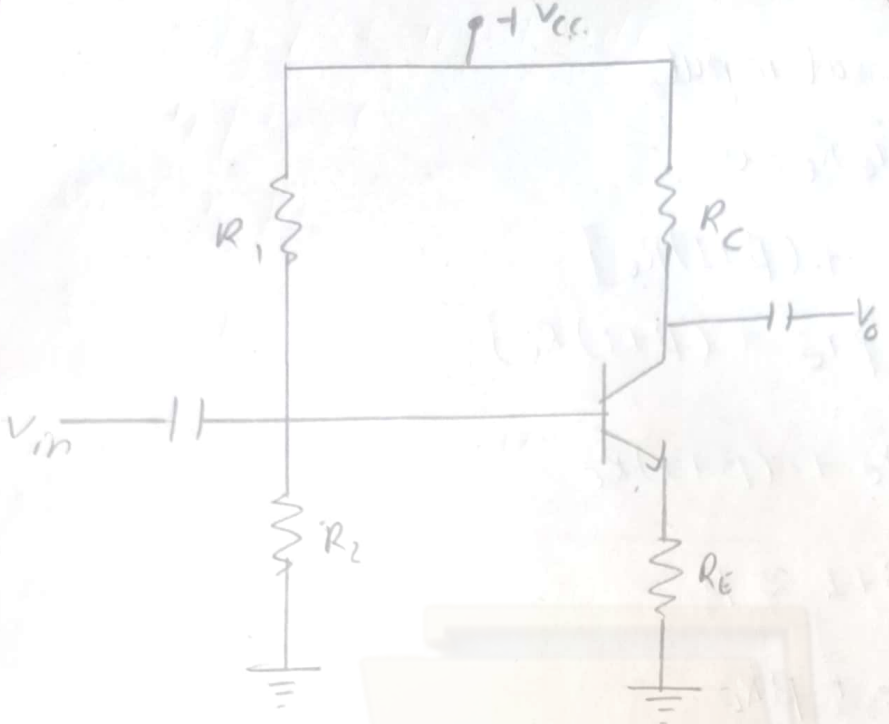
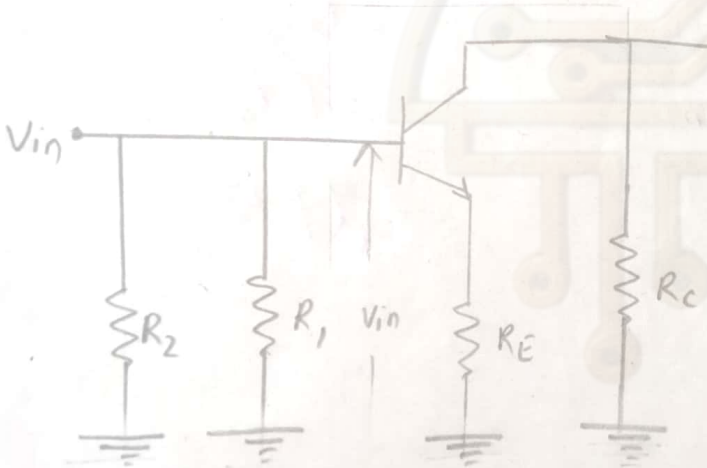
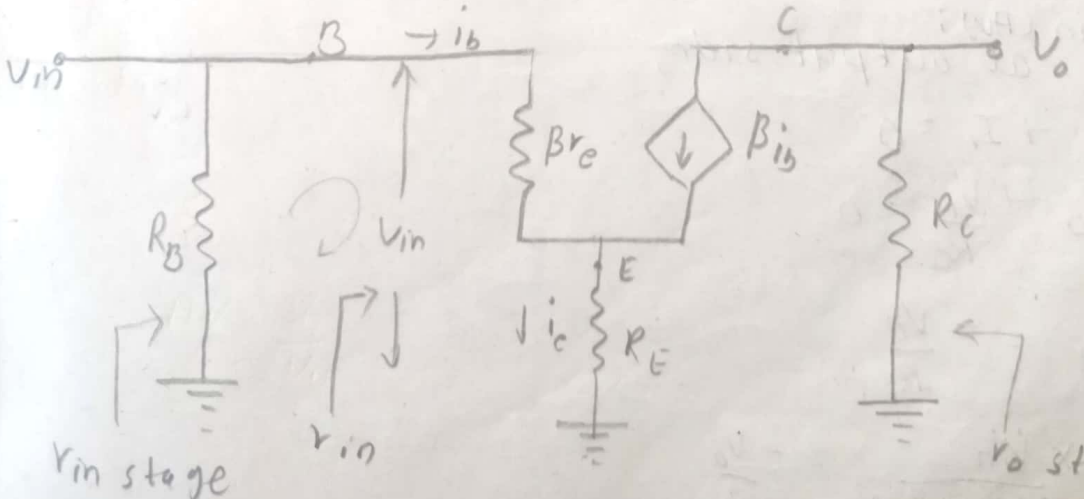
Voltage divider circuit with emitter bypass




For AC equivalent circuit

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(a) Input impedence:

Applying KVL at input,

$$V_{in} - i_b \beta r_e - i_e R_E = 0$$

or, $V_{in} = i_b \beta r_e + (\beta+1) R_E$

$$= i_b [\beta r_e + (\beta+1) R_E]$$

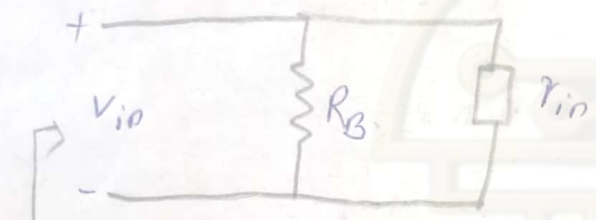
or, $\frac{V_{in}}{i_b} = \beta r_e + (\beta+1) R_E$

since, $\beta + 1 \approx \beta$

$$\frac{V_{in}}{i_b} = \beta r_e + \beta R_E$$

$$\therefore V_{in} = \beta (r_e + R_E)$$

$V_{in}$ stage $= R_B \| r_{in}$




$r_{in}$ stage

$i_e = (\beta+1) i_b$

$i_e = i_c + i_b$

$= \beta i_b + i_b$

$= (\beta+1) i_b$

(b) Output impedence.

$r_o$ stage $= R_C$

(c) Voltage gain ($A_V$):

Apply KCL at output side,

$$\beta i_b + I_1 = 0$$

or, $\beta i_b + \frac{V_o}{R_C} = 0$

or, $\beta i_b = -\frac{V_o}{R_C}$

or, $\beta \times \frac{V_{in}}{\beta (r_e + R_E)} = -\frac{V_o}{R_C}$

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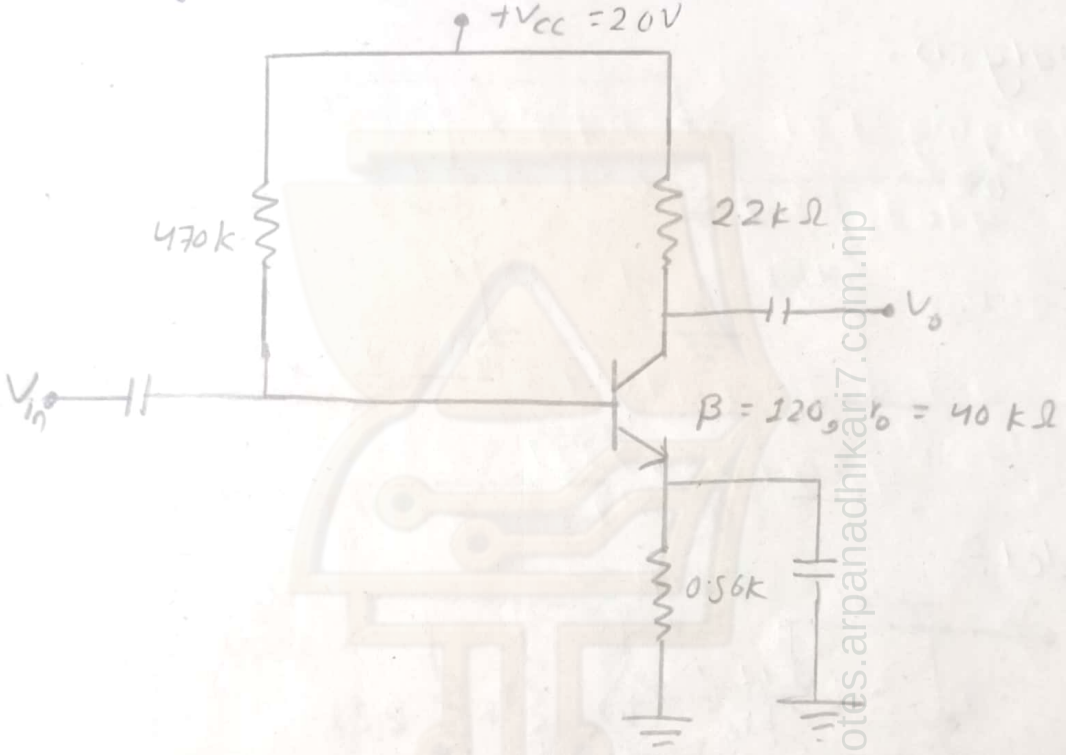
$$\Rightarrow \frac{V_o}{V_{in}} = -\frac{R_C}{r_e + R_E}$$

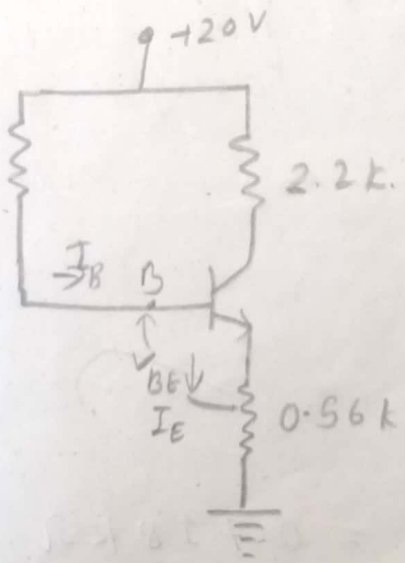
$$\therefore A_V = -\frac{R_C}{r_e + R_E} \quad \text{——(iii)}$$

## Numericals:

Qn.1. Find ~~$r_{in}$~~ stage, ~~$r_o$~~ stage, ~~$A_V$~~ for following circuit.




In order to find $r_e$, let us perform dc analysis.




Applying kvl at output,

$$20 - 470I_B - V_{BE} - 0.56I_E = 0$$

$$\text{or, } 20 - V_{BE} = 470I_B + 0.56(\beta+1)I_B$$

$$\text{or, } 20 - V_{BE} = I_B[470 + 0.56(\beta+1)]$$

$$\therefore I_B = \frac{20 - 0.7}{470 + 0.56 \times 121}$$

$$= 0.0358 \text{ mA}$$

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$I_E = (\beta+1) I_B$

$= (120+1) \times 0.0358$

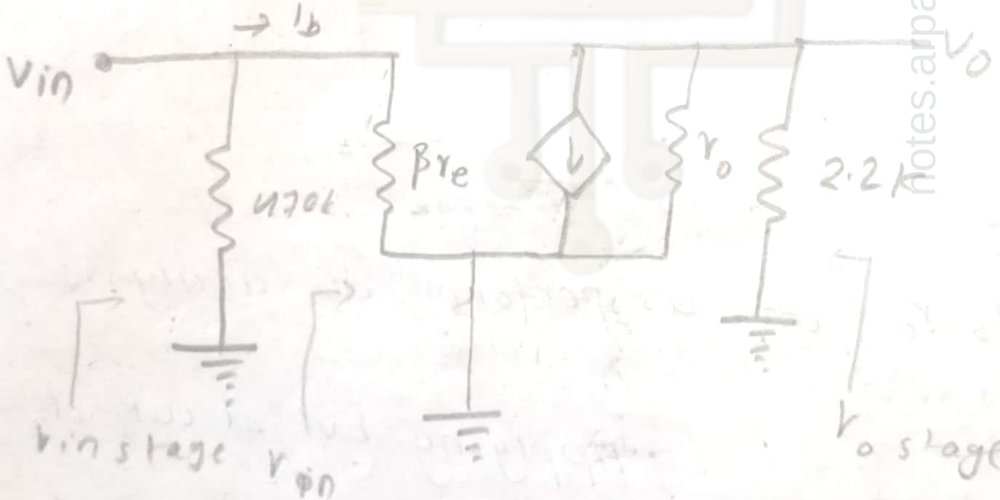
$= 4.331\ mA$

$\therefore r_e = \frac{26\ mV}{4.331\ mA}$

$= 6\Omega$

AC analysis:

$r_e$ model:




$r_{in} = \beta r_e = 120 \times 6$

$= 720\Omega$

$= 0.72\ k\Omega$

stage $= 470\ k \,//\, r_{in}$

$= \frac{470 \times 0.72}{470 + 0.72} = 0.718\ k\Omega$

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$r_o$ stage = $r_o \parallel R_c$

$$= \frac{40 \times 2.2}{40+2.2}$$

$$= 2.0853\ k\Omega$$

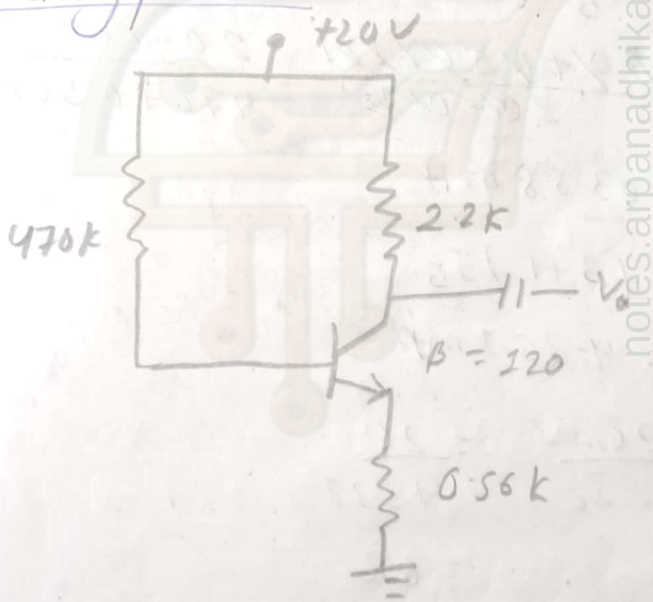
$$A_v = -\frac{(r_o \parallel R_c)}{r_e}$$

$$= -\frac{2.0853}{6} \times 1000$$

$$= -0.34755 \times 1000$$

$$= -347.55$$

For emitter unbypass




Solution,

In order to find $r_e$, let us perform dc analysis.

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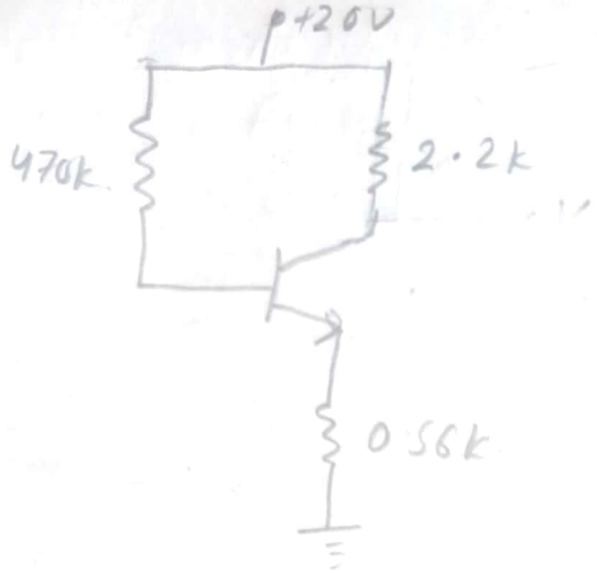



Applying KVL at input,

$20 - 470 I_B - V_{BE} - 0.56 I_E = 0$

$\Rightarrow 20 - V_{BE} = 470 I_B + 0.56 (\beta+1) I_B$

$\Rightarrow 20 - 0.7 = I_B (470 + 0.56 \times 121)$

$\therefore I_B = \frac{20-0.7}{470+0.56\times121}$

$\Rightarrow I_B = 0.0358\ mA$

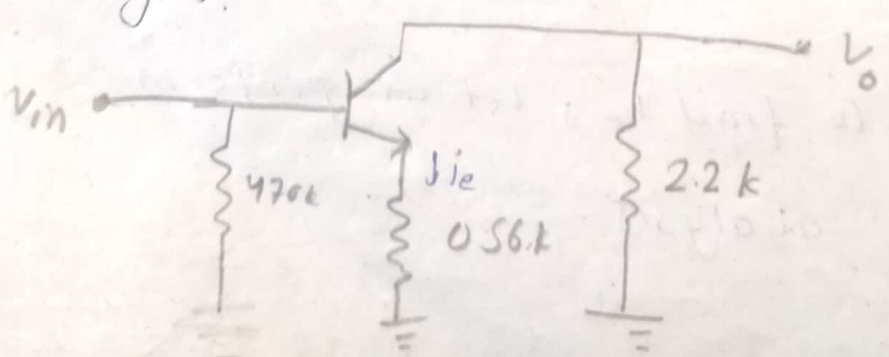
$I_E = (\beta+1) I_B$

$= 4.331\ mA$

$\therefore r_e = \frac{26mV}{I_E} = \frac{26}{4.331}$

$= 6\Omega$

AC analysis:

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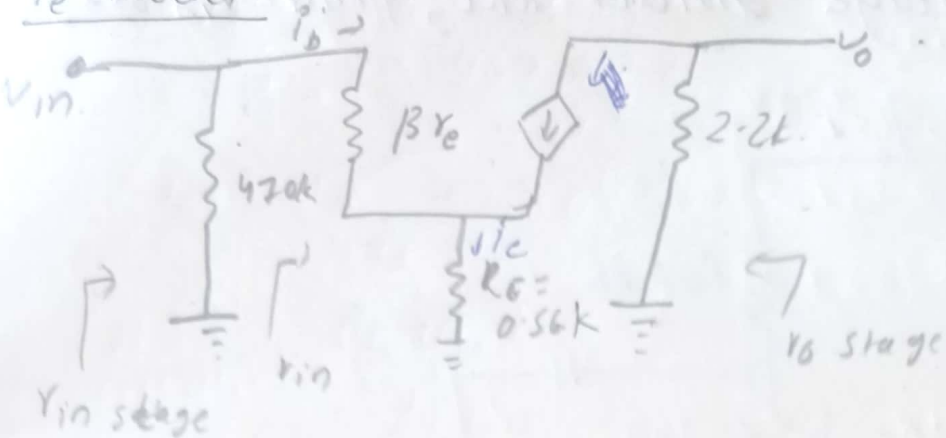
re model




$$r_{in} = \beta(r_e + R_E)$$

$$= 120\left(\frac{6}{1000} + 0.56\right)$$

$$= 120 \times 6.56$$

$$= 67.92\,k\Omega$$

$$r_{in\ stage} = R_B \,||\, r_{in}$$

$$= 470 \,||\, 67.92$$

$$= \frac{1}{\frac{1}{470} + \frac{1}{67.92}}$$

$$= 59.34\ k\Omega$$

$$r_o \text{ stage} = R_C$$

$$= 2.2k$$

$$A_V = \frac{-R_C}{r_e + R_E}$$

$$= -\frac{2.2}{0.006 + 0.56}$$

$$= 0.3353$$

$$= -3.8869$$

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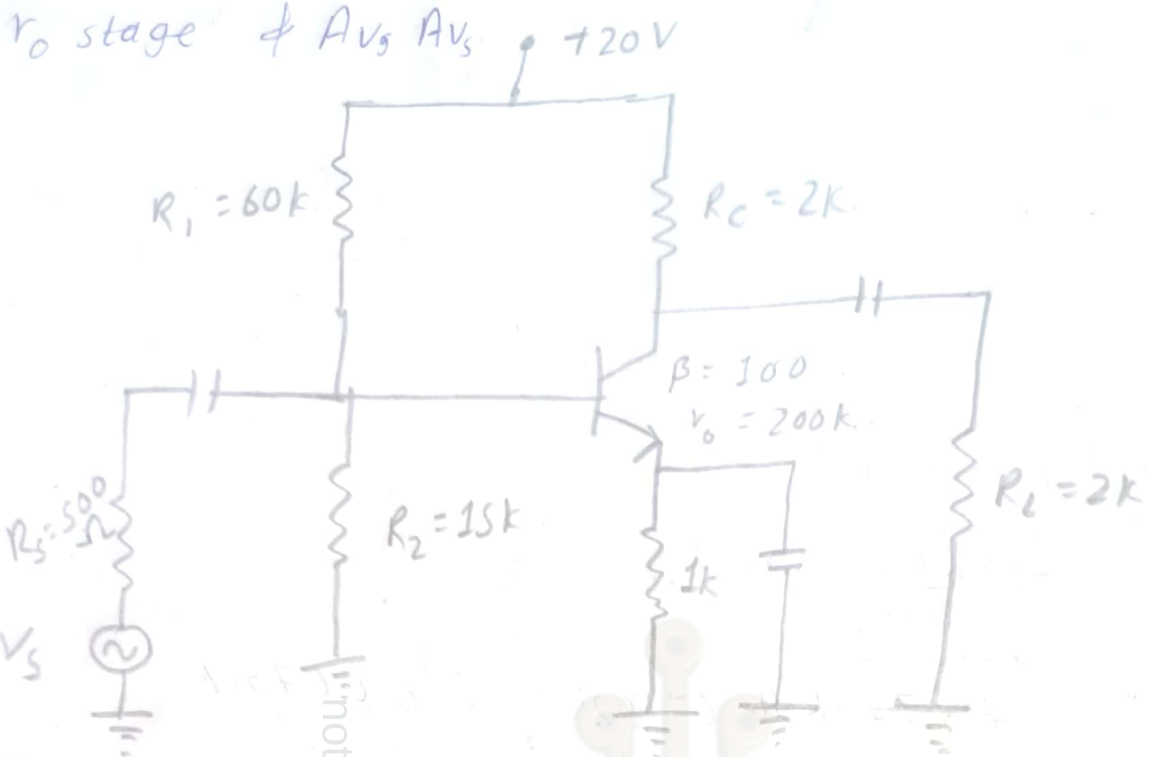
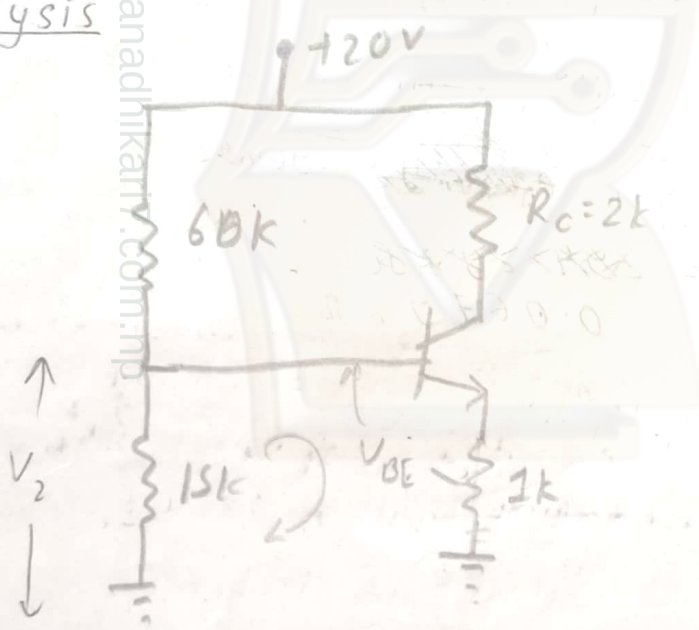
Qn 2. For the given voltage divider ... ;
$r_o$ stage & $A_{V_s}$ $A_{V_s}$




dc analysis




$$V_2 = \frac{V_{cc} \times R_2}{R_1 + R_2}$$

$$= \frac{20 \times 15}{60 + 15}$$

$$= 4V$$

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Applying KVL,

$$V_2 - V_{BE} - I_E R_E = 0$$

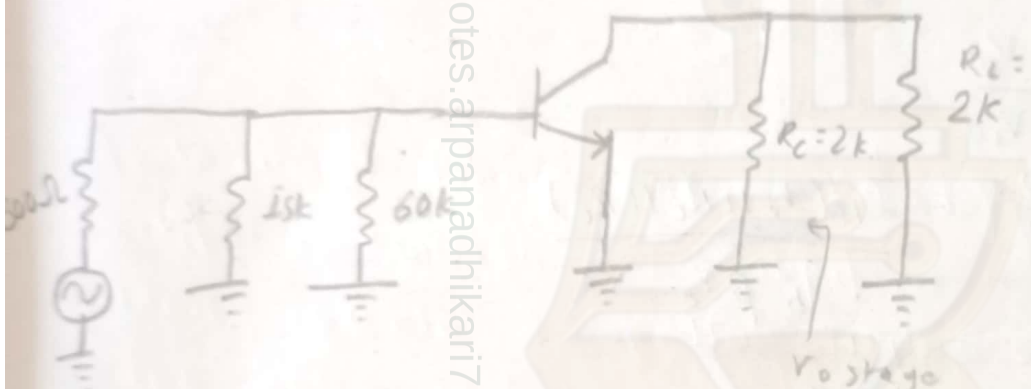
$$I_E = \frac{V_2 - V_{BE}}{R_E}$$

$$= \frac{4 - 0.7}{1k\Omega}$$

$$= 3.3\,mA$$

$$\therefore r_e = \frac{26mV}{I_E} = \frac{26mV}{3.3} = 7.87\Omega = 7.87 \times 10^{-3}$$

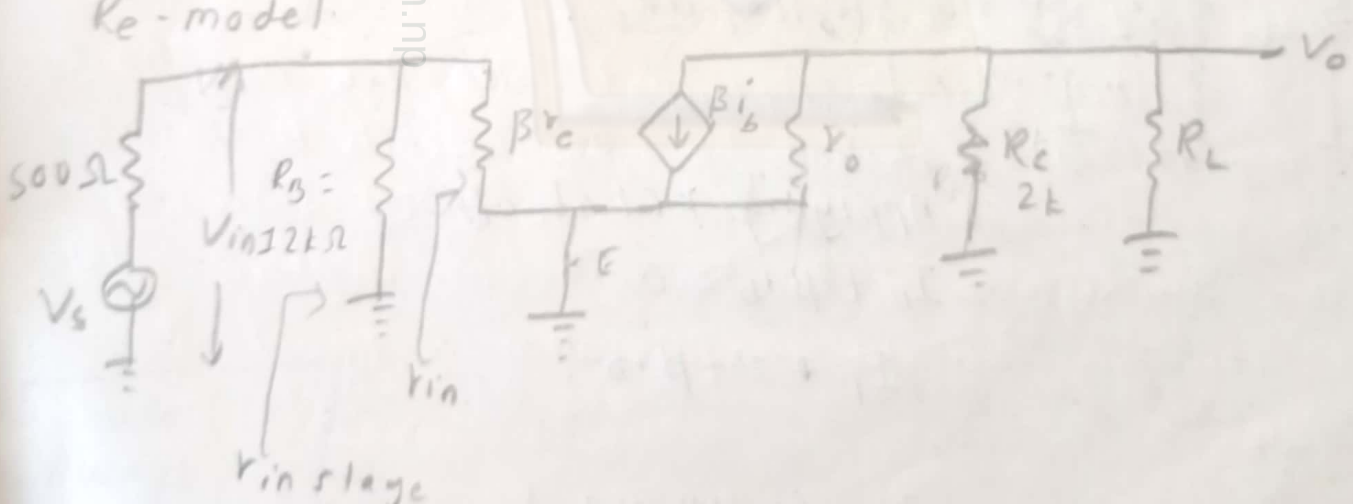
For ac analysis:




$$R_B = \frac{15 \times 60}{15 + 60}$$

$$= 12k\Omega$$

$r_e$ - model




$$V_{in} = \beta(r_e + R_E)$$

$$= \beta r_e$$

$$= 100 \times 7.87$$

$$= 787 = 0.787k$$

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$r_{in}$ stage $= R_B \| r_{in}$

$= \frac{12 \times 0.787}{12 + 0.787}$

$= 0.7385\ k\Omega$

$r_o$ stage: $r_o \| R_c$

$= \frac{200 \times 2}{200 + 2}$

$= 1.980\ k\Omega$

$A_V = \frac{R_c}{r_e} \frac{V_o}{V_{in}}$

$\frac{2000}{7.8} = \frac{I_1 (r_o \| R_c \| R_L)}{i_b \beta r_e}$

Applying KCl at o/p

$I_1 + \beta i_b = 0$

$\therefore I_1 = -\beta i_b$

$\therefore A_V = -\frac{\beta i_b (r_o \| R_c \| R_L)}{i_b \beta r_e}$

$= -\frac{(r_o \| R_c \| R_L)}{r_e}$

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$$A_V = -\frac{1}{\frac{1}{200} + \frac{1}{2} + \frac{1}{2}}$$

$$= -\frac{1}{0.00787}$$

$$= -\frac{0.9950}{0.00787}$$

$$= -126.43$$

$$A_{VS} = \frac{V_o}{V_S}$$

we know,

$$V_{in} = \frac{V_S}{R_S + r_{in\,stage}} \times r_{in\,stage}$$

$$\text{or, } \frac{V_{in}}{V_S} = \frac{r_{in\,stage}}{R_S + r_{in\,stage}} \quad \text{——— (1)}$$

Now, $A_{VS} = \frac{V_o}{V_S} = \frac{V_o}{V_{in}} \times \frac{V_{in}}{V_S}$

$$= A_V \times \frac{r_{in\,stage}}{R_S + r_{in\,stage}}$$

$$= -126.43 \times \frac{0.7385}{\frac{500}{1000} + 0.7385}$$

$$= -126.43 \times 0.5962$$

$$= -75.388$$

Note:
Always
$A_{VS} < A_V$

↓
overall voltage gain

The negative sign indicates 180° phase shift bet? input & output.

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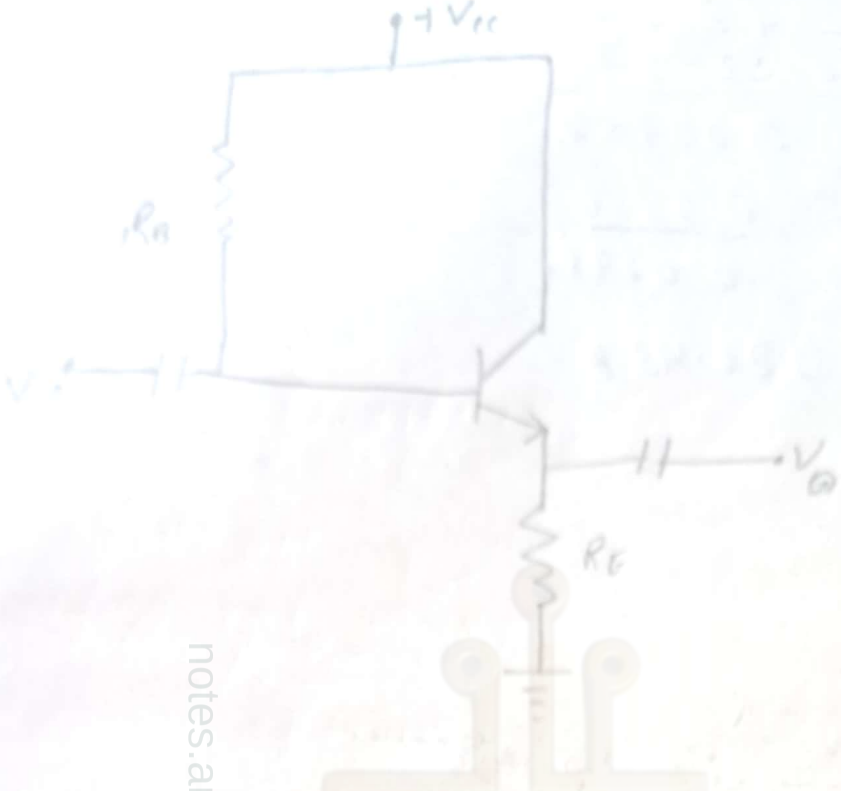
# Emitter - follower (common collector configuration)




When output is taken from the emitter terminal of transistor as shown in figure, the network is referred as emitter follower. The output voltage is always slightly less than the input signal but approximately voltage gain (Av) is unity (≈1).

The emitter voltage (ie output voltage) is in phase with the input voltage (ie both $V_o$ & $V_{in}$ attends their positive & negative peak at the same time). Since, $V_o$ follows the magnitude of $V_{in}$ with an inphase relationship, this common collector configuration is known as emitter follower.

The most common use of emitter follower configuration is in impedance matching application.

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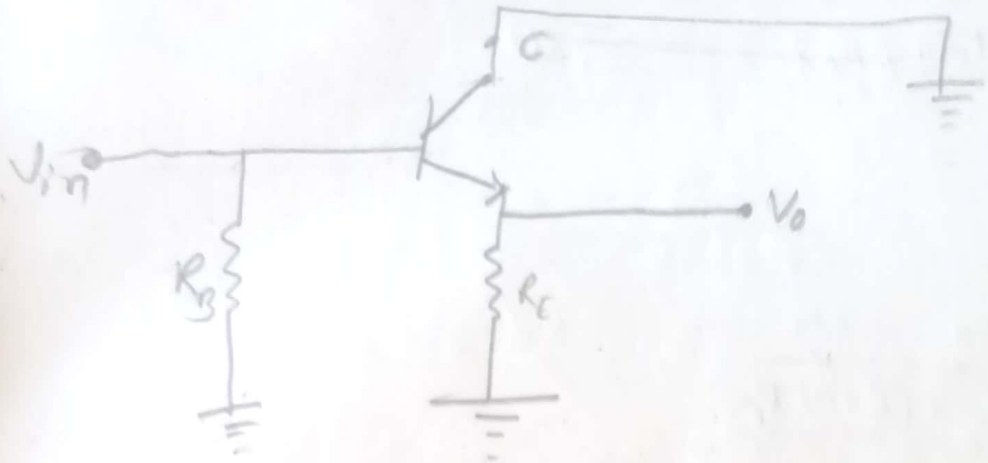
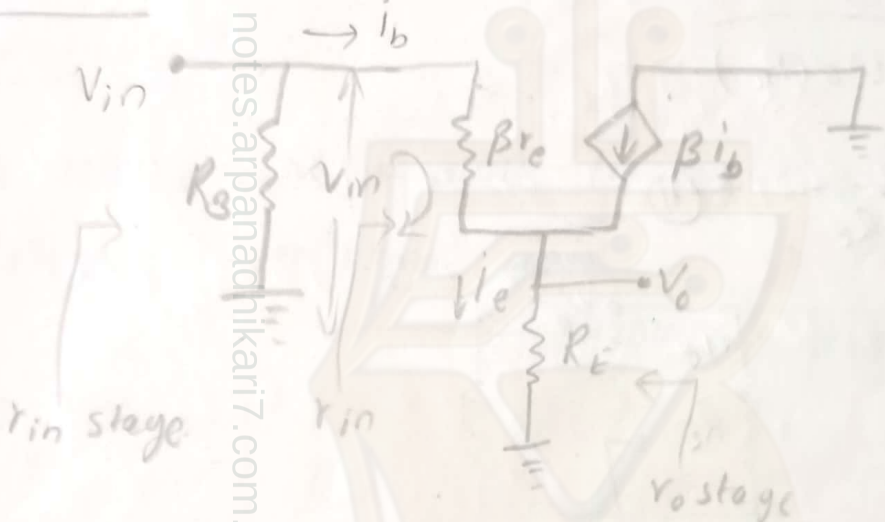
AC equivalent circuit




$R_e$ model:

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(a) Input impedence;

Applying KVL at input,

$$V_{in} - i_b \beta r_e - i_e R_E = 0$$

$$\text{or, } V_{in} - i_b \beta r_e - (\beta+1) i_b R_E = 0$$

$$\text{or, } V_{in} = i_b [\beta r_e + (\beta+1) R_E]$$

$$\text{or, } \frac{V_{in}}{i_b} = \beta r_e + (\beta+1) R_E$$

$$\text{or, } r_{in} = \beta r_e + (\beta+1) R_E$$

As $\beta+1 \approx \beta$

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$$r_{in} = \beta r_e + \beta R_E$$

$$\text{or, } r_{in} = \beta (r_e + \beta R_E) \quad \text{——} (i)$$

(b) Output impedence:

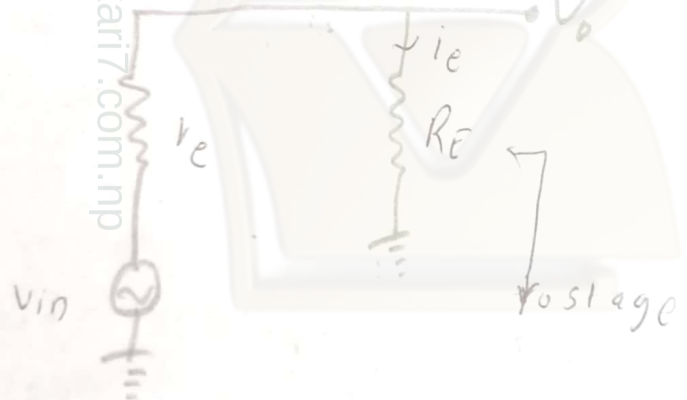
$$i_e = (\beta+1) i_b$$

$$= (\beta+1) \frac{V_{in}}{\beta r_e + (\beta+1) R_E}$$

Take $\beta+1 \approx \beta$ Then,

$$i_e = \beta \cdot \frac{V_{in}}{\beta (r_e + R_E)}$$

$$\text{or } i_e \quad \frac{V_{in}}{r_e + R_E} \quad \text{——} (ii)$$




for $r_o$ stage, $V_{in} = 0$

$$\therefore r_o \text{ stage} = (r_e \parallel R_E) \quad \text{——} (iii)$$

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① Voltage gain ($A_V$) = $\frac{V_o}{V_{in}}$

since, $V_o = \frac{V_{in}}{r_e + R_E} \times R_E$ [$\because V_o = i_e \times R_E$]

$\frac{V_o}{V_{in}} = \frac{R_E}{r_e + R_E}$

$\therefore A_V = \frac{R_E}{r_e + R_E}$ — (1)

note: (should be less than 1 in this type)

since, the o/p voltage follows the input voltage, an o/p is taken from emitter terminal. It is termed as emitter follower

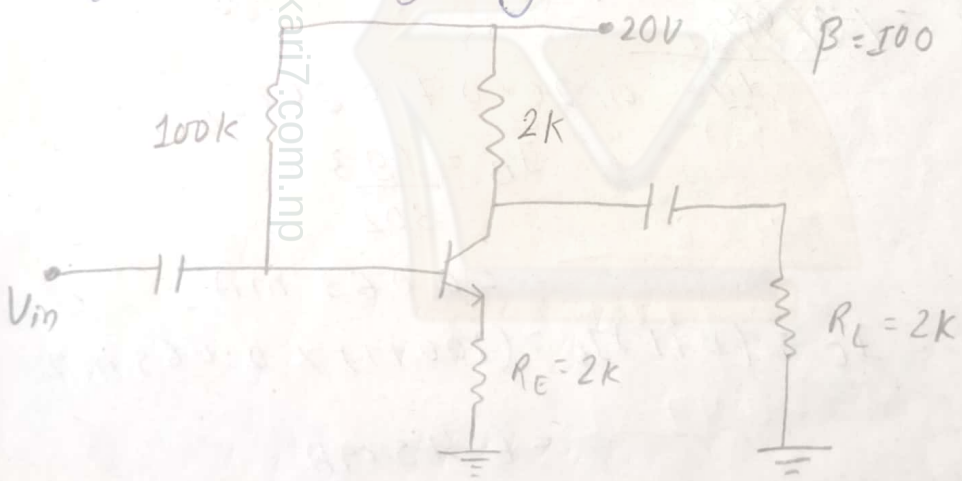
since, $r_e << R_E$

$\therefore A_V \approx 1 \quad \frac{V_o}{V_{in}} \approx 1$

$V_o \approx V_{in}$

## Numericals:

Qn-1: For the circuit shown in figure, find $r_{in}$ stage, $r_o$ stage & voltage gain.

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Solution

Since, the value of $r_e$ is not given in question, first compute $I_E$ from dc analysis and then find $r_e$ by using formular $r_e = \frac{26mV}{I_E}$.

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dc analysis.




Applying KVL at input

$$20 - 100 I_B - V_{BE} - I_E R_E = 0$$

$$I_E = \frac{20 - 100 I_B + V_{BE}}{R_E}$$

$$= \frac{20 - 100 I_B + 0.7}{2}$$

or, $20 - 100 I_B - 0.7 - 2(\beta+1) I_B = 0$

or, $20 - 0.7 = 100 I_B + 2 \times 101 I_B$

or, $20 - 0.7 = 302 I_B$

$$I_B = \frac{19.3}{302}$$

$$= 0.063 \text{ mA}$$

$$\therefore I_E = (\beta+1) I_B = (100+1) \times 0.063 \text{ mA}$$

$$= 6.45 \text{ mA}$$

$$\therefore r_e = \frac{26 \text{ mV}}{6.45 \text{ mA}}$$

$$= 4.03 \, \Omega.$$

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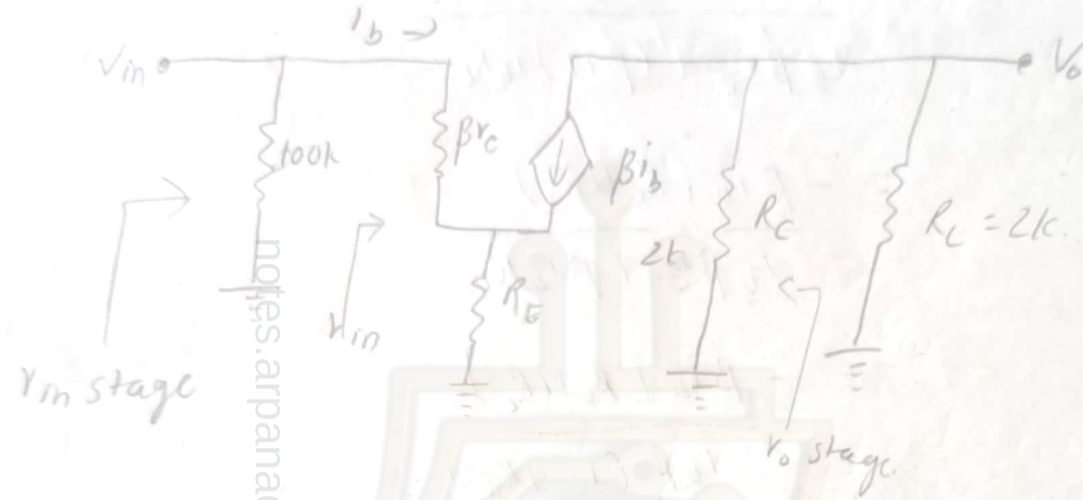
# AC analysis




We know,

$$r_{iD} = \beta(r_e + R_E)$$

$$= 100\left(\frac{4.031}{1000} + 2\right)$$

$$= 200.403 \text{ k}\Omega$$

$$r_{in\ stage} = (100 \,||\, r_{iD})$$

$$= 100 \,||\, 200.403$$

$$= \frac{1}{\frac{1}{100} + \frac{1}{200.403}}$$

$$= 66.71 \text{ k}\Omega$$

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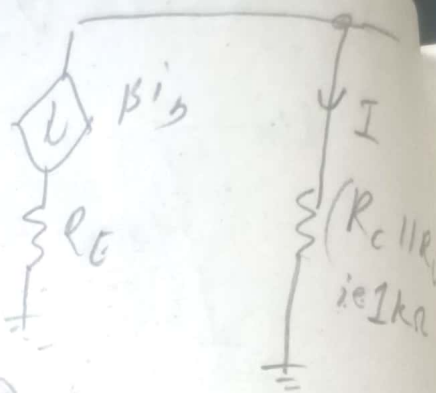
AC/F

$r_o$ stage $= R_c$

$= 2k$




$$A_V = \frac{V_o}{V_{in}} = \frac{I(R_c \| R_L)}{i_b \beta r_e + i_e R_E}$$

$I + \beta i_b = 0$

$I = -\beta i_b$

$$= \frac{-\beta i_b (R_c \| R_L)}{i_b \beta r_e + (\beta+1) i_b R_E}$$

$$= \frac{-\beta i_b (R_c \| R_L)}{i_b \beta r_e + \beta i_b R_E}$$

$$= \frac{-\beta i_b (R_c \| R_L)}{i_b \beta (r_e + R_E)}$$

$$A_V = \frac{-(R_c \| R_L)}{(r_e + R_E)}$$

$$= \frac{-1000}{4.031 + 2000}$$

$$= -0.4989$$

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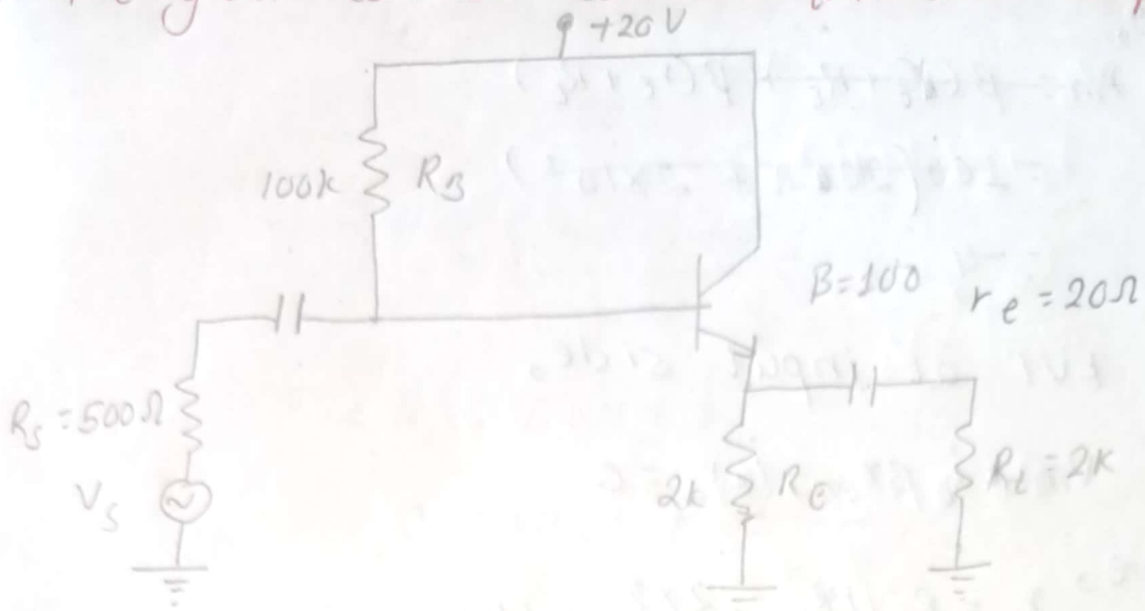
For the given common collector transistor amplifier.



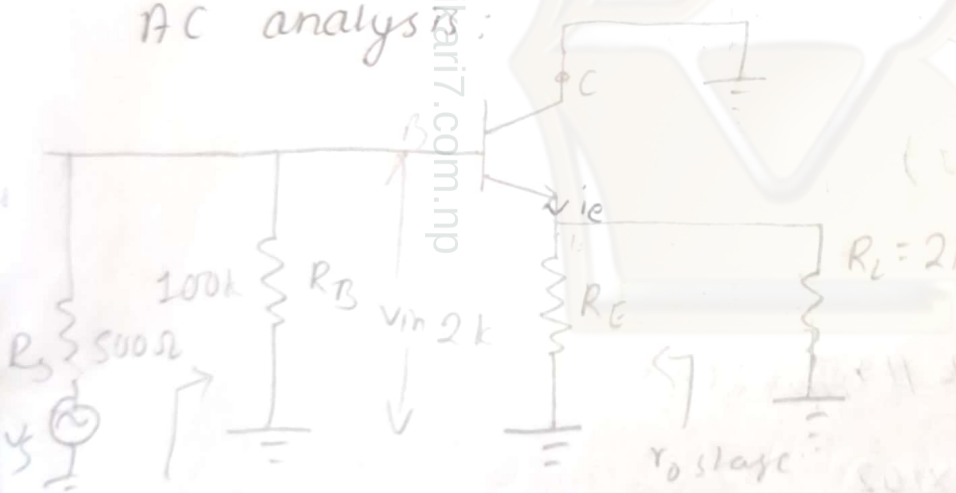

Find $R_{in}$ stage, $R_o$ stage, Av and Avs.

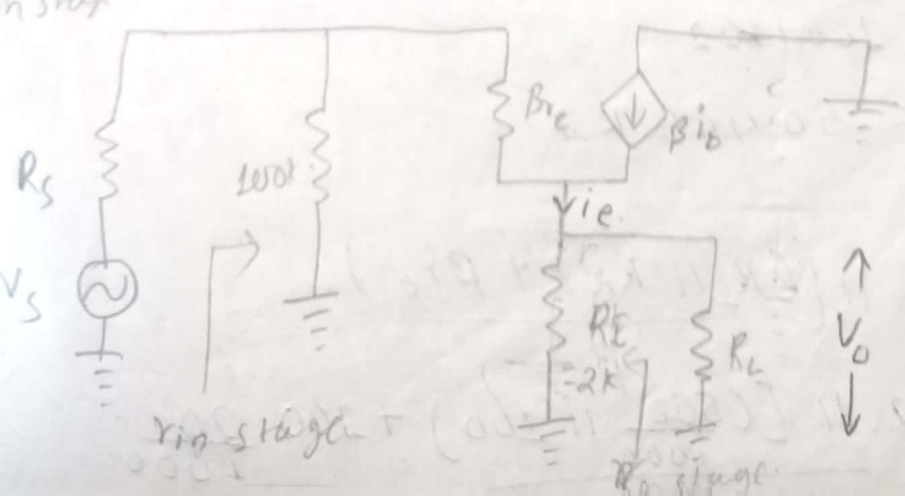
Solution

Given,

$r_e = 20\ \Omega$

AC analysis:

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We know,

$r_m = \beta(r_e + R_E) \quad \beta(r_e + R_E)$

$= 100(20\Omega + 2\times10^3)$

$= 4$

Applying KVL at input side,

$V_{in} - i_b \beta r_e - i_e r_L = 0$

where, $r_L = R_E \| R_L = \frac{2\times2}{2+2} = 1k\Omega$

Now,

$V_{in} - i_b \beta r_e - (\beta+1) i_b r_L = 0$

$\Rightarrow V_{in} = i_b \beta r_e + \beta i_b r_L \quad [\because \beta+1 \approx \beta] \quad \text{---(1)}$

$\Rightarrow \frac{V_{in}}{i_b} = \beta(r_e + r_L)$

$\Rightarrow r_{in} = 100\left(\frac{20}{1000} + 1\right)$

$\Rightarrow r_{in} = 102$

(I) $\therefore r_{in}$ stage $= 100k \| r_{in}$

$= \frac{100\times102}{100+102}$

$= 50.495\ k\Omega$

(II) $r_o$ stage $= R_E \| \left[\frac{(R_S \| R_3) + \beta r_e}{\beta}\right]$

$= 2 \| \frac{\left[\left(\frac{500}{1000} \| 100\right) + 100\frac{20}{1000}\right]}{100}$

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$r_o$ stage $= 2 \,||\, \left[\left[\frac{0.5 \,||\, 100}{100}\right] + 2\right]$

$= 2 \,||\, \left[\frac{0.4975}{100} + 2\right]$

$= 2 \,||\, 0.02497$

$= 0.02466\ K\Omega$

$= 24.66\ \Omega$

$A_V = \frac{V_o}{V_{in}}$

$= \frac{i_e \times r_L}{i_b \beta r_e + \beta i_b r_L}$ [$\because$ from (1)]

$= \frac{(i_b + i_c) \times r_L}{i_b (\beta r_e + \beta r_L)}$

$= \frac{r_L i_b + \beta i_b \times r_L}{i_b (\beta r_e + \beta r_L)}$

$= \frac{r_L i_b (1+\beta)}{i_b (\beta r_e + \beta r_L)}$

Another method,

$A_V = \frac{i_e r_L}{i_b \beta (r_e + r_L)}$

$= \frac{\beta i_b r_L}{i_b \beta (r_e + r_L)}$

$= \frac{r_L}{r_e + r_L}$

$= \frac{1+\beta r_L}{\beta r_e + r_L}$

$= \frac{1+100 \times 1}{100 \times \frac{20}{1000} + 1}$

$= \frac{101}{2+1}$

$= 33.666\ V$

$= \frac{1+100}{100 \times \frac{20}{1000} + 100 \times 1}$

$= 0.990$

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$$A_{VS} = \frac{V_o}{V_s}$$

$$= \frac{\cancel{I_c i_b (1+\beta)}}{}$$

$$= \frac{V_o}{V_{in}} \times \frac{V_{in}}{V_s}$$

$$= A_V \times \frac{r_{in(stage)}}{R_s + r_{in(stage)}}$$

$$= 0.990 \times \frac{50.49}{\frac{500}{1000} + 50.495}$$

$$= 0.9801$$

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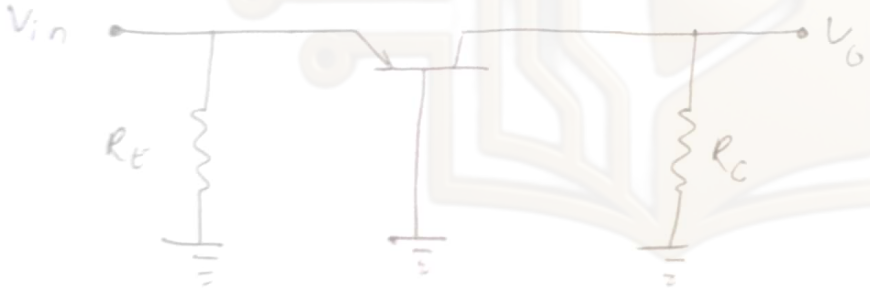
# Common Base Configuration

AC equivalent circuit:

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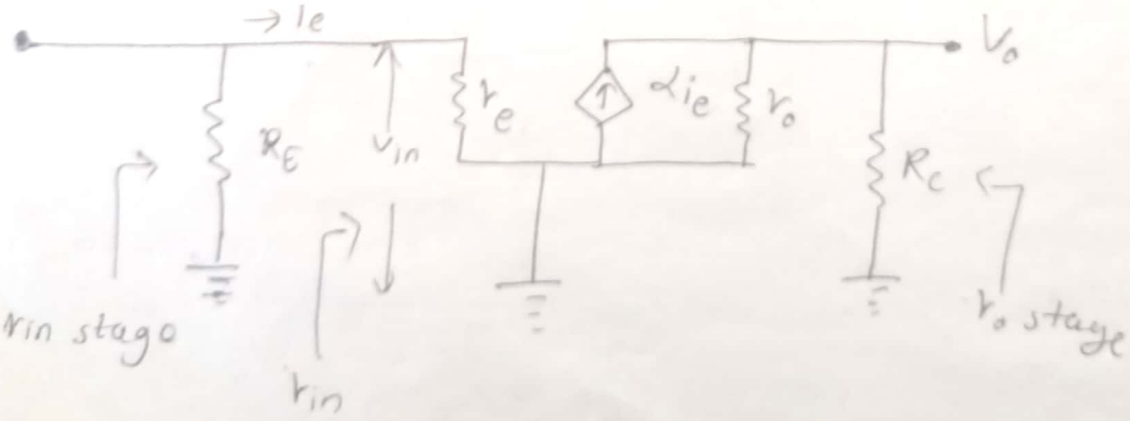


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Applying KVL at input,

$$V_{in} - i_e r_e = 0$$

$$\Rightarrow V_{in} = i_e r_e$$

$$\Rightarrow \frac{V_{in}}{i_e} = r_e$$

$$\therefore r_{in} = r_e \quad \text{---(i)}$$

$r_{in}$ stage $= R_E \| r_{in}$

$$= R_E \| r_e \quad \text{---(ii)}$$

$r_o$ stage $= r_o \| R_C$ ---(iii)

$$A_V = \frac{V_o}{V_{in}}$$

$$= \frac{\alpha i_e (r_o \| R_c)}{i_e r_e}$$

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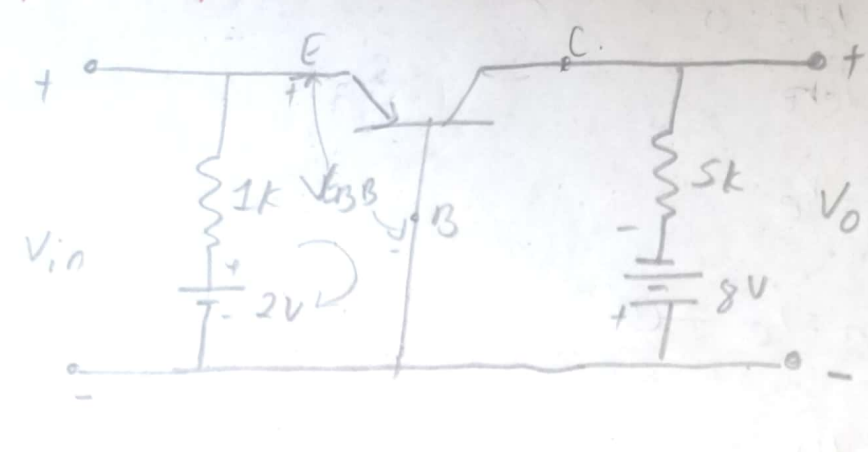
* Find input impedence, output impedance, gain.




Given

$\alpha = 0.98$

$r_o = 1M\Omega$

$= 1000 k\Omega$

DC analysis:

Since, $r_e$ is not given, so, from ~~AC analysis~~ AC Analysis, we find $r_e$,

Applying KVL at input side,

$$2 - I_E \times 1 - V_{EB} = 0$$

$$\Rightarrow I_E = \frac{2 - V_{EB}}{1}$$

$$= \frac{2 - 0.7}{1}$$

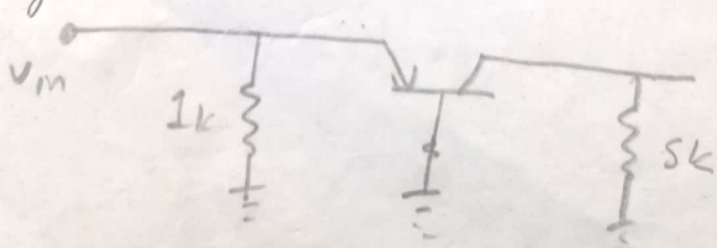
$$= 1.3 \text{ mA}.$$

$$\therefore r_e = \frac{26 \text{ mV}}{I_E}$$

$$= \frac{26}{1.3}$$

$$= 20\Omega$$

Ac analysis:

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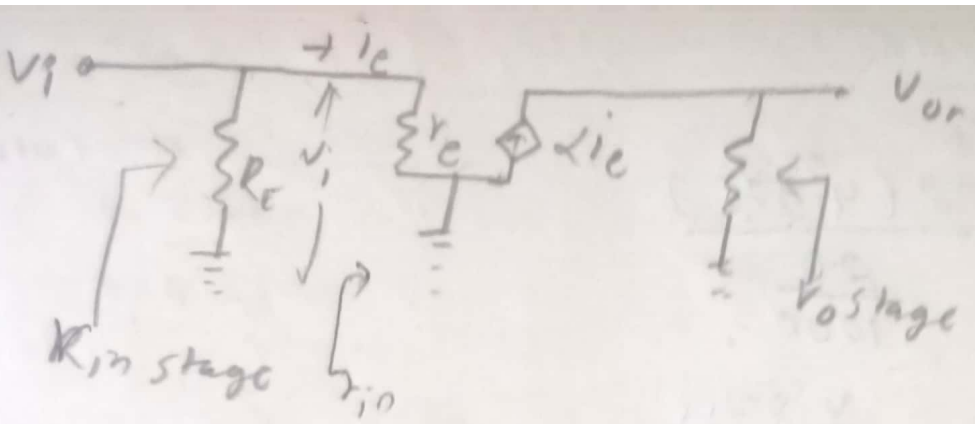


We know, ~~$r_{in} = r_e$~~

$\Rightarrow$ ~~$i_e \times r_e = r_e$~~

Applying KVL at input,

$$R_E i_e - i_e r_e = 0$$

$$\Rightarrow i_e (R_E - r_e) = 0$$

$$\Rightarrow 1 - \frac{20}{1000}$$

$$r_{in\ stage} = R_E \,||\, r_{in}$$

$$= 1 \,||\, r_e$$

$$= 1 \,||\, \frac{20}{1000}$$

$$= 1 \,||\, 0.02$$

$$= 0.0196\ k\Omega$$

$$r_{o\ stage} = r_o \,||\, R_c$$

$$= 1000 \,||\, 5$$

$$= 4.975\ k\Omega$$

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$$A_V = \frac{\alpha (r_o \| R_C)}{r_e}$$

$\alpha$ = current gain

$$= \frac{0.98 (4.975)}{\frac{20}{1000}}$$

$$= \frac{4.8755}{0.02}$$

$$= 243.775$$

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Qn-1. For the given common base configuration find $r_e$, $r_{in}$, $r_{in}$ stage, $r_o$ stage, Av and Avs.

Given: $\beta = 100$

$r_o = 200 k\Omega$

DC analysis.

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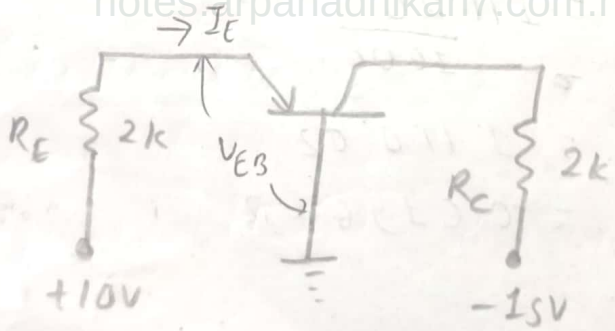


Applying KVL at input

$$10 - 2I_E - V_{EB} = 0$$

$$\Rightarrow 10 - 2I_E - 0.7 = 0$$

$$\text{or, } I_E = \frac{10 - 0.7}{2}$$

$$= 4.65\ mA$$

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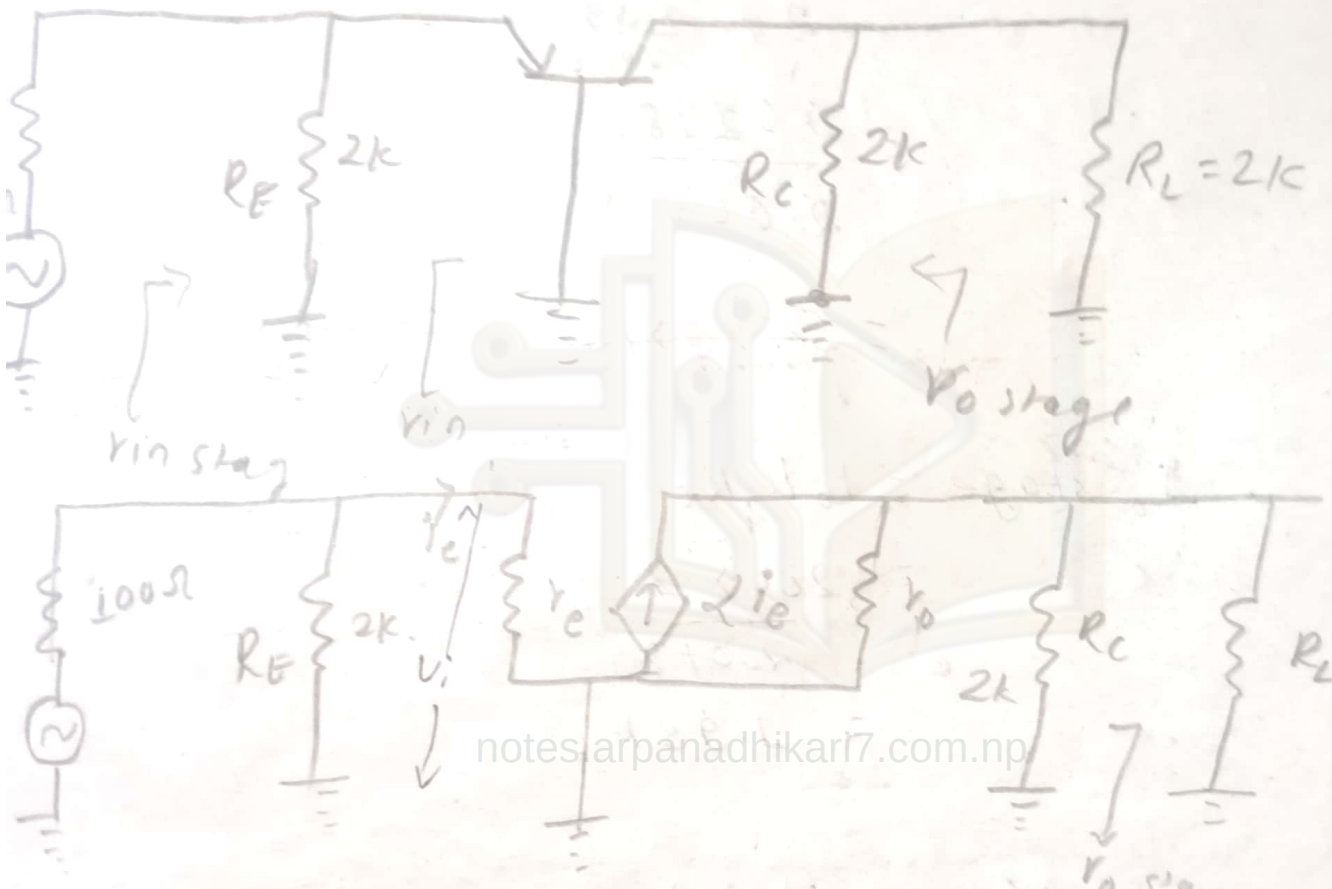
$\frac{26mV}{I_E}$

$= \frac{26mV}{4.65mA}$

$= 5.59\Omega$

~~$= 6\Omega$~~

analysis:



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we know,

$\alpha = \frac{\beta}{\beta + 1}$ [common Base]

$= \frac{100}{100+1}$ $\therefore \alpha$

$= 0.99$

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$\frac{V_i}{i_e}$ = $r_{in\ stage}$ = ~~$R_E$ // $r_e$~~

= 5.59 Ω

$r_{in\ stage}$ = $R_E$ // $r_e$

$= \frac{2 \times \frac{5.59}{1000}}{2 + \frac{5.59}{1000}}$

$= \frac{0.0118}{2.00559}$

= ~~$5.883 \times 10^{-3}$~~ $5.574 \times 10^{-3}$ kΩ

= 5.57 Ω

$r_{o\ stage}$ = $r_o$ // $R_C$

$= \frac{200 \times 2}{200 + 2}$

= 1.98 kΩ

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$A_V = \frac{\alpha_{ac}(r_o // R_C // R_L)}{i_e r_e}$

$= \frac{0.99(1.98 // 2)}{5.59}$

$= \frac{0.99(0.9949)}{\frac{5.59}{1000}}$

= 176.198

~~$A_{VS} = \frac{V_o}{V_S}$~~

=

Note: most $A_{VS} < A_V$

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$$A_{VS} = A_V \times \frac{r_{in\ stage}}{R_S + r_{in\ stage}}$$

$$= 176.19 \times \frac{5.57}{100 + 5.57}$$

$$= 9.295$$

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Configuration

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# Multistage Amplifier

In many application, the single stage amplifier can not provide all the gain that is required to drive a particular kind of load. In such case, multistage amplifier is used. The amplifier that creates voltage, current, and/or power gain through the use of two or more stage are called multistage amplifier. In multistage amplifier, the o/p of first stage becomes the input for second stage, the o/p of second stage becomes input for third stage and so on. The signal voltage is applied at the input of first stage & the o/p is available at the o/p terminal of last stage.

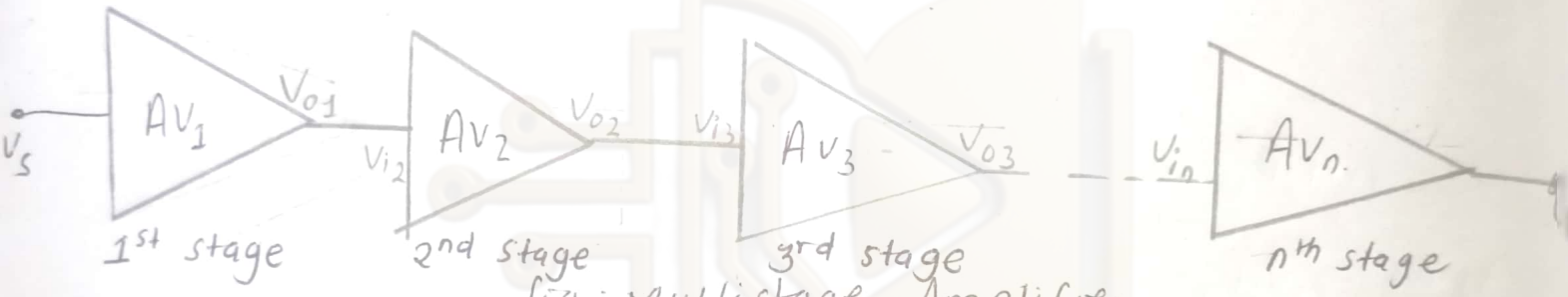


fig: Multistage Amplifier.

Let us consider a multistage amplifier with n stage as shown in figure above. Let $A_{V_1}$, $A_{V_2}$, $A_{V_3}$, ....., $A_{V_n}$ be the individual stage gain of first stage, 2nd stage, 3rd stage ....., $n^{th}$ stage respectively. The input voltage '$V_s$' is applied at the input of first stage and o/p voltage $V_o$ is available at the o/p of $n^{th}$ stage.

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Output of 1st stage, $V_{o1} = A_{V_1} \times V_s$ —①

output of 2nd stage, $V_{o2} = V_{i2} \times A_{V_2}$

Since, $V_{i2} = V_{o1}$

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$\therefore V_{02} = V_{01} \times A_{V_2}$

$= A_{V_1} \times V_S \times A_{V_2}$ —(ii)

output of 3rd stage, $V_{03} = A_{V_3} \times V_{i3}$

since, $V_{i3} = V_{02}$

$\therefore V_{03} = A_{V_3} \times V_{02}$.

$= A_{V_1} \times A_{V_2} \times A_{V_3} \times V_S$ —(iii)

Similarly,

$V_o = A_{V_n} \times V_{in}$

$= A_{V_n} \times V_{o(n-1)}$

$\therefore V_o = A_{V_1} \times A_{V_2} \times A_{V_3} \times \ldots \times A_{V_n} \times V_S$ —(iv)

$\therefore$ overall gain $\left(\frac{V_o}{V_S}\right) = A_{V_1} \times A_{V_2} \times A_{V_3} \times \ldots \times A_{V_n}$ —(v)

$\therefore A_{V_{Total}} = A_{V_1} \times A_{V_2} \times A_{V_3} \times \ldots \times A_{V_n}$ —(vi)

Therefore, the overall voltage gain of n stage cascade (joining in series) amplifier is equal to the product of gain of individual stage.

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## Gain in dB (decibel)

When gains are expressed in dB, the overall gain of multistage amplifier is equals to the sum of individual stage gain expressed in decibel.

Taking logarithm of base 10 in eqn (vi) and multiplying each term by 20. we get;

$$20\log_{10}(A_{V_{Total}}) = 20\log_{10}(A_{V_1} \times A_{V_2} \times A_{V_3} \times \cdots \times A_{V_n})$$

$$\Rightarrow 20\log_{10}(A_{V_{Total}}) = 20\log_{10}A_{V_1} + 20\log_{10}A_{V_2} + 20\log_{10}A_{V_3} + \cdots + 20\log_{10}A_{V_n}.$$

$$\Rightarrow A_{V(total)}\,(dB) = A_{V_1}(dB) + A_{V_2}(dB) + A_{V_3}(dB) + \cdots + A_{V_n}(dB) \quad \text{---(vii)}$$

## Expression for phase angle:
OR
## Total phase shift of n-stage cascade amplifier.

Let $\phi_1, \phi_2, \phi_3, \cdots, \phi_n$ be the phase shift betⁿ input & output of first stage, second stage, third stage, ....., $n^{th}$ stage respectively.

Then from eqn (vi)

$$(A_{V(Total)} \angle \phi_{Total}) = (A_{V_1} \angle \phi_1)(A_{V_2} \angle \phi_2)(A_{V_3} \angle \phi_3) \cdots (A_{V_n} \angle \phi_n)$$

$$(r_1 \angle \theta_1)\ (r_2 \angle \theta_2) \cdots$$

$$(r_1 \cdot r_2) \angle (\theta_1 + \theta_2)$$

$$\therefore A_{V\,Total} \angle \phi_{Total} = (A_{V_1} \cdot A_{V_2} \cdot A_{V_3} \cdots A_{V_n}) \angle (\phi_1 + \phi_2 + \phi_3 + \cdots + \phi_n) \quad \text{---(viii)}$$

Thus, the total phase shift of $n^{th}$ stage cascaded amplifier is the sum of phase shift introduced by individual stage.

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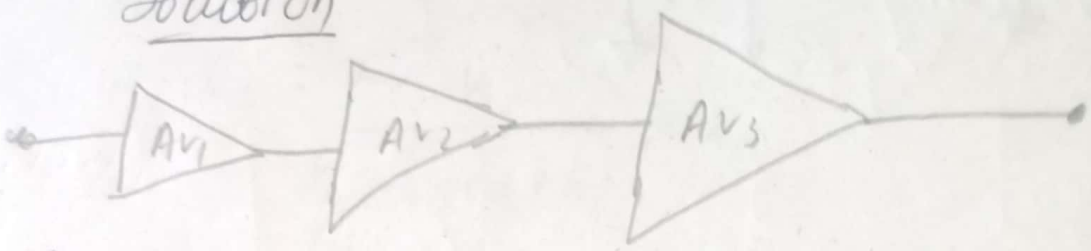
n:1 A three stage amplifier has a voltage gain of 50, 100 and 200 for the first, 2nd & third stage. Find the overall voltage of the amplifier. Also, find the gain of each stage & overall gain in decibel (dB).

Solution




Given,
Voltage gain of 1st stage $(A_{V_1}) = 50$
" " " 2nd " $(A_{V_2}) = 100$
" " " 3rd " $(A_{V_3}) = 200$

Now,
We know,

$$A_{V\,Total} = A_{V_1} \times A_{V_2} \times A_{V_3}$$
$$= 50 \times 100 \times 200$$
$$= 10^6$$

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Voltage gain of 1st stage in dB $= 20 \log_{10} A_{V_1}$
$= 20 \log_{10} 50$
$= 33.97$ dB.

Voltage gain of 2nd stage in dB $= 20 \log_{10} A_{V_2}$
$= 20 \log_{10} 100$
$= 40$ dB.

Voltage gain of 3rd stage in dB $= 20 \log_{10} A_{V_3}$
$= 20 \log_{10} 200$
$= 46.03$ dB.

$\therefore A_{V\,Total}$ (dB) $= 33.97 + 40 + 46.03$
$= 120$ dB.

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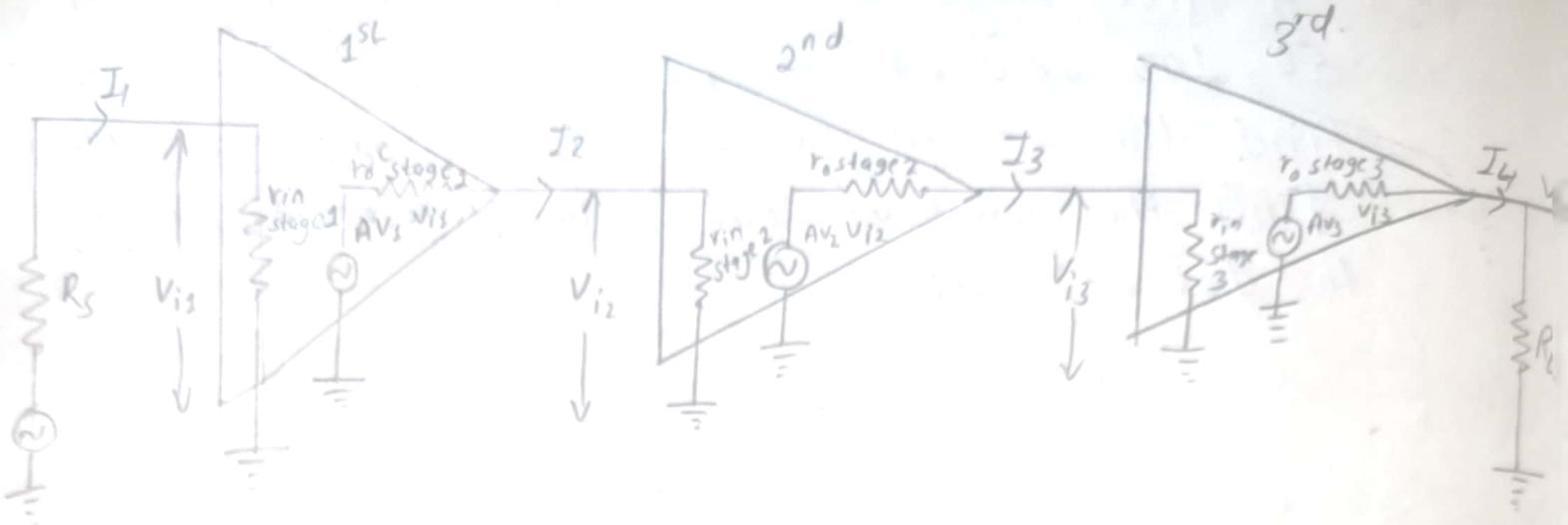
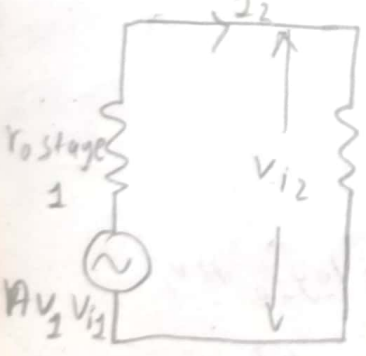
# Gain of Multistage Amplifier with loading effe



$$V_{i1} = r_{in\ stage\ 1} \times I_1$$

Now, $$V_{i1} = \frac{V_s}{R_s + r_{in\ stage\ 1}} \times r_{in\ stage\ 1} \quad \text{---(i)}$$




$$V_{i2} = I_2 \times r_{in\ stage\ 2}$$

$$= \frac{A_{V_1} V_{i1}}{r_{o\ stage\ 1} + r_{in\ stage\ 2}} \times r_{in\ stage\ 2}$$

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$$\therefore V_{i2} = A_{V_1} \times V_s \times \frac{r_{in\ stage\ 1}}{R_s + r_{in\ stage\ 1}} \times \frac{r_{in\ stage\ 2}}{r_{o\ stage\ 1} + r_{in\ stage\ 2}} \quad \text{---(ii)}$$

$$V_{i3} = I_3 \times r_{in\ stage\ 3}$$

$$= \frac{A_{V_2} V_{i2}}{r_{o\ stage\ 2} + r_{in\ stage\ 3}} \times r_{in\ stage\ 3}$$

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$$\text{or } V_{i3} = A_{V_2} \times A_{V_2} \times V_S \times \frac{r_{in\,stage1}}{R_S + r_{in\,stage1}} \times \frac{r_{in\,stage2}}{r_{o\,stage1} + r_{in\,stage2}}$$

$$\times \frac{r_{in\,stage3}}{r_{o\,stage2} + r_{in\,stage3}} \quad \text{---(iii)}$$

Again, $V_o = I_4 \times R_L$

$$\text{or } V_o = \frac{A_{V_3} V_{i3}}{r_{o\,stage3} + R_L} \times R_L$$

$$= A_{V_3} \times A_{V_2} \times A_{V_3} \times V_S \times \frac{r_{in\,stage1}}{R_S + r_{in\,stage1}} \times \frac{r_{in\,stage2}}{r_{o\,stage1} + r_{in\,stage2}}$$

$$\times \frac{r_{in\,stage3}}{r_{o\,stage2} + r_{in\,stage3}} \times \frac{R_L}{r_{o\,stage3} + R_L}$$

$$\therefore \frac{V_o}{V_S} = A_{V_1} \times A_{V_2} \times A_{V_3} \times \frac{r_{in\,stage1}}{R_S + r_{in\,stage1}} \times \frac{r_{in\,stage2}}{r_{o\,stage1} + r_{in\,stage2}} \times$$

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$$\frac{r_{in\,stage3}}{r_{o\,stage2} + r_{in\,stage3}} \times \frac{R_L}{r_{o\,stage3} + R_L} \quad \text{---(iv)}$$

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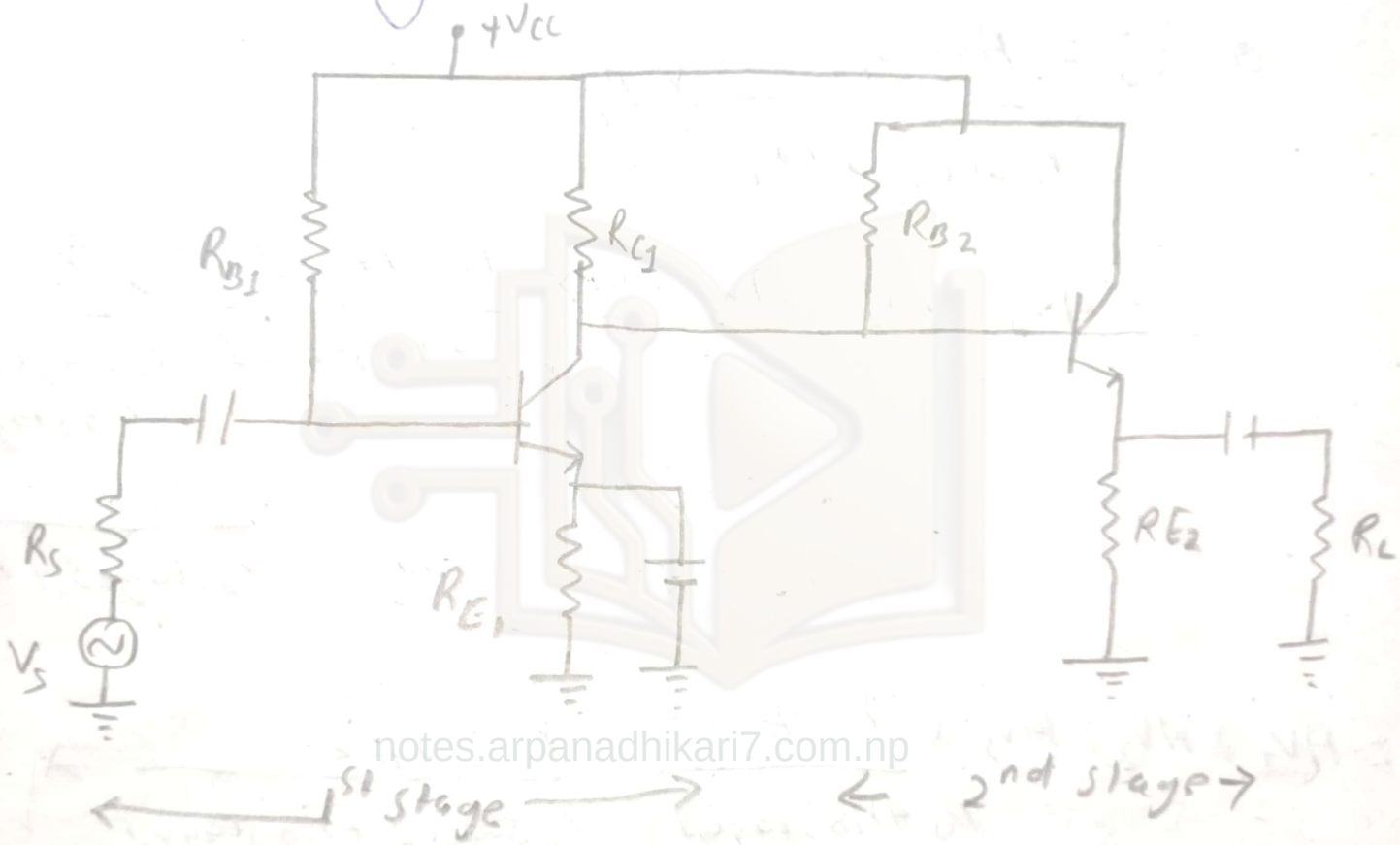
# Methods of Coupling:

1. Direct Coupling
2. R-C coupling
3. Transformer coupling
4. Impedance coupling

## 1. Direct Coupling:

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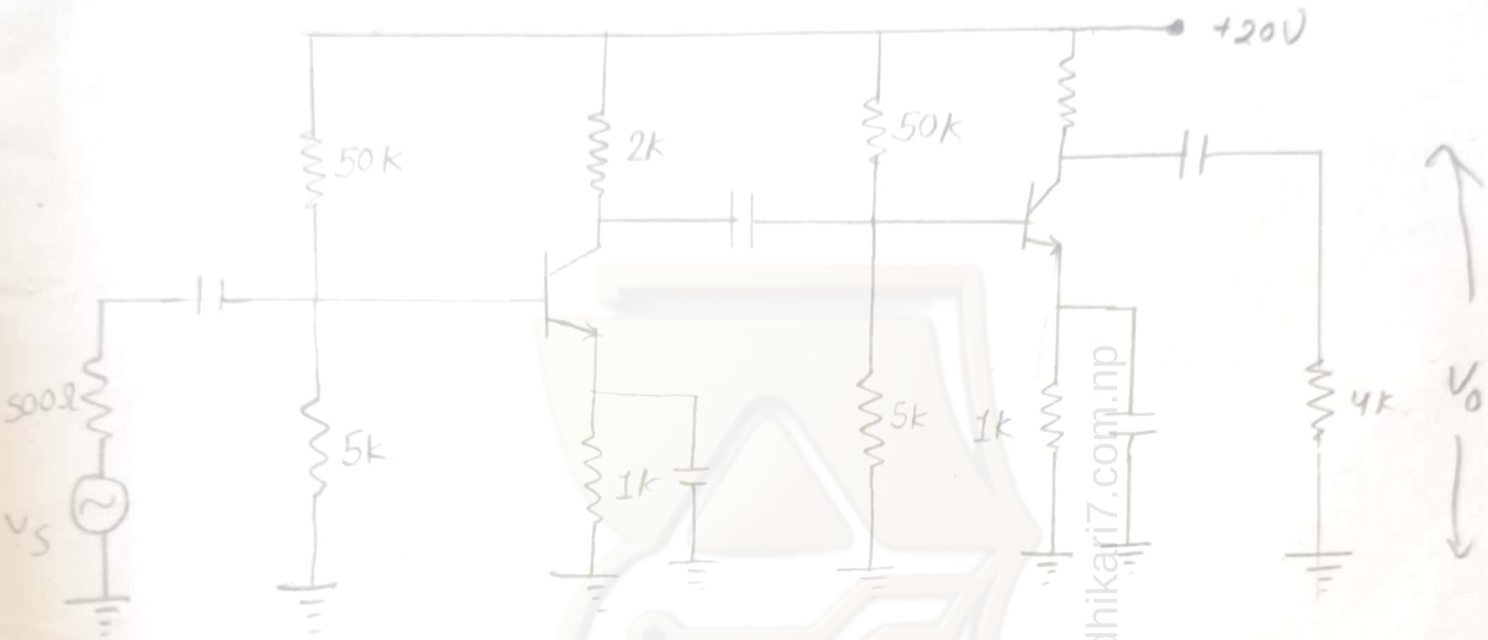
## Numericals:

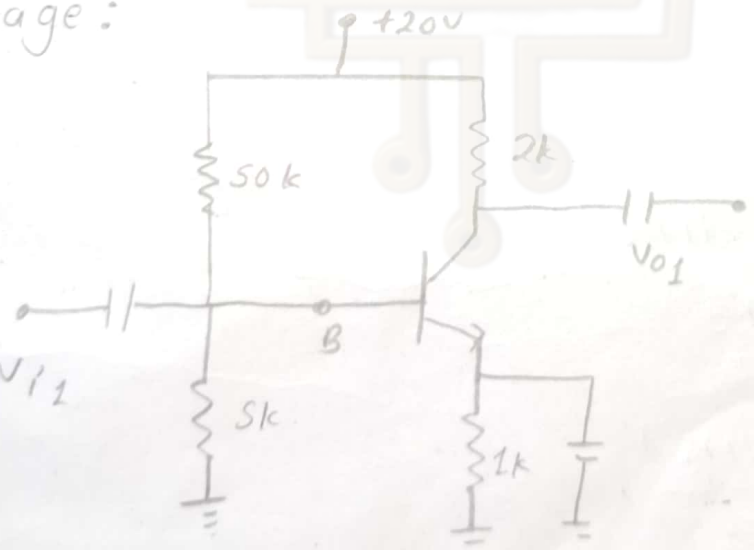
Qn.1. The compute the overall voltage gain for the two stage RC coupled amplifier as shown in figure.

Given: $\beta_1 = \beta_2 = 100$

$r_{e_1} = r_{e_2} = 2325\ \Omega$




1st stage:

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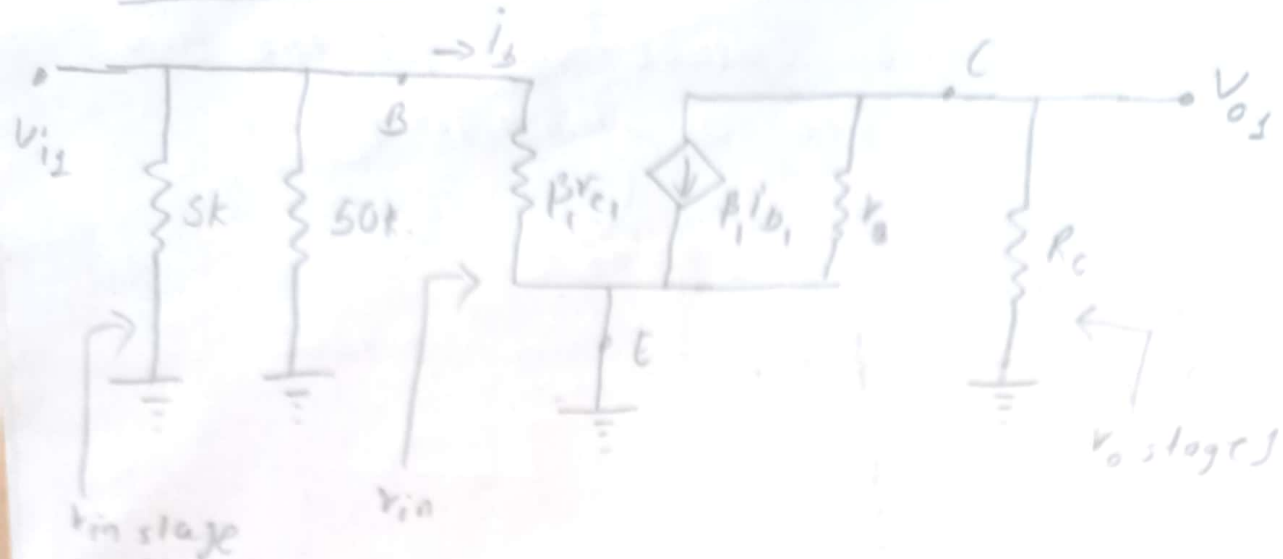
## $r_e$ - model




$$r_{in_1} = \beta r_e = 100 \times 23.25$$
$$= 2325\,\Omega$$
$$= 2.325\,k\Omega$$

$$V_{i_1} - i_b \beta r_e = 0$$
$$V_{i_1} = i_b \beta r_e$$
$$\frac{V_{i_1}}{i_b} = \beta r_e$$

$$r_{in\ stage_1} = (5k \,||\, 50k) \,||\, r_{in}$$
$$= \left(\frac{5 \times 50}{5+50}\right) || \, r_{in}$$
$$= 1.538\,k\Omega$$

$$r_{o\ stage\ 1} = (r_0 \,||\, R_c)$$
$$= R_c$$
$$= 2k$$

[$r_0$ if not given $r_0 = \infty$ ∴]

$$A_{v_1} = \frac{-(r_0 \,||\, R_c)}{r_{e_1}}$$
$$= -\frac{2 \times 1000}{23.25}$$
$$= -86.02$$

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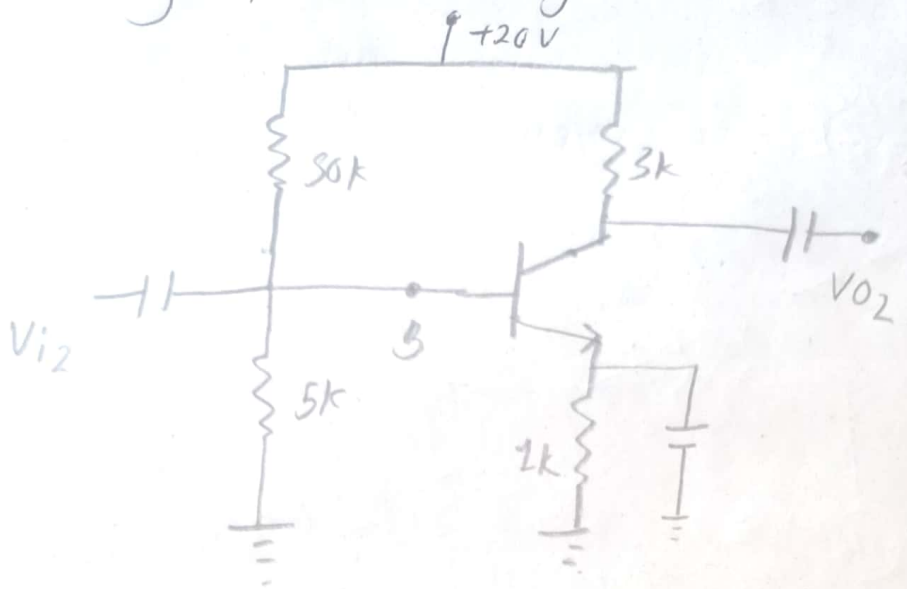
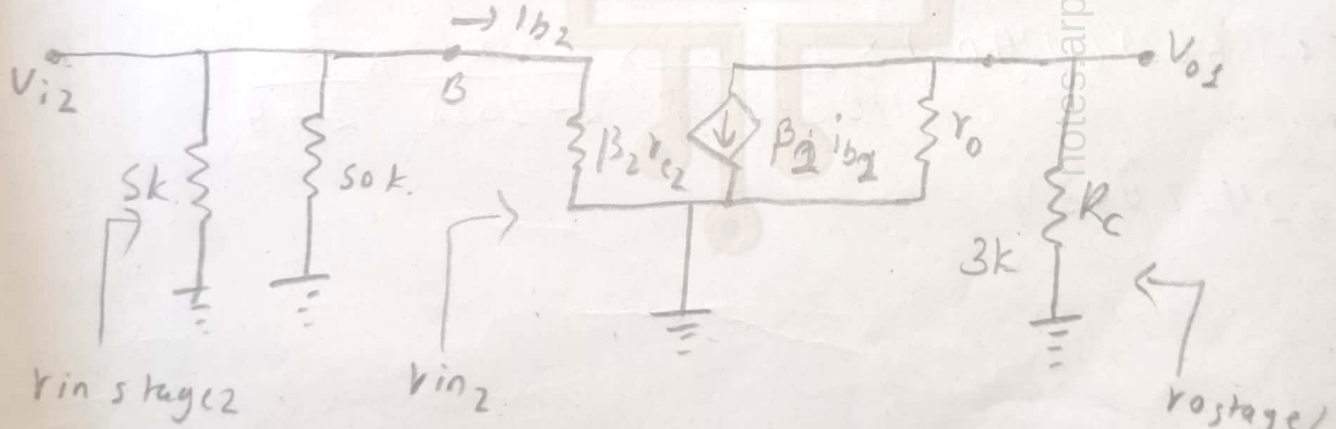
Similarly for 2nd stage.




$$A_{V_2} = -\frac{(r_o \,||\, R_c)}{r_{e_2}}$$

$$= -\frac{3 \times 1000}{23.25}$$

$$= -129.03$$




$$r_{in_2} = \beta r_{e_2} = 100 \times 23.25$$

$$= 2.325\ k\Omega.$$

$$r_{in\ stage_2} = (5k \,||\, 50k) \,||\, r_{in_2}$$

$$= \frac{5 \times 50}{5+50} \,||\, r_{in_2}$$

$$= 1.538\ k\Omega$$

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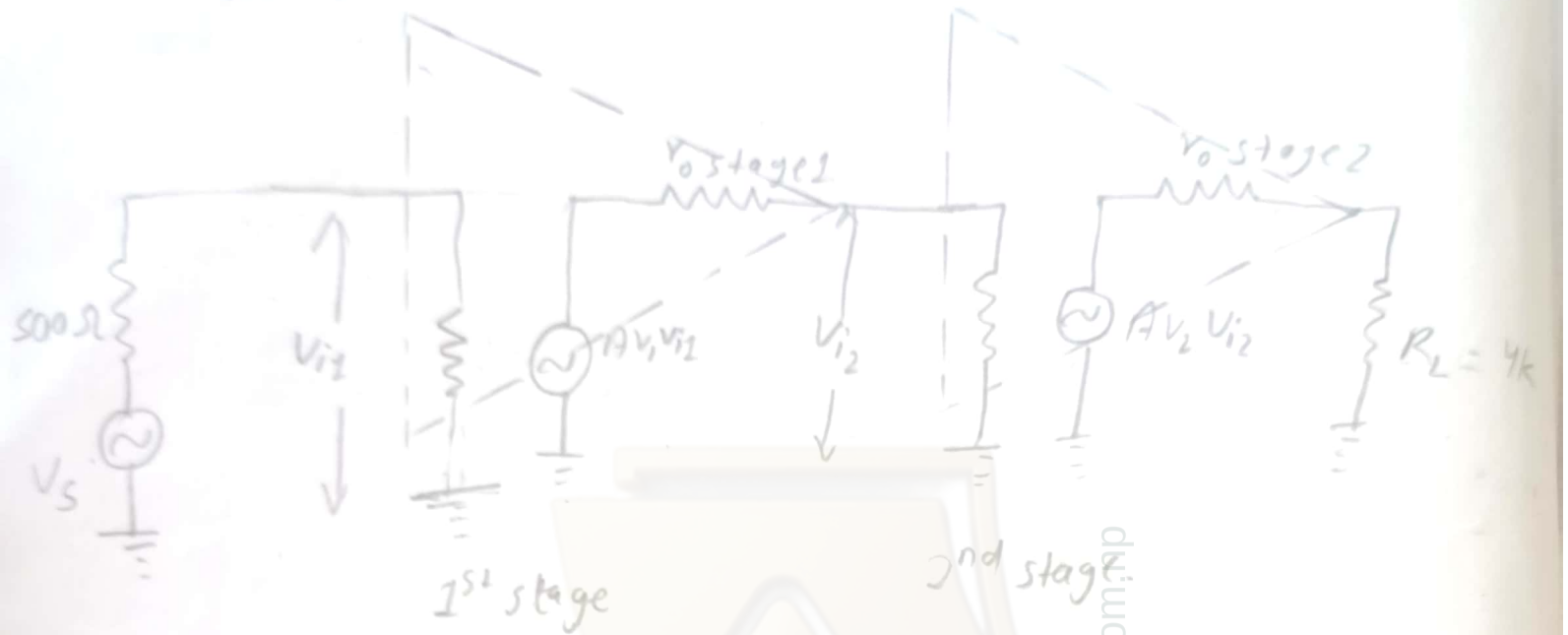
$r_o$ stage 2 = ($r_o$ || $R_c$)

= $R_c$

= 3k




$$\frac{V_o}{V_S} = Av_1 \times Av_2 \times \frac{r_{in\,stage\,1}}{R_s + r_{in\,stage\,1}} \times \frac{r_{in\,stage\,2}}{r_{o\,stage\,1} + r_{in\,stage\,2}} \times \frac{R_L}{r_{o\,stage\,2} + R_L}$$

$$= 0 - (86.02)(-129.03) \times \frac{1.538}{0.5 + 1.538} \times \frac{1.538}{2 + 1.538} \times \frac{4}{2 + 4}$$

$$= 2427.445\ V$$

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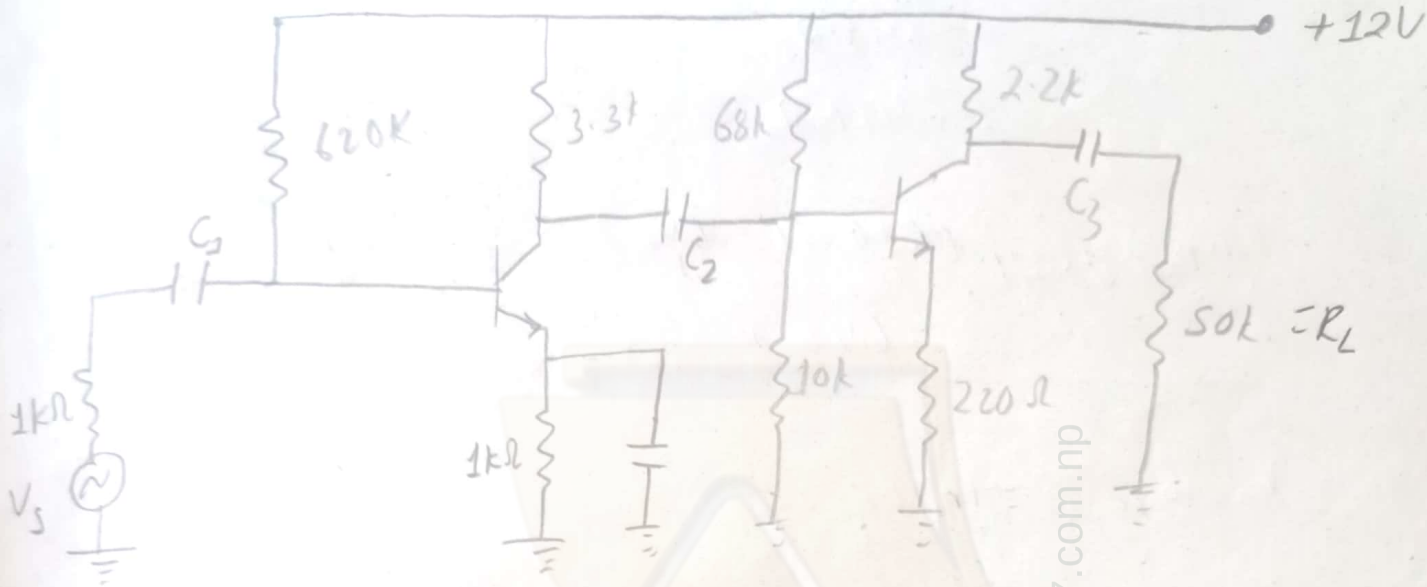
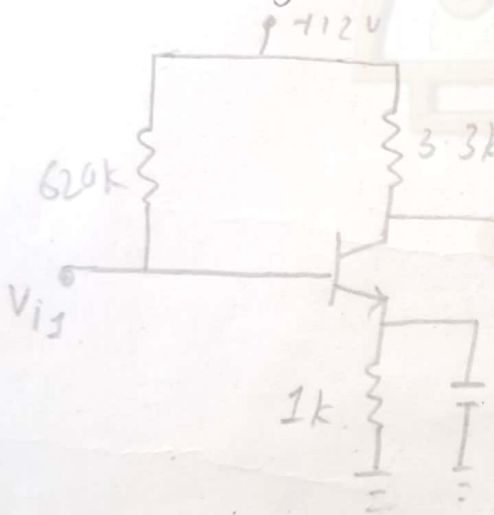
Qn2. Find overall voltage gain. Also find the O/P voltage $V_o$ if $V_s = 1mV$ $V_o = ?$

Given: $\beta_1 = \beta_2 = 100$

$r_{c1} = r_{c2} = 1M\Omega$




For 1st stage:




DC analysis:

$12 - 620 I_B - V_{BE} - 1 I_E = 0$

$\Rightarrow 12 - 620 I_B - V_{BE} - (\beta + 1) I_B = 0$

$\Rightarrow 721 I_B = 12 - V_{BE}$

$\Rightarrow I_B = \frac{12 - 0.7}{721} = 0.015\ mA$

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$I_E = (\beta + 1) I_B$

$= 1.58\ mA$.

$\therefore r_{e1} = \frac{26\ mV}{I_E}$

$= \frac{26}{1.58}$

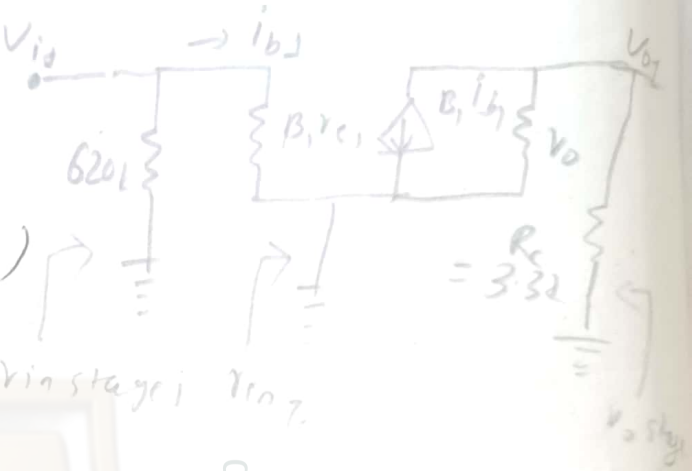
$= 16.45\ \Omega$.

AC analysis:

$r_{in1} = \beta_1 r_{e1} = 100 \times 16.45$

$= 1645\ \Omega$.

$= 1.645\ k\Omega$.




$r_{in\ stage\ 1} = (620 \parallel 1.645)$

$= 1.64\ k\Omega$.

$r_{o\ stage\ 1} = (r_o \parallel R_c)$

$= \left(\frac{r_{c1}}{\beta} \parallel R_c\right)$

$= \left(\frac{10^6}{100} \parallel 3.3 \times 1000\right)$

$= 10^4 \parallel 3300$

$= 2481\ \Omega$

$= 2.481\ k\Omega$

$A_{V_1} = -\frac{(r_o \parallel R_c)}{r_{e1}}$

$= -\frac{2481 \times 1000}{16.45}$

$= -150.82$

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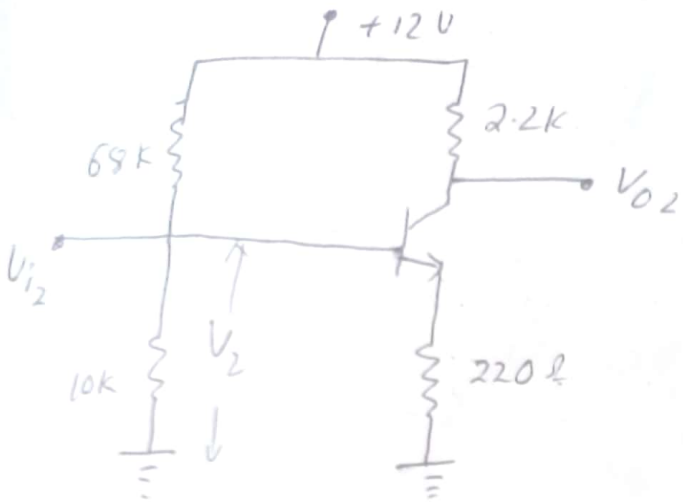
Similarly; For 2nd stage:




DC analysis:

$$V_2 = \frac{12}{68+10} \times 10$$

$$= 1.538\ V$$

$$V_2 - V_{BE} - 220 I_E = 0$$

$$I_E = \frac{1.538 - 0.7}{220}$$

$$= 3.80 \times 10^{-3} A$$

$$I_E = 3.80\ mA.$$

$$\therefore r_{e2} = \frac{26mV}{3.80\ mA}$$

$$= 6.84\ \Omega.$$

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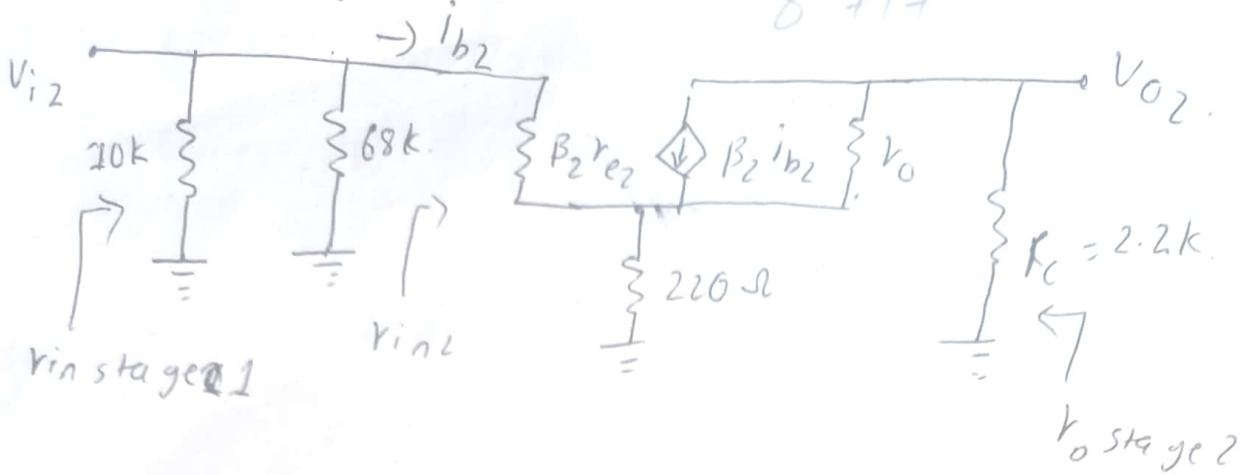
AC analysis:




$$r_{in2} = \beta_2 (r_{e_2} + R_E)$$

$$= 100 (6.84 + 220)$$

$$= 22684$$

$$= 22.684 \text{ k}\Omega$$

$$r_{in\ stage\ 2} = (10 \parallel 68) \parallel r_{in2}$$

$$= \left(\frac{1}{\frac{1}{10} + \frac{1}{68}}\right) \parallel 22.684$$

$$= 6.3 \text{ k}\Omega$$

$$r_{o\ stage\ 2} = (r_o \parallel R_C)$$

$$= \frac{r_{C2}}{\beta} \parallel R_C$$

$$= \left(\frac{10^6}{100} \parallel 2.2 * 1000\right)$$

$$= 10^4 \parallel 2200$$

$$= 1803.27$$

$$= 1.803 \text{ k}\Omega$$

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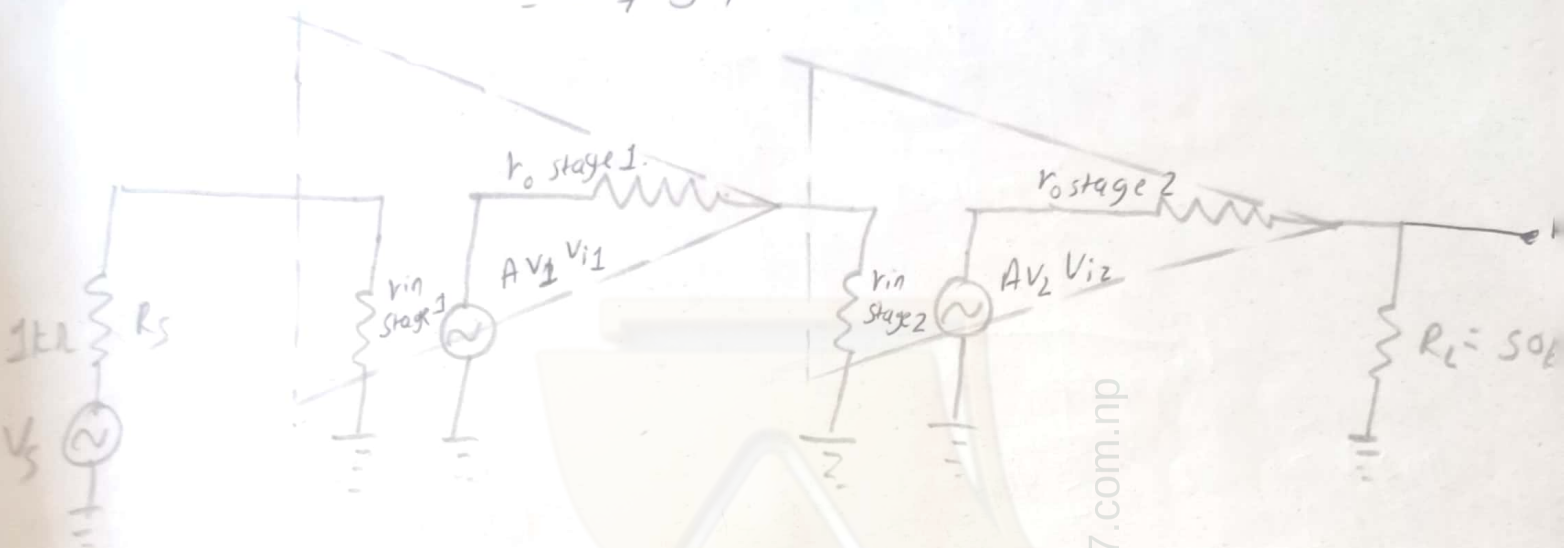
$$A_{V_2} = -\frac{(r_o \parallel R_c)}{(r_{e_2} + R_E)}$$

$$= -\frac{1.803 \times 1000}{6.84 + 220}$$

$$= -7.94$$




$$\frac{V_o}{V_S} = A_{V_1} \times A_{V_2} \times \frac{r_{in\ stage1}}{R_S + r_{in\ stage1}} \times \frac{r_{in\ stage2}}{r_{o\ stage1} + r_{in\ stage2}} \times \frac{R_L}{r_{o\ stage2} + R_L}$$

$$\Rightarrow \frac{V_o}{V_S} = (-150.82)(-7.94) \times \frac{1.64k\Omega}{1 + 1.64} \times \frac{6.3}{2.481 + 6.3} \times \frac{50}{1.803 + 50}$$

$$= 515.146\ V$$

$$V_o = 515.146 \times V_S$$

$$= 515.146 \times 1 \times 10^{-3}$$

$$= 0.514\ V$$

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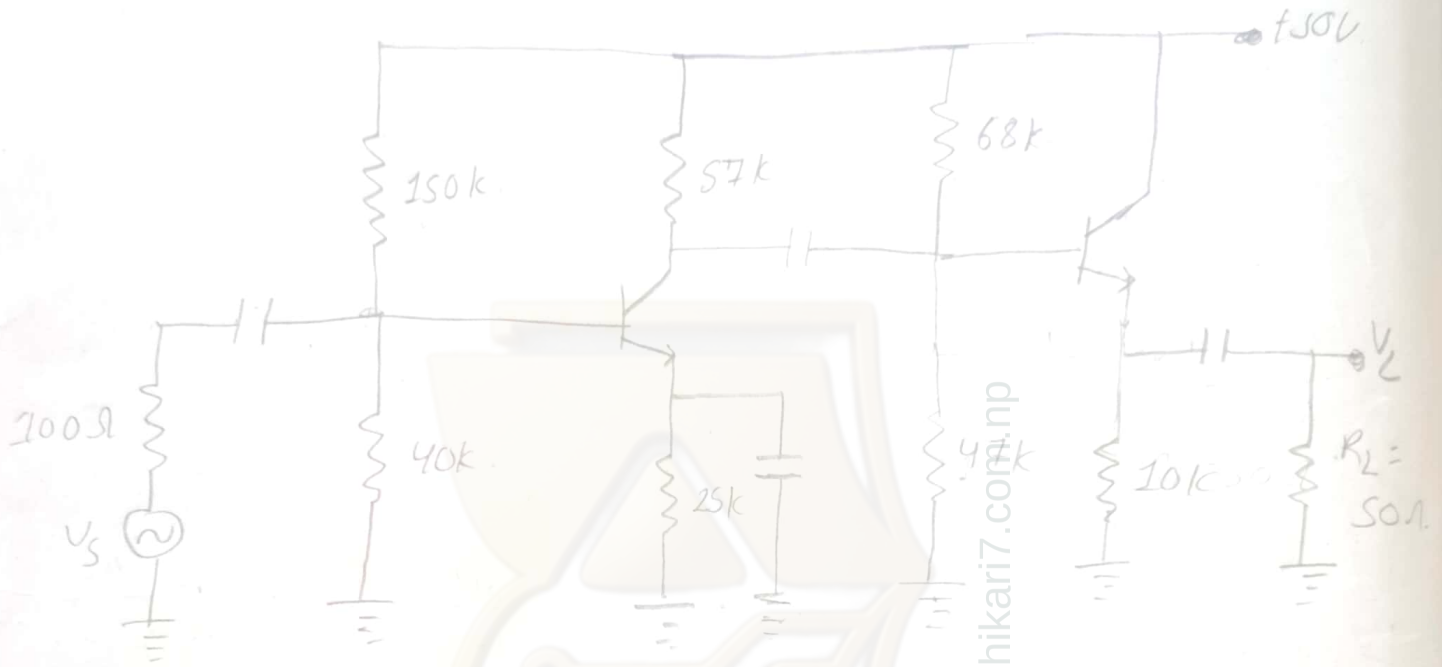
Qn.3: For the circuit, shown in figure. Find the overall voltage gain $\left(\frac{V_L}{V_S}\right)$.

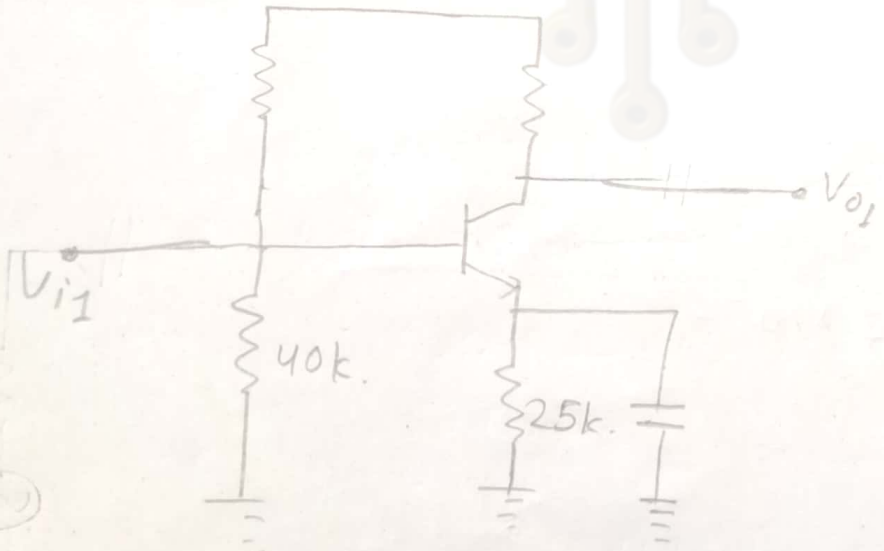
Given: $Q_1$ : $r_{e1} = 15\,\Omega$, $\beta_1 = 180$, $r_{o1} = \infty$

$Q_2$ : $r_{e2} = 25\,\Omega$, $\beta_2 = 180$, $r_{o2} = \infty$




Solution.

For 1st stage.

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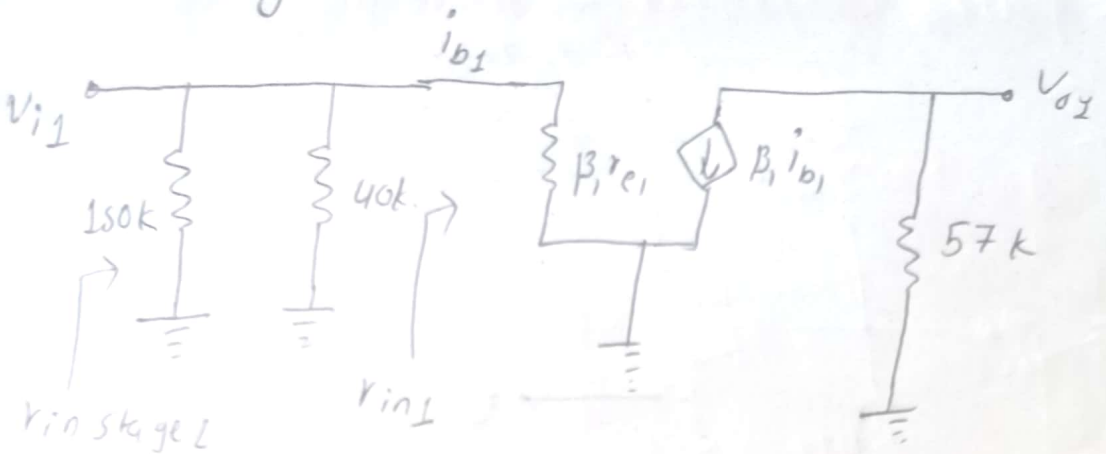
AC analysis:




$$r_{in1} = \beta_1 r_{e1}$$

$$= 180 \times 15$$

$$= 2700$$

$$= 2.7\ k\Omega$$

$$r_{in\ stage\ 1} = (150 \parallel 40) \parallel r_{in}$$

$$= \left(\frac{150 \times 40}{150 + 40}\right) \parallel 2.7$$

$$= 31.578 \parallel 2.7$$

$$= 2.487\ k\Omega.$$

$$r_{o\ stage\ 1} = (r_o \parallel R_c)$$

$$= R_c$$

$$= 57\ k\Omega.$$

$$A_{v1} = -\frac{(r_o \parallel R_c)}{r_{e1}}$$

$$= -\frac{R_c}{r_{e1}} = -\frac{57000}{15}$$

$$= -3800$$

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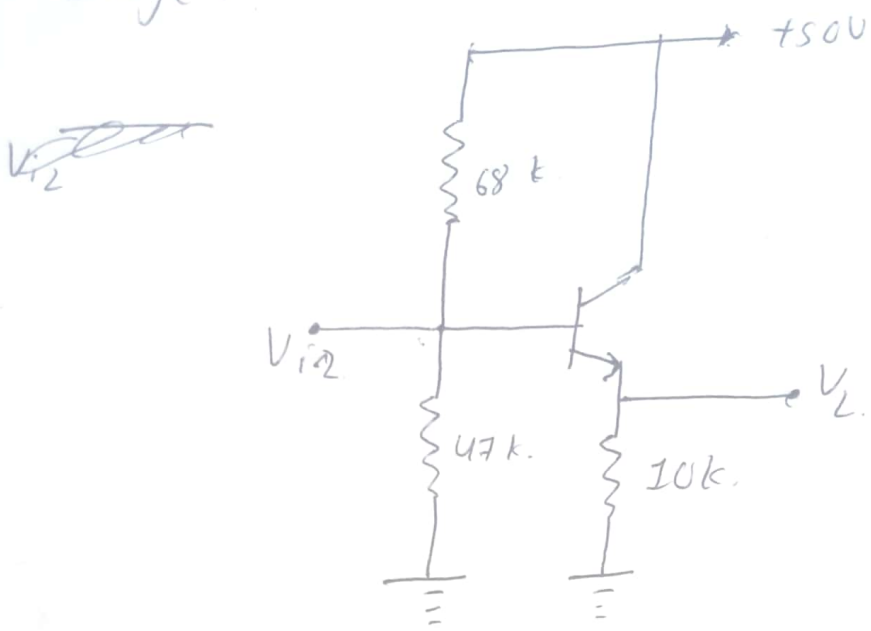
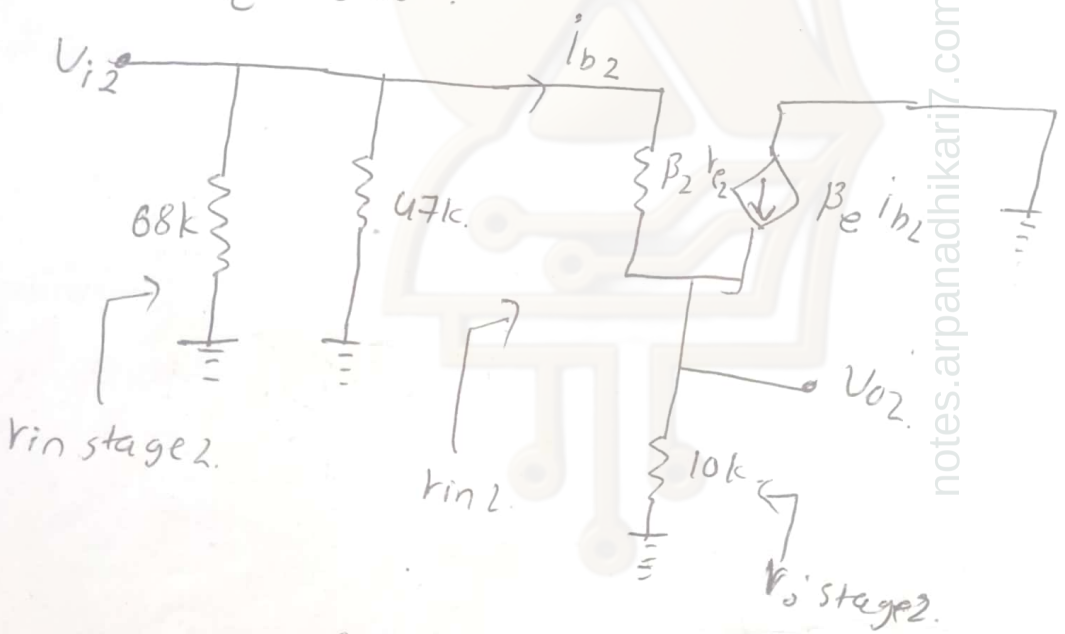
For stage 2




~~AC equivalent~~

Re model:




$r_{in_2} = \beta_2 (r_{e_2} + R_E)$

$= 180 \left( \frac{25}{1000} + 10 \right)$

$= 1804.5 \text{ k}\Omega$

$r_{in \text{ stage } 2} = (68 \,||\, 47) \,||\, r_{in_2}$

$= \frac{1}{\frac{1}{68} + \frac{1}{47}} \,||\, 1804.5$

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$r_{in\ stage2} = 27.791 \parallel 1804.5$

$= 27.369$

$r_{o\ stage2} = r_{e2} \parallel R_E$

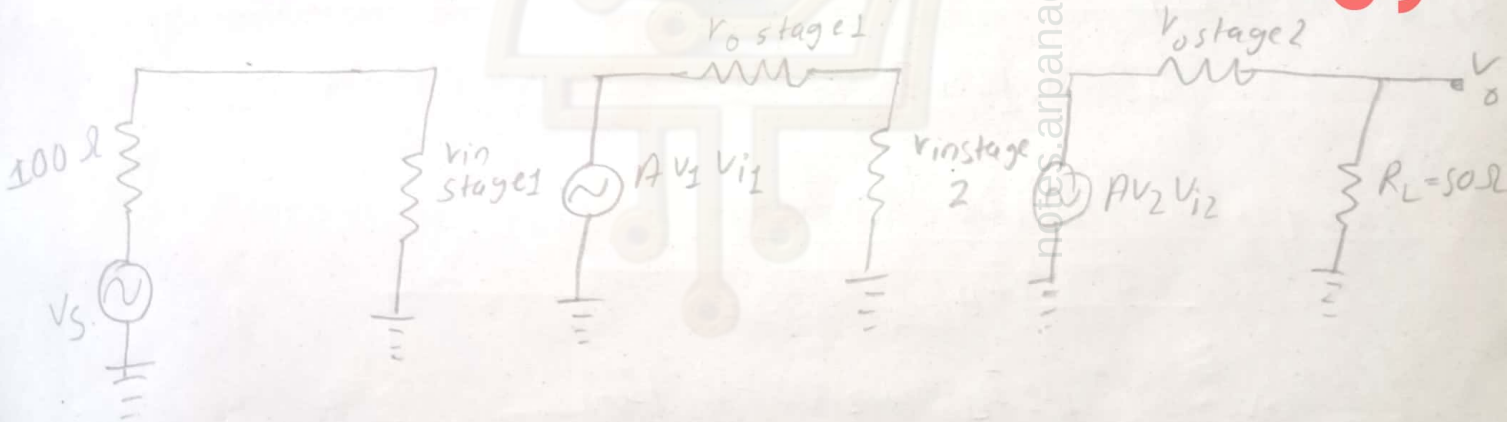
$= 25 \parallel 10000$

$= 24.937\ \Omega.$

$A_{V_2} = \dfrac{R_E}{r_{e2} + R_E}$

$= \dfrac{10}{\frac{25}{1000} + 10}$

$= 0.9975$




$\dfrac{V_L}{V_S} = A_{V_1} \times A_{V_2} \times \dfrac{r_{in\ stag1}}{R_S + r_{in\ stag1}} \times \dfrac{r_{in\ stage2}}{r_{o\ stage2} + r_{in\ stage2}} \times \dfrac{R_L}{r_{o\ stage2} + R_L}$

$= -3800 \times 0.997 \times \dfrac{2.48}{\frac{100}{1000} + 2.48} \times \dfrac{27.37}{57 + 27.37} \times \dfrac{50}{24.93 + 50}$

$= 788.818$

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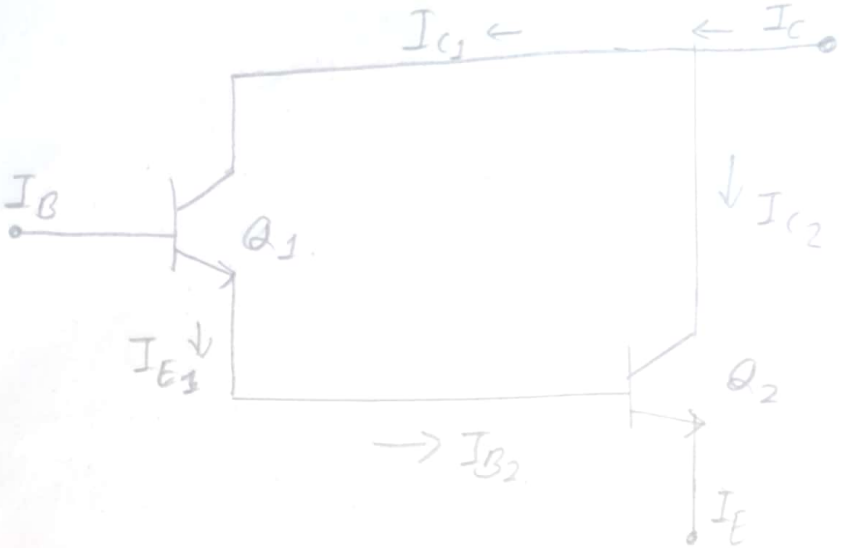
# Darlington pair Amplifier




The circuit diagram of Darlington pair Amplifier is shown above.
Let $\beta_1$ & $\beta_2$ be the current gain of transister $Q_1$ and $Q_2$ respectively

$$I_C = I_{C_1} + I_{C_2}$$
$$= \beta_1 I_B + \beta_2 I_{B_2}$$
$$= \beta_1 I_B + \beta_2 I_{E_1}$$
$$= \beta_1 I_B + \beta_2 (1+\beta_1) I_B$$
$$= \beta_1 I_B + \beta_2 I_B + \beta_1 \beta_2 I_B$$
$$= I_B (\beta_1 + \beta_2 + \beta_1 \beta_2)$$

$$\therefore \frac{I_C}{I_B} = \beta_1 + \beta_2 + \beta_1 \beta_2$$

$$\beta_{effective} = \beta_1 + \beta_2 + \beta_1 \beta_2 \quad \text{—— (1)}$$

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if $\beta_1 = \beta_2 = 100$

$\beta_{eff} = 100 + 100 + 100 \times 100$

$= 10200$

Since current gain of Darlington pair Amplifier is very high, it is termed as super beta transistor

The main characteristic of Darlington pair amplifier are:

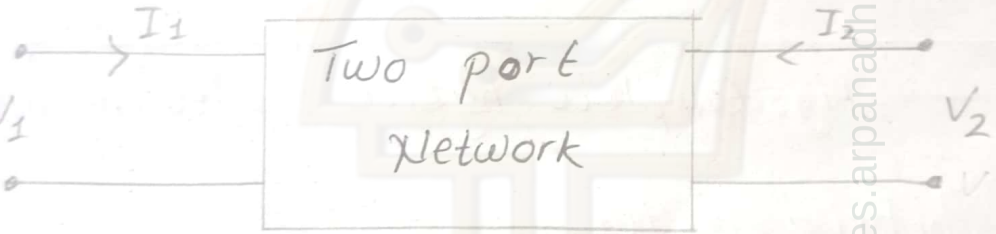
(i) Its input impedence is very high

(ii) Its ouput impedence is very low.

(iii) Its voltage gain is approximately equals to unity.

(iv) Its current gain is very high.

## Hybrid Model




$$V_1 = h_{11} I_1 + h_{12} V_2 \quad \text{——(I)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \quad \text{——(II)}$$

The four parameters in equation (I) & (II) i.e $h_{11}, h_{12}, h_{21}$ and $h_{22}$ are known as h parameters.

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$h_{11} = \left.\frac{V_1}{I_1}\right|_{V_2=0} = h_i = \text{input resistance}$$

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$h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0} = h_f$ = forward transfer current ratio.

$h_{12} = \left.\frac{V_1}{V_2}\right|_{I_1=0} = h_r$ = Reverse transfer voltage ratio.

$h_{22} = \left.\frac{I_2}{V_2}\right|_{I=0} = h_o$ = output conductance.

Transistor can be represented by the hybrid parameters as shown in figure.

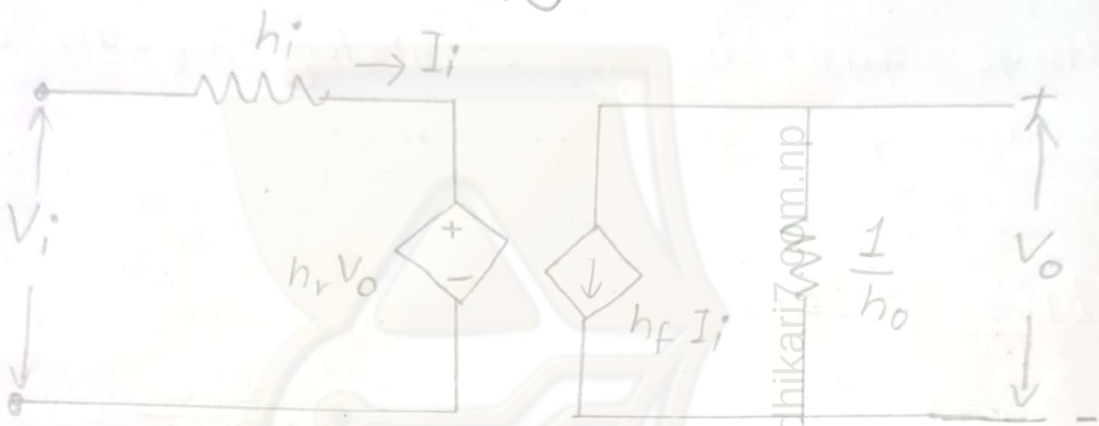


fig: Complete hybrid equivalent model

Imp

## Complete hybrid Model

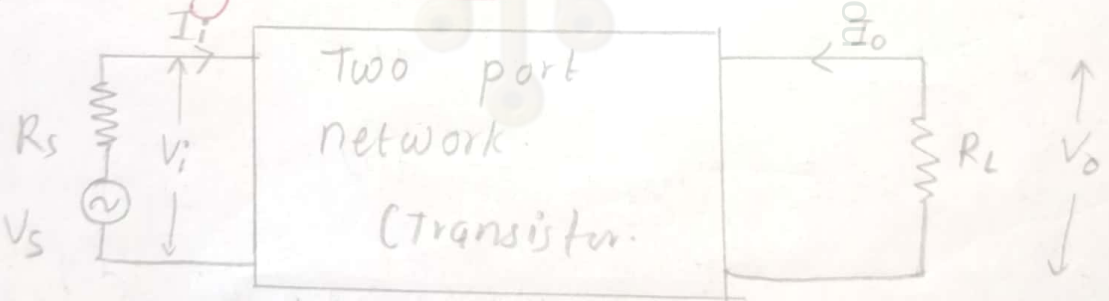


fig: Two port Network

Let us consider a two port n/w ~~of t~~ or transistor ckt as shown in figure above. Now, replace two port transistor n/w by its hybrid equivalent circuit.

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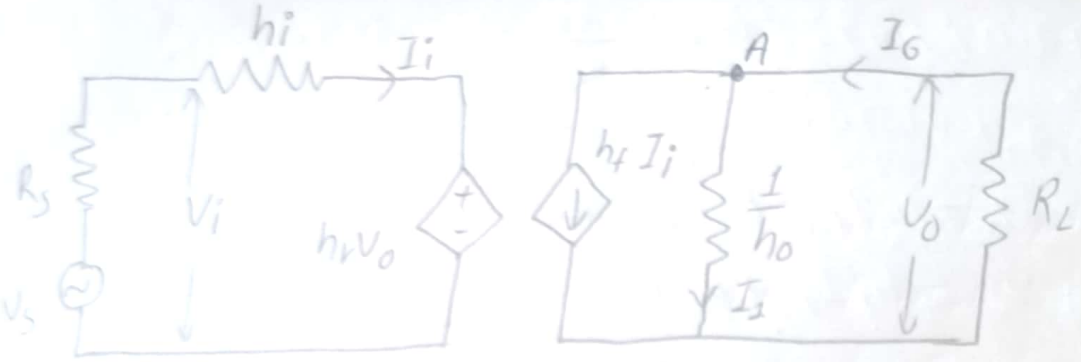


1. Current gain ($A_I = \frac{I_o}{I_i}$)

Applying KCL at node A;

$$I_o - I_1 - h_f I_i = 0$$

$$\Rightarrow I_o = \frac{V_o}{1/h_o} + h_f I_i$$

$$\Rightarrow I_o = h_o V_o + h_f I_i$$

$$\Rightarrow I_o = h_o (-I_o R_L) + h_f I_i$$

$$\Rightarrow I_o + h_o R_L I_o = h_f I_i$$

$$\Rightarrow I_o (1 + h_o R_L) = h_f I_i$$

$$\Rightarrow \frac{I_o}{I_i} = \frac{h_f}{1 + h_o R_L}$$

$$\Rightarrow \boxed{A_I = \frac{h_f}{1 + h_o R_L}} \quad \text{---(1)}$$

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2. Voltage gain ($A_V$): $A_V = \frac{V_o}{V_i}$

Applying KVL at input side,

$V_i - h_i I_i - h_r V_o = 0$

$\Rightarrow V_i = h_i I_i + h_r V_o$

$\Rightarrow V_i = h_i \left[ I_o \left( \frac{1 + h_o R_L}{h_f} \right) \right] + h_r V_o$

$\Rightarrow V_i = h_i \left[ \frac{-V_o}{R_L} \left( \frac{1 + h_o R_L}{h_f} \right) \right] + h_r V_o \quad \left[ \because I_o = \frac{-V_o}{R_L} \right]$

$\Rightarrow V_i = -V_o \left[ \frac{h_i (1 + h_o R_L)}{h_f R_L} - h_r \right]$

$\Rightarrow \frac{V_i}{V_o} = \frac{1}{\left[ \frac{h_i (1 + h_o R_L)}{h_f R_L} - h_r \right]}$

$\Rightarrow \frac{V_i}{V_o} = \frac{1}{\left[ \frac{h_i + h_i h_o R_L - h_r h_f R_L}{h_f R_L} \right]}$

$\Rightarrow \frac{V_o}{V_i} = \frac{h_f R_L}{h_i + h_i h_o R_L - h_r h_f R_L}$

$\Rightarrow A_V = \frac{h_f R_L}{h_i + (h_i h_o - h_r h_f) R_L}$

$\therefore A_V = \frac{-h_f R_L}{h_i + h_i h_o R_L - h_r h_f R_L}$

$\therefore A_V = \frac{-h_f R_L}{h_i + (h_i h_o - h_r h_f) R_L}$ (ii)

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3. Input impedence $\left(Z_i = \frac{V_i}{I_i}\right)$

Applying KVL at input side,

$$V_i - h_i I_i - h_r V_o = 0$$

$$\Rightarrow V_i = h_i I_i + h_r V_o$$

Since, $V_o = -I_o R_L$

$$V_i = h_i I_i + h_r (-I_o R_L)$$

$$\Rightarrow V_i = h_i I_i - h_r R_L I_o$$

$$\Rightarrow V_i = h_i I_i - h_r R_L A_I I_i \quad [\because A_I = I_o/I_i]$$

$$\Rightarrow V_i = I_o [h_i - h_r R_L A_I]$$

$$\Rightarrow \frac{V_i}{I_i} = h_i - h_r R_L A_I$$

$$\Rightarrow \boxed{Z_i = h_i - h_r R_L A_I} \quad \text{——(iii)}$$

$$Z_i = h_i - h_r R_L \cdot \frac{h_f}{1 + h_o R_L}$$

$$\boxed{Z_i = h_i - \frac{h_r h_f R_L}{1 + h_o R_L}}$$

(4) Output Impedence $\left(Z_o = \frac{V_o}{I_o}\right)$

The output impedence of an amplifier is defined as the ratio of output voltage to the output current with $V_s = 0$ for input circuit ($V_s = 0$)

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$V_S - I_i R_S - I_i h_i - h_r V_0 = 0$

$$\Rightarrow \boxed{I_i = \frac{h_r V_0}{R_S + h_i}} \quad \text{——} (*)$$

Also, Applying KCL at node A,

$$I_0 = h_f I_i + I_1$$

$$\Rightarrow I_0 = h_f I_i + \frac{V_0}{1/h_0}$$

$$\Rightarrow I_0 = h_f I_i + h_0 V_0$$

Substituting value of $I_i$

$$I_0 = h_f \left(-\frac{h_r V_0}{R_S + h_i}\right) + h_0 V_0$$

$$\Rightarrow I_0 = -\frac{h_f h_r V_0}{R_S + h_i} + h_0 V_0$$

$$\Rightarrow I_0 = V_0 \left(h_0 - \frac{h_f h_r}{(R_S + h_i)}\right)$$

$$\Rightarrow \frac{V_0}{I_0} = \frac{1}{h_0 - \frac{h_f h_r}{(R_S + h_i)}}$$

$$\Rightarrow Z_0 = \frac{R_S + h_i}{R_S h_0 + h_i h_0 - h_f h_r}$$

$$\Rightarrow \boxed{Z_0 = \frac{1}{h_0 - \frac{h_f h_r}{R_S + h_i}}} \quad \text{——} (iv)$$

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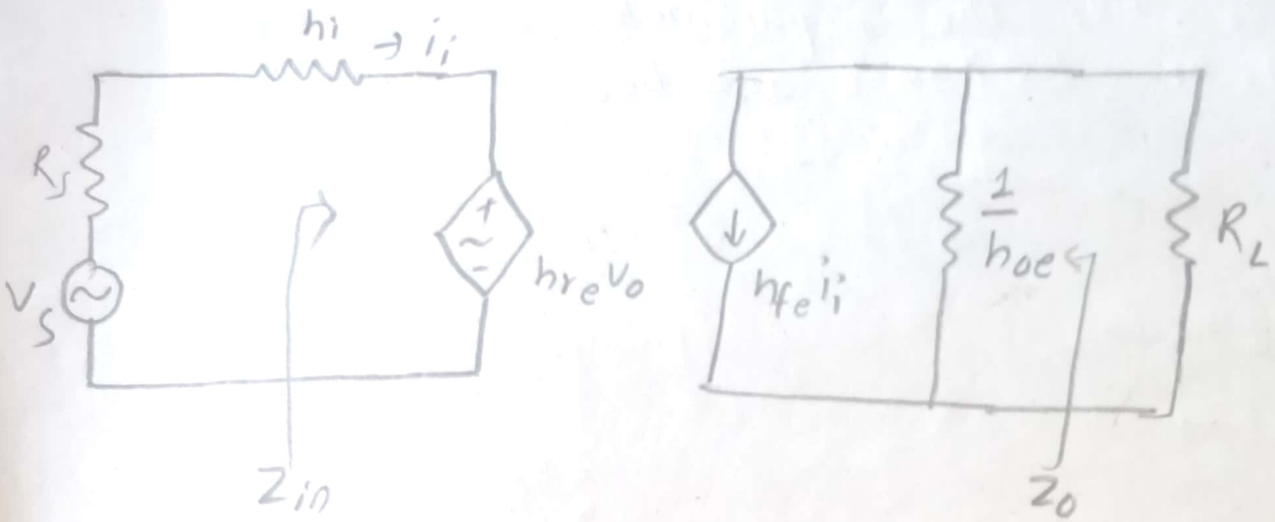
Note:




$$\frac{I_o}{I_i} = A_I = \frac{h_f}{1 + h_o R_L}$$

$$A_v = \frac{-h_f R_L}{h_i + (h_i h_o - h_r h_f) R_L}$$

$$Z_{in} = h_i - \frac{h_r h_f R_L}{1 + h_o R_L}$$

$$Z_o = \frac{1}{h_o - \frac{h_f h_r}{R_s + h_i}}$$

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Q.N.1. For the circuit system shown in figure, find $A_{vs}$, $Z_{in}$, $A_v$, $Z_o$, $A_{vs}$. The h-parameters are $h_{ie} = 2K\Omega$, $h_{fe} = 49$, $h_{re} = 1.6 \times 10^{-4}$ and $h_{oe} = 50\ \mu A/V$

Solution




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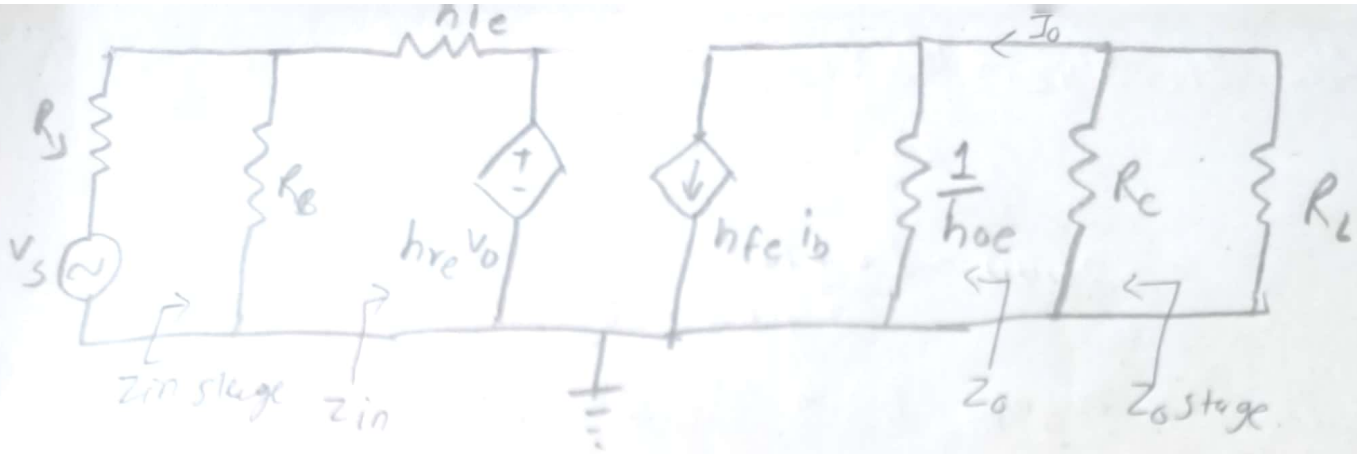


$$R_B = \frac{R_1 \times R_2}{R_1 + R_2}$$

$$= \frac{80 \times 20}{80 + 20}$$

$$= 16\,K\Omega$$

We have,

$$\frac{I_o}{I_i} = \frac{h_{fe}}{1 + h_{oe} Z_L}$$

OR,

$$\frac{I_L}{I_i} = -\frac{h_{fe}}{1 + h_{oe} Z_L}$$

where, $Z_L = R_C \parallel R_L$

$$= \frac{2 \times 2}{2 + 2}$$

$$= 1\,k\Omega$$

$$A_I = \frac{I_L}{I_i} = \frac{-h_{fe}}{1 + h_{oe} Z_L}$$

$$= \frac{-49}{1 + 50 \times 10^{-6} \times 1000}$$

$$= -46.67$$

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$$Z_{in} = h_{ie} - \frac{h_{re}\, h_{fe}\, Z_L}{1 + h_{oe} Z_L}$$

$$= 2000 - \frac{1.6 \times 10^{-4} \times 49 \times 1000}{1 + 50 \times 10^{-6} \times 1000}$$

$$= 1992.53\ \Omega$$

$$\approx 2\ k\Omega$$

$$Z_{in\ stage} = Z_{in} \,||\, R_B$$

$$= \frac{2 \times 16}{2 + 16}$$

$$= 1.77\ k\Omega$$

$$A_V = \frac{-h_{fe} Z_L}{h_{ie} + (h_{ie} h_{oe} - h_{re} h_{fe}) Z_L}$$

$$= \frac{-49 \times 1000}{2000 + (2000 \times 50 \times 10^{-6} - 1.6 \times 10^{-4} \times 49) \times 1000}$$

$$= -23.420$$

$$Z_o = \frac{1}{h_{oe} - \frac{h_{fe}\, h_{re}}{(R_S \,||\, R_B + h_{ie})}}$$

$$= \frac{1}{50 \times 10^{-6} - \frac{49 \times 1.6 \times 10^{-4}}{\left(\frac{0.6 \times 16}{0.6 + 16} + 2000\right)}}$$

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CE

$$Z_o = \frac{1}{50 \times 10^{-6} - \frac{49 \times 1.6 \times 10^{-4}}{570 + 2000}}$$

$$= 21650.35\ \Omega$$

$$= 21.299\ k\Omega$$

$$Z_{o\,stage} = R_c \| Z_o$$

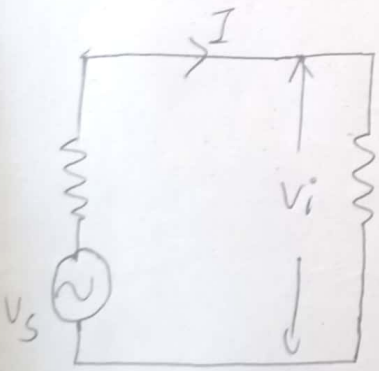
$$= \frac{2 \times 21.299}{2 + 21.299}$$

$$= 1.828$$

$$A_{VS} = A_V \times \frac{Z_{in\,stage}}{R_S + Z_{in\,stage}}$$

$$= \frac{-23.42 \times 1.77}{0.6 + 1.77}$$

$$= -17.49$$




$$V_i = I \times Z_{in\,stage}$$

$$= \frac{V_S}{R_S + Z_{in\,stage}} \times Z_{in\,stage}$$

$$\frac{V_i}{V_S} = \frac{Z_{in\,stage}}{R_S + Z_{in\,stage}}$$

$$A_{VS} = \frac{V_o}{V_S} = \frac{V_o}{V_i} \times \frac{V_i}{V_S}$$

$$= A_V \times \frac{Z_{in\,stage}}{R_c + Z_{in\,stage}}$$

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# Approximate hybrid Model

For the given circuit, find $A_i$, $A_v$, $Z_i$, $Z_{in}$ stage, $Z_o$ and $Z_o$ stage.

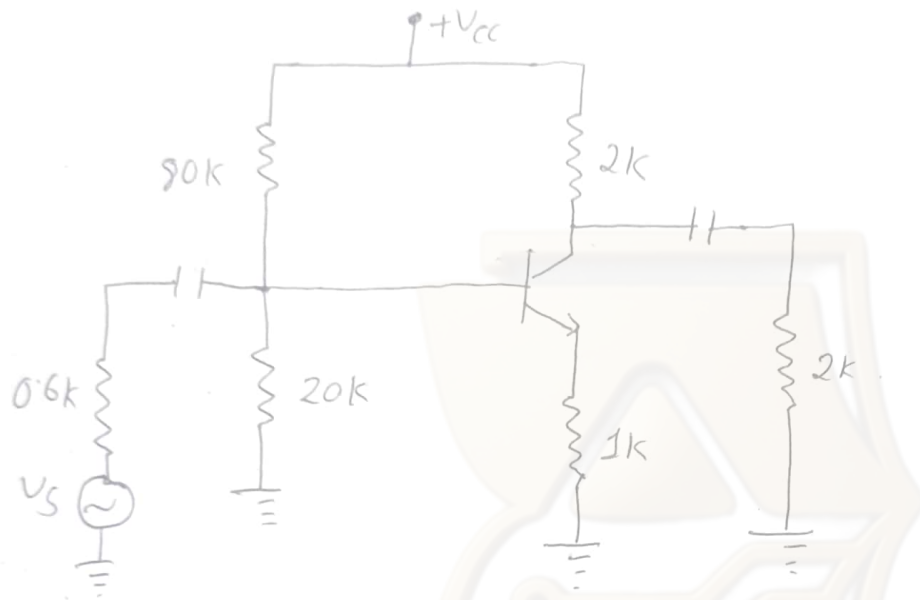
Note: If value of $h_{re}$ not given then use approximate hybrid model

Given,

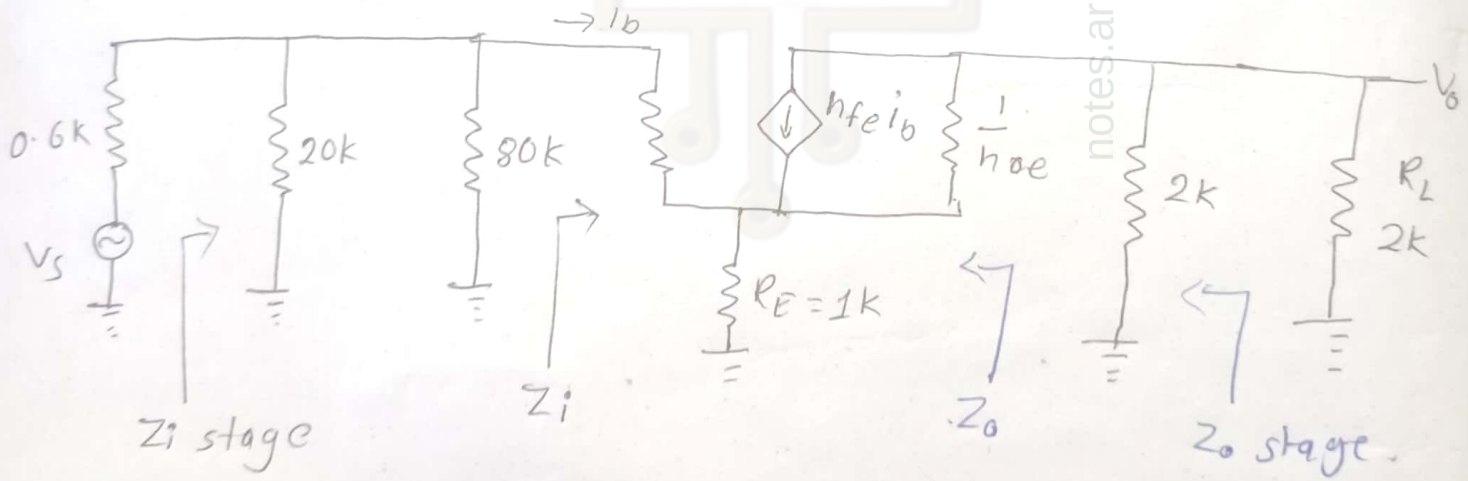
$h_{ie} = 2k\Omega$

$h_{fe} = 49$

$h_{oe} = 50\mu A/V$




## Solution




## For $Z_i$

Applying KVL at input,

$$V_i = i_b h_{ie} - (i_b + h_{fe} i_b) R_E = 0$$

$$\Rightarrow V_i = i_b h_{ie} + i_b (1 + h_{fe}) R_E$$

$$\Rightarrow \frac{V_i}{i_b} = h_{ie} + (1 + h_{fe}) R_E$$

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$\therefore Z_i = h_{ie} + (1+h_{fe}) R_E$

$= 2 + (1+49)1$

$= 52\ k\Omega$

$Z_i$ stage $= (20 \| 80) \| Z_i$

$= \frac{20 \times 80}{20+80} \| 52$

$= 16 \| 52$

$= 12.23$

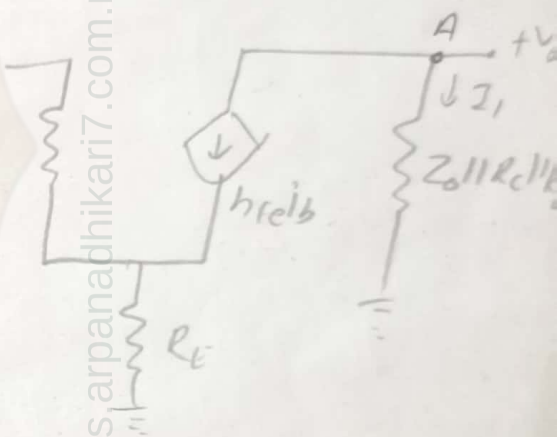
$Z_o \approx \frac{1}{h_{oe}} = \frac{1}{50 \times 10^{-6}} = 2 \times 10^4\ \Omega$

$= 20 K\Omega$

$Z_o$ stage $= (Z_o \| R_C)$

$= \frac{20 \times 2}{20+2}$

$= 1.818\ K$




$A_V = \frac{V_o}{V_i}$

~~Applying kcl at point A.~~

$= - \frac{h_{fe} i_b (Z_o \| R_C \| R_L)}{i_b h_{ie} + i_b (1+h_{fe}) R_E}$

$= - \frac{h_{fe} \times 0.95}{h_{ie} + (1+h_{fe}) R_E}$

$= - \frac{49 \times 0.95}{2 + (1+49) R_E 1}$

$= -0.895$

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$$A_I = \frac{I_L}{I_i}$$

$$A_I = \frac{-h_{fe}}{1 + h_{oe} Z_L} \qquad [\because Z_L = R_C \| R_L]$$

$$= \frac{-49}{1 + 50 \times 10^{-6} \times 1000}$$

$$= -46.66$$

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Chapter- Power Amplifier: (1 long question) + short note (1)

(or Large signal Amplifier)

Large signal or power amplifier are the amplifiers that provides sufficient power to the output load. A power amplifier converts the dc power into the ac power in the presence of input ac signal. A power amplifier is required to produce large voltage & current in the load.

On the basis of the position of operating point or by biasing techniques, the power amplifier are classified as:

1. Class A power Amplifier
   - (i) Direct coupled load
   - (ii) Capacitor coupled load
   - (iii) Transformer coupled load ✓
2. Class B power Amplifier
   - (i) Transformer coupled load class B push pull.
   - (ii) Complementry symmetry class B push pull.
3. Class AB
4. Class C
5. Class D

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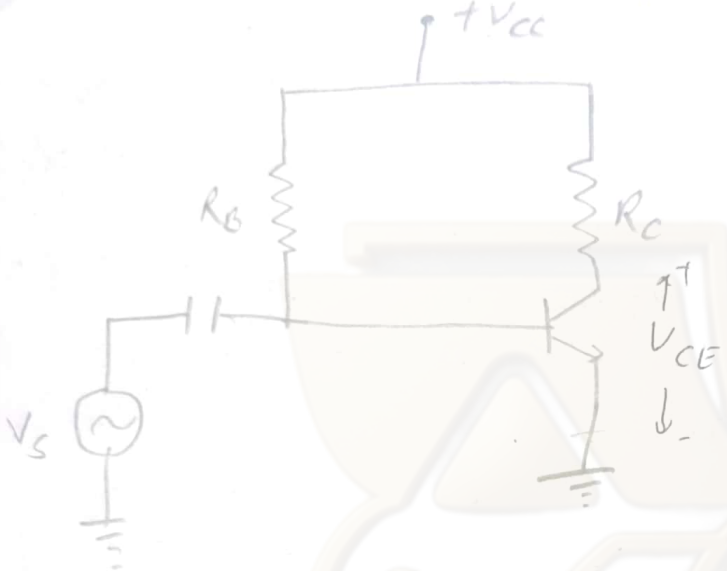
Imp

## 1. Class A power Amplifier

In this type of amplifier, the transistor is biased in such a way that the operating point lies at the center of active region. There is no distortion in the output signal. However, amplitude distortion may occur if the input signal is very large.

### (i) Direct Coupled load class A power Amplifier:




Power amplifier

The direct coupled load class A, is shown in figure above. The transistor is biased in such a way that it operating point lies in the active region during a complete cycle. The output never saturates or reaches cut off.

Applying KVL at output s/dc,

$$V_{CC} - I_C R_C - V_{CE} = 0$$

put $V_{CE} = 0$, $I_C = \frac{V_{CC}}{R_C}$ $\left(0, \frac{V_{CC}}{R_C}\right)$

put $I_C = 0$ $V_{CE} = V_{CC}$ $(V_{CC}, 0)$

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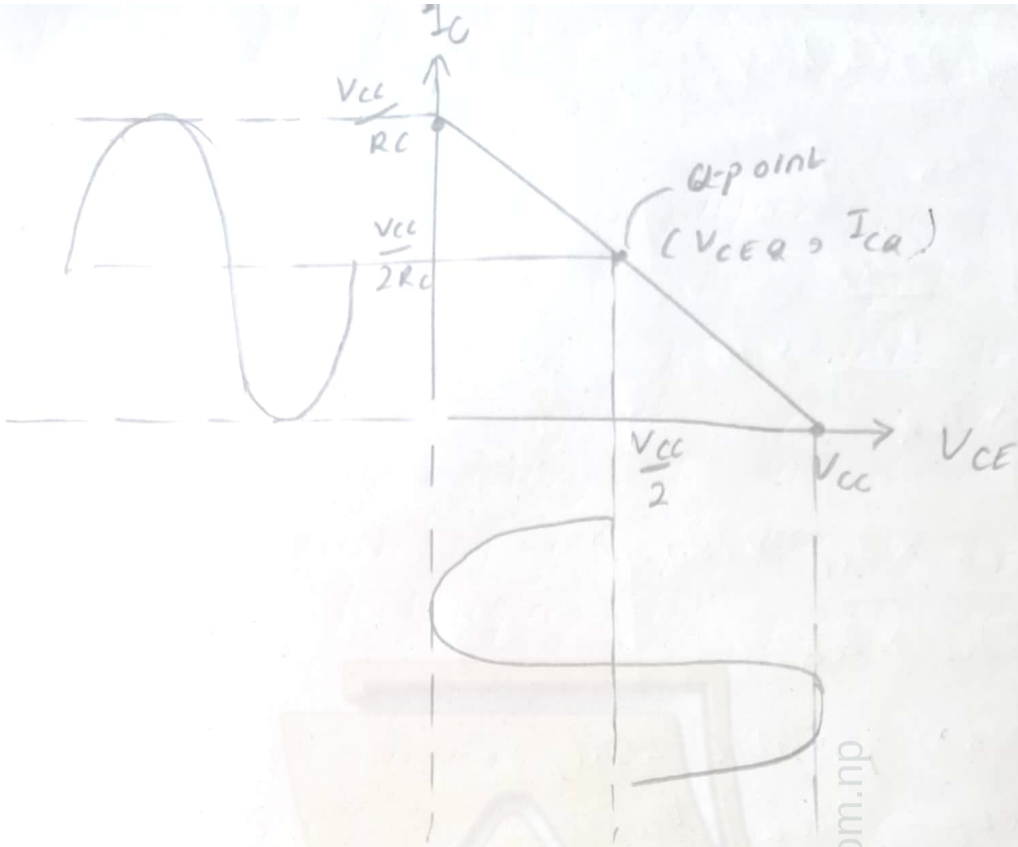
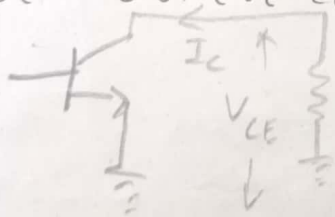



fig:

When an input ac signal is applied to the amplifier the output will vary from its dc biased operating voltage & current.

DC power drawn from supply $(P_{dc}) = V_{cc} I_{CQ}$ —(i)

Ac power delievered to load $(P_{ac}) = (V_{CE})_{rms} \times (I_C)_{rms}$ —(ii)




DC power drawn from supply $(P_{dc}) = V_{cc} I_{CQ}$ —(i)

Ac power delievered to load $(P_{ac}) = (V_{CE})_{rms} \times (I_C)_{rms}$ —(ii)

Efficiency $(\eta) = \frac{\text{AC power delieverd to load } (P_{ac})}{\text{DC " drawn from supply } (P_{dc})} \times 100$

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$$= \frac{(V_{CE})_{rms} \times (I_c)_{rms}}{V_{CC} I_{CQ}} \times 100$$

$$= \frac{\frac{V_{CEP}}{\sqrt{2}} \times \frac{I_{CP}}{\sqrt{2}}}{V_{CC} \times I_{CQ}} \times 100$$

$$\eta = \frac{V_{CEP} \times I_{CP}}{2 V_{CC} \times I_{CQ}} \times 100$$

For maximum possible case,

$$V_{CEP} = V_{CEQ} = \frac{V_{CC}}{2}$$

$$I_{CP} = I_{CQ} = \frac{V_{CC}}{2R_C}$$

$$\therefore \eta = \frac{\frac{V_{CC}}{2} \times I_{CQ}}{2 \times V_{CC} \times I_{CQ}} \times 100$$

$$= 25\%.$$

Hence, the maximum possible efficiency for direct coupled load class A power amplifier is 25%.

Collector efficiency:

$$\eta_c = \frac{\text{Average (ac) power delivered to load}}{\text{Average (dc) power dissipated at the collector}} \times 100$$

$$= \frac{(V_{CE})_{rms} \times I_{C(rms)}}{V_{CEQ} \times I_{CQ}} \times 100$$

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$$\eta_c = \frac{\frac{V_{CEP}}{\sqrt{2}} \times \frac{I_{CP}}{\sqrt{2}}}{(V_{CC} - I_{CQ} R_C) I_{CQ}} \times 100$$

For maximum possible case,

$$V_{CEP} = \frac{V_{CC}}{2}$$

$$I_{CP} = I_{CQ} = \frac{V_{CC}}{2R_C}$$

$$\eta_c = \frac{\frac{V_{CC}}{2\sqrt{2}} \times \frac{I_{CQ}}{\sqrt{2}}}{\left(V_{CC} - \frac{V_{CC}}{2R_C} \times R_C\right) I_{CQ}} \times 100$$

$$= \frac{V_{CC} \times I_{CQ}}{4\left(V_{CC} - \frac{V_{CC}}{2}\right) I_{CQ}}$$

$$= \frac{V_{CC}}{4 \times \frac{V_{CC}}{2}} \times 100$$

$$= 50\%$$

Thus, the collector efficiency of direct coupled load class A power Amplifier is 50%.

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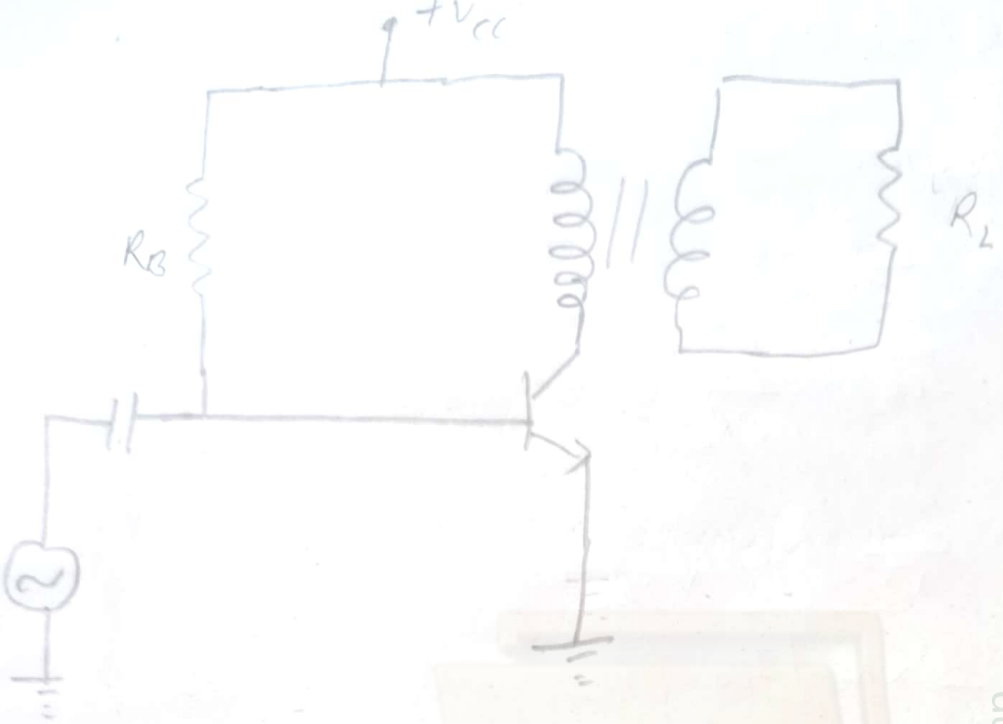
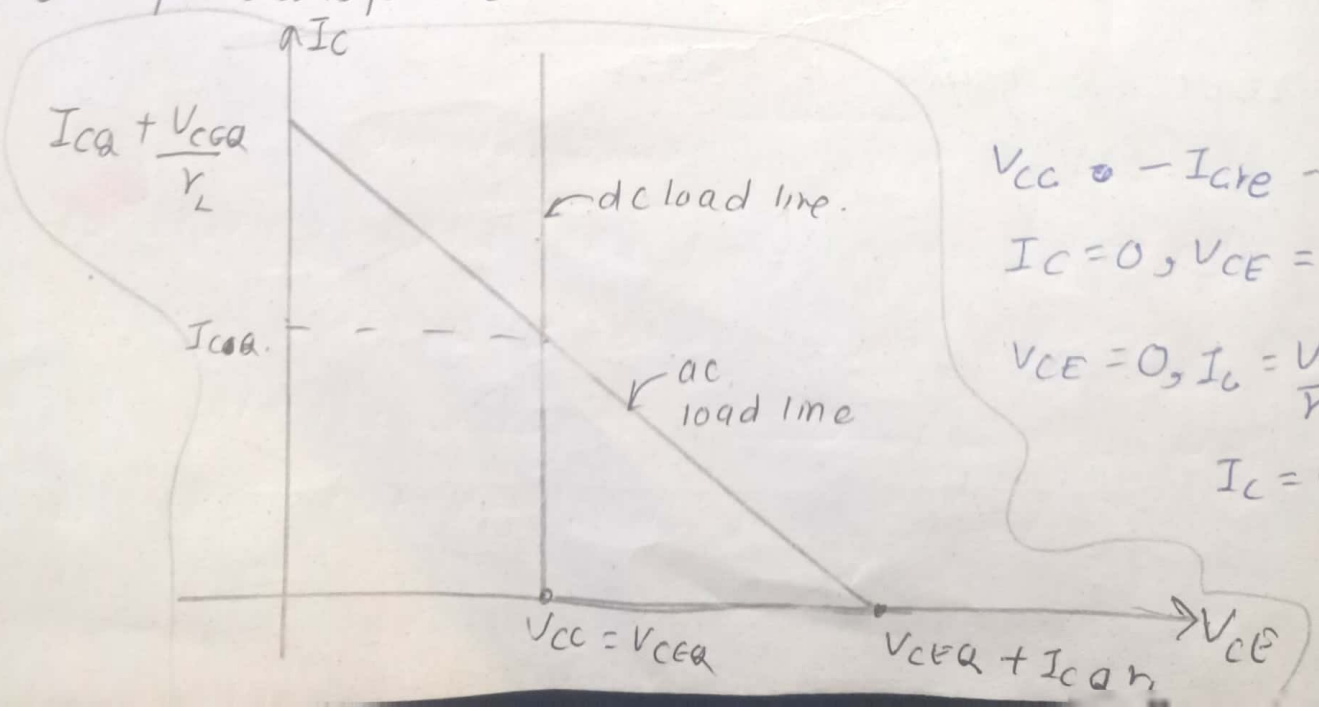
(ii) Transformer Coupled load class A power Amplifier




A large power is wasted across the collecter resistance $R_C$ when dc current flows through it in direct coupled load class A amplifier. In a Transformer coupled load, a transformer is used to couple power amplifier to their loads. An ac pow is magnetically coupled across the transformer into the load & there is no dc power loss in primar side of transformer.




$V_{CC} - I_C R_C - V_{CE} = 0$

$I_C = 0, V_{CE} = V_{CC}$ (V

$V_{CE} = 0, I_C = \frac{V_{CC}}{R_C}$ if R

$I_C = \infty$ (O

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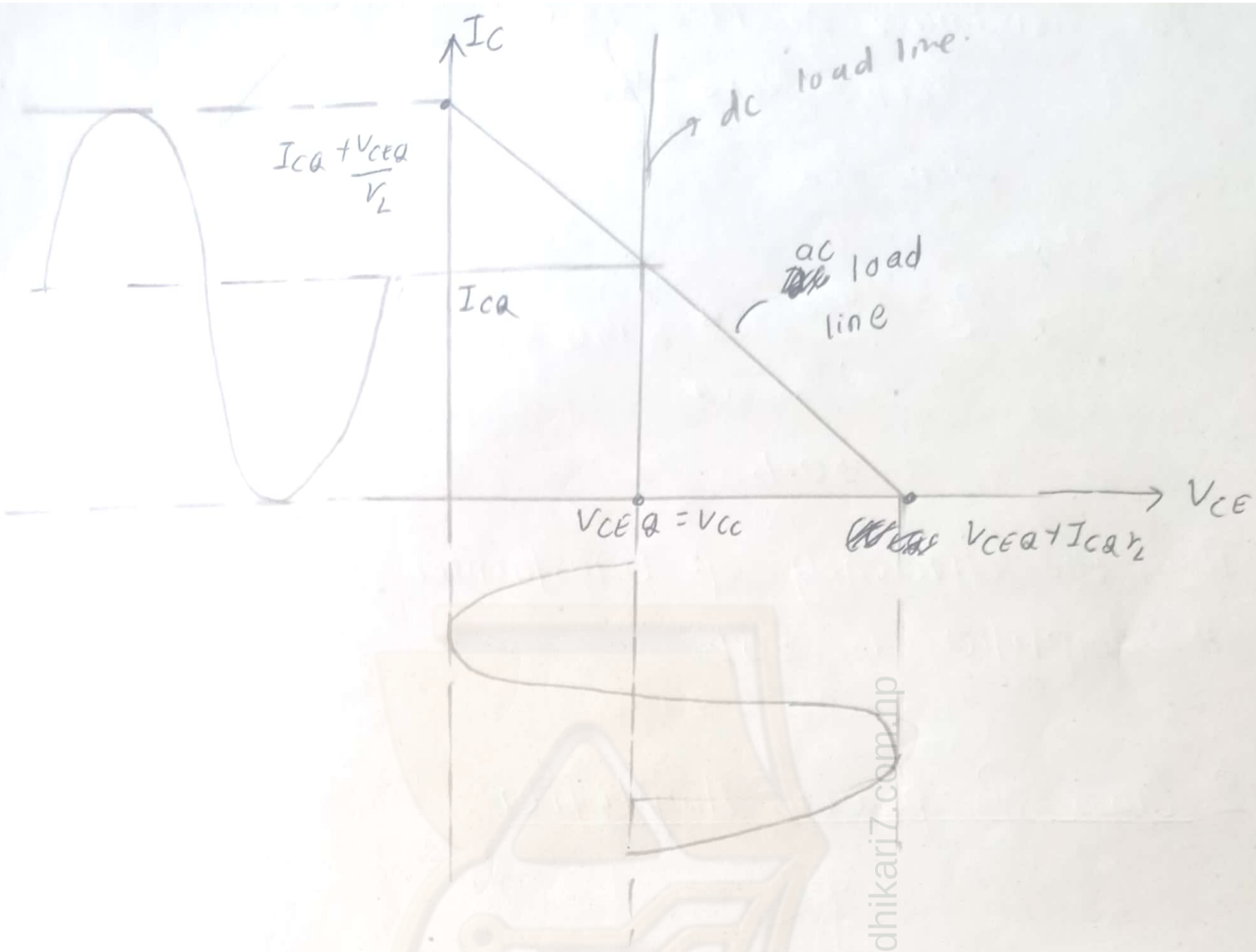


Now,

$$\text{Efficiency } (\eta) = \frac{\text{Average (ac) power delievered to load}}{\text{Average dc power drawn from soure.}} \times 100$$

$$= \frac{(V_{CE})_{rms} \times (I_C)_{rms}}{V_{CC} \times I_{CQ}} \times 100$$

$$= \frac{\frac{V_{CEP}}{\sqrt{2}} \times \frac{I_{CP}}{\sqrt{2}}}{V_{CC} \times I_{CQ}} \times 100$$

$$= \frac{V_{CEP} I_{CP}}{2 V_{CC} I_{CQ}} \times 100.$$

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For maximum possible case,

$$V_{CEP} = V_{CEQ} = V_{CC}$$

$$I_{CP} = I_{CQ}$$

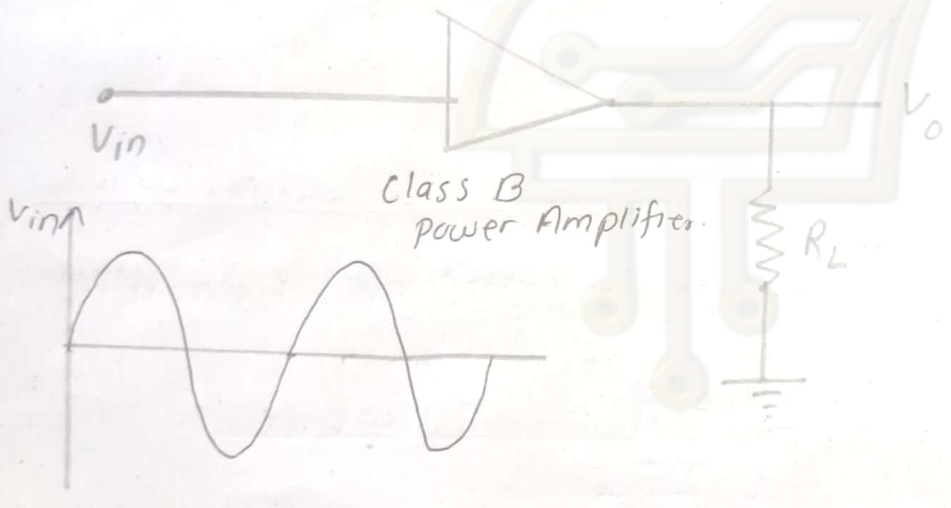
Now,

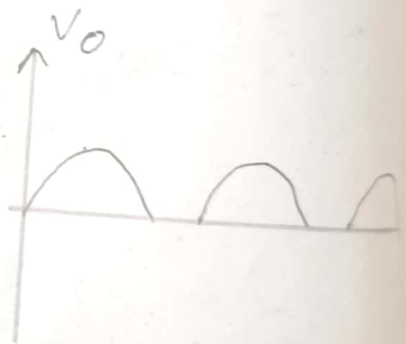
$$\eta = \frac{V_{CC} I_{CQ}}{2 V_{CC} I_{CQ}} \times 100\%$$

$$= 50\%$$

Thus, the efficiency of transformer coupled load class A amplifier is 50%.

## 2. Class B power Amplifier:




In class B power amplifier, the o/p current varies during only one half cycle of ~~side~~ sine wave inpal. The transistor bias is so adjusted that 0 signal collector current is 0. (ie the Q-point is set to the cut off region).

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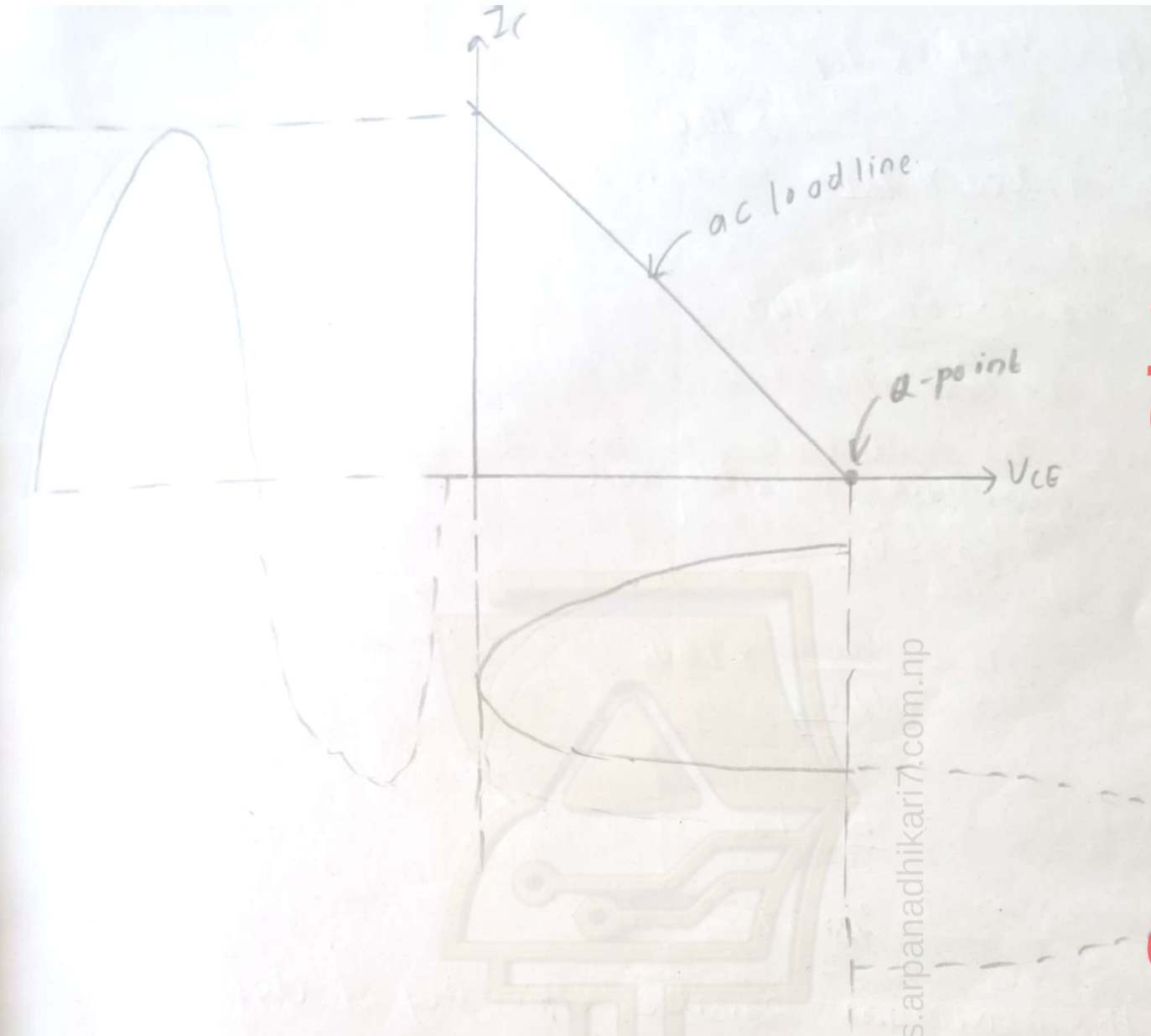


Since, the collector current flows for a half cycle only the o/p waveform is highly distorted. However, class B power amplifier provides higher power o/p & collector efficiency. Thus, these amplifiers are mostly used for power amplification in push-pull arrangement. Now, the efficiency is given by,

$$\text{Efficiency } (\eta) = \frac{\text{Average (ac) power delievered to load}}{\text{Average dc power drawn from source.}} \times 100$$

$$= \frac{(V_{CE})_{rms}(I_C)_{rms}}{V_{cc} \times I_{dc}} \times 100$$

← because half cycle → $I_{dc}$ because $I_{CQ} = 0$.

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$$\eta = \frac{\frac{V_{CEP}}{2} \times \frac{I_{CP}}{2}}{V_{CC} \times \frac{I_{CP}}{\pi}} \times 100$$

$$= \frac{V_{CEP}}{\frac{4V_{CC}}{\pi}} \times 100$$

For maximum possible case,

$$V_{CEP} = V_{CC}$$

$$\therefore \eta = \frac{V_{CC}}{\frac{4V_{CC}}{\pi}} \times 100$$

$$= \frac{\pi}{4} \times 100$$

$$= 78.5\%$$

Thus, the maximum possible efficiency of class B power amplifier is 78.5%.

## Push-pull Amplifier

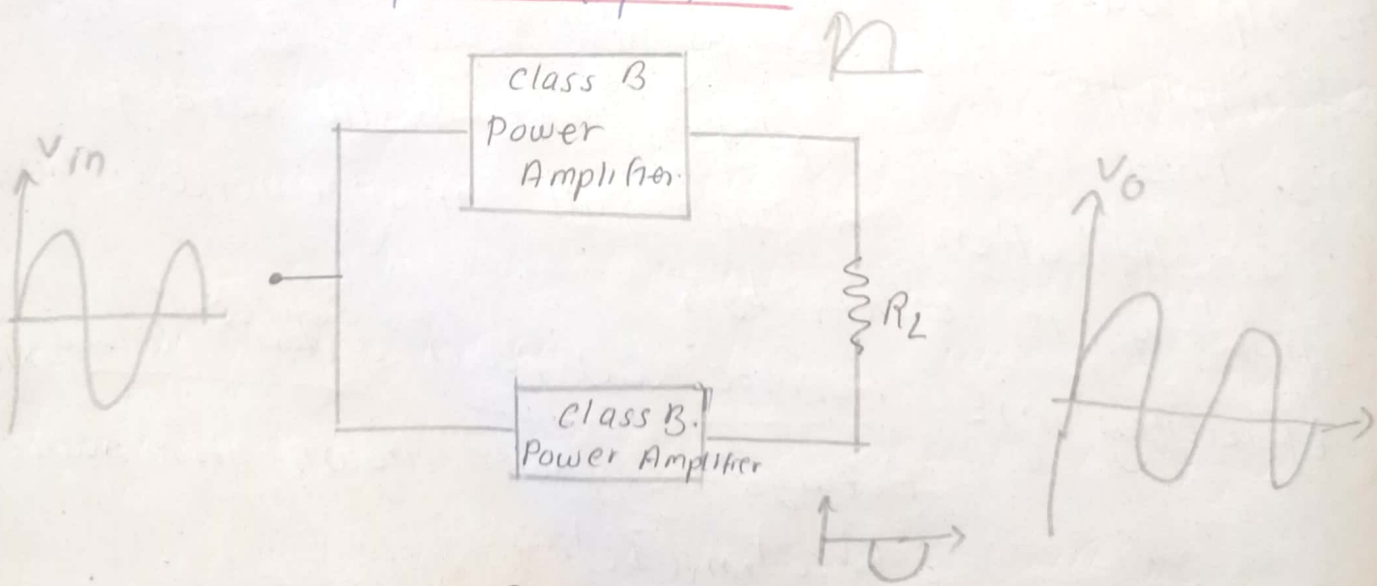


fig: Block diagram of push pull amplifier.

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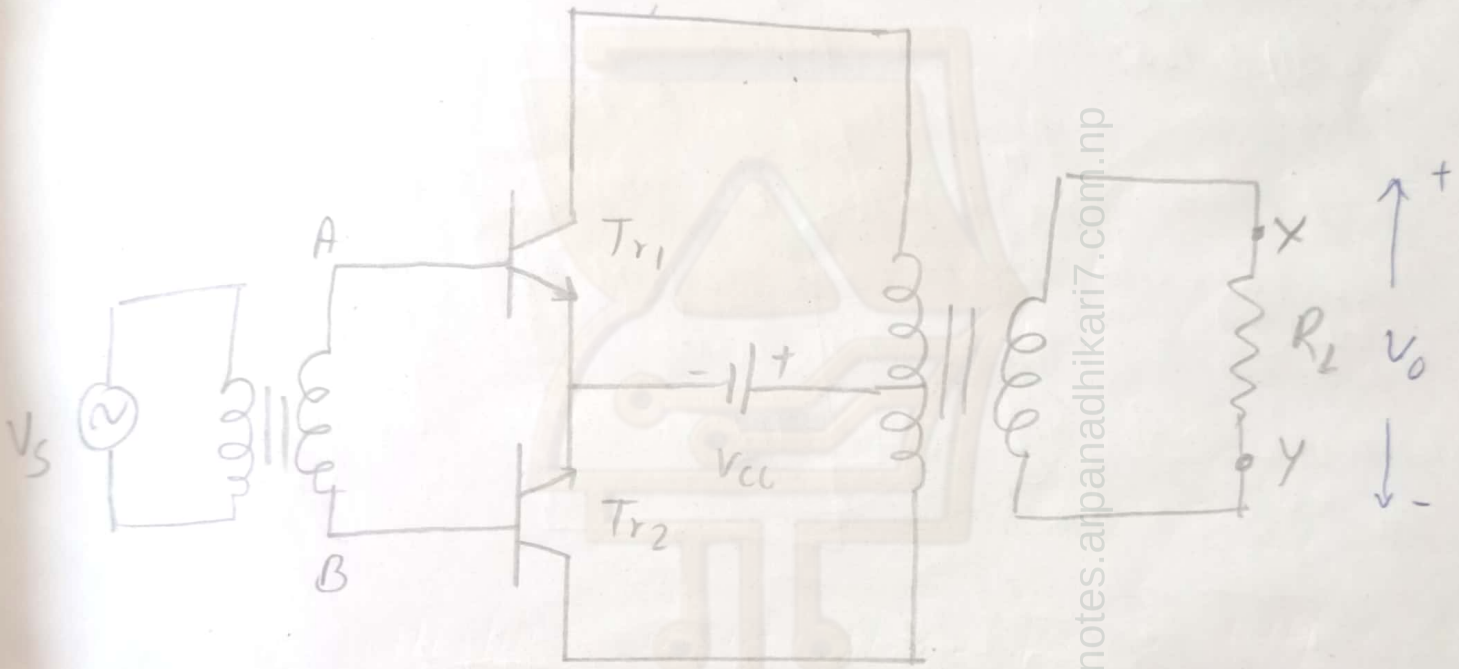
The push pull amplifier is power amplifier where two transistor operating in class B operation are employed. Each transistor conducts on opposite half cycles. Since, one part of the circuit push the signal high during one half cycle and other part pulls the signal low during the other half cycle, the circuit is refered as push-pull circuit.

(i) Transformer coupled load pushpull amplifier:

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Note:

For +ve half.

$Tr_1 \rightarrow$ ON

$Tr_2 \rightarrow$ OFF

current in $R_L$ flows from X to Y.

For -ve half.

$Tr_1 \rightarrow$ OFF

$Tr_2 \rightarrow$ ON

current in $R_L$ flows from Y to X

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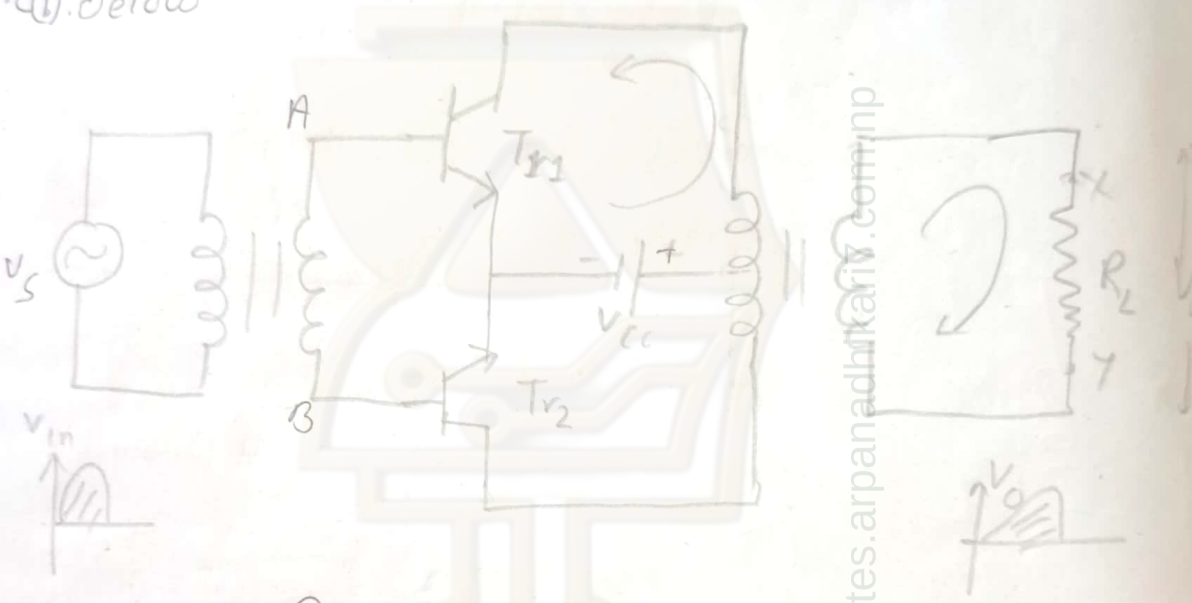
The transformer coupled load class B push pull amplifier is shown in figure. It consists of two npn transistor $T_{r1}$ and $T_{r2}$ and load is connected to the circuit through a transformer.

Suppose during the +ve half of the input signal, the end A becomes +ve wrt B. Thus, the during +ve half cycle $T_{r1}$ will turn ON and $T_{r2}$ will turn OFF.
For +ve half cycle, current flows from $V_{cc}$ to $T_{r1}$ as shown in figure(i). below




fig(i) During +ve half of input.

Now, During -ve half of the input, polarity of end A & B will be reversed (i.e end A becomes -ve & end B becomes +ve). Thus, transistor $T_{r1}$ will turn off & $T_{r2}$ will be turn on. In this condition, the current flows through lower half as shown in figure(ii)
The direction of current through $R_L$ will be reversed as compared to the direction during +ve half.

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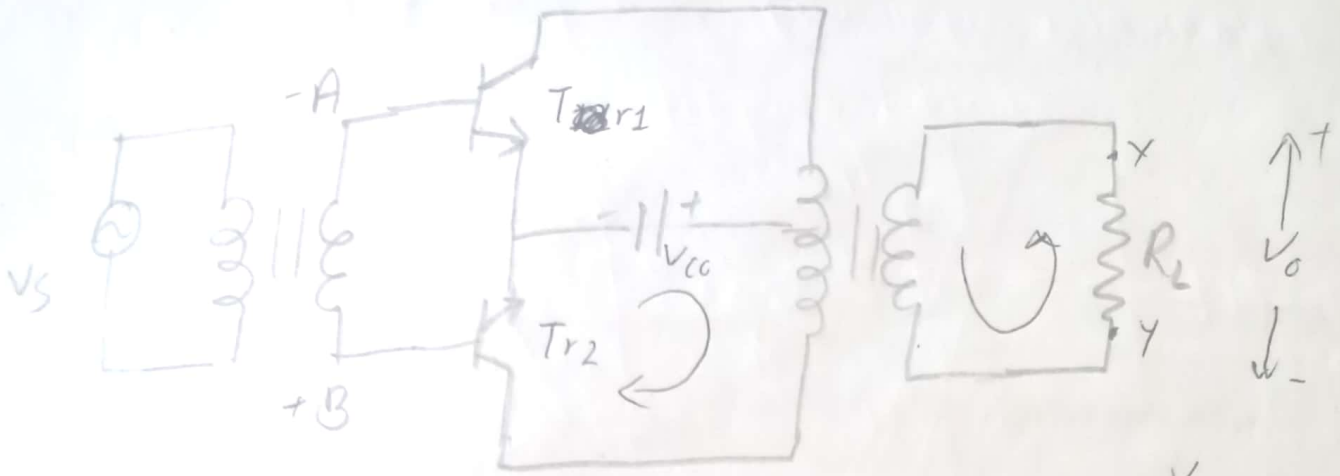
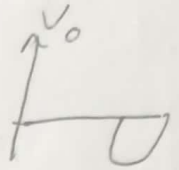


fig: During -ve half of I/P.




Efficiency ($\eta$) = $\frac{\text{Average ac power delivered to load}}{\text{Average dc power drawn from source}} \times 100$

$$= \frac{(V_{CE})_{rms} \times (I_C)_{rms}}{V_{CC} \times I_{dc}}$$

$$= \frac{\frac{V_{CEP}}{\sqrt{2}} \times \frac{I_{CP}}{\sqrt{2}}}{V_{CC} \times \frac{2I_{CP}}{\pi}} \times 100$$

$$\frac{V_{CEP}}{\frac{4}{\pi} V_{CC}} \times 100 \cdot 1.$$

At maximum ~~power~~ possible case,

$$V_{CEP} = V_{CC}.$$

$$\therefore \eta = \frac{\frac{V_{CC} \times I_{CP}}{2}}{V_{CC} \times \frac{2I_{CP}}{\pi}} \times 100$$

$$= 78.5\%.$$

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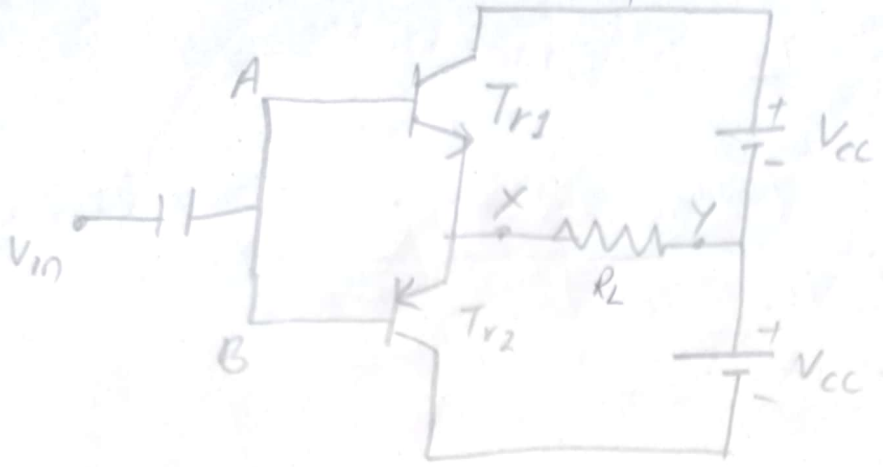
(ii) Complementry Symmetry push pull amplifier.




The need of the transformer in push-pull operation is eliminated in complementry symmetry push-pull amplifier. It employs one npn transistor & one pnp transistor. During +ve half cycle of input, the transister $Tr_1$ (npn transistor) will turn on & transistor $Tr_2$ (pnp transitor) will turn off. So, current flows in upper half only. The direction of current in $R_L$ will be X to Y.

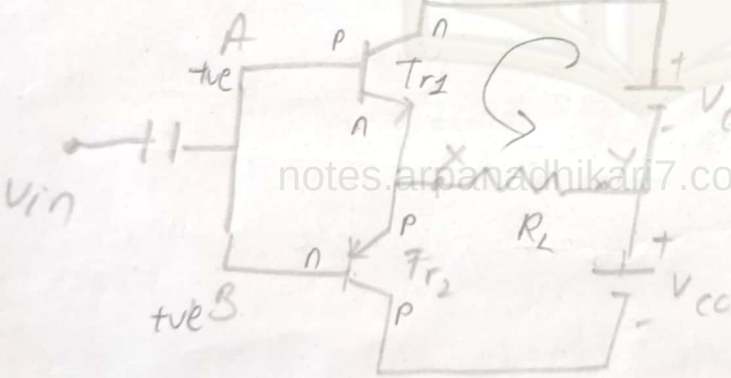



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Fig: During +ve half of I/P.

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During -ve half of I/P, the transistor $Tr_1$ will turn off & $Tr_2$ will turn on. So, current flows in lower half only. So, current flows from Y to X in $R_L$.

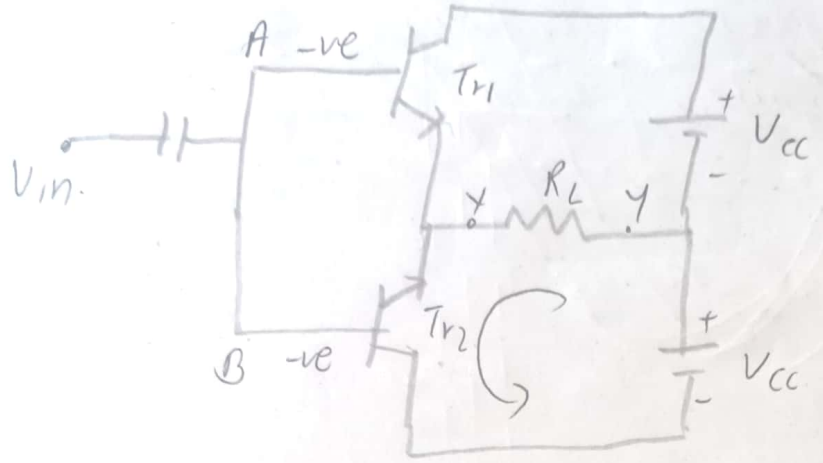


Fig: During -ve half of I/P

$$\text{Efficiency } (\eta) = \frac{\text{Average (ac) power delivered to load}}{\text{Average (dc) power drawn from source}} \times 100$$

$$= \frac{(V_{CE})_{rms}\,(I_C)_{rms}}{V_{CC} \times I_{dc}} \times 100$$

$$= \frac{\frac{V_{CEP}}{\sqrt{2}} \times \frac{I_{CP}}{\sqrt{2}}}{V_{CC} \times 2\frac{I_{CP}}{\pi}} \times 100$$

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$$= \frac{V_{CEP}}{4\frac{V_{CC}}{\pi}} \times 100\%$$

For maxm possible case: $V_{CEP} = V_{CC}$

$$\therefore \eta = \frac{V_{CC}}{4\frac{V_{CC}}{\pi}} \times 100\%$$

$$= \frac{\pi}{4} \times 100\%$$

$$= 78.5\%$$

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Imp Crossover distortion short q^n

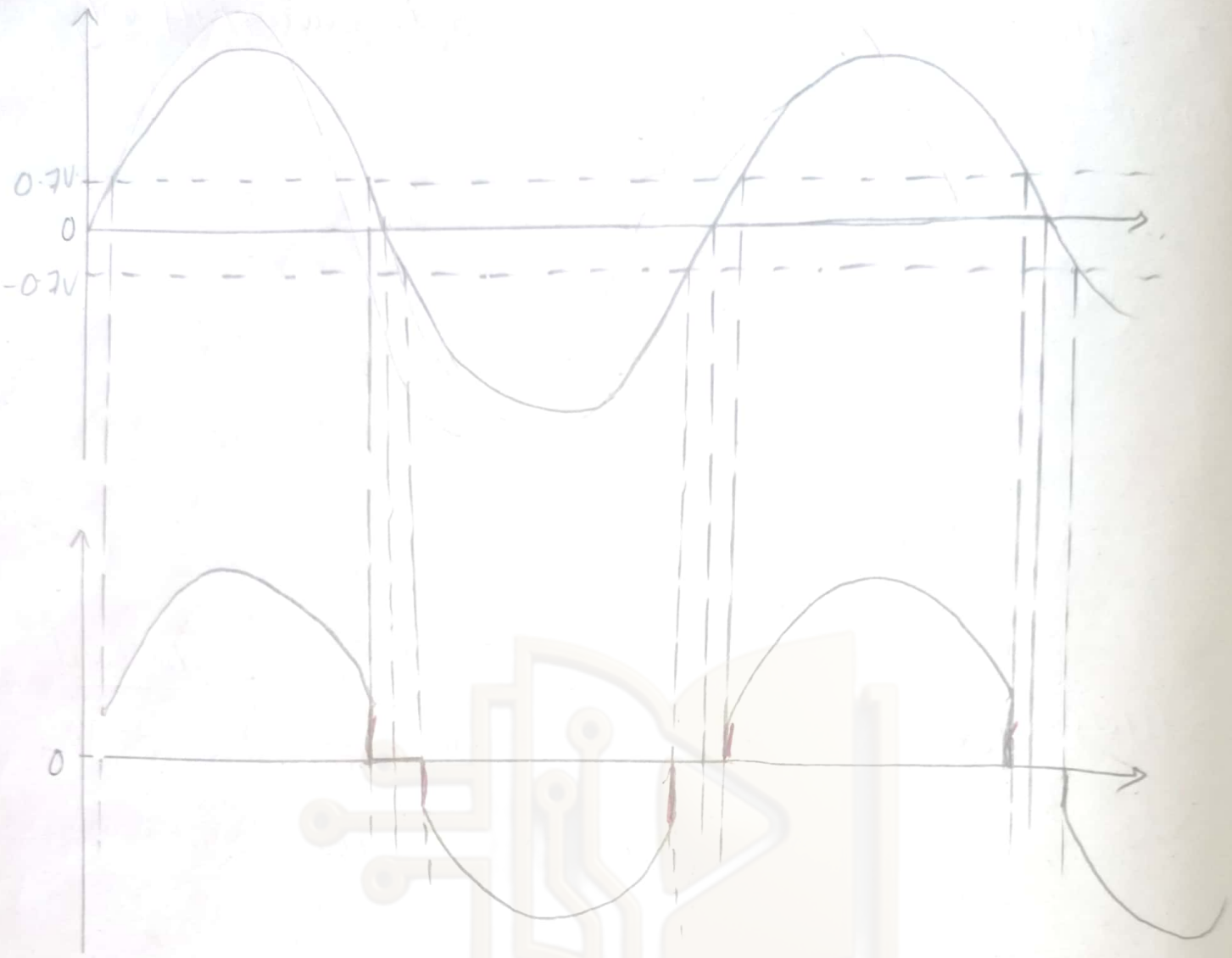


fig: Crossover Distortion

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In class B push pull operation, there is several distortion at very low signal level because signal level because base of transistor do not turn at 0V but at 0.3V for germanium & 0.7V for silicon. During the period, until the transistor turns on, there will be no output. This results that the waveform will not be pure sine but distorted form. This type of distortion is called crossover distortion. The crossover distortion

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distortion as a result of one transistor cutting off before the other transistor being conducting.

The crossover distortion can be eliminated by applying slight forward biased to each emitter diode so that Q-point of each transistor will located slightly above cut off region. Therefore, one transistor operates for more the one half cycle since, each transistor operates slightly more than half cycle, its results in AB operation.

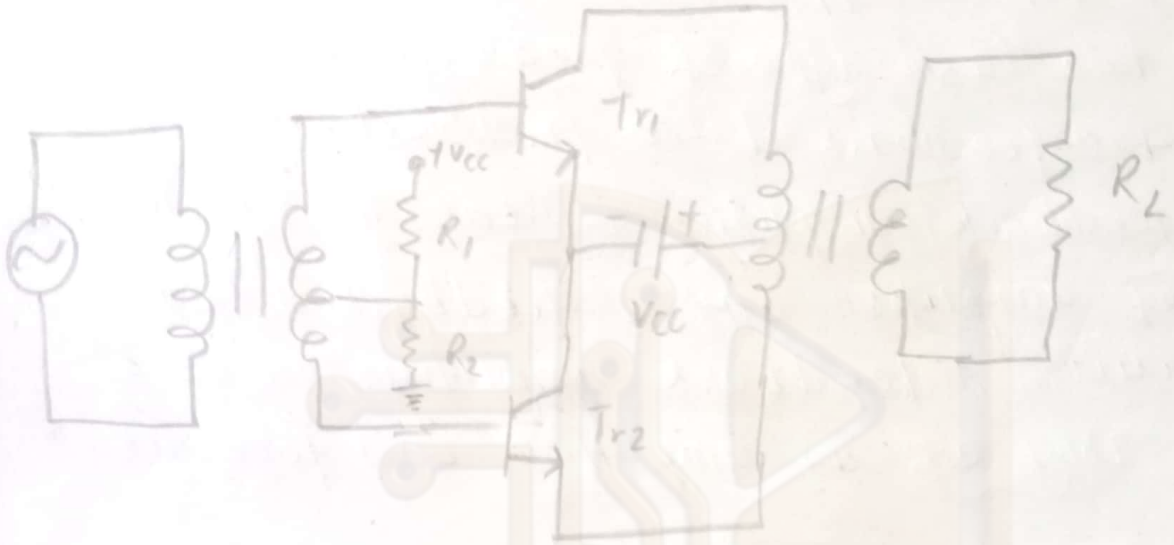


fig: Class AB power Amplifier.

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Power dissipation &

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Chapter-8

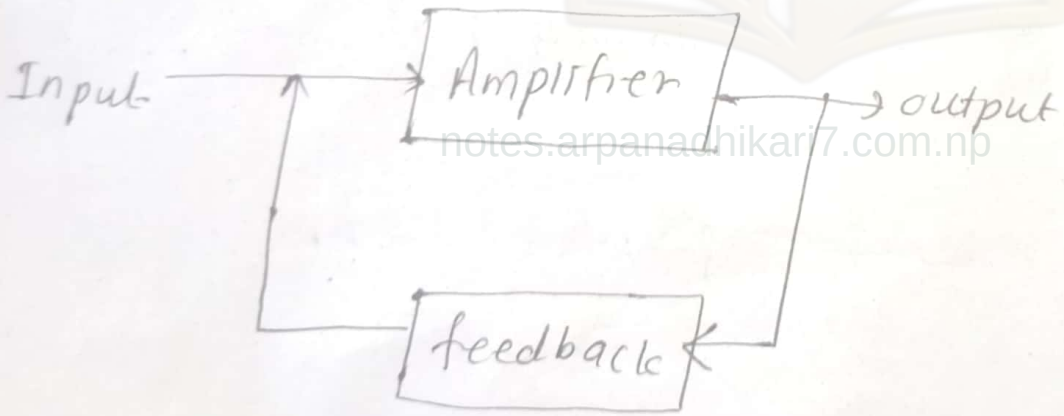
(7-8 marks)

# Feedback Amplifier

The process of injecting a fraction of o/p energy of some device back to the input is known as feedback

Depending upon whether the feedback energy aids (supports) or oppose the input signal. There are two basic type of feedback in amplifier. They are +ve feedback & -ve feedback.

A feedback amplifier is one in which a fraction of amplifier o/p is feedbacked to the input circuit. A feedback amplifier consists of two parts. They are an amplifier and feedback ckt.

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## 1. Positive feedback:

When the feedback energy is in phase with the input signal & thus aids it, it is called positive feedback. The positive feedback increases the gain of an amplifier. However, it has the disadvantage of increased distortion & instability. It is seldom used in amplifier but it is used in oscillator circuits.

Let us consider an amplifier with open loop gain (A). Let positive feedback with feedback ratio $\beta = \frac{V_f}{V_o}$ is applied as shown in figure.

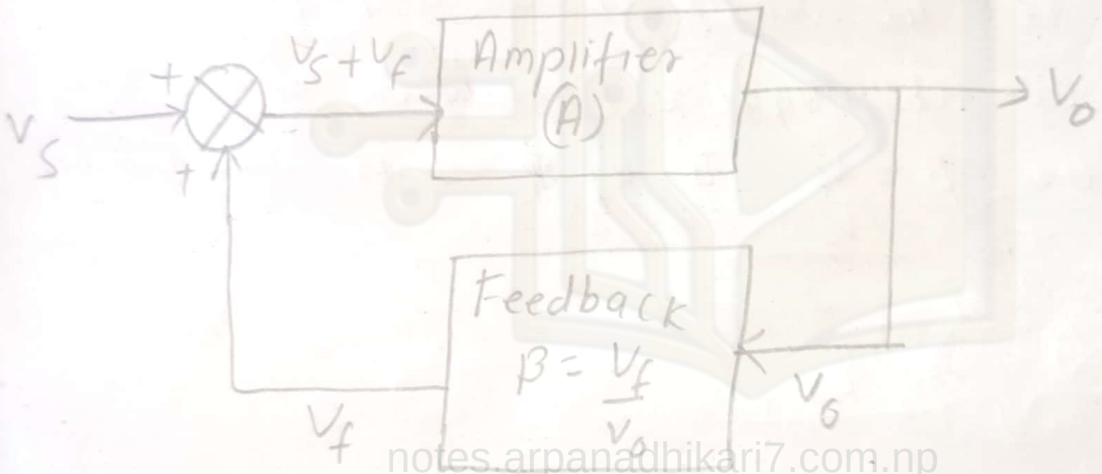


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fig: Positive feedback Amplifer

$$V_o = A(V_s + V_f)$$

$$= AV_s + AV_f$$

$$= AV_s + A \cdot \beta V_o$$

$$\Rightarrow V_o - \beta A V_o = AV_s$$

$$\Rightarrow V_o(1 - \beta A) = AV_s$$

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$$\frac{V_o}{V_s} = \frac{A}{1-\beta A} \quad \text{= without feedback}$$

$$\boxed{A_f = \frac{A}{1-\beta A}} \quad -① \quad \text{, with feedback}$$

Thus, the overall gain ~~of~~ ($A_f$) of feedback amplifier will increases with positive feedback.

2. Negative feedback:

When the feedback energy is in out of phase with the input signal & thus it opposes it, it is called negative feedback. ~~The negative feedback~~

Let us consider an amplifier with open loop gain. let negative feedback with feedback ratio $\beta = \frac{V_f}{V_o}$ is applied as shown in fig

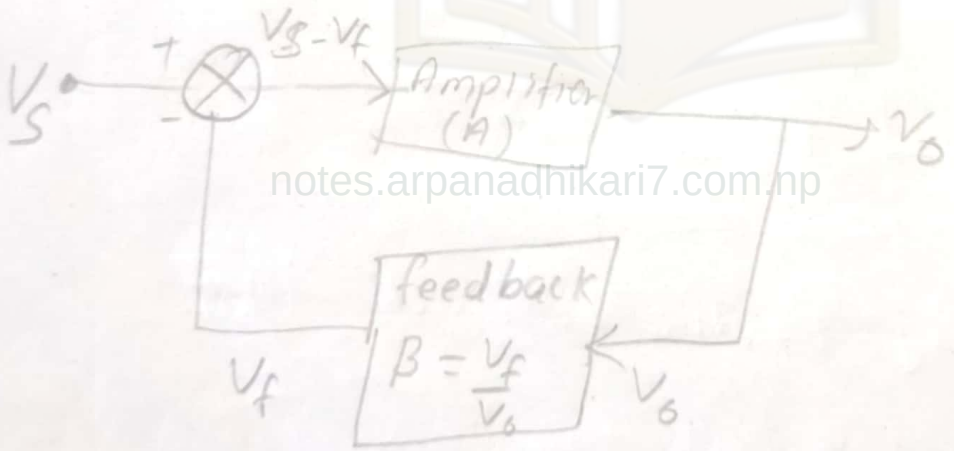


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fig: Negative feed back Amplifier

$$V_o = A(V_s - V_f)$$

$$V_o = AV_s - AV_f$$

$$V_o = AV_s - A\cdot\beta V_o$$

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$$V_o + \beta A V_o = A V_s$$

$$V_o (1 + \beta A) = A V_s$$

$$\frac{V_o}{V_s} = \frac{A}{1 + \beta A}$$

$$\boxed{A_f = \frac{A}{1 + \beta A}} \quad -(1)$$

Thus, the overall gain ($A_f$) of feedback amplifier will decreaces with negative ~~am~~ feedback.

## Advantages of negative feedback.

(i) Highly stabilized gain ie ~~improved frequency respond~~

(ii) Increased band width ie ~~improved frequency respond~~

(iii) less amplitude distortion.

(iv) less harmonic distortion.

(v) less frequency distortion.

(vi) less phase ,,

(vii) Reduce noice.

(viii) Input & output impedance can be modified as designed.

(ix) Mode linear operation

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# Effect of negative feedback on gain stability:

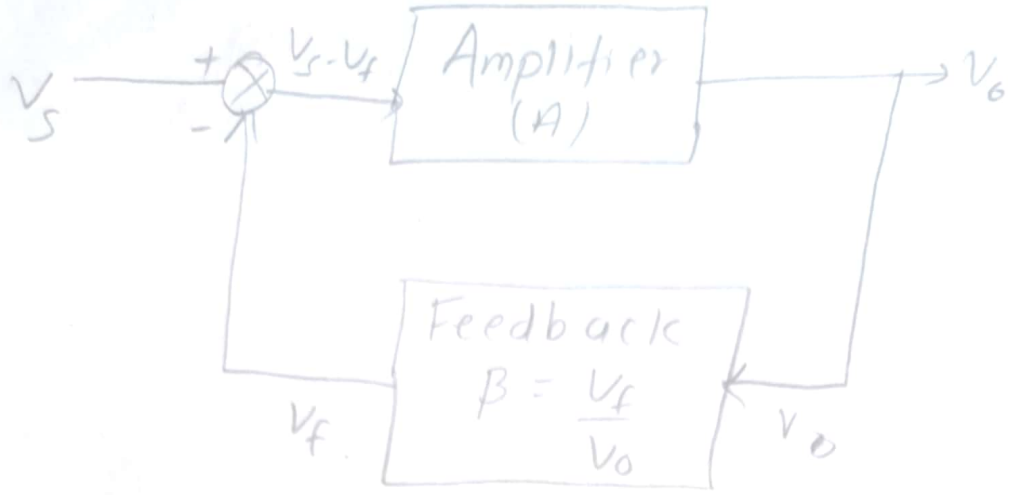


fig: Negative feedback Amplifier.

$$A_f = \frac{A}{1+\beta A} \quad \text{—(1)}$$

The gain of an amplifier with negative feedback is given by,

$$A_f = \frac{A}{1+\beta A} \quad \text{—(1)}$$

Now, Diff eqn (1) wrt A.

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$$\frac{dA_f}{dA} = \frac{d\left(\frac{A}{1+\beta A}\right)}{dA}$$

$$\frac{d\left(\frac{u}{v}\right)}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

$$= \frac{(1+\beta A)\frac{dA}{dA} - A\frac{d(1+\beta A)}{dA}}{(1+\beta A)^2}$$

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$$\frac{dA_f}{dA} = \frac{(1+\beta A) - A(0+\beta)}{(1+\beta A)^2}$$

$$= \frac{1+\beta A - \beta A}{(1+\beta A)^2}$$

$$= \frac{1}{(1+\beta A)^2}$$

$$dA_f = \frac{dA}{(1+\beta A)^2}$$

Now, dividing both side by $A_f$

$$\frac{dA_f}{A_f} = \frac{dA}{A_f(1+\beta A)^2}$$

$$\Rightarrow \frac{dA_f}{A_f} = \frac{dA}{\frac{A}{1+\beta A}(1+\beta A)^2}$$

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$$\Rightarrow \frac{dA_f}{A_f} = \frac{dA}{A(1+\beta A)^2}$$

$$\Rightarrow \frac{dA_f}{A_f} = \frac{\frac{dA}{A}}{(1+\beta A)} \quad \text{——— (ii)}$$

From eqn (ii), it is clear that the percentage change in gain with negative feedback i.e $\frac{dA_f}{A_f}$ is less than the percentage change in gain without feedback. i.e $\frac{dA}{A}$. Thus the negative feedback

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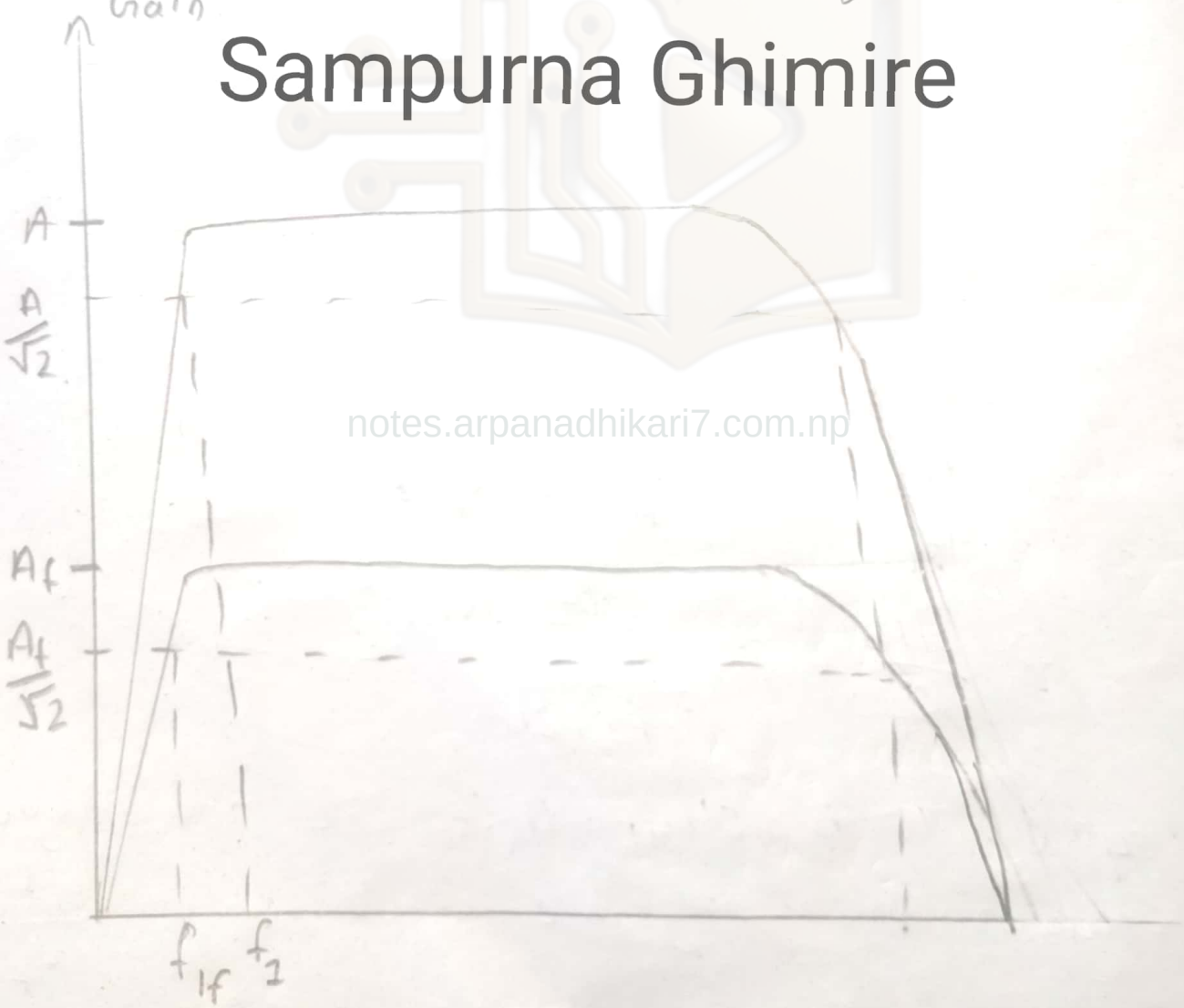
improves the gain stability of amplifier.

## Effect of negative feedback on bandwidth.

For the lower frequency range, the gain of an amplifier increases with frequency. At mid frequency range, the gain will remains almost constant and at higher frequency range, the gain decreases with increase in frequency. The gain of an amplifier will decrease with negative feedback. The frequency response of an amplifier with out feed back and with negative feed back is shown in figure below.

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Let $f_1$ & $f_2$ be the lower cut off frequency & upper cut off frequency resply of an amplifier without feedback. Therefore, bandwidth of an amplifier without feedback is given by,

$$\text{Bandwidth } (B.W) = f_2 - f_1$$

Since, $f_1$ is very small. So, $f_2 - f_1 \approx f_2$

$$\therefore BW = f_2 \quad \text{——— (1)}$$

After negative feedback, the mid-band gain will decrease by $1+\beta A$, lower cut off frequency decrease by $1+\beta A$ times and upper cut off frequency increase by $(1+\beta A)$.

∴ After negative feedback,

lower cut off frequency $f_{1f} = \dfrac{f_1}{(1+\beta A)}$

upper cut off frequency $f_{2f} = f_2 (1+\beta A)$.

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$$\therefore (B.W)_f = f_{2f} - f_{1f}$$

$$= f_2 (1+\beta A) - \frac{f_1}{(1+\beta A)}$$

$$= f_2 (1+\beta A)$$

$$(B.W)_f = (B.W)(1+\beta A) \quad \text{——— (11)}.$$

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Thus, the bandwidth of an amplifier increases by $(1+\beta A)$ times with negative feedback.

## Numericals:

Qn.1. An amplifier with voltage gain of 1000 uses $\frac{1}{20}$ of its output in negative feedback. Calculate the gain with feedback in dB.

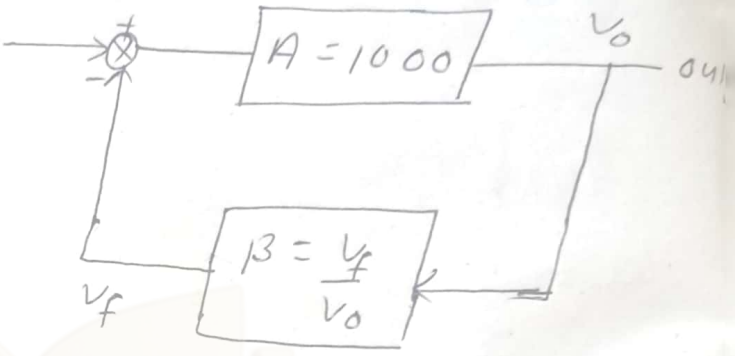
Solution

Given

Open loop gain $(A) = 1000$

feedback ratio $(\beta) = \frac{1}{20}$

Closed loop gain $(A_f) = ?$




We have,

$$A_f = \frac{A}{1+\beta A}$$

$$= \frac{1000}{1+\frac{1}{20}\times 1000}$$

$$= 19.60$$

$$V_f = \frac{1}{20} V_o$$

$$\frac{V_f}{V_o} = \frac{1}{20}$$

$$\beta = 1/20$$

gain in dB $= 20 \log_{10} A_f$

$= 20 \log_{10} (19.60)$

$= 25.84$ dB.

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Qn. An amplifier with negative feed back provides an output voltage of 5V with an input voltage of 0.2V. On removal of the feedback, it needs only 0.1V input to give the same output. Determine.

(i) Gain without feedback

(ii) " with "

(iii) feedback ratio.

Solution

With feedback

input voltage ($V_s$) = 0.2V

output " ($V_o$) = 5V

Without feedback

input voltage ($V_s$) = 0.1V

o/p " ($V_o$) = 5V

(i) Gain without feedback $(A) = \frac{V_o}{V_s} = \frac{5}{0.1} = 50$

(ii) Gain with " $(A_f) = \frac{V_o}{V_s} = \frac{5}{0.2} = 25$

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(iii) feedback ratio $(\beta) = ?$

since, $A_f = \frac{A}{(1+\beta A)}$

or, $A_f(1+\beta A) = A$

or, $\beta A = \frac{A}{A_f} - 1$

or, $\beta = \frac{1}{A}\left(\frac{50}{25} - 1\right)$

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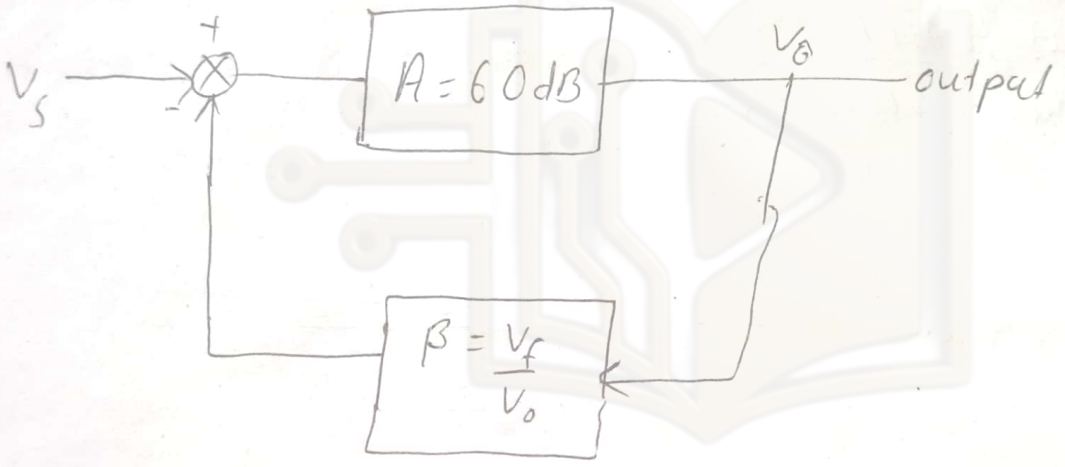
$\Rightarrow \beta = \frac{1}{50} \quad (2-1)$

$\Rightarrow \beta = \frac{1}{50}$

Q0.3 A negative feedback of 0.2% is applied to an amplifier gain of 60 dB. Calculate the % change in the overall voltage gain of the feed amplifier if the internal amplifier is subjected to a gain reduction of 15%.

Solution

Given,

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open loop gain $(A) = 60\,dB$.

feed back ratio $(\beta) = 0.002$

$V_f = 0.2\%$ of $V_o$

$\frac{V_f}{V_o} = \frac{0.2}{100}$

$\Rightarrow \beta = 0.002$

Change in open loop gain

$\left(\frac{dA}{A}\right) = 15\%$

Change in closed loop gain $\frac{dA_f}{A_f} = ?$

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Since,

$$\frac{dA_f}{A_f} = \frac{dA/A}{1+\beta A}$$

As, gain in dB $= 20 \log_{10} A$

or, $60 = 20 \log_{10} A$

or, $\log_{10} A = 3$

or, $A = 1000$

$$\therefore \frac{dA_f}{A_f} = \frac{15}{1+0.002\times10^3}$$

$$= 5\cdot1$$

Qn·4 An amplifier gain changes by ±10·1 using the negative feedback. The amplifier is said to be modified to yield a gain of 100 with ± 0·1·1. variation. Find the required open loop gain of the amplifier & the amount of the negative feedback.

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Solution,

Given,

$$\frac{dA}{A} = 10\cdot1$$

$$\frac{dA_f}{A_f} = 0\cdot1\cdot1$$

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$A_f = 100$

Since,

$$\frac{dA_f}{A_f} = \frac{\frac{dA}{A}}{1+\beta A}$$

or, $0.1 = \frac{10}{1+\beta A}$

or $1+\beta A = 100$ ——— (1)

Then,

$$A_f = \frac{A}{1+\beta A}$$

or $100 = \frac{A}{100}$

or $A = 10000$

Then putting value of A in eqn (1).

$$1+\beta \times 10000 = 100$$

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$$\beta = \frac{100-1}{10000}$$

$$\beta = \frac{99}{10000}$$

$$\therefore \beta = 9.9 \times 10^{-3}$$

$$= 0.0099$$

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Qn-5. The ~~are~~ An RC coupled amplifier has $A_m = 50,000$, $f_H = 20\,KH_z$, $f_L = 30H_z$, $\beta = 5 \times 10^{-5}$ a resist~~ive~~ voltage negative feedback is added such that $\beta = 5 \times 10^{-5}$. Find $A_{mf}$, $f_{Hf}$ and $f_{Lf}$.

Solution,

$A_m = 50,000$

$f_H = 20\,KH_z$

$f_L = 30H_z$

$\beta = 5 \times 10^{-5}$

We know,

$$A_{mf} = \frac{A_m}{1 + \beta A_m} \quad (14285.7)$$

$$= \frac{50000}{1 + 5 \times 10^{-5} \times 50000}$$

$$= 14285.71$$

$$f_{Hf} = (1 + \beta A_m) f_H$$

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70 KHz

$$= (1 + 5 \times 10^{-5} \times 50000)\, 20$$

$$= 70\,kH_z$$

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$$f_{Lf} = \frac{f_L}{1+\beta A_m}$$

$$= \frac{30}{1+5\times10^{-5}\times50000}$$

$$= 8.571 \text{ Hz}$$

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An Amplifier has a midband gain of 1500 and a bandwidth of 4 MHz. The midband gain reduces 150 when a negative feedback is applied. Determine the value of feedback factor & the bandwidth.

Solution

Given,

$A_m = 1500$

$A_{mf} = 150$

B.W $= 4 \text{ MHz} = 4\times10^6 \text{ Hz}$

feedback factor $(\beta A_m) = ?$

Bandwidth after -ve feedback $(B.W)_f = ?$

$$A_{mf} = \frac{A_m}{1+\beta A_m}$$

$$\text{or, } 150 = \frac{1500}{1+\beta A_m}$$

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$$\text{or } 150 + 150\beta A_m = 1500$$

$$\text{or } \beta A_m = \frac{1500 - 150}{150}$$

$$\text{or } \beta A_m = 9$$

$$(B.W)_f = (1+\beta A_m) B.W$$

$$= (1+9)\ 4\times10^6$$

$$= 10 \times 4 \times 10^6$$

$$= 4 \times 10^7 \text{ Hz}$$

$$= 40 \text{ MHz}$$

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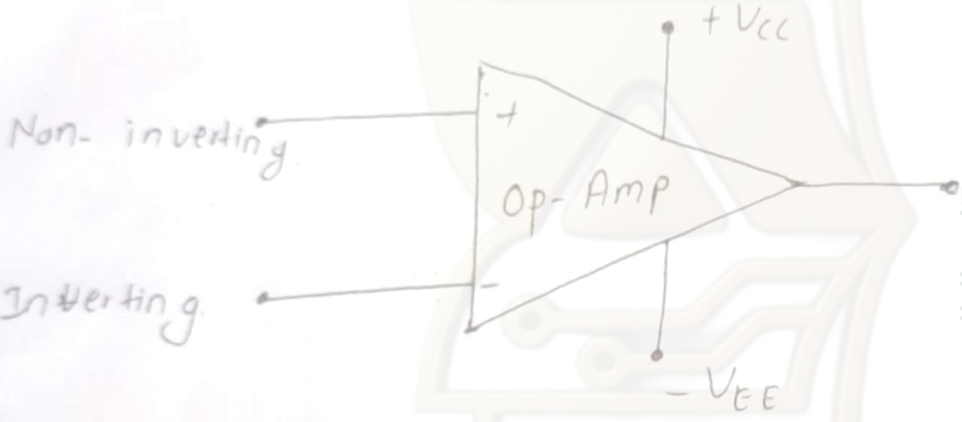
chapter-9.

# Operational Amplifier and Oscillators.

## Operational Amplifier (op-Amp)

An operational Amplifier is a very high gain differential amplifier with high input impedence & low o/p impedance. Typical uses of op-Amp are to provide voltage amplitude change, oscillators, filter circuits and many types of instrumentation circuit.




The figure shows the basic operational amplifier with two input & one output. Each input results in either the same or opposite polarity output depending on whether the signal is applied to the non-inverting or inverting input respectively.

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## Characteristics of op-Amp.

| | Parameters | Ideal | Practical. |
|---|---|---|---|
| 1. | Open loop voltage gain | $\frac{V_o}{V_{in}}$ $\infty$ | High |
| 2. | Input impedence | $\infty$ | High |
| 3 | Output impedence | 0 | low |
| 4 | ~~output~~ offset voltage | 0 | low |
| 5. | Bandwidth | $\infty$ | High |
| 6. | CMRR (common mode rejection ratio) | $\infty$ | High |
| 7 | Slew rate (rate of change of o/p) | $\infty$ | High |

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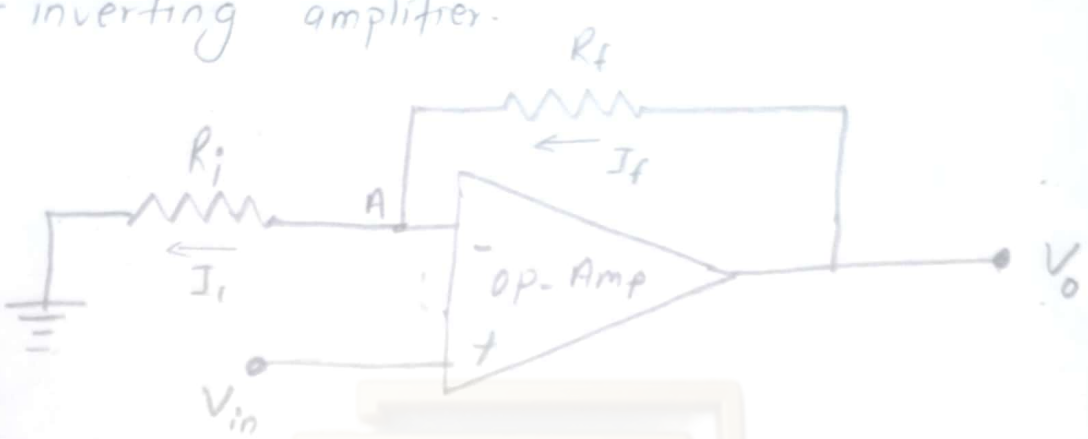
# Non-inverting Amplifier

When input is applied to non-inverting terminal and inverting terminal connected to the ground, it is known as non-inverting amplifier.




Applying KCL at node A,

$$I_1 = I_f$$

$$\frac{V_A}{R_i} = \frac{V_o - V_A}{R_f} \quad \text{---(I)}$$

since, $V_{in} - V_A = 0$

$$V_A = V_{in} \quad \text{---(II)}$$

from (I) & (II)

$$\frac{V_{in}}{R_i} = \frac{V_o - V_{in}}{R_f}$$

$$\frac{V_{in}}{R_i} + \frac{V_{in}}{R_f} = \frac{V_o}{R_f}$$

$$V_o = R_f \left( \frac{V_{in}}{R_i} + \frac{V_{in}}{R_f} \right)$$

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$$V_o = V_{in}\left(1 + \frac{R_f}{R_i}\right) \quad \text{— (iii)}$$

$$\frac{V_o}{V_{in}} = \left(1 + \frac{R_f}{R_i}\right) \quad \text{— (iv)}$$

When input is applied to the inverting terminal with non-inverting terminal connected to the ground, it is known as inverting amplifier.

p.d. 180°

Applying KCL at node A,

$$I_1 = I_f \quad \text{— (i)}$$

$$\frac{V_{in} - V_A}{R_i} = \frac{V_A - V_o}{R_f}$$

Since, $0 - V_A = 0$

$$V_A = 0 \quad [\because \text{virtual ground}]$$

$$\frac{V_{in}}{R_i} = -\frac{V_o}{R_f}$$ (– represents phase diff = 180°)

$$V_o = -\frac{R_f}{R_i} \times V_{in} \quad \text{—(ii)}$$

$$A_v = \left(\frac{V_o}{V_{in}}\right) = -\frac{R_f}{R_i} \quad \text{— (iii)}$$

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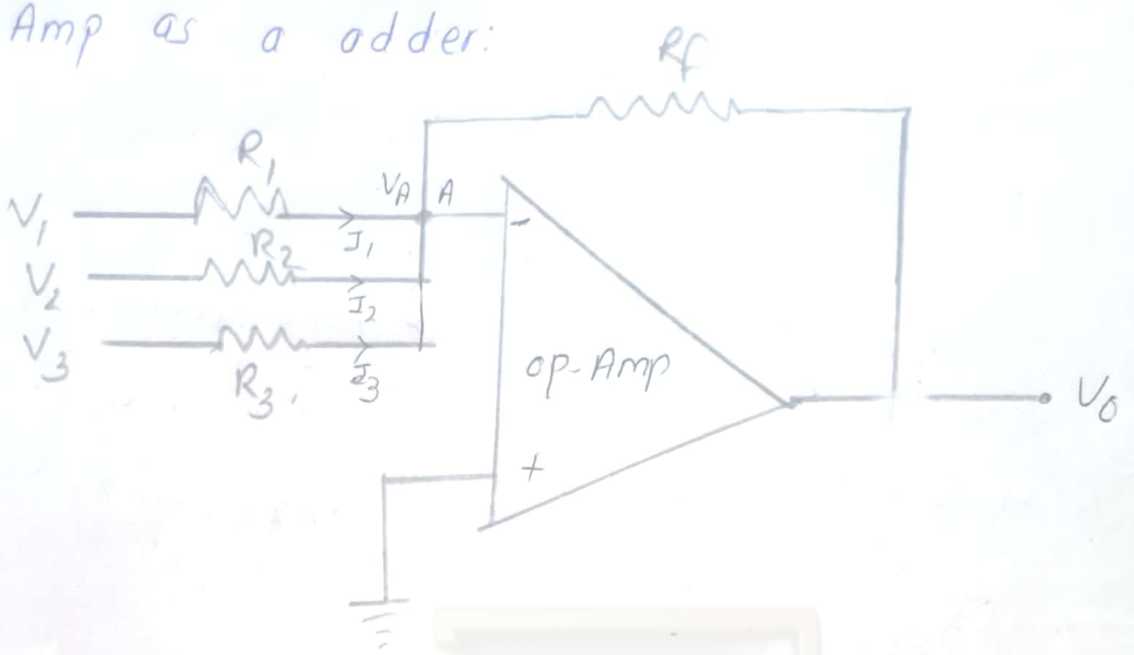
# Application of op-Amp

V.V imp

## (1) op-Amp as a adder:




It provides the output voltage proportional to or equal to the algebric sum of two or more input volt each multiplied by a constant gain factor.

Applying KCL at node A,

$$I_1 + I_2 + I_3 = I_f \quad \text{——(i)}$$

$$\Rightarrow \frac{V_1 - V_A}{R_1} + \frac{V_2 - V_A}{R_2} + \frac{V_3 - V_A}{R_3} = \frac{V_A - V_o}{R_f}$$

since, $V_A = 0$ [virtual ground].

$$\therefore \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = \frac{-V_o}{R_f}$$

$$V_o = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3\right) \quad \text{——(ii)}$$

If $R_1 = R_2 = R_3 = R_f$

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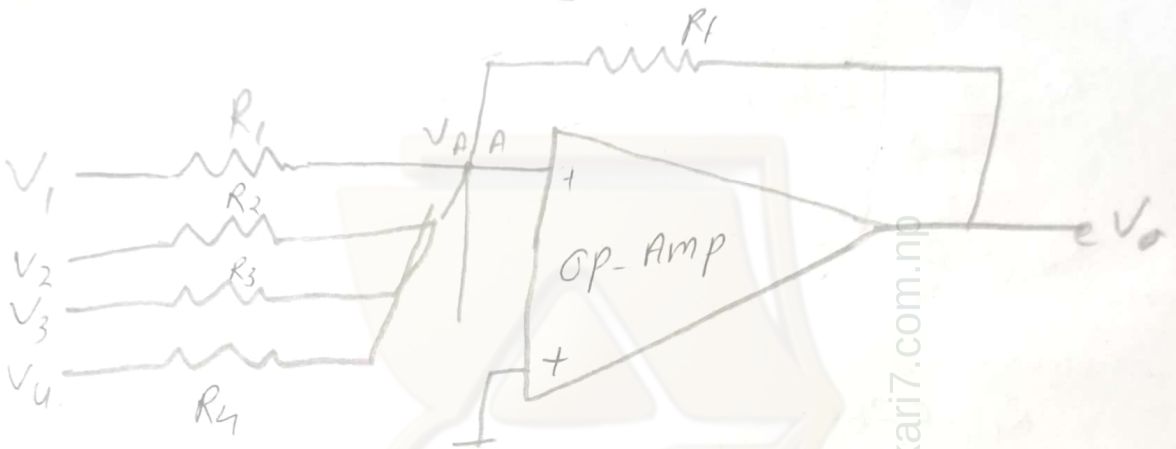
Then,

$$V_o = -(V_1 + V_2 + V_3) \quad \text{—(iii)}$$

Thus, the o/p voltage is proportional to the algebric sum of 3 input voltage.

Qn.1. Design an operational amplifier ckt to produce o/p $V_o$ = $V_o = -8V_1 + 5V_2 - 0.5V_3 - 4V_4$.




Given, $V_o = -8V_1 + 5V_2 - 0.5V_3 - 4V_4$

output voltage expression for given circuit is,

$$V_o = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3 + \frac{R_f}{R_4}V_4\right) \quad \text{—(i)}$$

Now, Expressing the given o/p voltage in the form as in eqn (i)

$$V_o = -\{8V_1 + 5(-V_2) + 0.5V_3 + 4V_4\} \quad \text{—(ii)}$$

Comparing eqn (i) & (ii),

$$\frac{R_f}{R_1} = 8$$

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$\frac{R_f}{R_2} = 5$ $\frac{R_f}{R_3} = 0.5$ $\frac{R_f}{R_4} = 4$.

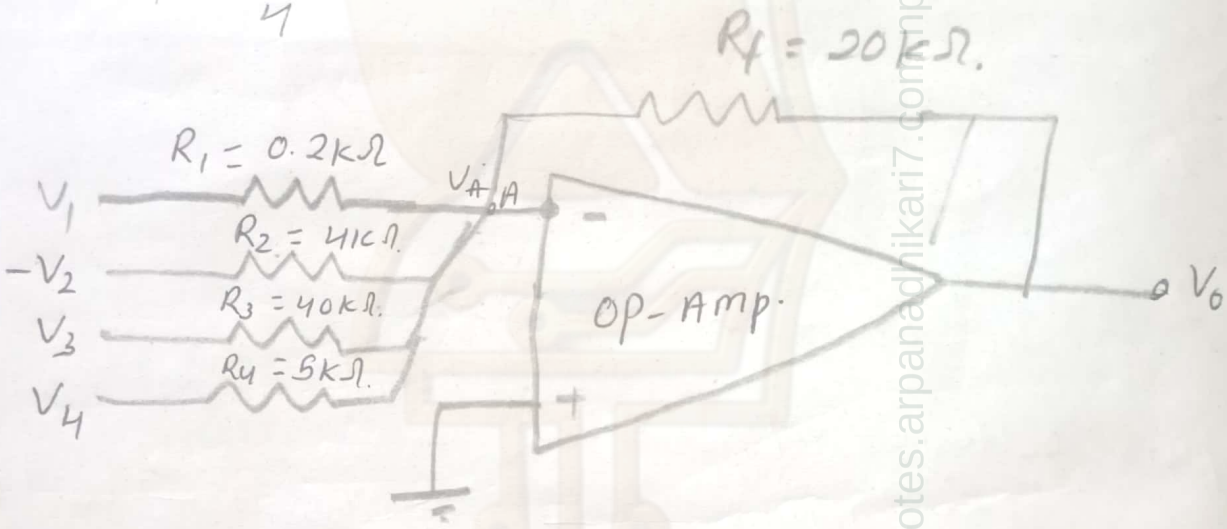
Assume $R_f = 20k\Omega$

$\therefore R_1 = \frac{20}{8} = 0.2k\Omega$

$R_2 = \frac{20}{5} = 4k\Omega$

$R_3 = \frac{20}{0.5} = 40k\Omega$

$R_4 = \frac{20}{4} = 5k\Omega$.

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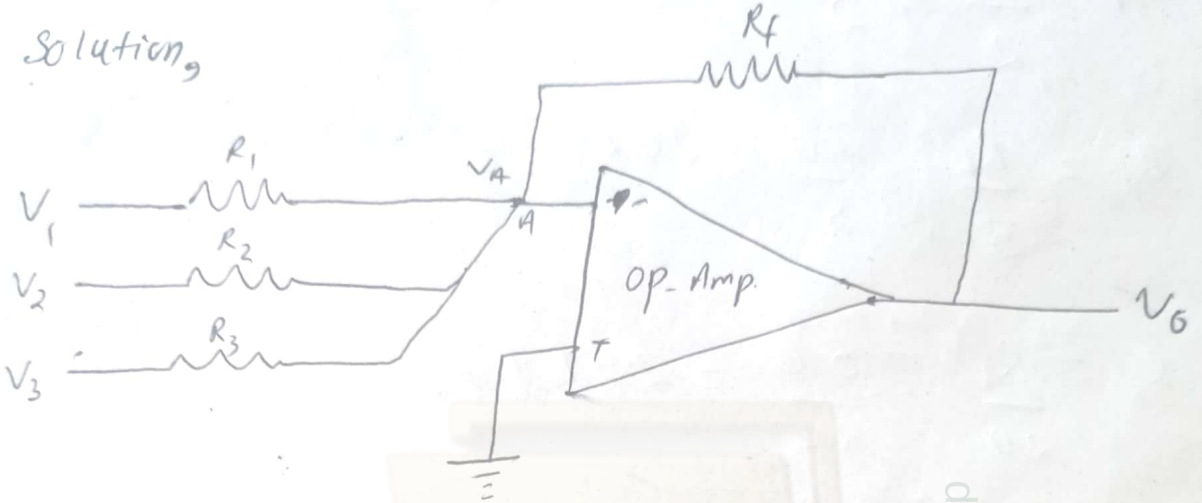
Qn.2 Design a summing ~~opera~~ op-Amp circuit that will produce an output voltage $(V_{out}) = -2V_1 + 5V_2 - 20V_3$ where $V_1$, $V_2$ & $V_3$ are the inputs. Assume $R_f = 50k\Omega$

Solution,




Given, $V_{out} = -2V_1 + 5V_2 - 20V_3$.

Now, Vout expression for given circuit is.

$$V_o = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right) \quad \text{—(i)}$$

Now, Expressing the given O/P voltage in the form as eq$^n$ (i)

$$V_o = -(2V_1 + 5(-V_2) + 20V_3) \quad \text{—(ii)}.$$

Comparing eqn (i) & (ii)

$$\frac{R_f}{R_1} = 2$$

$$\frac{R_f}{R_2} = 5$$

$$\frac{R_f}{R_3} = 20$$

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Assuming $R_f = 50k\Omega$

Then,

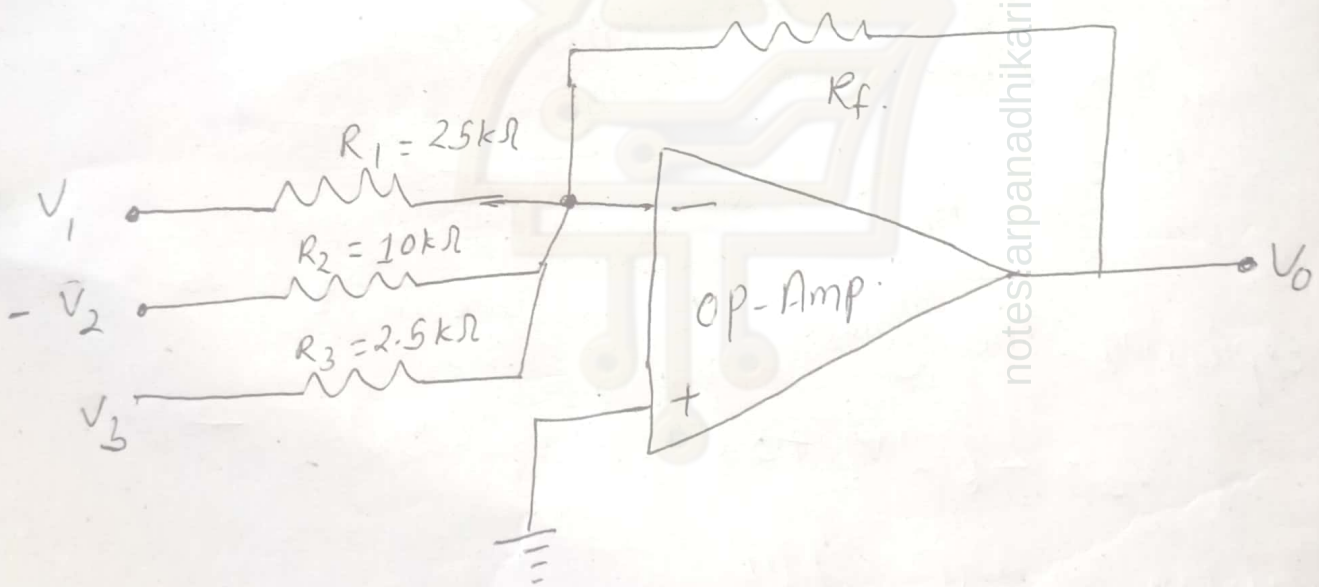
$$R_1 = \frac{R_f}{2} = \frac{50}{2} = 25k\Omega$$

$$R_2 = \frac{R_f}{5} = \frac{50}{5} = 10k\Omega$$

$$R_3 = \frac{R_f}{20} = \frac{50}{20} = 2.5k\Omega$$

Then,

Given circuit can be shown as.

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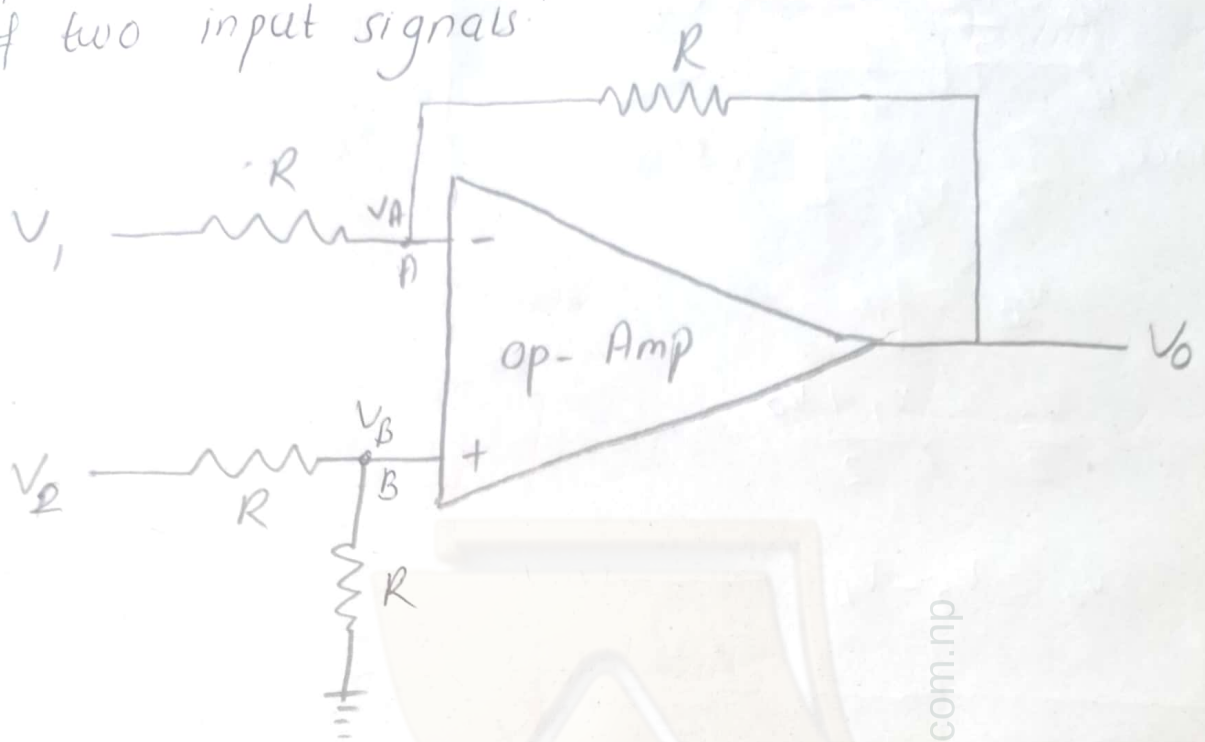
## 2. Op-Amp as a substractor / Difference Amplifier

It provides an output proportional to or equals to difference of two input signals




Applying KCL at node A,

$$I_1 = I_f$$

$$\Rightarrow \frac{V_1 - V_A}{R} = \frac{V_A - V_0}{R}$$

$$\Rightarrow V_1 - V_A = V_A - V_0$$

$$\Rightarrow V_1 + V_0 = 2V_A$$

$$\text{or, } V_A = \frac{V_1 + V_0}{2}$$

Applying KCL at node B,

$$I_3 = I_4$$

$$\Rightarrow \frac{V_2 - V_B}{R} = \frac{V_B - 0}{R}$$

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$$V_2 = 2V_B$$

$$V_B = \frac{V_2}{2} \quad \text{——— (ii)}$$

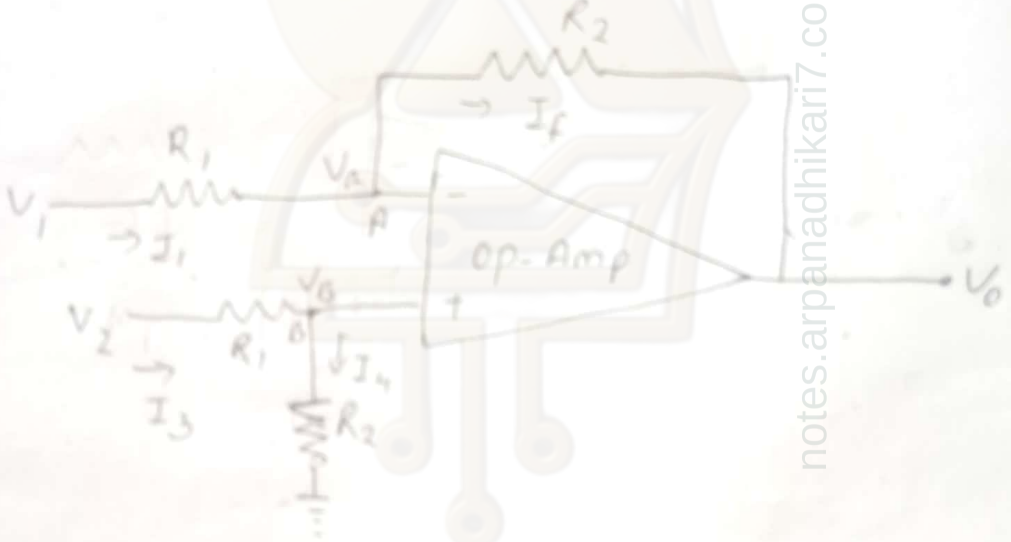
Since,

$$V_B - V_A = 0$$

$$V_B = V_A$$

$$\frac{V_2}{2} = \frac{V_1 + V_0}{2}$$

$$V_0 = V_2 - V_1 \quad \text{——— (iii).}$$




Applying KVL at node A,

$$I_1 = I_f$$

$$\frac{V_1 - V_A}{R_1} = \frac{V_A - V_0}{R_2}$$

$$\frac{V_1}{R_1} - \frac{V_A}{R_1} = \frac{V_A}{R_2} - \frac{V_0}{R_2}$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_0}{V_2} = V_A \left(\frac{1}{R_1} + \frac{1}{R_2}\right) \quad \text{——— (i)}$$

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Applying KCL at node B

$$I_3 = I_4$$

$$\Rightarrow \frac{V_2 - V_B}{R_1} = \frac{V_B - 0}{R_2}$$

$$\Rightarrow \frac{V_2}{R_1} = V_B \left( \frac{1}{R_1} + \frac{1}{R_2} \right) \quad \text{---(ii)}$$

Since,

$$V_B - V_A = 0$$

$$V_B = V_A$$

$$V_B \left( \frac{1}{R_1} + \frac{1}{R_2} \right) = V_A \left( \frac{1}{R_1} + \frac{1}{R_2} \right).$$

$$\frac{V_1}{R_1} + \frac{V_0}{R_2} = \frac{V_2}{R_1}$$

$$\frac{V_0}{R_2} = \frac{V_2}{R_1} - \frac{V_1}{R_1}$$

$$V_0 = \frac{R_2}{R_1} (V_2 - V_1) \quad \text{---(iii)}$$

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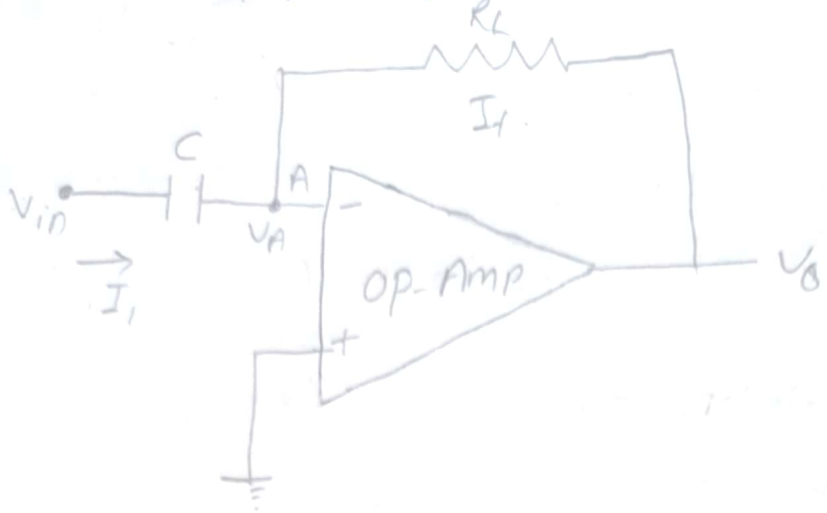
3. Op-Amp as a differentiator:




Thus, output voltage is proportional to the derivative of input voltage with having scale factor is $R_f \times C$.

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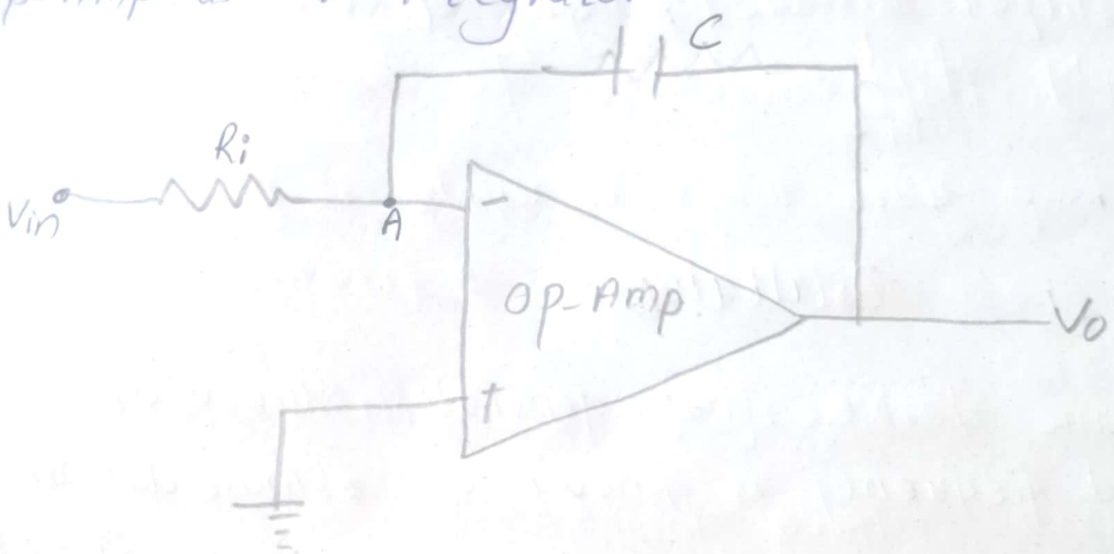
4. Op-Amp as a Integrator.




It provides an output voltage which is proportional to the integral of input voltage.

Applying KCL at node A.

$$I_1 = I_f \quad \text{—— (1)}$$

$$\Rightarrow \frac{V_{in} - V_A}{R_i} = \frac{dQ}{dt}$$

$$\Rightarrow \frac{V_{in} - V_A}{R_i} = C\frac{d(V_A - V_o)}{dt}$$

since, $V_A = 0$ [virtual ground]

$$\frac{V_{in}}{R_i} = C\frac{dV_o}{dt}$$

$$\Rightarrow -\frac{V_{in}}{CR_i} = \frac{dV_o}{dt}$$

Integrating both sides we get,

$$V_o = \frac{-1}{R_i C}\int V_{in}\, dt \quad \text{—— (11.)}$$

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Thus, the output is the integral of the input with in the scale multiply of $\frac{1}{R_i C}$

## Oscillators

An electronic device that generates sinosuidal oscillation of desired frequency is known as sinusoidal oscillators. Oscillations are produced without any external signal source.

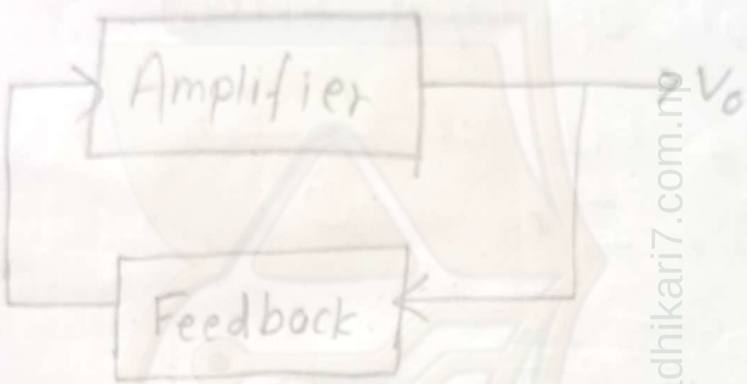


fig: Oscillators

Importance of Positive feedback in oscillator circuits

For positive feedback circuit, overall gain is given by

$$A_f = \frac{A}{1-\beta A} \quad \text{——(1)}$$

When $\beta = 1$, $A_f = \infty$. So, even a small noise signal can provide a measurable output voltage and the circuit acts as an oscillator.

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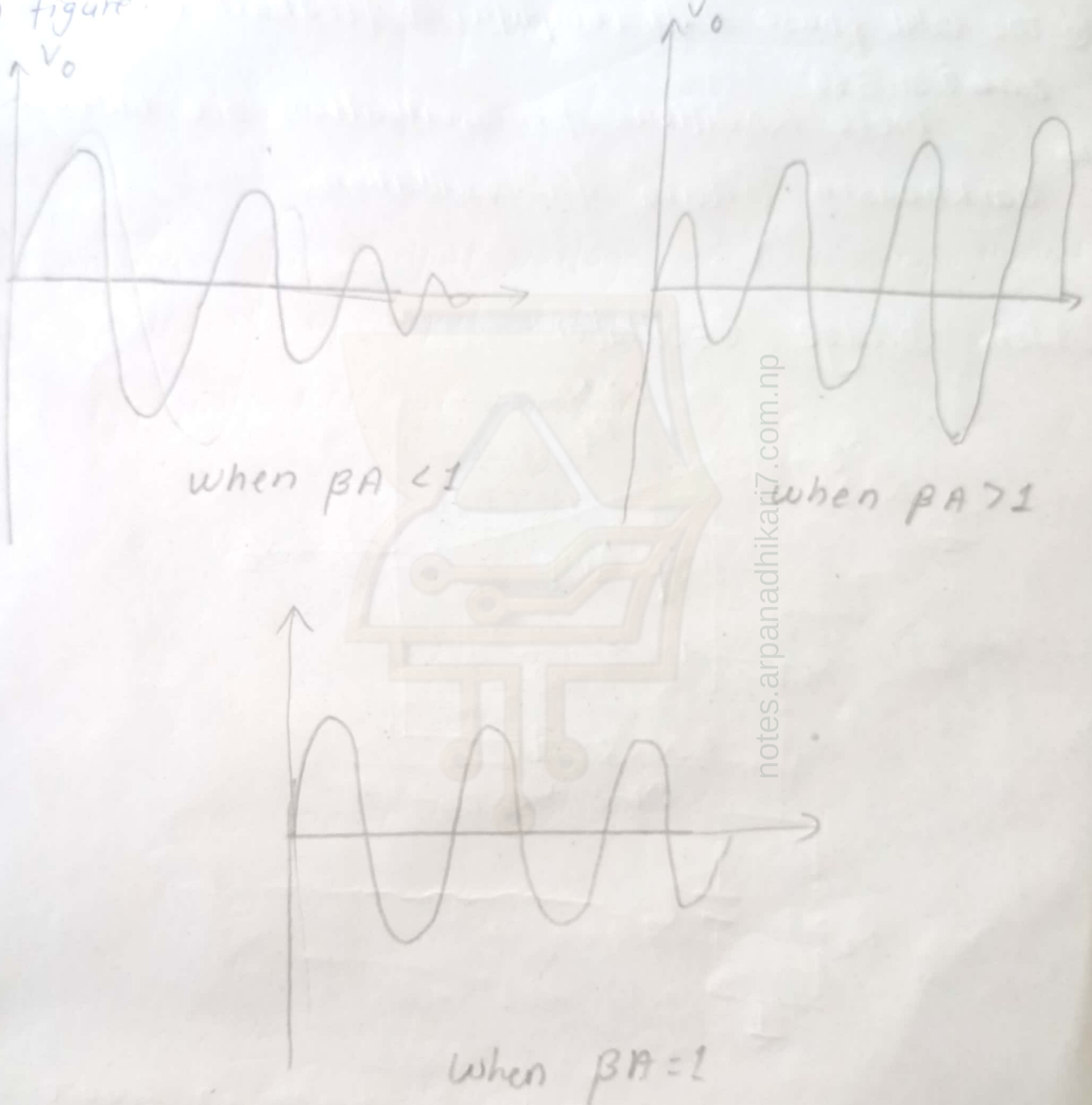
If $\beta A < 1$, the output signal will slowly vanishes. If $\beta A > 1$, the output signal will build up & if $\beta A = 1$ is obtained. The output signal a steady output signal for various value of $\beta A$ is shown in figure.

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Imp Essential condition for oscillation.

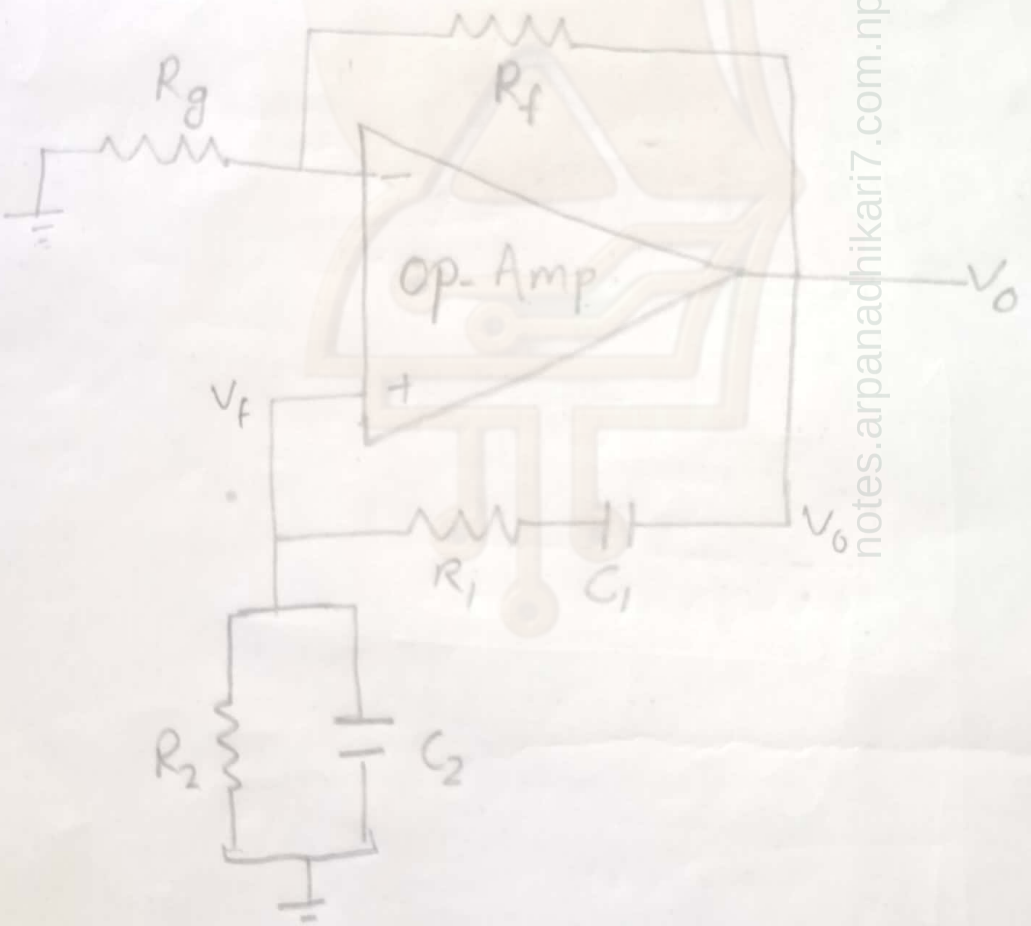
The essential conditions for oscillations are.

(i) $\beta A$ i.e loop gain must be unity i.e $\beta A = 1$.

(ii) The total phase shift (amplifier & feedback ckt) must 0° or 360°

These conditions for oscillations are called barkhausen criteria for oscillation.

1. Wein bridge Oscillator:




The wein bridge oscillator is shown in figure wein bridge oscillator, resistor $R_1$ & $R_2$ and capa $C_1$ & $C_2$ forms the frequency adjustment eleme and resistor $R_g$ and $R_f$ form the part of fe

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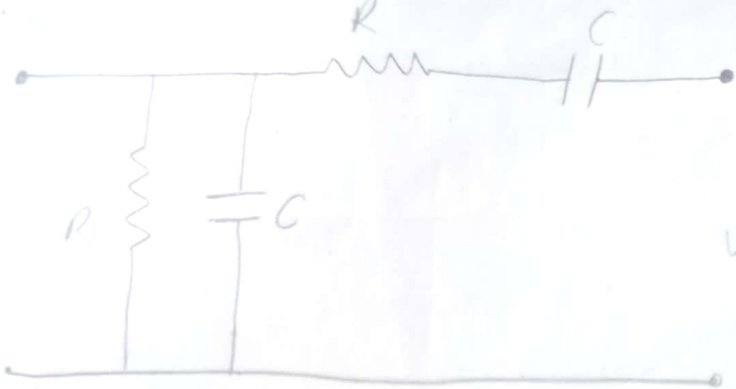
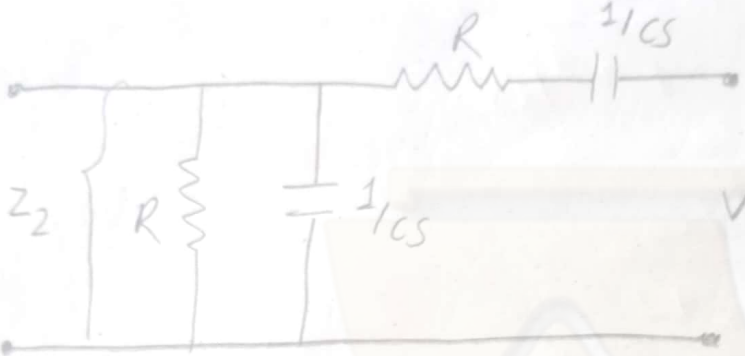
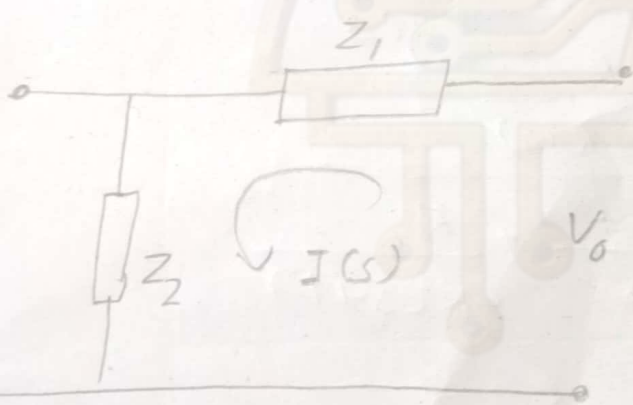
path.




$$Z_1 = R + \frac{1}{Cs} \qquad Z_2 = \left(R \,||\, \frac{1}{Cs}\right)$$




Now,

$$\beta = \frac{V_f}{V_o}$$

$$= \frac{I(s) \times Z_2}{I(s)\,[Z_1 + Z_2)}$$

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$$\text{or } \beta = \frac{Z_2}{Z_1 + Z_2}$$

$$\text{or } \beta = \frac{\frac{R \cdot \frac{1}{CS}}{R + \frac{1}{CS}}}{\left(R + \frac{1}{CS}\right) + \frac{R/CS}{R + 1/CS}}$$

$$\text{or } \beta = \frac{\frac{R/CS}{\frac{RCS+1}{CS}}}{\left(\frac{RCS+1}{CS}\right) + \left(\frac{R/CS}{\frac{RCS+1}{CS}}\right)}$$

or

$$\text{or } \beta = \frac{\frac{R}{1+RCS)}}{\frac{RCS+1}{CS} + \frac{R}{CR(CS+1)}}$$

$$\text{or } \beta = \frac{\frac{R}{(1+RCS)}}{\frac{CR(CS+1)^2 + RCS}{CS(1+RCS)}}$$

$$\text{or } \beta = \frac{RCS}{R^2C^2S^2 + 2RCS + 1 + RCS}$$

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$$\Rightarrow \beta = \frac{RCs}{R^2C^2s^2 + 3RCs + 1}$$

$$= \frac{RCs}{Cs\left(R^2Cs + 3R + \frac{1}{Cs}\right)}$$

$$\Rightarrow \beta = \frac{R}{R^2Cs + 3R + \frac{1}{Cs}}$$

$$\therefore \boxed{\beta = \frac{R}{R^2Cs + 3R + \frac{1}{Cs}}}$$

Substitute $s = j\omega$,

$$\beta = \frac{R}{3R + jR^2\omega C + \frac{1}{j\omega C}}$$

$$= \frac{R}{3R + jR^2\omega C - j\left(\frac{1}{\omega C}\right)} \qquad \left[\because \frac{-j^2}{j\omega C} = -\frac{1}{\omega C}\right]$$

$$\boxed{\beta = \frac{R}{3R + j\left(R^2\omega C - \frac{1}{\omega C}\right)}} \quad \text{——— (x)}$$

In order to have 0° phase shift $\beta$ must b
real number (i.e $R^2\omega C - \frac{1}{\omega C} = 0$)

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$$\text{or, } R^2\omega C = \frac{1}{\omega C}$$

$$\Rightarrow \omega^2 = \frac{1}{R^2C^2}$$

$$(2\pi f)^2 = \frac{1}{R^2C^2}$$

$$2\pi f = \frac{1}{RC}$$

$$f = \frac{1}{2\pi RC}$$

$\therefore$ frequency of oscillation $\boxed{(f) = \frac{1}{2\pi RC}}$ —— (i)

$\therefore \beta = \frac{1}{3}$ [since, imaginary term = 0].

Also,

$$\beta A = 1$$

$$\frac{1}{3} \times A = 1$$

$$A = 3$$

$$1 + \frac{R_f}{R_g} = 3$$

$$\frac{R_f}{R_g} = 2.$$

$$\therefore \boxed{R_f = 2R_g.} \quad \text{—— (ii)}$$

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Design a wein bridge oscillator that oscillates at 25kHz

Solution,

Given, $R_1 = R_2 = R$

$C_1 = C_2 = C$

$f = 25kH_2 = 25 \times 10^3 H_2$

We have,

$$f = \frac{1}{2\pi RC}$$

Let $C = 0.001 \mu F = 0.001 \times 10^{-6} F$

Then,

$$f = \frac{1}{2 \times \pi \times R \times 0.001 \times 10^{-6} F}$$

or $$R = \frac{1}{2 \times \pi \times 25 \times 10^3 \times 0.001 \times 10^{-6}}$$

$$= 6.366 \times 10^3 \Omega$$

$$= 6.366 k\Omega.$$

Also, $R_f = 2R_g$

let $R_g = 10k$

$\therefore R_f = 2 \times 10$

$= 20k.$

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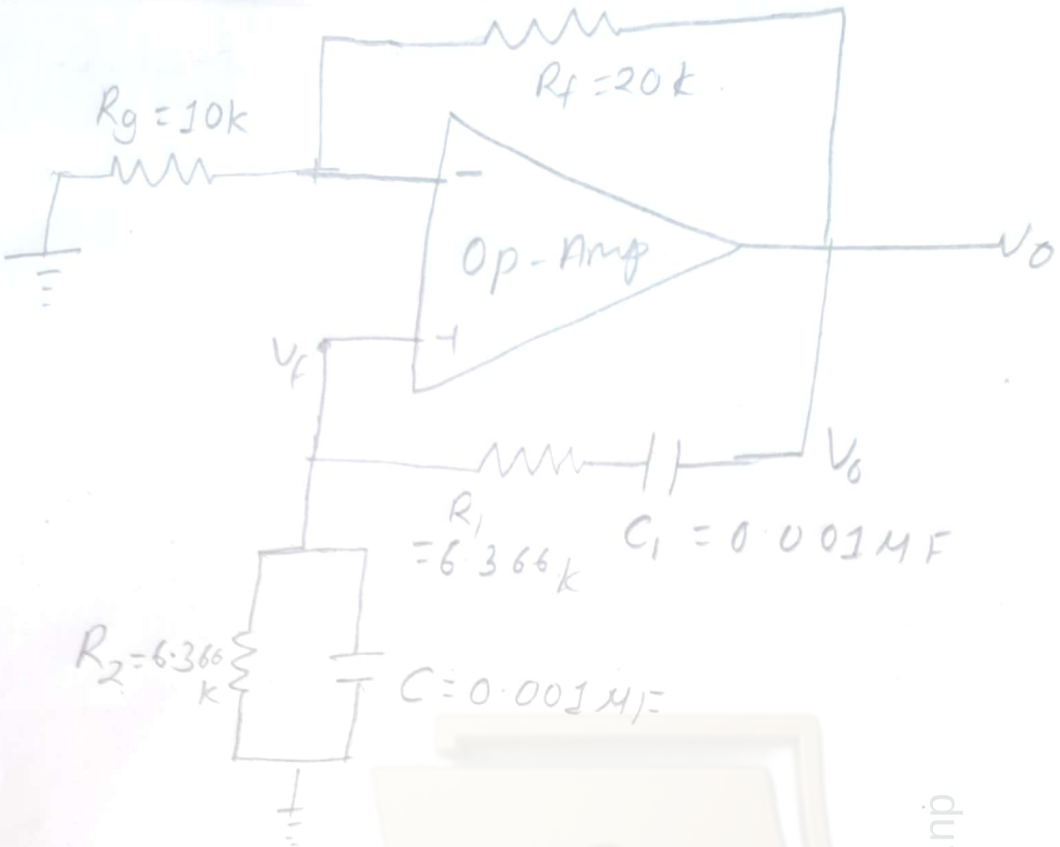
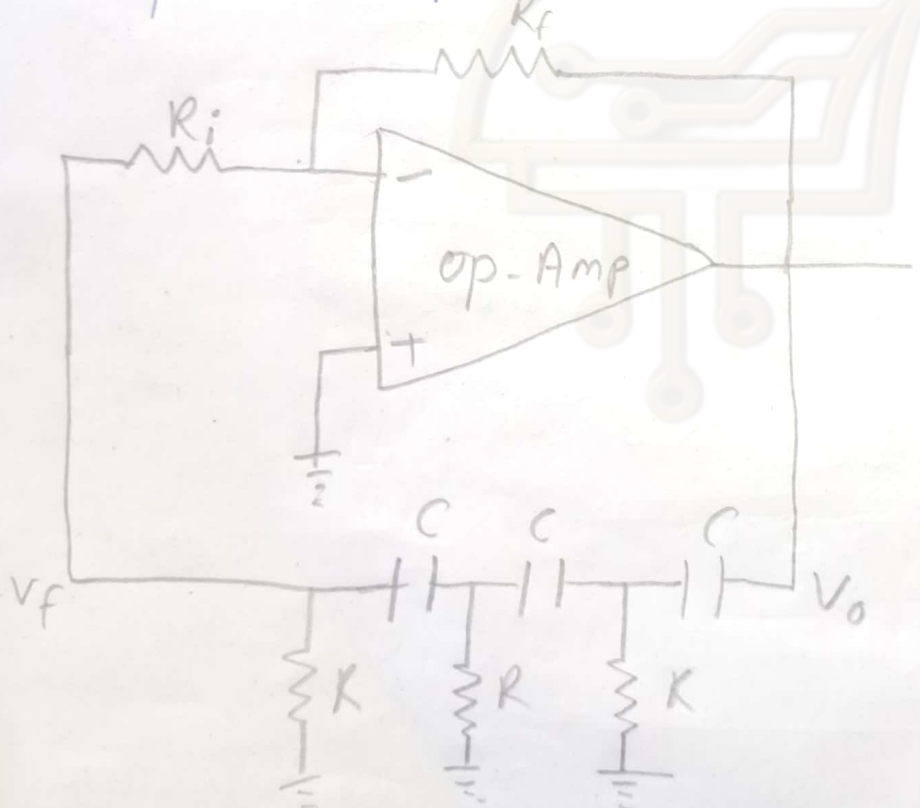



2. RC phase shift oscillator.




RC phase shift oscillator circuit consists of negative gain operational amplifier and a 3 section RC network that produce 180° phase sh

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The phase shift network is connected from the Op-Amp output back to ~~the~~ its inverting input and ~~soon~~ shown.

An inverting op-Amp provides 180° phase shift and 3 RC network provides 180° phase shift. Therefore, the total phase shift becomes 360° which is one of the condition for oscillation.

For 3 stage RC n/w, the feedback ratio $\beta$ can be expressed as $\beta = \frac{R^3}{(R^3 - 5RX_C)^2 + j(X_C^3 - 6R^2X_C)}$ ——— (1)

For oscillation, $\beta$ must be real number. ie

$$X_C^3 - 6R^2X_C = 0$$

$$\Rightarrow X_C^3 = 6R^2X_C$$

$$\Rightarrow X_C^2 = 6R^2 \quad \text{——— (ii).}$$

$$\Rightarrow \left(\frac{1}{2\pi fC}\right)^2 = 6R^2$$

$$\Rightarrow \frac{1}{2\pi fC} = R\sqrt{6}$$

$$\Rightarrow f = \frac{1}{2\pi RC\sqrt{6}}$$

$\therefore$ frequency of oscillation,

$$\boxed{f = \frac{1}{2\pi RC\sqrt{6}}} \quad \text{——— (*)}$$

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$$\beta = \frac{R^3}{R^3 - 5RX_C^2} \quad [\text{since imaginary term} = 0]$$

$$= \frac{R^3}{R^3 - 5R \times 6R^2}$$

$$= \frac{R^3}{R^3 - 30R^3}$$

$$= \frac{R^3}{-29R^3}$$

$$\therefore \beta = -\frac{1}{29} \quad \text{---(IV)}$$

Since, $\beta A = 1$

$$-\frac{1}{29} A = 1$$

$$\Rightarrow A = -29$$

$$\Rightarrow -\frac{R_f}{R_i} = 29$$

$$\therefore R_f = 29 R_i \quad \text{---(V)}$$

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